

COMPACT MODULI OF NUMERICALLY POLARIZED DEGREE TWO ENRIQUES SURFACES

by

DANIEL GARZA

(Under the Direction of Valery Alexeev)

ABSTRACT

We construct and describe a modular compactification $\overline{F}_{\text{En},2}$ of the moduli space of degree 2 numerically polarized Enriques surfaces by KSBA stable pairs compactifications. Our approach blends explicit lattice-theoretic methods – rooted in Nikulin’s classification of 2-elementary lattices and the arithmetic of the transcendental polarizing lattice T_{En} – with the Hodge-theoretic theory of period maps and the analytic theory of compactification for Type IV locally symmetric spaces of orthogonal type. We provide a detailed classification of $\partial\overline{F}_{\text{En},2}$ via orbits of primitive isotropic sublattices in T_{En} , connecting the five 0-cusps of the Baily–Borel compactification $\overline{F}_{\text{En},2}^{\text{BB}}$ to explicit degenerations of Enriques surfaces. Combining combinatorial, geometric, and computational techniques, we construct explicit divisorial log terminal (dlt) models for all stable limits in $\partial\overline{F}_{\text{En},2}$, relating them to semitoroidal compactifications $\overline{F}_{\text{En},2}^{\mathcal{F}\bullet}$ defined by semifans at each boundary component in the Baily–Borel compactification $\overline{F}_{\text{En},2}^{\text{BB}}$. The key innovation is the use of “folding” procedures, arising from involutions of the K3 lattice L_{K3} – as the primary source of combinatorial data to construct models for all strata in $\partial\overline{F}_{\text{En},2}$. The irreducible components appearing include ADE *surfaces*, as well as new, non-simply-laced types *B* and *C*.

Our main theorem establishes an isomorphism between the normalization of the KSBA compactification and an explicit semitoroidal compactification $\overline{F}_{\text{En},2}^{\mathcal{F}\bullet}$ determined by the arithmetic of T_{En} . This provides the first explicit and functorial identification of these two approaches for Enriques surfaces.

INDEX WORDS: Algebraic Geometry, Enriques Surfaces, Moduli Spaces, KSBA Compactifications, Semitoroidal Compactifications, Stable Pairs, Lattices, Reflection Groups, Coxeter Diagrams.

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Part I

**Foundations and Geometric
Setup**

Throughout this text, we work over $k = \mathbf{C}$. A **K3 surface** is a smooth, projective, geometrically integral surface X such that $\omega_X \cong \mathcal{O}_X$ is trivial, and $q(X) := h^{0,1}(X) := H^0(\mathcal{O}_X) = 0$. An **Enriques surface** is a non-rational algebraic surface Z of Kodaira dimension $\kappa(Z) = 0$ for which $h^1(\mathcal{O}_Z) = h^2(\mathcal{O}_Z) = 0$ and K_Z is nontrivial 2-torsion in $\text{Pic}(Z)$. Equivalently, they are quotients $Z = X/\iota$ of K3 surfaces X by fixed-point-free involutions ι , satisfying $2K_Z \sim 0$ and $q(Z) = 0$. Enriques surfaces were among the first examples of non-rational surfaces with $p_g = 0$. They have Kodaira dimension $\kappa(Z) = 0$ and are minimal. We are interested in the study of moduli spaces of such surfaces, and in particular compactifications of $F_{\text{En},2}$, the moduli space of degree 2 numerically polarized Enriques surfaces, which we will describe shortly.

For typical moduli spaces $\mathcal{M}$, one often has many choices for what a compactification $\overline{\mathcal{M}}$ should mean: GIT quotients $\overline{\mathcal{M}}^{\text{GIT}}$, Baily-Borel compactifications $\overline{\mathcal{M}}^{\text{BB}}$, toroidal compactifications $\overline{\mathcal{M}}^{\Sigma^\bullet}$, the semitoroidal compactifications $\overline{\mathcal{M}}^{\mathcal{F}^\bullet}$ of [Loo85] which simultaneously generalize $\overline{\mathcal{M}}^{\text{BB}}$ and $\overline{\mathcal{M}}^{\Sigma^\bullet}$, KSBA compactifications $\overline{\mathcal{M}}$, quotient stack constructions, derived quotient stacks, and so on. A disadvantage of many naively constructed compactifications $\overline{\mathcal{M}}$ is that they do not a priori have modular interpretations, i.e. $\partial\overline{\mathcal{M}}$ may not parameterize singular limits of smooth interior points in any canonically meaningful way. However, the theory developed by KSBA (see [Kol23]) provides a functorial, geometrically meaningful compactification which can be defined in a canonical way, provided one can make a natural choice of divisor on the varieties in question (e.g. a polarization or a ramification divisor), and $\partial\overline{\mathcal{M}}$ can be understood as stable pairs (X, R) : a variety X with slc singularities and an ample divisor R . Our goal for Part I is thus to establish the setting of the main theorem, along with the lattice-theoretic background and computational tools needed to contextualize and prove it.

CHAPTER 1

INTRODUCTION

The central object needed to define such a compactification is a **KSBA stable pair**: a pair $(Z, \epsilon R_Z)$ where Z is a connected projective variety and R_Z is an effective $\mathbf{Q}$ -divisor such that the pair $(Z, \epsilon R_Z)$ is semi-log canonical for some $0 < \epsilon \leq 1$, and $K_Z + \epsilon R_Z$ is ample. We will then define KSBA compactifications as closures of spaces of particular types of surface pairs in the space of stable pairs. When constructing such compactifications, we will often take the divisor R_Z to be the ramification divisor corresponding to a 2-divisible ample line bundle coming from a branched double-cover construction, i.e. an involution. We note, however, that explicitly determining or classifying the boundary strata of a given KSBA compactification and the exact stable pairs that appear is nontrivial. In [Ale+23], we classify the strata of $\overline{\partial F_{\text{En},2}}$, the KSBA compactification of $F_{\text{En},2}$, and find that the irreducible components of stable limits of surface pairs are described by $ADE + BC$ **diagrams**, which we mean the classical Dynkin diagrams corresponding to the semisimple complex Lie algebras of types A_n, D_n, E_6, E_7, E_8 , which we refer to as simply-laced, and the diagrams of types B_n and C_n . The latter can be obtained from the former by a classically well-known operation called **folding**.

By the work of [AET19a], to each such diagram one can associate a pair (Y, C) where Y is a surface, which in many cases is toric, and C is a reduced boundary divisor such that (Y, C) is an lc pair and $-2(K_Y + C)$ is an ample Cartier divisor providing a natural polarization. This provides a natural association of a classical $ADE + BC$ diagram (decorated with extra combinatorial parity data) to, in many cases, an explicit projective toric variety. We refer to such surfaces as $ADE + BC$ **surfaces**. This thesis details the construction

of an isomorphism between the normalization of the KSBA stable pair compactification $\overline{F}_{\text{En},2}$ and a semitoroidal compactification $\overline{F}_{\text{En},2}^{\mathcal{F}_\bullet}$ for the moduli space of Enriques surfaces with *numerical polarization* of degree 2. The main theorem establishes an isomorphism $\overline{F}_{\text{En},2}^{\mathcal{F}_\bullet} \rightarrow \overline{F}_{\text{En},2}$ where $\mathcal{F}_\bullet = \{\mathcal{F}_k\}_{k=1,2,3,4,5}$ is a collection of semifans determined by the five 0-dimensional boundary components of the Baily-Borel compactification $\overline{F}_{\text{En},2}^{\text{BB}}$. Our goal is to prove the following:

Theorem 1.0.0.1 ([Ale+23, Thm. 1.1]). *Let $F_{\text{En},2}$ be the moduli space of numerically polarized degree 2 Enriques surfaces, and let $\overline{F}_{\text{En},2}$ be its KSBA compactification. There is a morphism*

$$\overline{F}_{\text{En},2}^{\mathcal{F}_\bullet} \xrightarrow{\sim} \overline{F}_{\text{En},2}^\nu,$$

where $(-)^{\nu}$ denotes the normalization, the left-hand side is the semitoroidal compactification corresponding to an explicit collection $\mathcal{F}_\bullet = \{\mathcal{F}_1, \Sigma_2, \mathcal{F}_3, \Sigma_4, \mathcal{F}_5\}$ of semifans, one for each 0-cusp of the Baily-Borel compactification $\overline{F}_{\text{En},2}^{\text{BB}}$, and the right-hand side is the KSBA compactification. The semifans Σ_2, Σ_4 are fans, while $\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3$ are strict semifans.

We note that the normalization is a technical condition that is often applied in the setting of KSBA compactifications, since the KSBA compactification is not guaranteed to be normal in general. Roughly speaking, this is due to the fact that its construction involves taking a Zariski closure, which can introduce non-normal points where degenerations are identified, leading to a non-separated stack. Since the normalization morphism is finite, birational, and relatively smooth in codimension one, this replacement restricts the worst singularities to lie in high codimension sub-loci and is thus a desirable tradeoff.

A standard construction in the study of del Pezzo and Enriques surfaces involves analyzing the invariant and coinvariant sublattices of a lattice L acted on by an involution L . In this situation, we take $L = L_{K3}$, consider three involutions $I_\star$, and study the invariant sublattices $T_\star = L_{K3}^{I_\star=1}$ – these are the lattices into which the transcendental lattices T_Z of Enriques surfaces Z primitively embed. With a transcendental lattice identified, because these lattices have very particular signatures, we are placed in a setting where two combinatorial compactifications are accessible: the Baily Borel compactification of [BB66], and the semitoroidal compactifications of [Loo85], which simultaneously generalize both the Baily Borel and toroidal compacti-

fications. Their boundaries are stratified by *cusps*, and understanding how maps of moduli spaces induce maps on cusps is the first step toward classifying the boundary strata. To finally attain a comparison with KSBA compactifications, we then leverage the main result of [AE23a], that the (normalization) of the KSBA compactification of stable K3 pairs $(X, \varepsilon R)$ for a **recognizable** divisor R (see Chapter 4 and Chapter 5) is isomorphic to a semitoroidal compactification. Then Theorem 1.0.0.1 follows from [AEH21, Thm. 3.26], which shows that certain ramification divisors are recognizable.

The standard approach to understanding boundaries of semitoroidal compactifications (including Baily-Borel and toroidal compactifications as special cases) involves several difficult intermediate computational problems, among which is describing the faces of certain hyperbolic polytopes corresponding to fundamental domains of actions by reflection groups. Typical methods include the use of Vinberg’s algorithm [Vin72], which although constructive, is computationally intensive for high rank lattices (and quickly becomes intractable) and is not generally known to be a halting procedure. The approach we take in [Ale+23] largely bypasses many of these computational difficulties, in favor of more easily constructible divisorial log terminal (dlt) models which can be described in the finitary data of an integral affine 2-sphere, which we refer to as an IAS^2 throughout this work.

These folding and involution-based methods recover the work of [Ste91], while the more general theory established in [AET23a; AEH22; AE23b; AET23a; AE23a; ABE22] has been shown to recover results from e.g. [Sca87] and others. The classification data is explicit and combinatorial, making it amenable to computation and reproducibility.

1.1 Historical Context and Related Work

The systematic study of compactifications of moduli spaces began with the foundational work of [DM69], who compactified the moduli space of curves $\mathcal{M}_g$ by incorporating stable curves, i.e., one allows certain controlled degenerations with prescribed singularities. For moduli of varieties described as arithmetic quotients of bounded symmetric domains, [BB66] developed a different approach, introducing a compactification by adjoining Hodge-theoretic boundary components to the period domain.

Subsequent developments included the work of [AW71] on stable reduction and semistable models for families of curves. Although their results focused on curves, the methods – particularly the use of semistable models and an emphasis on controlling degenerations – provided the groundwork for extending these ideas to higher-dimensional varieties, including surfaces. In particular, the techniques and perspectives developed for understanding degenerations of curves influenced later approaches to constructing and compactifying moduli spaces of surfaces, where the behavior of singular fibers and degenerations is even more complex. The development of higher-dimensional minimal model theory (see, for example, [KM98b]) built directly on these ideas, offering new tools for analyzing the compactification of moduli spaces for surfaces and varieties of general type, especially through the theory of log-canonical pairs.

Enriques surfaces were initially studied by Federico Enriques at the beginning of the twentieth century as part of his broader classification of algebraic surfaces; see [Enr06]. The modern viewpoint that an Enriques surface arises as a quotient of a K3 surface by a fixed-point-free involution was developed much later in the work of [Hor78] and [Nam85]. The study of the moduli of Enriques surfaces advanced through the work of [CD89], who developed a classification theory for linear systems on Enriques surfaces, laying the groundwork for the study of moduli of polarized Enriques surfaces. [Ste91] constructed the first complete (Baily–Borel) compactification of the moduli space $F_{\text{En},2}$ by exploiting the lattice-theoretic structure inherited from the universal K3 cover.

The modern approach to compactifying moduli spaces of surfaces has been dominated by the theory of stable pairs established by [KS88; Ale96b]. This machinery produces projective compactifications by allowing degenerations to semi-log-canonical pairs, and we refer to [Kol23] for details. KSBA theory has been especially effective for surfaces of general type and for Calabi–Yau surfaces.

For Enriques surfaces, whose canonical class is numerically trivial, there is not always a canonical choice of polarization or divisor. Recent advances include the classification of ADE surfaces (surfaces with at worst rational double points of Dynkin type A , D , or E) by [AT19], which establishes a foundation for describing boundary points corresponding to simply-laced singularities. More recently, this framework has been extended [Ale+23] to cover the case of non-simply-laced sin-

gularities (types B and C), which naturally appear in the boundaries of moduli spaces as degenerating families of Enriques surfaces.

1.2 The K3 and Enriques Lattices L_{K3} and L_{En}

The **K3 lattice** is given by

$$L_{K3} = U^3 \oplus E_8^2 \xrightarrow{\sim} \text{II}_{3,19},$$

where U is the hyperbolic plane (??) and E_8 is the negative-definite E_8 root lattice (see ??). It is the unique even unimodular lattice of signature $(3, 19)$ up to isometry, which we write as $\text{II}_{3,19}$. For any K3 surface X , there exists a **marking** $H^2(X, \mathbf{Z}) \xrightarrow{\sim} L_{K3}$ that identifies the cohomology lattice with this fixed reference lattice. We refer to a K3 surface with a marking as a **marked K3 surface**. Moduli spaces of polarized and marked K3 surfaces are typically governed by the following two lattices:

$$S_X := \text{NS}(X) \cong \text{Pic}(X) \hookrightarrow H^2(X; \mathbf{Z}), \quad T_X := S_X^{\perp_{H^2(X; \mathbf{Z})}},$$

where $\text{NS}(X) := \text{Pic}(X)/\text{Pic}^0(X)$ is the Neron-Severi lattice, and we take the orthogonal complement with respect to the bilinear form induced by the intersection pairing in cohomology. If the signature of $H^2(X; \mathbf{Z})$ is (p, q) , then we must have $\text{sig}(S_X) + \text{sig}(T_X) = (p, q)$ since $T_X \oplus S_X \hookrightarrow H^2(X; \mathbf{Z})$ embeds as a finite-index submodule. In particular, if one chooses S such that $\text{sig}(S_X) = (1, \rho(X) - 1)$, this forces $\text{sig}(T_X) = (p - 1, q - \rho(X) - 1)$. When $p = 3$ one obtains a lattice T_X of signature $(2, n)$. For such lattices, there is a well-defined Type IV Hermitian symmetric domain D_{T_X} , and for any choice of neat arithmetic subgroup $\Gamma \leq \text{O}^*(T_X)$ of the *stable orthogonal group* of T_X , there is a well-defined arithmetic quotient

$$F_\Gamma := D_{T_X}/\Gamma, \quad \Gamma \leq \text{O}^*(T_X).$$

The S and T lattices are dual in many senses – for projective surfaces, S_X contains ample line bundles, and hence the algebraic/projective structure of X (as determined by its polarizations) are governed by S_X , while its complement T_X governs transcendental data such as periods and variations of Hodge structure, which are typically orthogonal to the polarization. For K3 surfaces, by the global Torelli theorem, its

isomorphism is determined by its polarized weight 2 Hodge structure on $H^2(X; \mathbf{C})$, and since period domains of the form D_{T_X} classify such Hodge structures, they can be used to construct coarse moduli spaces of S_X -polarized marked K3 surfaces.

1.2.1 Unpolarized Enriques Surfaces and F_{En}

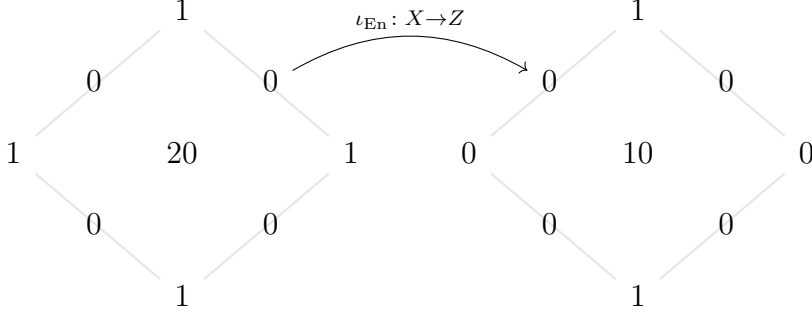
As quotients of K3 surfaces, Enriques surfaces Z admit similar Hodge-theoretic moduli spaces, and we are thus naturally lead to study the same lattice-theoretic techniques used in the construction D_{T_X} for K3 surfaces X , along with the involutions involved in double cover constructions $X \rightarrow Z$. By the classification theory for Enriques surfaces (see [CD89, Cor. 1.2.3]), ω_Z is a generator of the 2-torsion in $\text{Pic}(Z)$. The Néron–Severi group satisfies

$$\text{NS}(Z) \cong H^2(Z; \mathbf{Z}) \cong \mathbf{Z}^{10} \oplus \mathbf{Z}_2.$$

Let $H^2(Z; \mathbf{Z})_f$ denote the free part. The intersection pairing endows $H^2(Z; \mathbf{Z})_f$ with the structure of a lattice, i.e. a free $\mathbf{Z}$ -module with a integral symmetric bilinear form, and there are isometries

$$L_{\text{En}} := H^2(Z; \mathbf{Z})_f \xrightarrow{\sim} U \oplus E_8 \xrightarrow{\sim} \text{II}_{1,9},$$

the unique even unimodular lattice up to isometry (see Section 2.2.3), where U and E_8 are as above. We similarly define a **marking** for an Enriques surface Y as an isometry $H^2(Y, \mathbf{Z})_f \cong L_{\text{En}}$, and refer to an Enriques surface with a marking as a **marked** Enriques surface. For any Enriques surface Z , since ω_Z is 2-torsion, there exists an étale double cover $\pi : X \rightarrow Z$ such that X is a K3 surface. This is the universal cover, and the fundamental group $\pi_1(Z) \cong \mathbf{Z}_2$ is generated by the covering involution, which acts freely and properly discontinuously on X . The Hodge diamonds of K3 and Enriques surface are well known, and visibly reflect the structure of the double cover:



In particular, $h^{1,1}(Z) = 10$, $h^{2,0}(Z) = h^{0,2}(Z) = 0$, confirming $p_g(Z) = 0$, and all other Hodge numbers vanish except $h^{0,0}(Z) = h^{2,2}(Z) = 1$. Conversely, any K3 surface X with a fixed-point-free involution ι_{En} is said to have an *Enriques involution*, gives rise to an Enriques surface $Y := X/\iota_{\text{En}}$ as its quotient. The pullback $\pi^*H^2(Z; \mathbf{Z})$ under the covering map can be identified with the lattice $L_{\text{En}}(2)$, i.e., L_{En} with the intersection form multiplied by 2. There is a unique primitive embedding $L_{\text{En}}(2) \hookrightarrow L_{\text{K3}}$ up to isometry, and so we regard it as a primitive sublattice and set

$$S_{\text{En}} := L_{\text{En}}(2) = U(2) \oplus E_8(2), \quad T_{\text{En}} \cong S_{\text{En}}^{\perp L_{\text{K3}}} = U \oplus U(2) \oplus E_8(2) \cong U \oplus S_{\text{En}},$$

which have satisfy $\text{sig } S_{\text{En}} = (1, 9)$ and $\text{sig } T_{\text{En}} = (2, 10)$ respectively. The computation of T_{En} can be found, for example, in [Ste91], although it was known earlier. We note that when X is a K3 surface that admits an Enriques involution ι_{En} , letting I_{En} denote the induced isometry on $H^2(X; \mathbf{Z})$, we obtain

$$S_{\text{En}} \oplus T_{\text{En}} \xrightarrow{\sim} L_{\text{K3}}^{I_{\text{En}}=1} \oplus L_{\text{K3}}^{I_{\text{En}}=-1} \hookrightarrow L_{\text{K3}},$$

realizing these as the invariant and coinvariant lattices respectively.¹ This similarly determines a Type IV period domain $D_{T_{\text{En}}}$, and the quotient

$$F_{\text{En}} := D_{T_{\text{En}}}/\Gamma_{\text{En}}, \quad \Gamma_{\text{En}} := \text{O}(T_{\text{En}})$$

is a 10-dimensional rational quasi-projective variety. The global Torelli theorem for Enriques surfaces (see [Hor78; Nam85]) states that two Enriques surfaces Z_1, Z_2 are isomorphic if and only if there exists an isometry $\phi : H^2(Z_1; \mathbf{Z}) \rightarrow H^2(Z_2; \mathbf{Z})$ preserving the intersection pairing and Hodge structure. This implies that the natural map from the set $\mathcal{M}_{\text{En}}$ of isomorphism classes of Enriques surfaces to

¹ We note that there are mixed conventions in the literature regarding whether S or T should be the invariant sublattice – we choose here the convention used in [Ale+23].

F_{En} is injective on the open subset parameterizing smooth surfaces. Thus, F_{En} serves as the coarse moduli space of (unpolarized) Enriques surfaces.

1.2.2 Numerically Polarized Enriques Surfaces and $F_{\text{En},2}$

Let Z be an Enriques surface. The first Chern class induces an isomorphism

$$c_1: \text{Pic}(Z) \xrightarrow{\sim} H^2(Z; \mathbf{Z})$$

so that the free part $H^2(Z; \mathbf{Z})_f$ is identified with the group of numerical divisor classes, $\text{Num}(Z)$. A **numerical polarization** $[h]$ on Z is the numerical class of $h := c_1(\mathcal{L})$ for an ample line bundle $\mathcal{L} \in \text{Pic}(Z)$, which we often write $[\mathcal{L}]$. For a fixed degree $2d$, one considers the moduli space $F_{\text{En},h}$ of numerically polarized Enriques surfaces of degree $2d$. The set of primitive classes h with $h^2 = 2d$, up to the action of $\text{O}(L_{\text{En}})$, may consist of more than one orbit, except in the case $d = 1$, where the orbit is unique. Therefore, the moduli space $F_{\text{En},2}$ of numerically polarized Enriques surfaces of degree 2 is distinguished by this uniqueness property. The moduli space $F_{\text{En},2}$ parameterizes pairs $(Z, [\mathcal{L}_Z])$, where Z is an Enriques surface, possibly with ADE singularities, and $[\mathcal{L}_Z] \in \text{Pic}(Z)/\mathbf{Z}_2 \cong \text{Num}(Z)$ is the numerical class of an ample polarization of degree 2. Equivalently, $F_{\text{En},2}$ may be described as the moduli space of pairs $(Z, \mathcal{M})$, where $\mathcal{M} = \mathcal{L}_Z^{\otimes 2} \in \text{Pic}(Z)$ is a 2-divisible polarization of degree 8. Given an ample line bundle $\mathcal{L}$ on Z , set $\mathcal{M} = \mathcal{L}^{\otimes 2}$, which has degree 8. By the classification of big and nef linear systems on Enriques surfaces (see [Cos83]), the linear system $|\mathcal{M}|$ is basepoint-free and defines a morphism $\rho: Z \rightarrow W$ where W is a quartic del Pezzo surface with singularities of type $4A_1$ or $A_3 + 2A_1$. The morphism ρ is a double cover of W which is branched along a divisor $B \subset W$ where the corresponding ramification divisor $R_Z = \rho^{-1}(B)$ is ample, $\mathbf{Q}$ -Cartier, and in the linear system $|\mathcal{M}|$. The pair $(Z, \varepsilon R_Z)$ is log-canonical for sufficiently small $\varepsilon > 0$, and thus $F_{\text{En},2}$ admits a KSBA compactification by stable pairs, which we will simply denote $\overline{F_{\text{En},2}}$ throughout this work.

To put us in the setting of the main theorem, fix a basis of L_{K3} in the decomposition above, so in coordinates we have $(u_1, u_2, u_3, \alpha_1, \alpha_2) \in U^3 \oplus E_8^2$, where each u_i is in a copy of U and each α_i is in a copy of E_8 .

Consider three morphisms on L_{K3} , acting on vectors $(u_1, u_2, u_3, \alpha_1, \alpha_2) \in U^3 \oplus E_8^2$ in this basis in the following way:

$$\begin{aligned} I_{\text{dP}}(u_1, u_2, u_3, \alpha_1, \alpha_2) &= (-u_1, u_3, u_2, -\alpha_1, -\alpha_2) \\ I_{\text{En}}(u_1, u_2, u_3, \alpha_1, \alpha_2) &= (-u_1, u_3, u_2, \alpha_2, \alpha_1) \\ I_{\text{Nik}}(u_1, u_2, u_3, \alpha_1, \alpha_2) &= (u_1, u_2, u_3, -\alpha_2, -\alpha_1) \end{aligned}$$

These arise as the lattice involutions on L_{K3} induced by three types of geometric involutions, ι_{dP} , ι_{En} , and ι_{Nik} respectively, on a K3 surface X . A direct computation shows that the group $\langle I_{\text{dP}}, I_{\text{En}}, I_{\text{Nik}} \rangle$ is isomorphic to $\mathbf{Z}_2^2$, and thus these involutions mutually commute. For each such involution $I_\star$, we write $S_\star := L_{K3}^{I_\star=1}$ and $T_\star = L_{K3}^{I_\star=-1}$ for the invariant and co-invariant sublattices under the group actions $\langle I_\star \rangle \curvearrowright L_{K3}$. Similarly direct computations yield the invariant and coinvariant sublattices shown in the table below, where the triples $(r, a, \delta)_{n_+}$ are the invariants shown by [Nik80] to classify 2-elementary lattices T which admit a primitive embedding $T \hookrightarrow L_{K3}$. Concretely, $r := \text{rank}_{\mathbf{Z}}(T)$ is the rank, a is the *length* of L , which can be expressed as $\dim_{\mathbf{F}_2}(A_T)$ where $A_T := T^\vee/T$ is the *discriminant group* of T , the integer $\delta \in \{0, 1\}$ is the *coparity*, and the subscript n is used to track the rank of a maximal positive-definite sublattice, which can be used to recover the signature as $(n_+, r - n_+)$.

Table 1.2.1: (Co)Invariant Lattices for the Three Involutions

L	Isometry Class	$\text{rank}_{\mathbf{Z}}(L)$	$\text{sig}(L)$	$(r, a, \delta)_n$	A_L
S_{dP}	$U(2)$	2	(1, 1)	$(2, 2, 0)_1$	$\mathbf{Z}_2^2$
T_{dP}	$U \oplus U(2) \oplus E_8^2$	20	(2, 18)	$(20, 2, 0)_2$	$\mathbf{Z}_2^2$
S_{En}	$U(2) \oplus E_8(2)$	10	(1, 9)	$(10, 10, 0)_1$	$\mathbf{Z}_2^{10}$
T_{En}	$U \oplus U(2) \oplus E_8(2)$	12	(2, 10)	$(12, 10, 0)_2$	$\mathbf{Z}_2^{10}$
L_{Nik}^+	$U^3 \oplus E_8(2)$	14	(3, 11)	$(14, 8, 0)_3$	$\mathbf{Z}_2^8$
L_{Nik}^-	$E_8(2)$	8	(0, 8)	$(8, 8, 0)_0$	$\mathbf{Z}_2^8$

1.3 Structure and Overview

The thesis is organized into three parts, establishing the foundations and proving the main isomorphism theorem:

- Part I develops the combinatorial and lattice-theoretic foundations:
 - Chapter 1 introduces the main theorem, historical context, and key concepts including KSBA stable pairs and ADE + BC surfaces.
 - Chapter 2 establishes lattice-theoretic foundations, particularly Nikulin’s theory of 2-elementary lattices, with applications to the invariant lattices L_{K3} , len , T_{En} , and T_{dP} . This includes the classification of primitive embeddings and the structure of discriminant groups.
 - Chapter 3 develops the theory of Enriques surfaces and their K3 covers, analyzing the period domains $F_{\text{En},2}$ and $F_{(2,2,0)}$, and classifying the five 0-cusps of $\partial \overline{F_{\text{En},2}}^{\text{BB}}$ through explicit lattice-theoretic computations.
- Part II constructs the compactifications:
 - Chapter 4 presents the theory of KSBA, Baily-Borel, toroidal, and semitoroidal compactifications. It establishes the framework for stable pairs and recognizable divisors that enables the comparison between algebraic and analytic compactifications.
 - Chapter 5 develops integral affine geometry and the theory of Kulikov models. It describes how Type III degenerations correspond to integral affine spheres with singularities (IAS^2), and how these yield divisorial log terminal (dlt) models through explicit algorithms involving Coxeter polytopes and monodromy invariants.
- Part III proves the main theorem and provides explicit computations:
 - Chapter 6 establishes the isomorphism $\overline{F_{\text{En},2}} \cong \overline{F_{\text{En},2}}^{\mathcal{F}\bullet}$ through:
 1. Embedding $F_{\text{En},2}$ into the moduli space $F_{(2,2,0)}$ of $(2, 2, 0)$ -polarized K3 surfaces as a Noether-Lefschetz locus
 2. Showing the normalized closure B^ν inherits a semitoroidal structure from $\overline{F_{(2,2,0)}}$

3. Constructing the universal family of Enriques pairs via quotient by the Enriques involution
 4. Proving the classifying map $B^\nu \rightarrow \overline{F_{\text{En},2}}$ is finite using geometric constraints on double curves
 5. Applying Zariski's Main Theorem to conclude the isomorphism
- Chapter 7 provides computational examples, constructing explicit dlt models for degenerations at each of the five 0-cusps of $\partial \overline{F_{\text{En},2}}^{\text{BB}}$. It demonstrates the folding procedure that relates K3 degenerations to Enriques degenerations through the action of commuting involutions on IAS^2 structures.

CHAPTER 2

LATTICE THEORY

Compactifications of $F_{\text{En},2}$, and more generally moduli spaces of K3 surfaces, rely on the arithmetic and combinatorial structure of even indefinite (and often hyperbolic) lattices. This chapter discusses the lattice-theoretic techniques used in such problems, emphasizing both theoretical results and explicit computational techniques.

Of particular importance is the structure theory of *2-elementary* lattices: even integral lattices whose discriminant group is an $\mathbf{F}_2$ -module. In the context of $F_{\text{En},2}$, the sublattice T_{En} associated to the Enriques involution ι_{En} on a K3 cover X is 2-elementary of signature $(2, 10)$. The invariants (r, a, δ) , recording the rank, length of the discriminant group, and *coparity*, completely determine the isometry classes of such lattices, as well as their embedding properties. The results of [Nik79a] yield existence and uniqueness of the relevant period domains $D_{T_{\text{En}}}$, their arithmetic quotients $F_{\text{En},2}$, and the structure of boundary strata in various compactifications.

A major theme is the connection between the geometry of the period domain and the arithmetic of T_{En} : the construction of $F_{\text{En},2}$ relies on an arithmetic subgroup $\Gamma_{\text{En},2}$, and the $\Gamma_{\text{En},2}$ -orbits of primitive isotropic vectors η and primitive isotropic planes I in T_{En} index the rational boundary components of the Baily-Borel compactification $\overline{F_{\text{En},2}}^{\text{BB}}$, called *cusps*. It has 0-cusps and 1-cusps indexed by such η and I respectively. This chapter details the lattice-theoretic methods classically used to find the five 0-cusps of $F_{\text{En},2}$. We begin with definitions and standard constructions:

- Properties of even lattices, dual lattices, discriminant groups, and genus;

- The hyperbolic plane U , indefinite root lattices A_n, D_n, E_8 and their twists, and particularly their duals and discriminant groups;
- Criteria for primitively embedding one lattice into another, including Nikulin's theorems on extensions, embeddings, and overlattices, as well as his classification of 2-elementary lattices.

Special focus is given Nikulin's work: explicit determination of invariants, criteria for primitive embeddings into even unimodular lattices such as L_{K3} and len , and explicit discriminant group techniques that can be used to reduce the aforementioned orbit problems to the study of finitary, computable objects. In later chapters, we will see how the explicit classification of isotropic sublattices of T_{En} determines the structure of $\partial \overline{F_{\text{En},2}^{\text{BB}}}$ and consequently that of semitoroidal compactifications $\overline{F_{\text{En},2}^{\mathcal{F}^\bullet}}$, and by the main theorem, provide an inroad into studying the boundary $\partial \overline{F_{\text{En},2}}$ of the KSBA compactification.

2.1 Foundations

2.1.1 Bilinear and Quadratic Forms

Pairings

Definition 2.1.1.1 (Symmetric Bilinear Form). Let L be a $\mathbf{Z}$ -module. A **bilinear form** β on L is a morphism $\beta : L \otimes_{\mathbf{Z}} L \rightarrow \mathbf{Q}$. We often write $v \cdot w$ or vw for $\beta(v, w)$. A bilinear form β is:

- **ε -symmetric** for $\varepsilon \in \mathbf{Q}$ if $\beta(a, b) = \varepsilon \cdot \beta(b, a)$.
- **Symmetric** if $\varepsilon = 1$.
- **Skew-symmetric** if $\varepsilon = -1$.
- **Alternating** if $\beta(a, a) = 0$ for all $a \in L$.
- **Integral** if its image $\beta(L, L)$ is contained in $\mathbf{Z}$.
- **Nondegenerate** if the map $L \rightarrow \text{Hom}_{\mathbf{Z}}(L, \mathbf{Z})$ given by $v \mapsto \beta(v, \cdot)$ is injective.

Definition 2.1.1.2 (Quadratic Form and Associated Bilinear Form). A **quadratic form** on a $\mathbf{Z}$ -module L is a map of sets $q : L \rightarrow \mathbf{Q}$ such

that $q(\lambda v) = \lambda^2 q(v)$ for all $v \in L, \lambda \in \mathbf{Q}$, and such that its **polar form** β_q is a symmetric bilinear form on L :

$$\begin{aligned}\beta_q : L \otimes_{\mathbf{Z}} L &\rightarrow \mathbf{Q} \\ (v, w) &\mapsto \beta_q(v, w) := q(v + w) - q(v) - q(w)\end{aligned}$$

We say q is **integral** if $q(L) \subseteq \mathbf{Z}$. The pair (L, q) is called a **quadratic $\mathbf{Z}$ -module**.

Lemma 2.1.1.3 (Correspondence between Bilinear and Quadratic Forms). *Every $\mathbf{Q}$ -valued bilinear module (L, β) determines a $\mathbf{Q}$ -valued quadratic module (L, q_β) by $q_\beta(v) := \beta(v, v)$, which depends only on the symmetric part of β . Conversely, every $\mathbf{Q}$ -valued quadratic module (L, q) determines a symmetric bilinear module (L, β_q) via the polar form.*

Lemma 2.1.1.4 (Bijection for Even Lattices). *There is a bijection between even symmetric integral forms and integral quadratic forms:*

$$\begin{aligned}\{\beta \in \text{Sym}_{\mathbf{Z}}^2(L^\vee) \mid \beta \text{ is even}\} &\rightleftharpoons \text{Quad}_{\mathbf{Z}}(L) \\ \beta &\mapsto q(v) := \frac{1}{2}\beta(v, v) \\ \beta_q &\leftarrow q\end{aligned}$$

In other words, the polar form of any integral quadratic form is an even symmetric integral form. Conversely, every even symmetric integral form β is the polar form of the integral quadratic form $q(v) := \frac{1}{2}\beta(v, v)$.

Lattices

Definition 2.1.1.5 (Lattice). A **lattice** is a pair (L, β) where L is a free $\mathbf{Z}$ -module of finite rank and β is a nondegenerate symmetric bilinear form, which is typically integral but may be $\mathbf{Q}$ -valued. A **quadratic lattice** is a pair (L, q) where q is a nondegenerate quadratic form. A lattice (L, β) is **even** if $\beta(v, v) \in 2\mathbf{Z}$ for all $v \in L$, and **odd** otherwise.

Gram Matrices

Definition 2.1.1.6 (Gram Matrix). Given a basis $B_L = (e_i)_{1 \leq i \leq n}$ for a bilinear module (L, β) , the **Gram matrix** of β is $G_\beta := (\beta(e_i, e_j))_{i,j} \in \text{Mat}_{n \times n}(\mathbf{Q})$. For vectors $v = \sum a_j e_j$ and $w = \sum b_j e_j$, we have $\beta(v, w) = v^t G_\beta w$. Similarly, for a quadratic module (L, q) , a **Gram**

matrix G_q is any matrix such that $q(v) = v^t G_q v$. We define the **discriminant** $\text{disc}(L)$ of L as $\det(G_\beta)$ in any choice of basis. We note that for any sublattice $S \leq L$, we have the formula

$$\text{disc}(S) = [L : S]^2 \text{disc}(L),$$

so the discriminant typically *increases* when passing to a sublattice. We say L is **unimodular** if $\text{disc}(L) = \pm 1$. The Gram matrix reflects properties of the form: β is symmetric $\iff G_\beta^t = G_\beta$, skew-symmetric $\iff G_\beta^t = -G_\beta$, and an integral lattice is even $\iff G_\beta$ is an integer matrix with diagonal entries in $2\mathbf{Z}$.

Proposition 2.1.1.7 (Primitive Isotropic Vectors, Divisibility, and Hyperbolic Splittings). *Let S be any unimodular lattice admitting a primitive embedding into a nondegenerate lattice L . Then $L \cong S \oplus T$ where*

$$T := S^\perp = \left\{ v \in L \mid \beta(v, S) = 0 \right\}$$

*is the **orthogonal complement** of S in L . Moreover, if S is unimodular, then so is T .*

Proof. This follows from [PS24, Lem. 1.3.1]: we can write $\text{disc}(S) = [S \oplus T : L] \cdot c_S$ for some $c_S \in \mathbf{Z}$. By unimodularity, $\text{disc}(S) = \pm 1$ forces $c_S = \pm 1$ and $[S \oplus T : L] = 1$, yielding the first claim. For the second, we note that $\text{disc}(S \oplus T) = \text{disc}(S) \cdot \text{disc}(T)$ by standard properties of determinants, forcing $\text{disc}(T) = \pm 1$. $\square$

Definition 2.1.1.8 (Rank and Signature). Let (L, β) be a nondegenerate lattice. We say it is:

- **positive definite** if $\beta(v, v) > 0$ for all nonzero $v \in L$.
- **positive semidefinite** if $\beta(v, v) \geq 0$ for all $v \in L$.
- **negative definite** if $\beta(v, v) < 0$ for all nonzero $v \in L$.
- **negative semidefinite** if $\beta(v, v) \leq 0$ for all $v \in L$.
- **indefinite** if it is neither positive nor negative semidefinite.

Letting $(L_{\mathbf{R}}, \beta_{\mathbf{R}})$ be the extension of L to $\mathbf{R}$, the diagonalization of the Gram matrix p positive terms and q negative terms. We define the following:

- The **rank** of L is $\text{rank}(L) = p + q$.
- The **signature** of L is $\text{sig}(L) = (p, q)$.
- The **index** of L is $\tau(L) = p - q$.

We note that L is positive-definite if $q = 0$, negative-definite if $p = 0$, and indefinite if both $p, q > 0$. We say L is **hyperbolic** if its signature is $(1, n - 1)$ or $(n - 1, 1)$.

Definition 2.1.1.9 (Scaled Lattices). For any lattice (L, β) and positive integer m , the **scaled lattice** $L(m)$ is the same $\mathbf{Z}$ -module L equipped with the bilinear form $\beta_m(v, w) = m \cdot \beta(v, w)$. The signature of $L(m)$ is the same as L , but the discriminant scales as $\text{disc}(L(m)) = m^{\text{rank}(L)} \cdot \text{disc}(L)$. If L is unimodular and $m > 1$, $L(m)$ is not unimodular.

Example 2.1.1.10 (Diagonal and Hyperbolic Lattices). The **diagonal lattice** $\langle a_1, \dots, a_n \rangle$ is $\mathbf{Z}^n$ with the bilinear form $\beta(x, y) = \sum a_i x_i y_i$ and diagonal Gram matrix $\text{diag}(a_1, \dots, a_n)$. In the special case $a_1, \dots, a_p = 1$ and $a_{p+1}, \dots, a_n = -1$, we write this lattice as $I_{p,q}$, due to its distinguished nature as the unique nondegenerate odd unimodular lattice of signature p, q . The **hyperbolic lattice** U is the free $\mathbf{Z}$ -module $\mathbf{Z}^2$ with basis e, f such that $\beta(e, e) = \beta(f, f) = 0$ and $\beta(e, f) = 1$. It is an even, integral, rank 2 lattice with Gram matrix

$$G_U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

2.1.2 Sublattices, Embeddings, and Isometries

Definition 2.1.2.1 (Lattice Morphisms and Embeddings). A **morphism** between lattices (L_1, β_1) and (L_2, β_2) is a morphism of $\mathbf{Z}$ -modules $f : L_1 \rightarrow L_2$ that preserves the bilinear form: $\beta_1(v, w) = \beta_2(f(v), f(w))$ for all $v, w \in L_1$. We write Lat for the category of integral lattices, and $\text{Hom}_{\text{Lat}}(L_1, L_2)$ for their spaces of morphisms. An **embedding** is an injective morphism. An embedding is **primitive** if its cokernel, $\text{coker}(f)$, is a torsion-free $\mathbf{Z}$ -module. An **isometry** is a morphism of lattices that is also an isomorphism of $\mathbf{Z}$ -modules. Two lattices L_1, L_2 are **isometric**, written $L_1 \xrightarrow{\sim} L_2$, if an isometry exists between them. The **orthogonal group** of a lattice L is its group of self-isometries $O(L) := \text{Aut}_{\text{Lat}}(L)$. In terms of a Gram matrix G_β , the orthogonal

group has the characterization:

$$\mathcal{O}(L) = \{M \in \mathrm{GL}_n(\mathbf{Z}) \mid MG_\beta M^t = G_\beta\}$$

Definition 2.1.2.2 (Primitive and Saturated Sublattices). A sublattice $S \leq L$ is **primitive** (or **saturated**) if the quotient module L/S is torsion-free. An element $v \in L$ is **primitive** if the sublattice $\langle v \rangle_{\mathbf{Z}}$ is primitive.

Proposition 2.1.2.3 (Characterization of Primitive Sublattices). *For a sublattice $S \subseteq L$, the following are equivalent:*

- S is a primitive sublattice of L .
- The inclusion $S \hookrightarrow L$ is a primitive embedding.
- S is saturated in L , meaning $S = \mathrm{Sat}_L(S) := \{v \in L \mid nv \in S \text{ for some } n \in \mathbf{Z} \setminus \{0\}\}$.
- S is a direct summand of L as a $\mathbf{Z}$ -module (i.e., $L \cong S \oplus T$ for some submodule T).
- Any $\mathbf{Z}$ -basis of S can be extended to a $\mathbf{Z}$ -basis of L .
- $S_{\mathbf{Q}} \cap L = S$.
- $S = (S^{\perp L})^{\perp L}$.
- Every integral linear functional on S can be lifted to an integral linear functional on L .

Definition 2.1.2.4 (Equivalence of Embeddings). Two primitive embeddings $\iota_1: S \hookrightarrow L_1$ and $\iota_2: S \hookrightarrow L_2$ are **equivalent** if there exists an isometry $f \in \mathrm{Isom}(L_1, L_2)$ such that $f \circ \iota_1 = \iota_2$, so the following diagram commutes

$$\begin{array}{ccc} S & \xrightarrow{\iota_1} & L_1 \\ & \searrow \iota_2 & \downarrow f \\ & & L_2 \end{array}$$

If $L_1 = L_2 = L$ is a single lattice, two primitive embeddings $\iota_1, \iota_2: S \hookrightarrow L$ are **equivalent** if they are equivalent by an element of $\mathcal{O}(L)$ as above. The set of equivalence classes of primitive embeddings of S into L is denoted $\mathrm{Emb}(S, L)$.

Definition 2.1.2.5 (Isotropic Submodules and Vectors). For a lattice (L, β) , a submodule $W \subseteq L$ is **isotropic** if $\beta|_W = 0$ (equivalently, $W \subseteq W^{\perp L}$). An **isotropic vector** is an element $v \in L$ with $\beta(v, v) = 0$. The **Witt index** $\text{WI}(L)$ is the maximal rank of an isotropic sublattice.

Definition 2.1.2.6 (Divisibility). Let L be a lattice with form β , and let $v \in L$. The **divisibility** of v in L , denoted $\text{div}_L(v)$, is the positive generator of the ideal $\left\{ \beta(v, w) \mid w \in L \right\} \subseteq \mathbf{Z}$.

Proposition 2.1.2.7 (Divisibility and Discriminant for Primitive Vectors). *Let L be a nondegenerate lattice, and $v \in L$ an arbitrary (not necessarily isotropic) vector. The element $v^* := v / \text{div}_L(v) \in L^\vee$ is primitive in the dual lattice, and its image in the discriminant group $A_L := L^\vee / L$ has order $\text{div}_L(v)$. In particular, $\text{div}_L(v)$ divides the order of A_L , and hence $|\text{disc}(L)|$.*

Proof. Since the divisibility $\text{div}_L(v) = d$ is by definition the positive generator of the ideal $\{(v, w) \mid w \in L\}$, one has $v/d \in L^\vee$ and this vector is primitive in $L^\vee$. Indeed, if $v/d = my$ for some $m \in \mathbf{Z}$, $y \in L^\vee$, then $v = dmy \in L$, and primitivity of v in L implies $m = \pm 1$. The order of v^* in A_L is the minimal positive n such that $nv^* \in L$. This occurs precisely when n is divisible by d , so the order is $d = \text{div}_L(v)$. Since A_L is finite of order $|\text{disc}(L)|$, it follows in particular that $\text{div}_L(v)$ divides $|\text{disc}(L)|$. □

Lemma 2.1.2.8 (Divisibility of Isotropic Vectors in Unimodular Lattices). *Let L be a nondegenerate unimodular lattice, and let $v \in L$ be a primitive isotropic vector. Then there exists $w \in L$ such that $(v, w) = 1$. In particular, $\text{div}_L(v) = 1$ for all $v \in L$.*

Proof. Since L is unimodular, $L \xrightarrow{\sim} L^\vee$. The linear form $x \mapsto (v, x)$ is a nonzero element of $\text{Hom}(L, \mathbf{Z})$, and is surjective because v is primitive and L is unimodular. Hence there exists $w \in L$ with $(v, w) = 1$. □

2.1.3 Dual Lattices and Discriminant Forms

Definition 2.1.3.1 (Dual Lattice). For an integral lattice (L, β) , the **dual lattice** is the $\mathbf{Z}$ -module of linear functionals $L^\vee := \text{Hom}_{\mathbf{Z}}(L, \mathbf{Z})$.

Theorem 2.1.3.2 (Geometric Identification of the Dual Lattice). *For a nondegenerate integral lattice (L, β) , the dual lattice can be identified with a sublattice of $L_{\mathbf{Q}} := L \otimes_{\mathbf{Z}} \mathbf{Q}$ via the bijection:*

$$L^{\vee} \cong \{v \in L_{\mathbf{Q}} \mid \beta_{\mathbf{Q}}(v, L) \subseteq \mathbf{Z}\}$$

where a functional $\phi \in L^{\vee}$ corresponds to the unique vector $v_{\phi} \in L_{\mathbf{Q}}$ such that $\phi(w) = \beta(v_{\phi}, w)$ for all $w \in L$. Under this identification, we have the inclusions $L \subseteq L^{\vee} \subseteq L_{\mathbf{Q}}$.

Remark 2.1.3.3 (Properties of Dual Lattices). We summarize some standard properties of dual lattices

- Duality commutes with direct sums, $(L \oplus M)^{\vee} = L^{\vee} \oplus M^{\vee}$.
- If L has Gram matrix G_{β} in a basis B_L , then the dual basis satisfies $B_{L^{\vee}} = (B_L^t)^{-1}$, and the Gram matrix of the dual form is $G_{\beta^{\vee}} = G_{\beta}^{-1}$.
- The discriminant of the dual satisfies $\text{disc}(L^{\vee}) = 1/\text{disc}(L)$, and the dual of a scaled lattice is $(L(m))^{\vee} = L^{\vee}(1/m)$.

Definition 2.1.3.4 (Discriminant Group and Discriminant). For a nondegenerate lattice (L, β) , the **discriminant group** is the finite abelian group $A_L := L^{\vee}/L$. The order of the discriminant group equals the absolute value of the discriminant, i.e. $|A_L| = |\text{disc}(L)| = [L^{\vee} : L]$, so L is unimodular if and only if A_L is trivial.

Definition 2.1.3.5 (Discriminant Forms, [PS24, Def. 1.6.5]). For a nondegenerate *even* lattice (L, β) , the **discriminant bilinear form** is $\beta_{A_L} : A_L \times A_L \rightarrow \mathbf{Q}/\mathbf{Z}$, given by $\beta_{A_L}(\bar{x}, \bar{y}) = \beta(x, y) \bmod \mathbf{Z}$ for any lifts $x, y \in L^{\vee}$ and β the induced form on $L^{\vee}$. It admits an associated quadratic form $q_{A_L}(\bar{x}) = \beta(x, x) \bmod \mathbf{Z} \in \mathbf{Q}/\mathbf{Z}$ for any lift x . We note that there are two conventions in the literature, where one sometimes defines $q_{A_L}(\bar{x}) := \beta(x, x) \bmod 2\mathbf{Z} \in \mathbf{Q}/2\mathbf{Z}$; these agree by the isomorphism $\mathbf{Q}/\mathbf{Z} \xrightarrow{\sim} \mathbf{Q}/2\mathbf{Z}$ induced by multiplication by 2. We define its **orthogonal group** $O(A_L)$ as the automorphisms that preserve q_{A_L} . The **length** $\ell(L)$ of L is the minimal number of generators for the abelian group A_L .

Proposition 2.1.3.6 (Properties of Discriminant Forms). *The discriminant forms b_L and q_L of a nondegenerate lattice L are themselves nondegenerate. For any $[v], [w] \in A_L$, the bilinear and quadratic forms are*

related by $\beta_{A_L}([v], [w]) = \frac{1}{2}(q_{A_L}([v] + [w]) - q_{A_L}([v]) - q_{A_L}([w]))$, where multiplication by $\frac{1}{2}$ is interpreted as an isomorphism $\mathbf{Q}/2\mathbf{Z} \rightarrow \mathbf{Q}/\mathbf{Z}$.

Proposition 2.1.3.7 (Exact Sequence for Scaled Lattices). *For any lattice L and positive integer m , there is a short exact sequence:*

$$0 \rightarrow L/mL \hookrightarrow A_{L(m)} \rightarrow A_L \rightarrow 0$$

If L is unimodular, this implies $A_{L(m)} \cong L/mL$. Similarly, by functoriality, any isometry $f \in \mathrm{O}(L)$ lifts to an isometry of $L^\vee$ and thus induces an isometry on the discriminant group A_L . This defines a group homomorphism $\psi : \mathrm{O}(L) \rightarrow \mathrm{O}(A_L)$, which fits into an exact sequence:

$$0 \rightarrow \mathrm{O}^*(L) \rightarrow \mathrm{O}(L) \xrightarrow{\psi} \mathrm{O}(A_L) \rightarrow \mathrm{O}^*(A_L) \rightarrow 0$$

where $\mathrm{O}^*(L) := \ker(\psi)$ and $\mathrm{O}^*(A_L) := \mathrm{coker}(\psi)$ are the **stable orthogonal groups** of L and A_L . The cokernel, $\mathrm{O}^*(A_L) := \mathrm{coker}(\psi)$, measures the obstruction to lifting isometries from the discriminant form A_L to the lattice L . We note that ψ is surjective (i.e., $\mathrm{O}^*(A_L) = 0$) when L is indefinite and satisfies $\ell(A_L) + 2 \leq \mathrm{rank}(L)$, where $\ell(A_L)$ is the minimal number of generators of A_L . For unimodular lattices like U and E_8 , the discriminant group is trivial, so $\mathrm{O}^*(L) = \mathrm{O}(L)$.

Example 2.1.3.8 (U and $U(2)$). The hyperbolic lattice U is unimodular, so $U^\vee = U$ and its discriminant group A_U is trivial. For the scaled lattice $U(2)$, with Gram matrix $\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$, the dual is $U(2)^\vee = \frac{1}{2}U = U$ and we have $A_{U(2)} = U/U(2) \cong (\mathbf{Z}/2\mathbf{Z})^2 \cong U/2U$.

2.1.4 Torsion Forms over $\mathbf{Q}/\mathbf{Z}$

Definition 2.1.4.1 (Torsion Bilinear and Quadratic Forms). A **torsion bilinear form** is a pair (G, β) where G is a finitely generated torsion $\mathbf{Z}$ -module and $\beta : G \otimes_{\mathbf{Z}} G \rightarrow \mathbf{Q}/\mathbf{Z}$ is a bilinear form. A **torsion quadratic form** is a pair (G, q) where G is a torsion module and $q : G \rightarrow \mathbf{Q}/\mathbf{Z}$ is a quadratic form.

2.2 Fundamental Examples

2.2.1 The Hyperbolic Plane U

The **hyperbolic plane** U is the unique even, unimodular lattice of signature $(1, 1)$, with a standard basis given by vectors $\{e, f\}$ satisfying

$$e^2 = f^2 = 0, \quad e \cdot f = 1, \quad G_U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

A vector $v \in U$ is called **isotropic** if $v^2 = 0$. We have $\text{disc}(U) = -1$, and thus $U \cong U^\vee$, hence U is unimodular and $A_U = 0$. This can also be computed directly: if $v \in U^\vee$, then $v = ae + bf$ where $a, b \in \mathbf{Q}$. We then must have $vw \in \mathbf{Z}$ for all $w \in U$, and in particular this must hold for $w = e$ and $w = f$. We compute $ev = b$ and $fv = a$, so $a, b \in \mathbf{Z}$. Both e and f are primitive – for examples, the embedding $\text{gen } e \hookrightarrow U$ is primitive, since $U/\langle e \rangle \xrightarrow{\sim} \langle f \rangle$ is free. The full set of primitive isotropic vectors is $\{\pm e, \pm f\}$. To see this, let $v = ae + bf$ and compute $v^2 = (ae + bf)^2 = 2ab$. Thus, v is isotropic if and only if $ab = 0$, implying $v = ae$ or $v = bf$ for some $a, b \in \mathbf{Z}$. For v to additionally be primitive, we must have $a = \pm 1$ or $b = \pm 1$.

The scaled hyperbolic plane $U(2)$ has basis $\{e', f'\}$ such that

$$(e')^2 = (f')^2 = 0, \quad e' \cdot f' = 2, \quad G_{U(2)} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

We can similarly compute $U(2)^\vee$ by writing $v \in ae' + bf'$, then $e'v = 2b$ and $f'v = 2a$ are integral if and only if $a, b \in \frac{1}{2}\mathbf{Z}$ – thus $U(2)^\vee = \frac{1}{2}U$, and $A_{U(2)} = \frac{1}{2}U/U \cong U/2U \cong C_2^2$, as expected from the exact sequence in the previous section.

The orthogonal group of U is $O(U) = \{\pm \text{id}, \pm \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\} \cong C_2^2$, generated by the involutions $(e, f) \mapsto (-e, -f)$ and $(e, f) \mapsto (f, e)$. Explicitly, we have the following elements, closed under composition, each of order 2, generating a Klein four group:

$$\begin{aligned}
\sigma_0 = \text{id} : \quad e &\mapsto e, \quad f \mapsto f; \quad A_0 &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \\
\sigma_1 : \quad e &\mapsto -e, \quad f \mapsto -f; \quad A_1 &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \\
\sigma_2 : \quad e &\mapsto f, \quad f \mapsto e; \quad A_2 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \\
\sigma_3 : \quad e &\mapsto -f, \quad f \mapsto -e; \quad A_3 &= \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.
\end{aligned}$$

To see why this is true, since g must preserve primitive isotropic vectors and $ef = 1$, the images $g(e), g(f)$ must lie in $\{\pm e, \pm f\}$ and have the same pairing. Only four such transformations preserve $ef = 1$, as listed above. Direct computation verifies the relation $A_i^T G_U A_i = G_U$ for each i . Note however that the *rational* orthogonal group $O(U_{\mathbf{Q}})$ is much larger: it contains the one-parameter subgroups:

$$i_a = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}, \quad j_b = \begin{pmatrix} 0 & b \\ b^{-1} & 0 \end{pmatrix} \quad \text{for } a, b \in \mathbf{Q}^\times, .$$

Finally, we note that the lattices U and $I_{1,1} := \langle 1, -1 \rangle$ are not isometric over $\mathbf{Z}$, since U is even and $I_{1,1}$ is odd. However, both have signature $(1, 1)$ and discriminant -1 , so they become isometric over $\mathbf{Q}$ and $\mathbf{R}$. It is true, however, that U and $\Pi_{1,1}$ are isometric, where the latter is the unique *even* unimodular lattice discussed later in this section.

2.2.2 Root Lattices

We present here the classical definitions of the *positive-definite* root lattices of A, D, E types, noting that in applications in later chapters, we will typically choose the negative-definite variations of these lattices, as is the typical convention in algebraic geometry. We maintain the positive-definite convention until Section 2.7, where we give more precise constructions and critically use the positive-definite convention to carry out folding calculations in Euclidean spaces using standard linear algebra. After this, we will switch to the negative-definite convention that is typically more convenient for applications in algebraic geometry. We refer to Appendix A.0.2 for explicit computations, diagrams, and labeling conventions.

We summarize below all of the relevant information for the simply-laced ADE types, as well as the exceptional types E_6, E_7, E_8, F_4 , and G_2 . In all cases, the lattices L can be realized by a primitive embedding $L \hookrightarrow \mathbf{Q}^n$ for some n , where we identify $\mathbf{Z}^n \subseteq \mathbf{Q}^n$ with the abstract diagonal lattice $\langle 1, 1, \dots, 1 \rangle$. In these Euclidean embeddings, one can explicitly compute the Gram matrices G_L , the duals $L^\vee$ and discriminant groups A_L , and invariants such as $\text{disc}(L)$ using standard linear algebra. We let $R(L) \subseteq L$ denote the set of *roots* in each lattice (vectors $v \in L$ with $v^2 = 2$), $\Phi(L) \subseteq R(L)$ its corresponding *simple root system* – a subset of roots α_i such that, without loss of generality, $\mathbf{Z}_{\geq 0}\Phi(L) = R(L)$, i.e. every $v \in R(L)$ can be written as $v = \sum_{\alpha_i \in \Phi(L)} c_i \alpha_i$ with all $c_i \geq 0$ integral. This defines a sublattice $\mathbf{Z}\Phi(L) \hookrightarrow L$, and we say L is a **root lattice** if this is index 1. In any case, one can define a *Weyl group* $W(L) \leq \text{O}(L)$ of reflections generated in by hyperplanes $H_{\alpha_i} := \alpha_i^\perp$ for $\alpha_i \in \Phi(L)$. For root lattices, we can identify G_L with $G_{\mathbf{Z}\Phi(L)}$, the Gram matrix of intersections between the simple roots; these agree up to similarity since $\Phi(L)$ generates L .

Type A

Let $\mathbf{Z}^n$ denote the standard diagonal lattice $\langle 1, 1, \dots, 1 \rangle$ with generators $\{e_1, \dots, e_n\}$. The root lattice A_n is isometric to a root lattice in $v_{n+1}^\perp \subset \mathbf{Z}^{n+1}$ where $v_{n+1} := \sum_{i=1}^{n+1} e_i$, and thus concretely identified as a primitively embedded sublattice

$$A_n \hookrightarrow \left\{ x \in \mathbf{Z}^{n+1} \mid \sum_{i=1}^{n+1} x_i = 0 \right\}.$$

One defines $R(A_n)$ to consist of the roots $r_{i,j} := e_i - e_j$ in $v_{n+1}^\perp$ for $1 \leq i \neq j \leq n+1$, and so $|R(A_n)| = n \cdot (n+1)$. There are n simple roots forming $\Phi(A_n)$, which can be chosen as $\alpha_i := e_i - e_{i+1}$ for $1 \leq i \leq n$.

Type D

For any lattice L , one can consider the lattice $dL := \left\{ d \cdot v \mid v \in L \right\}$, noting that this differs from the twist $L(d)$. In particular, $d\mathbf{Z}^n$ forms a lattice for any n , and one can produce interesting lattices by taking the kernels of *reduced augmentation* morphisms $\varepsilon_d : L \rightarrow \mathbf{Z} \rightarrow \mathbf{Z}/d\mathbf{Z}$

given by $v \mapsto \sum v_i \pmod{d\mathbf{Z}}$. In particular, for $L = \mathbf{Z}^n$, we define $\mathbf{Z}_{\text{ev}}^n = \ker(\mathbf{Z}^n \xrightarrow{\varepsilon_2} \mathbf{Z}_2)$, which can be realized as

$$\mathbf{Z}_{\text{ev}}^n := \left\{ x = (x_1, \dots, x_n) \in \mathbf{Z}^n \mid \sum_{i=1}^n x_i \equiv 0 \pmod{2\mathbf{Z}} \right\} \hookrightarrow \mathbf{Z}^n$$

The D_n root lattice is the root sublattice $\mathbf{Z}_{\text{ev}}^n$ whose roots $R(D_n)$ consist of all vectors $r_{i,j} := \pm e_i \pm e_j$ in $\mathbf{Z}_{\text{ev}}^n$ for all $1 \leq i < j \leq n$. There are $n(n-1)/2$ unordered pairs of distinct indices, and for each such pair, four roots corresponding to all sign choices, and thus

$$|R(D_n)| = 4 \cdot \binom{n}{2} = 4 \cdot \frac{n(n-1)}{2} = 2n(n-1).$$

A set of simple roots for $\Phi(D_n)$ can be chosen as $\alpha_i := e_i - e_{i+1}$ for $1 \leq i \leq n-1$ and $\alpha_n := e_{n-1} + e_n$.

Type E The root lattice E_8 is the unique, even, unimodular, positive-definite lattice of rank 8, and admits an isometry

$$E_8 \xrightarrow{\sim} \left\{ x \in \mathbf{Z}^8 \cup \left(\mathbf{Z}^8 + \frac{1}{2}v_8 \right) \mid \sum_{i=1}^8 x_i \equiv 0 \pmod{2\mathbf{Z}} \right\} \hookrightarrow \mathbf{Q}^8$$

where $v_8 = (1, \dots, 1)$. It contains exactly $2^4 \cdot 3 \cdot 5 = 240$ roots:

- The difference vectors $r_{i,j}^{\pm,\pm} := \pm e_i \pm e_j$ for $1 \leq i \neq j \leq 8$, of which there are $\binom{8}{2} \cdot 4 = 112$, and
- The half-sum vectors $r_{\pm 8} := \frac{1}{2} \sum_{i=1}^8 \varepsilon_i e_i$ with $\varepsilon_i = \pm 1$ and $\prod \varepsilon_i = 1$, i.e. vectors of the form $\frac{1}{2}(\pm e_1 \pm e_2 \cdots \pm e_8)$ where the number of minus signs is even, of which there are $2^8/2 = 128$.

Of these, there are 8 simple roots comprising $\Phi(E_8)$, and there is a transitive action $W(E_8) \curvearrowright \Phi(E_8)$. The root lattice E_7 can be constructed as $\text{Ann}_{E_8}(L_1)$ for any sublattice $L_1 \leq E_8$ isometric to A_1 , usually taken to be generated by a specific vector denoted v_8 . Similarly, $E_6 = \text{Ann}_{E_8}(L_2)$ for any sublattice L_2 isometric to A_2 , usually chosen to be generated by a specific vector v_7 .

Types B, C, F_4 , G_2 .

The root system $R(B_n) \subset \mathbf{Z}^n$ consists of all vectors of the form $r_i := \pm e_i$ for $1 \leq i \leq n$ and $r_{ij} := \pm e_i \pm e_j$ for $1 \leq i < j \leq n$, of which

there are $2n + 2n \cdot (n - 1) = 2n^2$; of these, one can choose $\Phi(B_n)$ to consist of roots $\alpha_i := e_i - e_{i+1}$ for $1 \leq i \leq n - 1$ and $\alpha_n = e_n$. For type C , the root system $R(C_n)$ consists of the roots $2r_i$ and r_{ij} as in type B , again yielding $2n^2$, and one can choose all α_i in the same way, only replacing α_n with $2e_n$. It is worth pointing out the duality here: for B_n , there are $2n$ short roots r_i and $2n \cdot (n - 1)$ long roots r_{ij} ; in C_n , these switch roles, and there are $2n$ long roots $2r_i$ and $2n \cdot (n - 1)$ short roots r_{ij} .

The root lattice F_4 is realized as a root system in $\mathbf{Z}^4$ with $R(F_4)$ consisting of vectors $\pm e_i$ for $1 \leq i \leq 4$, $\pm e_i \pm e_j$ for $1 \leq i < j \leq 4$, and those of the form $\frac{1}{2}(\sum \pm e_i)$ where the number of minus signs is even. A system of simple roots can be taken to be

$$\begin{aligned}\alpha_1 &= e_2 - e_3, \\ \alpha_2 &= e_3 - e_4, \\ \alpha_3 &= e_4, \\ \alpha_4 &= \frac{1}{2}(e_1 - e_2 - e_3 - e_4).\end{aligned}$$

Finally, the root lattice G_2 embeds into $\mathbf{Z}_\Delta^3 \subseteq \mathbf{Z}^3$, i.e. those $x \in \mathbf{Z}^3$ with $\sum x_i = 0$. The root system $R(G_2)$ is generated by roots of the form $\pm(e_i - e_j)$ for $1 \leq i < j \leq 3$ and $\pm(2e_i - e_j - e_k)$ for i, j, k a permutation of $\{1, 2, 3\}$, yielding 12 total roots. A simple system can be taken to be

$$\begin{aligned}\alpha_1 &= e_1 - e_2, \\ \alpha_2 &= -e_1 + 2e_2 - e_3.\end{aligned}$$

The Plus Construction and Remarks

We also briefly note the classical “plus” construction: define two vectors in $\mathbf{Q}^n$ by

$$v_+ = \frac{1}{2}(1, 1, 1, \dots, 1), \quad v_- = \frac{1}{2}(-1, 1, 1, \dots, 1).$$

We then define the overlattices

$$D_n^+ := D_n \cup (v_+ + D_n), \quad D_n^- := D_n \cup (v_- + D_n).$$

If n is odd, then there are isometries

$$D_n^+ \xrightarrow{\sim} D_n^- \xrightarrow{\sim} D_n^\vee,$$

the dual lattice of D_n . If n is even, then $D_n^+ \not\xrightarrow{\sim} D_n^-$, and both contain D_n as an index 2 sublattice. Both D_n^+ and D_n^- are unimodular lattices, and are even integral lattices if and only if $n \equiv 0 \pmod{8}$. For $n = 8$, we have $D_8^+ \xrightarrow{\sim} E_8$.

We also recall that if $R := \mathbf{Z}\Phi$ is a lattice generated by a root system Φ , the dual lattice $R^\vee$ is referred to as the **weight lattice** associated to R , and the dual basis elements w_i defined by $w_i(\alpha_j) = \delta_{ij}$ are referred to as **weights** or **coroots**. For unimodular lattices such as E_8 , it is often useful to identify $E_8^\vee \cong E_8$ and work in a basis of coroots $w_1, \dots, w_8 := \alpha_1^\vee, \dots, \alpha_8^\vee$. The coordinates of each w_i in terms of the original roots α_i can be found explicitly; for example, note that

$$G_{E_8}^{-1} = \begin{bmatrix} -4 & -5 & -7 & -10 & -8 & -6 & -4 & -2 \\ -5 & -8 & -10 & -15 & -12 & -9 & -6 & -3 \\ -7 & -10 & -14 & -20 & -16 & -12 & -8 & -4 \\ -10 & -15 & -20 & -30 & -24 & -18 & -12 & -6 \\ -8 & -12 & -16 & -24 & -20 & -15 & -10 & -5 \\ -6 & -9 & -12 & -18 & -15 & -12 & -8 & -4 \\ -4 & -6 & -8 & -12 & -10 & -8 & -6 & -3 \\ -2 & -3 & -4 & -6 & -5 & -4 & -3 & -2 \end{bmatrix},$$

and thus one can write each w_i in terms of α_i basis using its columns:

$$\begin{aligned} w_1 &= -4\alpha_1 - 5\alpha_2 - 7\alpha_3 - 10\alpha_4 - 8\alpha_5 - 6\alpha_6 - 4\alpha_7 - 2\alpha_8 \\ w_2 &= -5\alpha_1 - 8\alpha_2 - 10\alpha_3 - 15\alpha_4 - 12\alpha_5 - 9\alpha_6 - 6\alpha_7 - 3\alpha_8 \end{aligned}$$

and so on. Finally, the lattice $E_8(2)$ frequently appears in moduli-theoretic applications. A direct determinant computation shows that $\text{disc}(E_8(2)) = 2^8$, and using the exact sequences of the previous chapter, we can write

$$E_8(2)^\vee \xrightarrow{\sim} \frac{1}{2}E_8 \implies A_{E_8(2)} \cong E_8/2E_8 \cong \frac{1}{2}E_8/E_8 \cong \mathbf{Z}_2^8,$$

and similarly $U(2)^\vee \xrightarrow{\sim} \frac{1}{2}U$, facts which were used in [Ste91] to compute $\overline{\partial F_{\text{En},2}}^{\text{BB}}$.

2.2.3 Unimodular lattices

Type I

For any positive integer n and non-negative integers $0 \leq p, q \leq n$ with $p + q = n$, let $L = \langle v_1, \dots, v_p, w_1, \dots, w_q \rangle \cong \mathbf{Z}^n$ be generated by $p + q$ elements where

$$\beta(v_i, v_j) = \delta_{ij}, \quad \beta(w_i, w_j) = -\delta_{ij}, \quad \beta(v_i, w_j) = 0 \quad \forall i, j.$$

The Gram matrix is diagonal with p copies of $+1$ and q copies of -1 :

$$G_\beta = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix} = \left[\begin{array}{c|c} \text{id}_{p \times p} & 0 \\ \hline 0 & -\text{id}_{q \times q} \end{array} \right] \in \text{Mat}_{p \times p}(\mathbf{Z}) \times \text{Mat}_{q \times q}(\mathbf{Z}).$$

We write this lattice as

$$I_{p,q} := \langle 1, 1, \dots, -1, -1 \rangle := \langle 1 \rangle^p \oplus \langle -1 \rangle^q.$$

This is an integral, odd, unimodular lattice of rank $p + q$ and signature (p, q) with discriminant $(-1)^q$. By the classification theorem of such lattices, if L is any odd indefinite unimodular lattice with $\text{sig}(L) = (p, q)$, then $L \xrightarrow{\sim} I_{p,q}$.

Type II

For $p, q \in \mathbf{Z}_{\geq 0}$, define

$$\Pi_{p,q} = \Pi_{p,q} = \begin{cases} E_8^{\frac{p-q}{8}} \oplus U^q, & p - q \geq 0 \text{ and } 8|(p - q) \\ E_8^{\frac{q-p}{8}} \oplus U^p, & p - q < 0 \text{ and } 8|(q - p) \end{cases}$$

where E_8 is the negative-definite E_8 lattice defined above and U is the hyperbolic lattice. This is an integral, even, unimodular lattice of rank $p + q$ and signature (p, q) with discriminant $(-1)^p$. We note that G_β^{-1} is generally nontrivial due to the E_8 factors, making the dual basis somewhat difficult to work with by hand. The following can be found in [Mil58; Ser73]: if L is an even indefinite unimodular lattice with $\text{sig}(L) = (p, q)$, then necessarily $\tau \equiv 0 \pmod{8\mathbf{Z}}$ and $L \xrightarrow{\sim} \Pi_{p,q}$.

We conclude with the major classification theorems:

Theorem 2.2.3.1 (Classification of indefinite unimodular lattices). *Any indefinite unimodular lattice is determined up to isometry by its rank, index, and parity. The same is true for definite unimodular lattices L with $\text{rank } L \leq 8$.*

Theorem 2.2.3.2 (Classification of small unimodular lattices). *Let L be any unimodular integral lattice, definite or indefinite, with $\text{rank}_{\mathbf{Z}} L \leq 4$. Then either*

- L is odd and $L \xrightarrow{\sim} I_{p,q}$ for some p, q , or
- L is even and either $L \xrightarrow{\sim} U$ or U^2 .

2.2.4 K3 and Enriques Lattices

The K3 lattice

$$L_{\text{K3}} := U^3 \oplus E_8^2 \xrightarrow{\sim} \text{II}_{3,19},$$

is the unique even unimodular lattice of rank 22 and signature $(3, 19)$ modeling $H^2(X; \mathbf{Z})$ of a K3 surface X . We define

$$L_{\text{K3},2d}^{(m)} := \langle -2d \rangle \oplus U^2 \oplus E_8^m, \quad L_{\text{K3},2d} := L_{\text{K3},2d}^{(2)},$$

the degree $2d$ K3 lattices that appear in the study the moduli spaces F_{2d} of K3 surfaces with a polarization of degree $2d$ and similar moduli problems. The lattice $L_{\text{K3},2d}$, corresponding to the $m = 2$ case, models the orthogonal complement in L_{K3} of a polarization h of degree $2d$. We also recall that the Enriques lattice is defined as

$$L_{\text{En}} := U \oplus E_8 \xrightarrow{\sim} \text{II}_{1,9},$$

the unique even unimodular lattice of rank 10 and signature $(1, 9)$, modeling (the free part of) $H^2(Z; \mathbf{Z})$ for an Enriques surface Z .

2.2.5 Nikulin's Lattices

Let $\mathbf{Z}_2$ denote the 2-adic integers. For $k \geq 0$, define

$$V_k := \left(\mathbf{Z}_2^2, \begin{bmatrix} 2^{k+1} & 2^k \\ 2^k & 2^{k+1} \end{bmatrix} \right)$$

For $k = 1$, we write $V := V_1$. This is an even, nondegenerate lattice of 2-adic rank 2 and signature $(2, 0)$ that we write as V . Similarly, we define

$$U_k := \left(\mathbf{Z}_2^2, \begin{bmatrix} 0 & 2^k \\ 2^k & 0 \end{bmatrix} \right)$$

over the 2-adic integers and write $U := U_1$. These lattices contribute to the indecomposable factors of general 2-elementary lattices, which we discuss in the next section.

2.3 Classification Results

Definition 2.3.0.1 (Genus of a Lattice). Two lattices L_1, L_2 belong to the same **genus** if $L_{1, \mathbf{Z}_p} \cong L_{2, \mathbf{Z}_p}$ for all primes p , where $\mathbf{Z}_p$ denotes the p -adic integers, and $L_{1, \mathbf{R}} \cong L_{2, \mathbf{R}}$. Lattices in the same genus have the same rank, signature, and determinant, but may not be isometric over $\mathbf{Z}$. We define the **class group** $\text{cl}(L)$ to be the set of isometry classes within a genus, and the **class number** as the size of $\text{cl}(L)$. We note that for indefinite even lattices L , one typically expects the class number of L to be one. For definite lattices, the situation is reversed, and having class number one is somewhat rare. By [Sca87, §3.4], if $\text{rank}(L) > 16 + \ell(L)$, then $\text{cl}(L) \geq 2$.

2.3.1 p-Elementary Lattices

A lattice (L, β) is **p-elementary** if its discriminant group A_L is a p -elementary abelian group, i.e., $A_L \cong (\mathbf{Z}/p\mathbf{Z})^a$ for some integer $a \geq 0$. We will be particularly concerned with the case $p = 2$. We call the exponent $a = \text{rank}_{\mathbf{F}_2} A_L$ the **2-rank** of the discriminant group. For a 2-elementary lattice L , let $L^\dagger := L^\vee(2)$. We say L is **co-even** if $L^\dagger$ is even and define the invariant $\delta = 0$, and **co-odd** otherwise and set $\delta = 1$. Equivalently,

$$\delta(L) := \begin{cases} 0 & \text{if } q_{A_L}(x) \in \mathbf{Z} \ \forall x \in A_L \\ 1 & \text{otherwise} \end{cases} \quad (2.1)$$

The following lattices are particularly important to the classification of 2-elementary lattices. We note, however, that Nikulin works over

$\mathbf{Z}_{\widehat{p}}$ and particularly over $\mathbf{Z}_{\widehat{2}}$, and so there are subtleties in choosing integral models that localize correctly.

(L, β)	(A_L, q)	Co-even/Co-odd
$\langle 2 \rangle \cong (\mathbf{Z}, [2])$	$p \cong \mathfrak{q}_1(2) \cong (C_2, [\frac{1}{2}])$	Co-odd
$\langle -2 \rangle \cong (\mathbf{Z}, [-2])$	$q \cong \mathfrak{q}_1(-2) \cong (C_2, [-\frac{1}{2}])$	Co-odd
$U(2) \cong (\mathbf{Z}_2^2, \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix})$	$u \cong \mathfrak{u}(2) \cong (C_2^2, \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix})$	Co-even
$V(2) \cong (\mathbf{Z}_2^2, \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix})$	$v \cong \mathfrak{v}(2) \cong (C_2 \times C_6, \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix})$	Co-even

Let L be an even, indefinite, 2-elementary lattice with $\text{rank}(L) \geq 4$. The classification of such lattices is determined by three invariants:

1. the rank $r = \text{rank}_{\mathbf{Z}}(L)$,
2. the 2-rank $a = \dim_{\mathbf{F}_2}(A_L)$, or equivalently $\ell(L)$, and
3. the coparity $\delta \in \{0, 1\}$ as defined above.

We denote by $(r, a, \delta)_{n_+}$ the 2-elementary lattice with these invariants and signature (n_+, n_-) . For each **admissible** triple (see [Nik80]) (r, a, δ) satisfying $r \geq 4$, there exists a unique isometry class of even, indefinite, 2-elementary lattice with these invariants. Equivalently, any two such lattices are isometric if and only if they have the same signature and isometric discriminant quadratic forms.

2.3.2 Decomposability of 2-elementary lattices

By [AE23b, §2.6] and [Nik79c, Prop. 1.8.1], if L is an even 2-elementary lattice, then A_L can be written as a finite direct sum of discriminant forms p, q, u , and v above, subject to the relations

$$\begin{aligned}
u^2 &= v^2 \\
p^4 &= q^4 \\
u \oplus p &= (p \oplus q) \oplus p \\
u \oplus q &= (p \oplus q) \oplus q \\
v \oplus p &= q^3 \\
v \oplus q &= p^3
\end{aligned}$$

It is interesting to note that if one freely imposes these relations on the free group generated by u, v, p, q , one obtains the group $\mathbf{Z} \times \mathbf{Z}/4\mathbf{Z}$ with generators v and q . However, it is not immediately clear if this coincides with the corresponding Grothendieck group K_0 of the category of such quadratic forms. In any case, the even 2-elementary lattices that admit a primitive embedding into L_{K3} are finite direct sums of the following lattices, whose discriminant forms are recorded as well:

Lattice L	Discriminant Form A_L	Co-even/Co-odd
A_1	$q := \mathbf{q}_1(-2)$	Co-odd
D_4	$v := \mathbf{v}(2)$	Co-even
D_6	$p^2 := \mathbf{q}_1(2)^2$	Co-odd
D_8	$u := \mathbf{u}(2)$	Co-even
E_7	$p := \mathbf{q}_1(2)$	Co-odd
E_8	0	Co-even
$E_8(2)$	$u^4 := \mathbf{u}(2)^4$	Co-even
$\langle 2 \rangle$	$p := \mathbf{q}_1(2)$	Co-odd
U	0	Co-even
$U(2)$	$u := \mathbf{u}(2)$	Co-even

2.3.3 Ordinary and Characteristic

For 2-elementary lattices, one always has $\text{div}_L(v) \in \{1, 2\}$. We set $v^* := v/\text{div}_L(v) \in A_L$. Letting $q_L : A_L \rightarrow \frac{1}{2}\mathbf{Z}/\mathbf{Z}$ be the induced quadratic form on A_L , we say v^* is **characteristic** if $q_L(x) = \beta_L(v^*, x) \bmod \mathbf{Z}$ for all $x \in A_L$, and is **ordinary** otherwise. We say that a primitive isotropic vector $e \in L$ is

- **odd** if $\text{div}_L(v) = 1$,
- **even ordinary** if $\text{div}_L(v) = 2$ and v^* is ordinary, or
- **even characteristic** if $\text{div}_L(v) = 2$ and v^* is characteristic.

Such vectors play a role in the mirror move algorithm of [1], which can be used to determine 0-cusps and 1-cusps of K3 compactifications.

Cite

2.4 Group Actions and Isometries

Define the space of **isometries** between two lattices (L_1, β_1) and (L_2, β_2) as

$$\text{Isom}(L_1, L_2) := \{f \in \text{Hom}_{\text{Lat}_{\mathbf{Z}}}(L_1, L_2) \mid f \text{ is an isomorphism of } \mathbf{Z}\text{-modules}\}$$

Two lattices are **isometric** ($L_1 \cong L_2$) if and only if this set is non-empty.

2.4.1 Reflections and Transvections

Let (L, β) be a nondegenerate integral lattice. For an anisotropic vector $\alpha \in L$ satisfying the condition $2\beta(\alpha, L) \subseteq \beta(\alpha, \alpha)\mathbf{Z}$, the **reflection in α** is the isometry:

$$s_\alpha(x) := x - 2\frac{\beta(\alpha, x)}{\beta(\alpha, \alpha)}\alpha$$

Reflections have determinant -1 and act as the identity on the orthogonal complement $\langle \alpha \rangle^{\perp L}$. The group generated by all such reflections is the **Weyl group** $W(L) \leq \text{O}(L)$. Letting $e \in L_{\mathbf{Q}}$ be an isotropic vector and let $a \in \langle e \rangle^{\perp L_{\mathbf{Q}}}$ be any vector satisfying $a \cdot e = 0$. The **Siegel-Eichler transformation** associated to the pair (e, a) is the isometry $E_{e,a} \in \text{O}(L_{\mathbf{Q}})$ defined by:

$$E_{e,a}(v) = v + \beta(e, v)a - \left(\beta(a, v) + \frac{1}{2}\beta(a, a)\beta(e, v) \right) e$$

If a is also isotropic, this simplifies to

$$E_{e,a}(v) = v + \beta(e, v)a - \beta(a, v)e.$$

The **Eichler transvection group** $E(L)$ is generated by all such transvections where $e \in L$ is a primitive isotropic vector with $\text{div}_L(e) = 1$, and $a \in \langle e \rangle^{\perp L}$. If L is even, then $E(L) \subseteq \text{O}^*(L)$.

2.4.2 Orbits of Isotropic Vectors

For a nondegenerate indefinite lattice L and any integer k , the set of $\text{O}(L)$ -orbits of primitive vectors of norm k , denoted $L[k]/\text{O}(L)$, is finite. In particular, let T be an even lattice of signature $(2, n)$ and

let $\Gamma \subseteq \mathrm{O}(T)$ be an arithmetic subgroup (e.g., a subgroup of finite index like $\mathrm{O}^*(T)$). The 0-cusps of the Baily-Borel compactification of the associated arithmetic quotient $F_\Gamma := D_T/\Gamma$ are in one-to-one correspondence with Γ -orbits of primitive rank 1 isotropic sublattices of T , i.e.

$$\partial_0 \overline{F_\Gamma}^{\mathrm{BB}} := \partial_0 \overline{D_T/\Gamma}^{\mathrm{BB}} \cong L[0]/\Gamma = \left\{ v \in L \mid v \text{ is primitive and } v^2 = 0 \right\} / \Gamma.$$

For $T := T_{\mathrm{En}}$, the appropriate arithmetic group $\Gamma_{\mathrm{En},2}$ has exactly 5 orbits of primitive isotropic vectors, yielding 5 cusps of $\overline{F_{\mathrm{En},2}}^{\mathrm{BB}}$. They are partitioned by divisibility: there is one orbit of divisibility 1 vectors, and 4 orbits of divisibility 2 vectors. We are thus interested in classifying orbits of primitive isotropic vectors. For lattices containing a double hyperbolic plane U^2 , we recount the following useful theorems:

Theorem 2.4.2.1. *There is an isometry*

$$\phi : U \oplus U \rightarrow (\mathrm{Mat}_{2 \times 2}(\mathbf{Z}), 2 \cdot \det).$$

Under ϕ , the group generated by Eichler-Siegel transformations corresponds to the image of $\mathrm{SL}_2(\mathbf{Z}) \times \mathrm{SL}_2(\mathbf{Z})$ acting on a matrix X by $(A, B) \curvearrowright X := AXB^{-1}$.

Proof. Write $L = U \oplus U'$ as two distinct copies of U . The map $\phi : L \rightarrow \mathrm{Mat}_{2 \times 2}(\mathbf{Z})$ defined by

$$ae + df + be' + cf' \mapsto \begin{pmatrix} a & b \\ -c & d \end{pmatrix}$$

is a $\mathbf{Z}$ -module isomorphism. To check that ϕ is an isometry for the quadratic form, observe that

$$(ae + df + be' + cf')^2 = 2ad + 2bc.,$$

which is precisely twice the determinant of the corresponding matrix. The group $\mathrm{SL}_2(\mathbf{Z}) \times \mathrm{SL}_2(\mathbf{Z})$ acts on $\mathrm{Mat}_{2 \times 2}(\mathbf{Z})$ by

$$(A, B) : X \mapsto AXB^{-1},$$

which preserves the determinant and thus the bilinear form. The action of $\mathrm{SL}_2(\mathbf{Z})$ by left multiplication corresponds to elementary row

operations (acting on the basis $\{e, f\}$), and the action on the right corresponds to elementary column operations (acting on the basis $\{e', f'\}$). We conclude using the fact that $SL_2(\mathbf{Z})$ is generated by elementary matrices. $\square$

Theorem 2.4.2.2 ([Sca87, Prop 3.7.3]). *Let L be an even lattice that decomposes as $L \xrightarrow{\sim} U^2 \oplus M$ where M is any lattice, and suppose $v, w \in L[k]$ for some fixed k . Then*

$$v \sim_{E(L)} w \iff [v^*] = [w^*] \in A_L.$$

Sketch. Suppose that $\phi(v) = w$ for some $\phi \in E(L)$. Then necessarily $\phi^*([v^*]) = [w^*]$ in A_L by restriction. However, $E(L) \subseteq O^*(L)$, so ϕ must induce the identity on A_L and thus $[v^*] = [w^*]$. Conversely, suppose that $[v^*] = [w^*]$ in A_L . Write $L = U_1 \oplus U_2 \oplus M$; then by the above proposition we may assume that $v, w \in L' := M \oplus U_1$ up to $E(U_1 \oplus U_2)$. Since $[v^*] = [w^*]$, in particular these elements have the same order in A_L , and so $\text{div}_L(v) = \text{div}_L(w) = d$ is a fixed positive integer. Thus there exist $x, y \in L'$ such that $\beta(x, v) = d$ and $\beta(y, w) = d$. We now claim that $z := \frac{v}{d} - \frac{w}{d} = \frac{v-w}{d}$ is an element of $M \oplus U_1$. This follows from the fact that $v^* \equiv w^* \pmod{L'}$, so $\frac{v}{d} \equiv \frac{w}{d} \pmod{L'}$. Thus $z := \frac{v}{d} - \frac{w}{d} \equiv 0 \pmod{L}$ is an element of L' . Using the fact that $dz^2 = 2vz$, a direct calculation shows that the following composition maps v to w :

$$E_{e,y} \circ E_{f,-z} \circ E_{e,-x}.$$

$\square$

2.4.3 Group Actions

For a faithful representation $G \hookrightarrow O(L)$ realizing a group G acting on L by isometries, the sublattice of **invariants** and **coinvariants** are defined as

$$L^G := \left\{ v \in L \mid g.v = v \ \forall g \in G \right\}, \quad L_G := (L^G)^{\perp L}.$$

If I is an involution ($I^2 = \text{id}$), these correspond to the eigenspaces for eigenvalues $+1$ and -1 , respectively, for the induced action on $L_{\mathbf{Q}}$. More generally, if $G \hookrightarrow O(L)$ is a faithful representation of a

finite cyclic group G , then L^G and L_G are primitive sublattices, and $L^G \oplus L_G$ has finite index in L . If $I \in \mathrm{O}^*(L)$ is an involution and L is 2-elementary, then both L^I and L_I are again 2-elementary lattices.

2.4.4 Applications

Definition 2.4.4.1 (Overlattices and Gluing Theory). An **overlattice** L of a lattice S is a lattice containing S as a finite-index sublattice. Nikulin's theory provides a classification of even overlattices of S in terms of isotropic subgroups of A_S up to $\mathrm{O}(S)$.

2.5 Nikulin's Embedding Theorems

Throughout this section, we adopt some simplifying notational conventions: for any lattice L , writing $\mathrm{sig}(L) = (p, q)$, we define $\tau(L) = p + q$, $\min \mathrm{sig}(L) = \min \{p, q\}$, and $\max \mathrm{sig}(L)$ similarly. Let $S \hookrightarrow L$ be an embedding of lattices. We say L is an **overlattice** of S if $\iota(S)$ is a finite index sublattice. We will be primarily interested in the case where $S \hookrightarrow L$ is a primitive embedding with $T := S^{\perp L} \hookrightarrow L$ its (primively embedded) orthogonal complement. In this situation, L is an overlattice of $S \oplus T$, and we would like to know when it is of index 1. When this happens, we will say that S **splits** L . We first note a basic tool: for any lattice L , there is a canonical primitive embedding

$$\begin{aligned} \delta : L(2) &\rightarrow L \oplus L \\ v &\mapsto (v, v) \end{aligned}$$

2.5.1 Scattone's Methods

Remark 2.5.1.1 (Scattone's Method: Enumerating Boundary Components via Niemeier Lattices). To motivate the detailed study of primitive embeddings, we note that the method of [Sca87] provides a concrete arithmetic approach to enumerating 0-cusps in the Baily–Borel compactification $\overline{F}_{2d}^{\mathrm{BB}}$ of the moduli space of degree- $2d$ polarized K3 surfaces.

By the classification of boundary components for arithmetic quotients of Hermitian symmetric domains ([BB66]), 0-cusps correspond

to Γ_{2d} -orbits of primitive isotropic lines $I \subset L_{2d}$, where

$$L_{K3,2d} := \langle -2d \rangle \oplus U^2 \oplus E_8^2$$

is the rank-21 lattice of signature $(2, 19)$ associated to degree- $2d$ polarized K3 surfaces.

Each such cusp corresponds to a degeneration of K3 surfaces with associated lattice $I^\perp/I$, an even lattice of signature $(1, 18)$, which encodes the limiting Hodge structure for degenerations to that cusp. [Sca87] classified boundary components by studying primitive embeddings $U \hookrightarrow L$ where L is one of the 24 **Niemeier lattices** – the even, negative-definite, unimodular lattices of rank 24. For each such embedding $U \hookrightarrow L$, the orthogonal complement $T := U^\perp_L$ is an even, negative-definite, unimodular lattice of rank 22. These are the possible isometry classes of lattices of the form $I^\perp/I$ at 0-cusps of $\overline{F}_{2d}^{\text{BB}}$. The enumeration of 0-cusps is thus reduced to counting the orbits of primitive embeddings $U \hookrightarrow L$ for each Niemeier lattice L up to $O(L)$, where [Sca87] establishes that each such orbit corresponds to a *distinct* 0-cusp, allowing for an explicit enumeration and thus an understanding of the entire cusp diagram for F_{2d} for a wide range of values of d . From this, we find that the class number $\text{cl}(T)$ directly influences the number of 0-cusps, and representatives of isometry classes can be used to provide an explicit indexing set.

2.5.2 Existence and Uniqueness

Theorem 2.5.2.1 (Nikulin’s Analog of Witt’s Theorem [Nik79a, Thm. 1.14.4]).

Let S be an even lattice and L an even unimodular lattice. Then

1. The weak case:

If $\text{sig}(L) > \text{sig}(S)$ and $\text{rank}(L) - \text{rank}(S) \geq 2 + \ell(A_S)$, then there exists a primitive embedding $S \hookrightarrow L$ that is unique up to isometry.

2. The generic case:

If $\text{sig}(L) > \text{sig}(S)$ and $\text{rank}(L) - \text{rank}(S) \geq 2 + \max_{p \neq 2} \ell(A_{S_p})$, then there exists a primitive embedding $S \hookrightarrow L$ that is unique up to isometry.

3. The exceptional 2-adic case:

If $\text{sig}(L) > \text{sig}(S)$ and $\text{rank}(L) - \text{rank}(S) = \ell(A_{S_2})$, then a

*primitive embedding exists **if and only if** there exists some q' such that q_S satisfies*

$$q_S \cong u_+^{(2)}(2) \oplus q' \quad \text{or} \quad q_S \cong v_+^{(2)}(2) \oplus q'.$$

We note that this is not how [Nik79a] originally states the theorem, but rather extracts the case that is more commonly used in applications for clarity. The generic case is both necessary and sufficient for the existence and uniqueness of a primitive embedding. The weak case, which uses the global invariant $\ell(A_S)$ is easier to check in practice, but only gives a sufficient condition and is thus strictly weaker than the generic case. This is because if A is any finite abelian group, one can consider the primary decomposition $A = \bigoplus_p A_p$, and there is an inequality $\max_p \ell(A_p) \leq \ell(A) \leq \sum_p \ell(A_p)$. Thus, using $\ell(A_S)$ in place of $\max_{p \neq 2} \ell(A_{S_p})$ can exclude embeddings that the generic case would allow.

The weak and generic embedding criteria for Nikulin's theorem coincide if and only if the minimal number of generators of the group equals the maximum of the minimal numbers of generators of its p -primary parts, i.e., $\ell(A) = \max_p \ell(A_p)$. When $\ell(A) > \max_p \ell(A_p)$, the weak criterion requires a larger difference in ranks than the generic criterion, and thus gives only a sufficient (not necessary) condition for embedding. This is best illustrated through concrete examples:

Let $A = \mathbf{Z}_6 \cong \mathbf{Z}_2 \times \mathbf{Z}_3$. Then $A_2 = \mathbf{Z}_2$ with $\ell(A_2) = 1$, and $A_3 = \mathbf{Z}_3$ with $\ell(A_3) = 1$. Since A is cyclic, we have $\ell(A) = 1$, and $\max\{\ell(A_2), \ell(A_3)\} = 1$. The weak and generic criteria coincide, since $\ell(A) = \max\{\ell(A_2), \ell(A_3)\}$.

Let $A = \mathbf{Z}_2 \oplus \mathbf{Z}_3$. Then $A_2 = \mathbf{Z}_2$ with $\ell(A_2) = 1$, $A_3 = \mathbf{Z}_3$ with $\ell(A_3) = 1$, and $\ell(A) = 2$, since both factors are needed to generate A . We have $\max\{\ell(A_2), \ell(A_3)\} = 1$, and the weak criterion now requires rank difference d to satisfy $d \geq 2 + 2$, while the generic criterion only requires $d \geq 2 + 1$. Thus, the weak case is strictly weaker and more restrictive: it may exclude embeddings allowed by the generic criterion. Similarly, let $A = \mathbf{Z}_2 \oplus \mathbf{Z}_2$. Then $A_2 = \mathbf{Z}_2 \oplus \mathbf{Z}_2$ with $\ell(A_2) = 2$, and $\ell(A) = 2$ with $\max\{\ell(A_2)\} = 2$. Here, the weak and generic criteria again coincide. Thus the weak criterion is **always** sufficient, but only necessary when $\ell(A) = \max_p \ell(A_p)$, and the actual difference between the two cases depends highly on the structure of the group in question. When $\ell(A) > \max_p \ell(A_p)$, the weak criterion

is strictly more restrictive and does not capture all cases allowed by the generic condition.

2.5.3 Finiteness

Proposition 2.5.3.1 (Finiteness of embeddings for even, unimodular lattices). *If S and L are even lattices and L is unimodular, then $\text{Emb}(S, L)$ is a finite set.*

Proof. By [Nik79c, Prop. 1.6.1], such a primitive embedding $\iota : S \hookrightarrow L$ is determined by an isometry $\gamma : A_S \xrightarrow{\sim} A_T(-1)$, two such primitive embeddings are equivalent if and only if γ_1 is conjugate to γ_2 under $\text{O}(A_T)$, and $\iota_1(S_1) \xrightarrow{\sim} \iota_2(S_2)$ are equivalent primitive sublattices if $\exists(\phi, \psi) \in \text{O}(S) \oplus \text{O}(T)$ such that $\gamma_1 \circ \phi|_{A_S} = \psi|_{A_T} \circ \gamma_2$. Since A_S, A_T are finite abelian groups, $\text{Isom}(A_S, A_T)$ is a finite set, as is $\text{O}(A_T)$. Moreover, noting that if $S_1 \xrightarrow{\sim} S_2$ then $A_{S_1} \xrightarrow{\sim} A_{S_2}$ and thus $\text{Emb}(S_1, L) \cong \text{Emb}(S_2, L)$, so $\text{Emb}(S, L)$ only depends on the isometry class of S . Since $\text{gen}(S)$ is a finite set, there are only finitely many isometry classes of S , so the class group $\text{cl}(S)$ is finite and thus $\text{Emb}(S, L)$ a finite set. $\square$

2.5.4 Splitting

Remark 2.5.4.1 (Discriminant form and anti-isometry). The discriminant group $A_S = S^\vee/S$ of S carries a finite quadratic form q_S . The structure of the orthogonal complement T and the existence of an overlattice are governed by the existence of an **anti-isometry** between (A_S, q_S) and $(A_T, -q_T)$. This anti-isometry encodes the gluing data for constructing L as an overlattice of $S \oplus T$. The following results collect the relevant facts for this aspect of the theory:

Theorem 2.5.4.2 (Primitive Embedding Theorem [PS24, Prop. 15.1.1]). *Let S be a primitive non-degenerate sublattice of a unimodular lattice L , and let $T = S^{\perp L}$ be its orthogonal complement. Then:*

1. $S \oplus T$ is a sublattice of L of finite index.
2. $[L : S \oplus T] = |A_S| = |A_T|$, where $A_S = S^*/S$ and $A_T = T^*/T$ are the discriminant groups.

3. There is a canonical isomorphism $\psi : A_S \rightarrow A_T$ such that the discriminant forms are related by an **anti-isometry**:

$$q_{A_T}(\psi(x)) = -q_{A_S}(x) \quad \forall x \in A_S.$$

2.5.5 Gluing and Overlattices

Throughout this section, for a discriminant group (G, q) , we write $G(n)$ for the group G with quadratic form $\tilde{q} := nq$. In particular, $G_1 \xrightarrow{\sim} G_2(-1)$ are isometric by a map f if and only if $G_1 \cong G_2$ as groups and $q_1(v) = -q_2(f(v))$ for all $v \in G_1$. Let L be as above; a primitive embedding $S \hookrightarrow L$ with $T := S^{\perp L}$ is uniquely determined by the choice of

1. A subgroup $H \leq A_L$, the *embedding subgroup*, and
2. An isometry $\gamma : H \xrightarrow{\sim} H' \subseteq A_S$, the *embedding isometry*, where H' is the image of H .

Letting Γ be the graph of γ in $A_L \oplus A_S(-1)$, one has $A_T = \Gamma^{\perp}/\Gamma$ and we note that there is a discriminant formula

$$|\text{disc } T| = \frac{|\text{disc } L| \cdot |\text{disc } S|}{(\#H)^2}.$$

Now let $\iota : S \hookrightarrow L$ be an embedding of even lattices where L is unimodular, and define $H_L := L/\iota(S)$. Using the chain of embeddings $S \hookrightarrow L \hookrightarrow L^{\vee} \hookrightarrow S^{\vee}$ to produce embeddings $H_L \hookrightarrow L^{\vee}/S \hookrightarrow A_S$, one can regard H_L as a subgroup of A_S . Conversely, for a subgroup $H \leq A_S$, write $\eta : S^{\vee} \rightarrow A_S$ and define a lattice $S_H := \eta^{-1}(H) \subseteq S^{\vee}$. We note that $S_H \supseteq S$, so S_H is an overlattice of S .

These constructions are mutually inverse and define a bijection:

$$\begin{aligned} \{ \text{Even overlattices } L \text{ of } S \} &\rightleftharpoons \{ \text{Isotropic subgroups } H \leq A_S \} \\ L &\mapsto H_L := L/S \\ L &:= S_H \leftarrow H \end{aligned}$$

We apply this to the following:

Proposition 2.5.5.1. *Let L be a unimodular lattice and $\iota : S \hookrightarrow L$ be a primitively embedded sublattice. Then $|\text{disc}(S)| = |\text{disc}(T)|$, and if S is unimodular, then $L \cong S \oplus T$.*

Proof. We have

$$|\operatorname{disc}(S)| = \sharp A_S = \sharp A_T = |\operatorname{disc}(T)|.$$

The isometry follows from Proposition 1.3.7: since $S \oplus T \leq L$ is a full-rank sublattice, T is also unimodular and thus

$$[L : S \oplus T]^2 = \frac{\operatorname{disc}(S \oplus T)}{\operatorname{disc}(L)} = \frac{\operatorname{disc}(S) \cdot \operatorname{disc}(T)}{\operatorname{disc}(L)} = 1.$$

Finally, a pair of isometries of S and T lifts to an isometry of L if and only if they preserve H_L , or equivalently commute with the glue map. Thus given $S \hookrightarrow L$ as above with $T := S^{\perp L}$, even if S does not split L , we still have a way to construct isometries on L : one first constructs isometries $f_S \in \operatorname{O}(S)$ and $f_T \in \operatorname{O}(T)$ such that the restricted action of f_S to A_S and that of f_T to A_T agree, using the anti-isometry $A_S \xrightarrow{\sim} A_T(-1)$, then produces a lift of $f_S \oplus f_T$ to an element of $f \in \operatorname{O}(L)$ that restricts to both f_S and f_T . In particular, f stabilizes both S and T , and thus defines isometries in the stabilizers $\operatorname{Stab}_{\operatorname{O}(L)}(S)$ and $\operatorname{Stab}_{\operatorname{O}(L)}(T)$. □

2.5.6 Applications to $F_{\operatorname{En},2}$

To see some of this theory applied to the moduli problem at hand, we take a small detour to prove that $F_{\operatorname{En},2}$ is the normalization of a closed subvariety of $F_{(2,2,0)}$ – an essential ingredient in the main theorem. The strategy is as follows:

1. Identify a morphism $\Psi : F_{\operatorname{En},2} \rightarrow F_{(2,2,0)}$ arising from a lattice embedding $\tilde{\Psi} : T_{\operatorname{En}} \hookrightarrow T_{\operatorname{dP}}$.
2. Establish a rigidity theorem at the level of lattice embeddings to assert that Ψ is well-defined and canonically determined.
3. Restrict Ψ to its scheme-theoretic image, i.e. the smallest closed subscheme of $F_{(2,2,0)}$ through which Ψ factors, to obtain $\Psi : F_{\operatorname{En},2} \rightarrow X$.
4. Since $F_{\operatorname{En},2}$ is known to be normal by the general theory of [BB66], we then appeal to Zariski's main theorem: since X is a closed subscheme of a normal variety, if Ψ is finite and birational, it satisfies the universal property of normalization.

We first claim there is a holomorphic, algebraic morphism of period domains

$$\tilde{\Psi} : D_{T_{\text{En}}} \rightarrow D_{T_{\text{dP}}}$$

This follows from defining an embedding of lattices by

$$\begin{aligned} \tilde{\Psi} : T_{\text{En}} &:= U \oplus U(2) \oplus E_8(2) \hookrightarrow T_{\text{dP}} := U \oplus U(2) \oplus E_8^2 \\ (u_1, u_2, v) &\mapsto (u_1, u_2, v, v) \end{aligned}$$

which in block form is $(\text{id}, \text{id}, \delta)$ where δ is the canonical doubling embedding. To see that this induces a well-defined morphism after passing from lattices to period domains, note that its construction is functorial: it involves tensoring to $\mathbf{C}$, projectivizing, restricting to a quadric, and then further restricting to a semialgebraic subset in both the source and the target, cut out by precisely the same conditions. The map $\tilde{\Psi}$ is holomorphic because it is the restriction of a linear map to an open set, and algebraic (and hence a morphism) because both varieties are quasiprojective and $\tilde{\Psi}$ is linear.

We now claim this morphism is well-defined and canonical in the following sense: let $A_i \rightarrow B_i \rightarrow C_i$ for $i = 1, 2$ be two sequences of primitive embeddings. We say two such sequences are **equivalent** if there exist isometries making the following diagram commute:

$$\begin{array}{ccccc} A_1 & \xrightarrow{f_1} & B_1 & \xrightarrow{g_1} & C_1 \\ \simeq \downarrow \phi_A & & \simeq \downarrow \phi_B & & \simeq \downarrow \phi_C \\ A_2 & \xrightarrow{f_2} & B_2 & \xrightarrow{g_2} & C_2 \end{array}$$

In particular, if $C_1 = C_2 = L$ is a fixed lattice, then $\phi_C \in \text{O}(L)$ and we say the sequences are *equivalent up to* $\text{O}(L)$. If $A_1 = A_2 = A$ and $B_1 = B_2 = B$ are also fixed and any two such sequences $A \rightarrow B \rightarrow L$ are equivalent, we say the sequence is **unique up to isometry** or **unique up to** $\text{O}(L)$. We observe several useful facts:

- $A \hookrightarrow B \hookrightarrow L$ is unique up to $\text{O}(L)$ if and only if $A^{\perp L} \hookrightarrow B^{\perp L} \hookrightarrow L$ is,
- $A \hookrightarrow B \hookrightarrow L$ is unique if $B \hookrightarrow L$ is unique and $A \hookrightarrow B$ is unique,

- $A(n) \hookrightarrow B(n)$ is unique if $A \hookrightarrow B$ is unique

The following is proved as [Ale+23, Lem. 2.4]:

Proposition 2.5.6.1. *The following sequence of primitive embeddings is unique up to isometry:*

$$\tilde{\Psi} : T_{\text{En}} \hookrightarrow T_{\text{dP}} \hookrightarrow L_{K3}.$$

Proof. Passing to orthogonal complements in L_{K3} yields the sequence

$$(S_{\text{En}} \hookrightarrow S_{\text{dP}} \hookrightarrow L) = (U(2) \hookrightarrow U(2) \oplus E_8(2) \hookrightarrow L).$$

We first claim $U(2) \hookrightarrow U(2) \oplus E_8(2)$ is unique. By untwisting, it suffices to show that $U \hookrightarrow U \oplus E_8$ is unique. Write $U = S = \text{II}_{1,1}$ and $U \oplus E_8 = L = \text{II}_{1,9}$, noting that both are the unique even unimodular lattices with those signatures. The existence of an embedding $S \hookrightarrow L$, is clear, since one can simply take $x \mapsto (x, 0)$ and check that the cokernel is isometric to E_8 and thus free. For uniqueness, let $T := S^{\perp L}$: then T is an even unimodular lattice of signature $(0, 8)$, and thus isometric to $E_8 = \text{II}_{0,8}$, which is unique up to isometry. So if $j_i : S_i \hookrightarrow L$ are any two primitive embeddings, there are decompositions $L \cong S_1 \oplus T_1$ and $L \cong S_2 \oplus T_2$ where $S_1 \cong S_2 \cong \text{II}_{1,1}$ and $T_1 \cong T_2 \cong \text{II}_{0,8}$ are both unique up to isometry. So there exist isometries $\phi_S : S_1 \rightarrow S_2$ and $\phi_T : T_1 \rightarrow T_2$, and thus an isometry $\phi_S \oplus \phi_T : S_1 \oplus T_1 \rightarrow S_2 \oplus T_2$. Since S_i, T_i are unimodular, ϕ_S and ϕ_T trivially act identically on the discriminant groups, and thus lift to an isometry $\phi \in \text{O}(L)$. So j_1 is equivalent to j_2 as an embedding.

We now claim that the second embedding $U(2) \oplus E_8(2) \hookrightarrow L_{K3} = U^2 \oplus E_8^3$ is unique. This follows from Nikulin's version of Witt's Theorem 2.5.2.1:

1. $\text{sig}(L_{K3}) = (3, 19) > \text{sig}(U(2) \oplus E_8(2)) = (1, 9)$,
2. $\text{rank}(L_{K3}) - \text{rank}(U(2) \oplus E_8(2)) = 22 - 10 \geq 2 + \ell(A_S) = 2 + 10$.

We conclude by the observations above. □

Proposition 2.5.6.2. *The map $\tilde{\Psi}$ descends to a well-defined algebraic morphism on arithmetic quotients:*

$$\Psi : F_{\text{En},2} = D_{T_{\text{En}}} / \Gamma_{\text{En},2} \rightarrow F_{(2,2,0)} = D_{T_{\text{dP}}} / \text{O}(T_{\text{dP}}).$$

Proof. It suffices to show that every isometry of T_{En} extends to an isometry of T_{dP} preserving T_{En} . This follows from a standard lattice-theoretic argument involving discriminant groups: let $S = T_{\text{En}} = U \oplus U(2) \oplus E_8(2)$ and $T = S_{\text{En}} = U(2) \oplus E_8(2) = S^{\perp L}$ where $L = L_{\text{K3}}$. We note that

$$A_S = A_{U(2)} \oplus A_{E_8(\textcircled{a})} = C_2^2 \oplus C_2^8 \cong C_2^{10} \cong A_T,$$

and that S and T are both even indefinite 2-elementary lattices. By [Nik79a, Thm. 3.6.3], the restrictions $O(S) \rightarrow O(A_S)$ and $O(T) \rightarrow O(A_T)$ are surjective, and thus of $f \in O(S)$, using the fact that $A_S \cong A_T$, the restricted isometry f_{A_S} induces an isometry f_{A_T} on A_T , which can be lifted to an isometry f_T on T . By construction, f_T and f_S act identically on A_S and A_T , and so lift to an isometry $f \in O(L)$ preserving both S and T . $\square$

Lemma 2.5.6.3 ([Ale+23, Lemma 2.8]). *There exists a closed subscheme $X \subset F_{(2,2,0)}$ such that $F_{\text{En},2}$ is canonically isomorphic to the normalization of X .*

Proof. The result follows by restricting the period morphism Ψ to its scheme-theoretic image X and replacing it by $\Psi : F_{\text{En},2} \rightarrow X$. The morphism Ψ is finite and birational. Birationality is established by the fact that Ψ is an open immersion over the locus of smooth, generic Enriques surfaces, and thus is birational onto its image. Finiteness holds since Ψ is proper and quasi-finite. Both $F_{\text{En},2}$ and $F_{(2,2,0)}$ are normal since they are complex analytic manifolds, following [BB66]. The closed subscheme X inherits normality as a subscheme of a normal variety.

By Zariski's Main Theorem, a finite birational morphism from a normal variety to an integral variety identifies the source with the normalization of the target. Therefore, $\Psi : F_{\text{En},2} \rightarrow X$ exhibits $F_{\text{En},2}$ as the normalization of X . $\square$

2.6 Coxeter-Vinberg Theory

2.6.1 Cusp Diagrams

Let T be an even nondegenerate lattice of signature $(2, n)$ and $\Gamma \leq \mathrm{O}^*(T)$ a finite index arithmetic subgroup. As we have seen and will explore in more detail in Chapter 4, the quotient $F_\Gamma := D_T/\Gamma$ admits a Baily–Borel compactification $\overline{F}_\Gamma^{\mathrm{BB}}$. The rational boundary components (or **cusps**) of $\partial\overline{F}_\Gamma^{\mathrm{BB}}$ are indexed by Γ -orbits of isotropic sublattices of L . In particular, **0-cusps** correspond to orbits of isotropic lines $\eta \subset L$, and **1-cusps** correspond to orbits of isotropic planes $I \subset L$. We can thus often combinatorially understand a compactification by first constructing the following:

Definition 2.6.1.1 (Cusp Diagrams). Let X be a complex analytic space equipped with a stratification $\partial X = \bigsqcup_{i \geq 0} \partial_i X$ by boundary strata indexed by codimension. The **cusp diagram** of X is the directed graph whose vertices index the irreducible components of $\partial_i X$, with a directed edge $e_{i \rightarrow j}$ corresponding to components V_i and V_j whenever V_j lies in the Zariski closure of V_i .

For a given 0-cusp defined by I , the orthogonal complement modulo I , which we refer to as the *boundary lattice* $\overline{T}_I := I^\perp/I$, is an even lattice of signature $(1, n-1)$ and admits the structure of a hyperbolic lattice. By studying reflective subgroups of $\mathrm{O}(T)$, one can determine 1-cusps adjacent $[I]$ using the Coxeter diagram associated to $W(M)$. As explained in e.g. [Sca87] and [Ste91], the 1-cusps adjacent to a given 0-cusp $[I]$ in the Baily–Borel compactification correspond bijectively to Γ -orbits of isotropic planes J such that $I \subset J$, or equivalently, to the set of **maximal parabolic subdiagrams** of the Coxeter diagram associated to the reflection group of the hyperbolic lattice $M := I^\perp/I$. Each such subdiagram determines a codimension-one face of the Coxeter polytope in $\mathbb{H}_M$, and hence a distinct Γ_I orbit of a 1-cusp, where Γ_I is the stabilizer of $[I]$ in Γ . We use this description in [Ale+23] to determine the boundary stratification of $\overline{F}_{\mathrm{En},2}$ using semitoroidal data, which in turn comes from Coxeter-theoretic data ranging over the 0-cusps of $\overline{F}_{\mathrm{En},2}^{\mathrm{BB}}$. We then reduce the analysis of $\partial\overline{F}_{\mathrm{En},2}$ to a combinatorial study of the poset of elliptic and parabolic subdiagrams of Coxeter diagrams that are “folded” from those of $F_{(2,2,0)}$. This yields integral affine structures and ultimately dlt models of stable degenerations of Enriques surfaces.

2.6.2 Root Systems and Weyl Groups

Definition 2.6.2.1 (Root System and k -Roots). Let L be a nondegenerate integral lattice and let $v \in L$ be a primitive vector. We recall that the reflection in v is defined by the formula

$$s_v(x) := x - 2 \frac{x \cdot v}{v^2} v, \quad s_v \in O(L_{\mathbf{Q}}),$$

which is generally only a rational isometry and not necessarily integral, and only well-defined if v is not isotropic. We note that a sufficient condition for integrality is that $2(v, L) \subseteq k\mathbf{Z}$. We note that this is precisely the formula for a reflection about the hyperplane $H_v := v^{\perp L}$, which we refer to as the **mirror** or **wall** associated to v . We define the **k -Weyl group** $W^k(L)$ as the group of reflections s_v such that $s_v \in O(L)$ is integral and $v^2 = -k$. More generally, we define the **full Weyl group** $W(L)$ as $\bigcup_k W^k(L)$, the group generated by integral reflections in vectors of any norm. We say v is a **k -root** if $s_v \in W^k(L)$, and more generally that v is simply a **root** if $s_v \in W(L)$, so its mirror supports a well-defined integral reflection isometry. For 2-elementary lattices L , roots are precisely those vectors $v \in L$ with either

$$v^2 = -2, \quad \text{or} \quad v^2 = -4, \text{div}_L(v) = 2$$

In light of this, we call roots of norm ± 2 **short roots**, and roots of norm ± 4 **long roots**. We say L is a **root lattice** if it is generated by a set of roots. We call any group generated by roots a **reflection group**.

Definition 2.6.2.2 (Discriminant Locus and Weyl Chambers). The **discriminant locus** is the union of all mirrors,

$$\Delta(L) := \bigcup_{v \in \Phi_2(L)} v^{\perp L}.$$

When $L \hookrightarrow L_{K3}$ is a primitively embedded lattice corresponding to a polarization h on a K3 surface X , this locus corresponds to points in the corresponding period domain D_T where X acquires extra algebraic cycles – if one identifies the holomorphic 2-form ω on X with its period point in D_L , the condition that ω on a mirror H_α is precisely that $\omega \cdot \alpha = 0$, so $\alpha \in \omega^{\perp H^2(X; \mathbf{C})}$ and thus $\omega \in H^{1,1}(X)$, which by the Lefschetz $(1, 1)$ -theorem for (-2) -curves on K3 surfaces makes α a Hodge class. The **Weyl chambers** are the connected components of $\mathfrak{C}_L^\circ := \mathfrak{C}_L \setminus \Delta(L)$ where $\mathfrak{C}_L$ is the cone of positive-norm vectors in L .

A **simple system** is a set of roots v such that the mirrors H_v form the bounding hyperplanes for a fundamental domain for the action of $W(L)$ on L .

2.6.3 Coxeter Groups and Polytopes

A **Coxeter group** is a group S with presentation

$$S = \left\langle s_1, \dots, s_n \mid s_i^2 = (s_i s_j)^{m_{ij}} = e, m_{ii} = 1 \forall i, m_{ij} = m_{ji} \geq 2 \forall i \neq j \right\rangle.$$

The **Coxeter diagram** G_S of S is a labeled graph whose vertices correspond to generators s_i , with edges between s_i and s_j for $i < j$ and $m_{ij} \geq 3$. We label the edges with m_{ij} when $m_{ij} \geq 3$. A general Coxeter group S encodes a bounding set of hyperplanes for a polytope P_S in a space $X = \mathbb{E}^n, S^n$, or $\mathbb{H}^n$, where $\mathbb{E}^n$ is the standard Euclidean space, S^n is the sphere, and $\mathbb{H}^n$ is hyperbolic n -space. The elements $s_i \in S$ encode reflections about hyperplanes H_i , and the orders m_{ij} encode the dihedral angles between H_i and H_j . The reflections H_i generate a reflection group W_S , and P_S is a fundamental domain for the action of W_S on X .

Let P be a polytope in a space $X = \mathbb{E}^n, S^n$, or $\mathbb{H}^n$ with finitely many faces. We say P is a **Coxeter polytope** if the following hold:

1. Each face of P lies in a hyperplane H_i that is the orthogonal complement of a vector α_i ,
2. The dihedral angles between adjacent faces are of the form π/m_{ij} for positive integers $m_{ij} \geq 1$, and
3. The reflection group $W = \langle s_{\alpha_i} \rangle$ generated by reflections about the faces of P acts properly discontinuously on X with fundamental domain P .

We often identify a Coxeter polytope P with a minimal set of bounding hyperplanes $\{H_i\}$ such that P is the intersection of the positive half-spaces of all of the H_i . We say P is **Euclidean**, **spherical**, or **hyperbolic** according to whether $X = \mathbb{E}^n, S^n$, or $\mathbb{H}^n$ respectively.

2.6.4 Hyperbolic Lattices and Polytopes

Let $\mathfrak{C}_L$ be the cone of positive norm vectors in $L_{\mathbf{R}}$, i.e. $\mathfrak{C}_L := \left\{ v \in L_{\mathbf{R}} \mid v^2 > 0 \right\}$.

There is a decomposition $\mathfrak{C}_L = \mathfrak{C}_L^+ \amalg \mathfrak{C}_L^-$, where

$$\mathfrak{C}_L^+ := \left\{ v \in \mathfrak{C}_L \mid v_0 > 0 \right\}, \quad \mathfrak{C}_L^- := \left\{ v \in \mathfrak{C}_L \mid v_0 < 0 \right\},$$

and note that we can equivalently characterize $\mathfrak{C}_L^+$ as those vectors $v \in L_{\mathbf{R}}$ such that $v \cdot h > 0$ an ordered basis where h is the first generator. Recall that a lattice L is called **hyperbolic** or **Lorentzian** if it is nondegenerate and has signature $(1, n+1)$ for $n \geq 1$, following the standard algebro-geometric convention. The standard example is the hyperbolic lattice U . The intersection pairing on L induces a hyperbolic metric $\rho(v, w) := \operatorname{arccosh}(v \cdot w)$ on $L_{\mathbf{R}}$. The group of orientation-preserving isometries of $\mathbb{H}_L^n$ is:

$$\operatorname{Isom}^+(\mathbb{H}_L^n) \cong \operatorname{O}^*(L_{\mathbf{R}}) := \operatorname{Stab}_{\operatorname{O}(L_{\mathbf{R}})}(\mathfrak{C}_L^+),$$

Moreover, the associated hyperbolic space $\mathbb{H}_L^n$ admits several standard models:

- The **unit hyperboloid model**:

$$\mathbb{H}_L^n := \left\{ v \in L_{\mathbf{R}} \mid v^2 = 1, v \in \mathfrak{C}_L^+ \right\},$$

- The **projective or Klein model**:

$$\mathbf{B}_L^n := \mathbf{P}(\mathfrak{C}_L) \subset \mathbf{P}(L_{\mathbf{R}}),$$

For a hyperbolic lattice L , consider the Poincaré ball model $\mathbf{B}_L^n$. For vectors $v, w \in L$ and their associated hyperplanes $H_v := v^\perp$ and $H_w := w^\perp$, the dihedral angle $\angle(H_v, H_w)$ between these hyperplanes is determined by vw . When $|vw| < 1$, the hyperplanes intersect in the interior of the ball and satisfy $-vw = \cos(\angle(H_v, H_w))$ where $\angle(H_v, H_w)$ denotes the dihedral angle. When $|vw| = 1$, the hyperplanes are *asymptotically parallel*, meeting at an *ideal point* in $\partial\mathbb{H}^n$. When $|vw| > 1$, the hyperplanes are *ultra-parallel* and satisfy $-vw = \cosh(\rho(H_v, H_w))$ where $\rho(H_v, H_w)$ denotes the hyperbolic distance between them.

2.6.5 Coxeter-Vinberg Diagrams

Let $v \in L$ be a simple root, the reflection s_v is in $O(L)$ precisely when $2(v, L)/v^2 \subseteq \mathbf{Z}$. Thus if L is 2-elementary, the roots of L are primitive vectors $v \in L$ such that either $v^2 = -2$, or $v^2 = -4$ and $\text{div}_L(v) = 2$. The mirrors $H_v := v^\perp \subset L_{\mathbf{R}}$ for $v \in \Phi(L)$ define a hyperplane arrangement. The intersection of the positive cone $\mathfrak{C}_L^+$ with the complement of all mirrors,

$$\mathfrak{C}_L^\circ := \mathfrak{C}_L^+ \setminus \bigcup_{v \in \Phi(L)} H_v,$$

decomposes into connected components called **Weyl chambers**, each of which is a fundamental domain P for the action of the Weyl group $W(L)$ on $\mathfrak{C}_L^+$. This is a hyperbolic Coxeter polytope, and thus the Gram matrix corresponding to its walls encodes a Coxeter diagram. Let P be a hyperbolic Coxeter polytope arising from a root system in L as above. Its **Coxeter diagram** is the colored undirected graph whose vertices correspond to simple roots r_i with $r_i^2 < 0$ and whose edges $e_{i,j}$ encode the angle $\angle(H_i, H_j)$ between $H_i := r_i^\perp$ and $H_j := r_j^\perp$ by the formula

$$g_{i,j} := \frac{r_i \cdot r_j}{\sqrt{r_i^2 r_j^2}} = \cos \left(\frac{\pi}{m_{i,j}} \right).$$

We let $e_{i,j}$ be

- an omitted edge if $m_{i,j} \leq 2$,
- a multiple edge if $3 \leq m_{i,j} < \infty$, so $H_i \cap H_j$ is in the interior of $\mathbb{H}^n$ and $g_{i,j} < 1$,
- a thick/bold edge if $H_i \cap H_j \in \partial \mathbb{H}^n$ and $g_{i,j} = 1$,
- dotted edge if the H_i, H_j do not intersect in the closure $\overline{\mathbb{H}^n}$ and $g_{i,j} > 1$.

In the 2-elementary setting, we additionally assign white vertices to Φ^2 and black vertices to Φ^4 . We refer to [Vin72; Vin75] for thorough treatments.

2.6.6 Elliptic/Parabolic Subdiagrams

Let G be the Coxeter diagram, for example corresponding to a Coxeter group or a Coxeter polytope arising from a reflection group. A subdiagram $D \subset G$ is called **elliptic** if the Gram matrix $(v_i \cdot v_j)$ is positive definite. Such a subdiagram defines a finite face of the Coxeter polytope. A subdiagram is **parabolic** if the Gram matrix is positive semidefinite of corank one. A parabolic subdiagram $D \subset G$ is **maximal** if it is not properly contained in a larger parabolic subdiagram. Such diagrams correspond bijectively to rank-2 isotropic subspaces $J \subset L$ with $I \subset J$, i.e., to 1-cusps adjacent to $[I]$, modulo Γ . The set of maximal parabolic subdiagrams of G thus indexes the 1-cusps contained in the closure of the 0-cusp corresponding to $[I]$. The incidence relations among boundary strata follow from containments among these diagrams.

2.6.7 Computing Cusps

Putting this together, we thus obtain a general, effective method for indexing rational boundary components for both the Baily–Borel and semitoroidal compactifications of modular varieties D/Γ of orthogonal type corresponding to lattices T of signature $(2, n)$:

1. Compute the 0-cusps of D/Γ by any means possible, yielding (orbits of) primitive isotropic vectors $\eta_i \in T$.
2. For each i , compute the lattice $\bar{T}_\eta := \eta_i^\perp$, which is of signature $(1, n-1)$ and thus hyperbolic.
3. Compute the “minimal stabilizing” subgroup Γ_{η_i} of η_i in Γ determined by the exact sequence

$$1 \rightarrow U_{\eta_i} \rightarrow \text{Stab}_\Gamma(\eta_i) \rightarrow \Gamma_{\eta_i} \rightarrow 1,$$

where U_{η_i} is the (maximal) unipotent subgroup acting trivially on $\bar{T}_\eta$.

4. Compute the reflection subgroups $W(\Gamma_{\eta_i})$, which are Coxeter groups with Coxeter-Vinberg diagrams $G(\bar{T}_\eta)$.
5. Enumerate orbits of maximal elliptic and parabolic subdiagrams of $G(\bar{T}_\eta)$ under diagram automorphisms to determine the the

face poset of the Coxeter polytope $P(\overline{T}_\eta)$, which corresponds bijectively to the incidence poset of $\partial\overline{D}/\Gamma^{\text{BB}}$.

Toward carrying out the program proposed above, let $\Gamma \subset O(L)$ be an arithmetic reflection subgroup generated by reflections in simple roots. The following algorithm due to [Vin75] constructs a fundamental chamber $P \subset \mathbb{H}^n$:

1. Choice of control vector:

Start with a generic positive-norm rational vector $v_0 \in \mathfrak{C}_L^+$, the **control vector**. The algorithm will construct the walls of the chamber P of $W(L)$ containing v_0 . Write $\tilde{P}$ for the cone formed by the intersection of the positive half-spaces of all currently obtained walls.

2. Enumerating roots:

Enumerate the set of all primitive vectors $v \in L$ such that $v^2 < 0$, $s_v \in \Gamma$, and the hyperplane H_v intersects $\tilde{P}$, to define a face of P containing v_0 . These vectors will define the simple roots of L .

3. Ordering/pruning:

List the simple roots in order of increasing *height* $h(v) := -2(v_0 \cdot v)/v^2$. For each candidate simple root v , check whether the associated reflection s_v defines a new bounding wall of P , i.e., whether H_v intersects $\tilde{P}$ transversely.

4. Constructing the diagram:

Whenever $\tilde{P}$ becomes a convex polytope, the Gram matrix $(v_i \cdot v_j)$ determines the angles between faces and the Coxeter diagram of P .

The fundamental polytope $P \subset \mathbb{H}_M^{n-1}$ constructed via Vinberg's algorithm is a rational polyhedral cone, possibly with compact and ideal faces. The algorithm terminates if and only if Γ has finite co-volume. This is not typically easy to determine a priori, and so it is difficult to tell if or when the algorithm terminates. For lattices of high rank, the search space grows extremely quickly, making this an ineffective strategy if applied naively. In practice, one often leverages specific properties of a given lattice to more intelligently enumerate the search space.

2.7 Lattice Folding

2.7.1 Examples and Conventions around Dynkin Diagrams

We quickly recall constructions of root systems of small rank, which easily generalize to larger rank but allow for explicit calculations. For each such lattice L , we recall the classical embeddings $L \hookrightarrow \mathbb{E}_L$ into Euclidean spaces, describe the simple roots $\Phi(L) = \{\alpha_i\}$ in coordinates in $\mathbb{E}_L$ and the Gram matrix G_L for $\Phi(L)$ in terms of the α_i by carrying out computations using Euclidean coordinates within $\mathbb{E}_L$. For this section, we explicitly write root lattices in the **positive-definite** convention, since we are working with roots in $\mathbb{E}_L$ directly. For the remainder of this work, we implicitly take the negative-definite variants; although this can create confusion, but makes the computations easily verifiable using linear algebra and the standard positive-definite Euclidean pairing on $\mathbf{Z}^n$. Thus throughout this section, if R is a root-lattice, it is assumed to be positive-definite, with negative-definite twist $R(-1)$. For explicit conventions, we refer to Appendix A.0.2.

2.7.2 Invariant and Coinvariant Sublattices

Let L be an even nondegenerate lattice containing a root system $\Phi \subset L$, and let $G \subset O(L)$ be a finite group of isometries preserving Φ and let L^G, L_G be the invariant and coinvariant sublattices respectively. Both are primitive and orthogonal in L . Suppose $G = \langle I \rangle \cong \mathbf{Z}/2\mathbf{Z}$ is generated by an involution I . Each $v \in L$ then decomposes orthogonally, up to a factor of 2, as:

$$2 \cdot v = v_+ + v_-, \quad v_+ = v + I(v) \in L^G, \quad v_- = v - I(v) \in L_G.$$

2.7.3 Folded Root Systems

Given a simple root $\alpha_i \in \Phi \subseteq L$, let $[\alpha_i] \subset \Phi$ denote its G -orbit. The associated **folded root** and **folded root system** are defined by

$$\beta_{[\alpha_i]} := \sum_{\alpha_j \in [\alpha_i]} \alpha_j \in L^G, \quad \Phi^G := \left\{ \beta_{[\alpha_i]} \in L^G \mid [\alpha_i] \in \Phi/G \right\}.$$

Each folded root, corresponding to a folded node in the corresponding Dynkin diagram, thus corresponds to a G -orbit of roots in the ambient

lattice L . For involutions, this simplifies to

$$\beta_{[\alpha_i]} := \alpha_i + I(\alpha_i) \in L^G.$$

If I fixes α_i , then $\beta_{[\alpha_i]} = 2\alpha_i$; otherwise, $\beta_{[\alpha_i]}$ has a different norm, thereby affecting the multiplicity of edges in the corresponding Dynkin diagram.

2.7.4 Examples of Classical Foldings

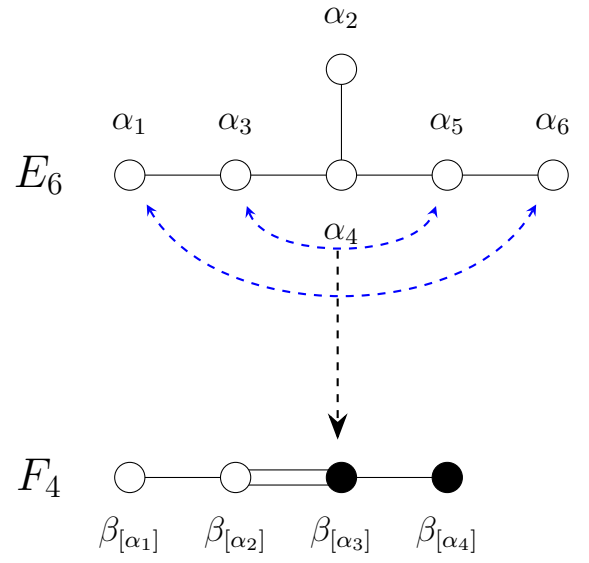
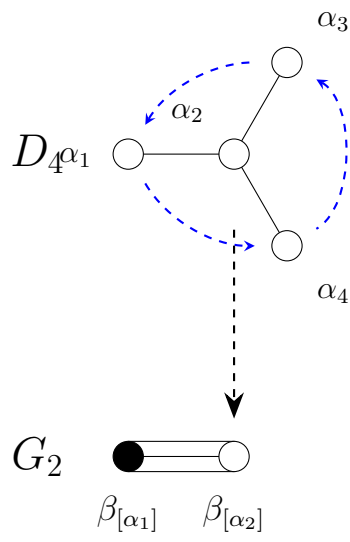
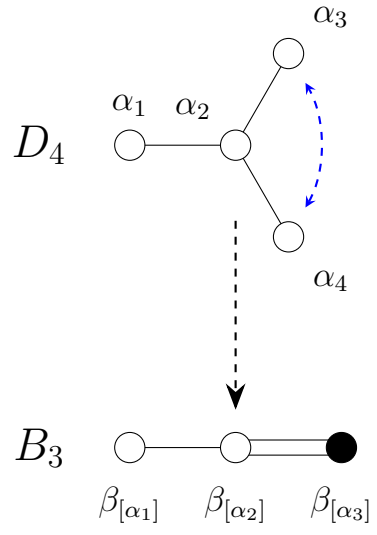
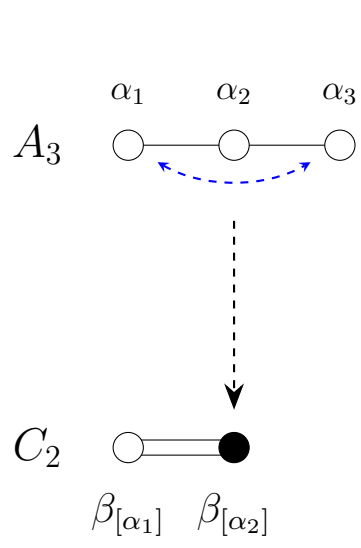
The process of folding nontrivial diagram automorphisms produces non-simply-laced root systems from simply-laced diagrams; the key condition is that roots in the same orbit under the folding group must be orthogonal for a valid quotient. We note that this is easy to check at the level of diagrams – noting that folding takes place at the level of lattices, i.e. computing $\Phi(L^G)$, such an automorphism group G can arise from any representation $G \rightarrow \text{Sym}(G_{\Phi(L)})$ on the Coxeter diagram of $\Phi(L)$. This uses the fact that for the lattices T that occur in our study, there is a decomposition $O(L) \cong W(L) \rtimes P$ where $P \leq \text{Sym}(G_{\Phi(L)})$ is some subgroup of diagram symmetries, which are simply graph isomorphisms that preserve vertex and edge weights. In good cases, P is the entire automorphism group, and so any representation $G \rightarrow \text{Sym}(G_{\Phi(L)})$ can be lifted along the inclusion $P \hookrightarrow O(L)$ to obtain a representation $G \rightarrow O(L)$, allowing one to apply the folding theory described above. In particular, this property holds for the classical root lattices, and we can thus construct foldings by selecting subgroups G of automorphisms of their Coxeter-Dynkin diagrams. Classical examples in the context of semisimple Lie algebras include:

Table 2.7.1: Foldings of simply-laced Dynkin diagrams

Folding	Symmetry Group	Mechanism
$A_{2n-1} \rightarrow C_n$	S_2	Horizontal reflection
$D_{n+1} \rightarrow B_n$	S_2	Vertical reflection
$D_4 \rightarrow G_2$	S_3	Rotation by $2\pi/3$
$E_6 \rightarrow F_4$	S_2	Horizontal reflection

In the study of $F_{\text{En},2}$, the folding procedure must be carried out more carefully at the level of lattices, explicitly tracking the explicit

twists that folding introduces, along with other parity-related data necessary for distinguishing irreducible components of the associated KSBA stable models, which we describe in the section on ADE surfaces. In order to make this theory concrete, we now show how to explicitly carry out the types of folding calculations that occur for $F_{\text{En},2}$, as well as for some of the exceptional types that do not occur in this specific case, noting the slightly new features that arise (like tracking twists) arise, causing some slight differences in conventions from the presentations of folding given in classical literature.



Case 1: $A_5 \rightarrow C_3(2)$ (**Horizontal Reflection**, $G = S_2$)

Let $L = A_5$, where we recall the simple roots and Gram matrix:

$$\Phi(A_5): \begin{cases} \alpha_1 = e_1 - e_2 \\ \alpha_2 = e_2 - e_3 \\ \alpha_3 = e_3 - e_4 \\ \alpha_4 = e_4 - e_5 \\ \alpha_5 = e_5 - e_6 \end{cases}, \quad G_{A_5} = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

Define $G = \langle I \rangle \cong S_2$ by

$$I: \begin{cases} \alpha_1 \mapsto \alpha_5 \\ \alpha_2 \mapsto \alpha_4 \\ \alpha_3 \mapsto \alpha_3 \\ \alpha_4 \mapsto \alpha_2 \\ \alpha_5 \mapsto \alpha_1, \end{cases} \implies I = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The invariant sublattice is

$$A_5^G \cong \{(x_1, x_2, x_3, x_3, x_2, x_1) \in \mathbf{Z}^6 : x_1 + x_2 + x_3 + x_3 + x_2 + x_1 = 0\}$$

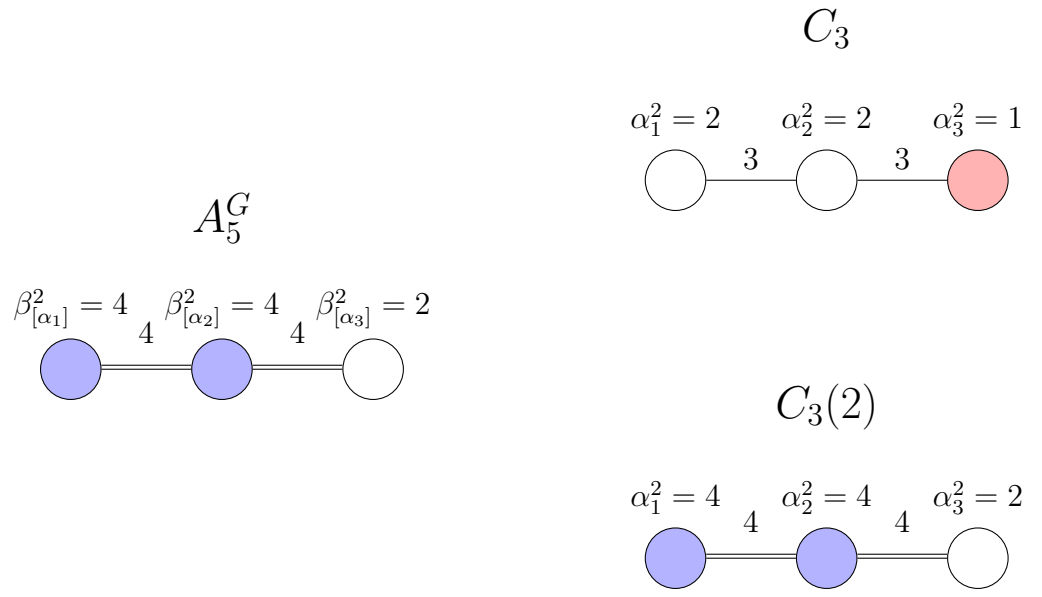
with orbits

$$G \cdot \alpha_i : \begin{cases} G \cdot \alpha_1 = \{\alpha_1, \alpha_5\} \\ G \cdot \alpha_2 = \{\alpha_2, \alpha_4\} \\ G \cdot \alpha_3 = \{\alpha_3\} \end{cases} \implies \Phi(A_5^G): \begin{cases} \beta_{[\alpha_1]} = \alpha_1 + \alpha_5 = e_1 - e_2 + e_5 - e_6 \\ \beta_{[\alpha_2]} = \alpha_2 + \alpha_4 = e_2 - e_3 + e_4 - e_5 \\ \beta_{[\alpha_3]} = \alpha_3 = e_3 - e_4, \end{cases}$$

and thus

$$G_{\Phi(A_5^G)} = \begin{pmatrix} 4 & -2 & 0 \\ -2 & 4 & -2 \\ 0 & -2 & 2 \end{pmatrix} \sim G_{C_3(2)} = 2 \cdot \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

which is the Gram matrix for C_3 up to uniform scaling, so $\Phi(A_5^G) \cong \Phi(C_3)$. One can see the equivalence of these root systems by noting that the Coxeter diagrams are isomorphic as weighted graphs with weighted nodes:



Case 2: $D_4 \rightarrow C_3$ (**Vertical Reflection**, $G = S_2$)

Let $L = D_4$, with simple roots and Gram matrix:

$$\Phi(D_4) = \begin{cases} \alpha_1 &= e_1 - e_2 \\ \alpha_2 &= e_2 - e_3 \\ \alpha_3 &= e_3 - e_4 \\ \alpha_4 &= e_3 + e_4 \end{cases}, \quad G_{D_4} = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}$$

and

$$I: \begin{cases} \alpha_1 \mapsto \alpha_1 \\ \alpha_2 \mapsto \alpha_2 \\ \alpha_3 \mapsto \alpha_4 \\ \alpha_4 \mapsto \alpha_3, \end{cases} \implies I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

The invariant sublattice is

$$D_4^G \cong \{(x_1, x_2, x_3, 0) \in \mathbf{Z}^4 : x_1 + x_2 + x_3 \in 2\mathbf{Z}\}$$

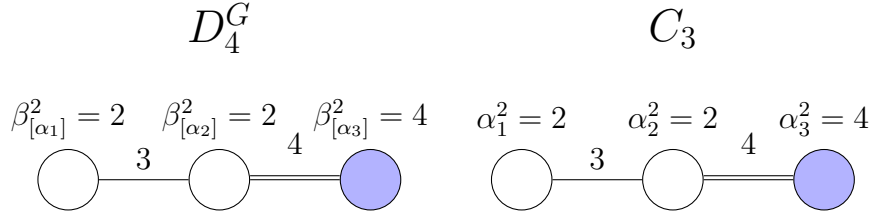
with orbits and folded root system

$$G \cdot \alpha_i : \begin{cases} G \cdot \alpha_1 = \{\alpha_1\} \\ G \cdot \alpha_2 = \{\alpha_2\} \\ G \cdot \alpha_3 = \{\alpha_3, \alpha_4\}, \end{cases} \implies \Phi(D_4^G) : \begin{cases} \beta_{[\alpha_1]} = \alpha_1 = e_1 - e_2 \\ \beta_{[\alpha_2]} = \alpha_2 = e_2 - e_3 \\ \beta_{[\alpha_3]} = \alpha_3 + \alpha_4 = 2e_3 \end{cases}$$

The Gram matrix is

$$G_{\Phi(D_4^G)} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -2 & 4 \end{pmatrix} = G_{C_3} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -2 & 4 \end{pmatrix},$$

which is precisely the Gram matrix for C_3 , so $\Phi(D_4^G) \cong \Phi(C_3)$. One can again see the equivalence by noting the isomorphism of Coxeter diagrams:



Case 3: $D_4 \rightarrow G_2$ (**Rotation by $2\pi/3$, $G = \mathbf{Z}/3\mathbf{Z}$**)

Let $L = D_4$ with σ acting by

$$\sigma : \begin{cases} \alpha_1 \mapsto \alpha_3 \\ \alpha_3 \mapsto \alpha_4 \\ \alpha_4 \mapsto \alpha_1 \\ \alpha_2 \mapsto \alpha_2 \end{cases} \quad \text{with orbits } G \cdot \alpha_i : \begin{cases} G \cdot \alpha_2 = \{\alpha_2\} \\ G \cdot \alpha_1 = \{\alpha_1, \alpha_3, \alpha_4\}. \end{cases}$$

Then

$$\Phi(D_4^G) : \begin{cases} \beta_{[\alpha_1]} = \alpha_1 + \alpha_3 + \alpha_4 = e_1 - e_2 + 2e_3 \\ \beta_{[\alpha_2]} = \alpha_2 = e_2 - e_3 \end{cases} \implies G_{\Phi(D_4^G)} = \begin{pmatrix} 2 & -3 \\ -3 & 6 \end{pmatrix} = G_{\Phi(G_2)},$$

yielding the isometry $\Phi(D_4^G) \cong \Phi(G_2)$ on the nose.

Case 4: $E_6 \rightarrow F_4$ (Horizontal Reflection, $G = S_2$)

We take the simple roots

$$\begin{cases} \alpha_1 = e_2 - e_3 \\ \alpha_2 = e_3 - e_4 \\ \alpha_3 = e_4 \\ \alpha_4 = \frac{1}{2}(e_1 - e_2 - e_3 - e_4), \end{cases} \quad G_{F_4} = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 1 & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & 1 \end{pmatrix}$$

and for E_6 , we fix a basis in the Bourbaki convention,

$$\begin{cases} \alpha_1 = \frac{1}{2}(e_1 - e_2 - e_3 - e_4 - e_5 - e_6 - e_7 + e_8) \\ \alpha_2 = e_1 + e_2 \\ \alpha_3 = -e_1 + e_2 \\ \alpha_4 = -e_2 + e_3 \\ \alpha_5 = -e_3 + e_4 \\ \alpha_6 = -e_4 + e_5, \end{cases} \quad G_{E_6} = \begin{pmatrix} 2 & 0 & -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 & 0 & 0 \\ -1 & 0 & 2 & -1 & 0 & 0 \\ 0 & -1 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

Writing the folding involution yields

$$I: \begin{cases} \alpha_1 \leftrightarrow \alpha_6 \\ \alpha_3 \leftrightarrow \alpha_5 \\ \alpha_2 \mapsto \alpha_2 \\ \alpha_4 \mapsto \alpha_4 \end{cases} \implies I = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Taking orbits,

$$G \cdot \alpha_i : \begin{cases} G \cdot \alpha_1 = \{\alpha_1, \alpha_6\} \\ G \cdot \alpha_2 = \{\alpha_2\} \\ G \cdot \alpha_3 = \{\alpha_3, \alpha_5\} \\ G \cdot \alpha_4 = \{\alpha_4\} \end{cases} \implies \Phi(E_6^G): \begin{cases} \beta_{[\alpha_1]} = \alpha_1 + \alpha_6 \\ \beta_{[\alpha_2]} = \alpha_2 \\ \beta_{[\alpha_3]} = \alpha_3 + \alpha_5 \\ \beta_{[\alpha_4]} = \alpha_4 \end{cases}$$

and thus

$$G_{\Phi(E_6^G)} = \begin{pmatrix} 4 & 0 & -2 & 0 \\ 0 & 2 & 0 & -1 \\ -2 & 0 & 4 & -2 \\ 0 & -1 & -2 & 2 \end{pmatrix} \sim G_{F_4(2)} = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 1 & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & 1 \end{pmatrix},$$

which coincide after applying the permutation $(1, 3, 4, 2)$ to the $\beta_{[\alpha_i]}$, so $\Phi(E_6^G) \cong \Phi(F_4(2))$. One can also additionally check that the computed Coxeter diagrams are isomorphic, or equivalently that the associated Cartan matrices are similar via a permutation matrix.

2.8 Applications to $F_{\text{En},2}$ and $F_{(2,2,0)}$

The main geometric application of folded diagrams is the construction of semitoroidal compactifications of moduli spaces F_Γ and their quotients via **generalized Coxeter semifans**. Unlike classical toroidal compactifications where all walls of the Coxeter chamber are preserved, semitoroidal compactifications allow certain **irrelevant roots** to be removed. In the case of interest to us, the semifan $\mathcal{F}_\bullet$ is constructed by:

1. Identifying the 0-cusps of $F_{(2,2,0)}$ and their corresponding Coxeter diagrams and semifans
2. Folding the semifans and Coxeter diagrams to determine the Coxeter diagrams for the 0-cusps of $F_{\text{En},2}$
3. Identifying the irrelevant roots for foldings as images of irrelevant roots of the corresponding $F_{(2,2,0)}$ cusp
4. Removing the corresponding walls to create a coarser fan structure for $F_{\text{En},2}$

This process constructs the compactification $\overline{F_{\text{En},2}}^{\mathcal{F}_\bullet}$ appearing in the main theorem, and the distinction between toroidal and strictly semitoroidal behavior depends on the order of the irrelevant root subgroup: toroidal cusps are those with finite irrelevant subgroups, while strictly semitoroidal cusps are those with infinite irrelevant subgroups. The folding procedure operates as follows: Let W be the reflection group generated by Φ inside $O(L)$, and W^G the reflection group generated by Φ^G where G acts as a diagram automorphism. If $\mathfrak{C} \subset L_{\mathbf{R}}$ is a G -invariant Weyl chamber for W , then the restriction

$$\mathfrak{C}^G := \mathfrak{C} \cap L_{\mathbf{R}}^G$$

is a fundamental chamber for W^G . The walls are given by hyperplanes orthogonal to the folded roots $\beta_{[\alpha_i]}$, where each folded root is either

a fixed root under G or the “average” of an orbit: the sum of roots in a G -orbit, satisfying orthogonality conditions. The relationship between $F_{(2,2,0)}$ (quartic hyperelliptic K3 surfaces) and $F_{\text{En},2}$ (degree 2 numerically polarized Enriques surfaces) is determined by the lattice involution $I = -I_{\text{En}}$ acting on the transcendental lattice $T_{\text{dP}} = U \oplus U(2) \oplus E_8^2$. The fixed sublattice under this involution is precisely the Enriques transcendental lattice $T_{\text{En}} = U \oplus U(2) \oplus E_8(2)$. At each 0-cusp η of $\overline{\partial F_{(2,2,0)}}^{\text{BB}}$, which are distinguished by primitive isotropic vectors $\eta \in T_{\text{dP}}$ of divisibility 1 or 2, there exists a stable reflection group Γ_η whose reflection group $W(\Gamma_\eta)$ and associated Coxeter diagram $G(\Gamma_\eta)$ encodes the incidence relations of points and curves in $\overline{\partial F_{(2,2,0)}}^{\text{BB}}$ at I . A technical result governing the folding is:

Lemma 2.8.0.1 (Root folding criterion). *Let $\Phi(T_{\text{dP}})$ be the root system of T_{dP} and I be the induced involution. The folded roots $\beta_{[v]} \in \Phi(T_{\text{dP}}^{(I)})$ arise in exactly one of the following ways:*

1. $v \in \Phi^2(T_{\text{dP}})$ and $\beta_{[v]} \in \Phi(T_{\text{En}})$,
2. $v \in \Phi^4(T_{\text{dP}})$ and $\beta_{[v]} \in \Phi(T_{\text{En}})$, or
3. $v \in \Phi^2(T_{\text{dP}}) \cap I(v)^{\perp T_{\text{dP}}}$, and so $\beta_{[v]} := v + I(v) \in \Phi^4(T_{\text{En}})$ is the sum of orthogonal roots in $\Phi^2(T_{\text{dP}})$.

Proof. Noting that $T_{\text{En}} = T_{\text{dP}}^{I=1}$ is a sublattice, any $v \in \Phi(T_{\text{dP}})$ which is I -invariant is also in $\Phi(T_{\text{En}})$ by definition, since its norm, primitivity, and divisibility are unchanged, and the folded root is defined by $\beta_{[v]} = v$. So suppose $I(v) \neq v$, so $\beta_{[v]} := v + I(v)$ is the orbit sum, which is I -invariant by construction. If $v^2 = -2$, noting that $I \in \text{O}(T_{\text{En}})$ is an isometry, we have

$$\beta_{[v]}^2 := (v + I(v))^2 = v^2 + 2v \cdot I(v) + I(v)^2 = 2v^2 + 2v \cdot I(v),$$

and thus if $v \perp I(v)$ then the norm is doubled, and if v is a (-2) -root then $\beta_{[v]}$ is a (-4) -root. The remaining cases are when

1. $I(v) \neq v$ but $v^2 = -4$, or
2. when $v^2 = -2$ but v is not orthogonal to $I(v)$

The first case can be immediately dispensed with, since folding doubles the norm. Suppose that $v \notin T_{\text{En}}$, so $v_I = v + I(v)$. Write $v \in T_{\text{dP}}$ in the block form as in Definition 2.3. Then one can write

$$\begin{aligned}
v = (u_1, u_2, u_3, \alpha_1, \alpha_2) &\implies I(v) = (u_1, u_2, u_3, -\alpha_2, -\alpha_1) \\
&= v - (0, 0, 0, \alpha_1 + \alpha_2, \alpha_1 + \alpha_2) \\
\implies v \cdot I(v) &= v^2 - (\alpha_1 + \alpha_2)^2, \\
\implies v_I^2 &= (v + I(v))^2 \\
&= 2(v^2 + v \cdot I(v)) \\
&= 2(2v^2 - (\alpha_1 + \alpha_2)^2) \\
&= 4v^2 - 2(\alpha_1 + \alpha_2)^2 < 0,
\end{aligned}$$

Now since $v \neq I(v)$, we have $0 \neq v - I(v) = (0, 0, 0, \alpha_1 + \alpha_2, \alpha_1 + \alpha_2)$ and thus $\alpha_1 + \alpha_2 \neq 0$ in E_8 , forcing $(\alpha_1 + \alpha_2)^2 < 0$ since E_8 is negative-definite. But then $v \cdot I(v) = v^2 + c$ where $c := -(\alpha_1 + \alpha_2)^2 > 0$, so $v \cdot I(v) > v^2$. Let $\lambda := (\alpha_1 + \alpha_2)^2$; there are now two cases:

Case 1: $v^2 = -2$

In this case, we have

$$v_I^2 = 4(-2) - 2\lambda = -8 - 2\lambda < 0$$

We want v_I to be a root with the correct divisibility and primitiveness in T_{En} . The only possibility is $v \cdot I(v) = 0$, and we thus have

$$\begin{aligned}
0 = v \cdot I(v) &= v^2 - \lambda \\
\implies \lambda &= -2 \\
\implies v_I^2 &= 4v^2 - 2\lambda \\
&= -8 - 2(-2) \\
&= -4
\end{aligned}$$

and since $\text{div}_{T_{E_n}}(v_I) = 2$, v_I is a well-defined root of T_{En} . However, note that $\text{div}_{T_{dP}}(v_I) \neq 2$ – otherwise, $\alpha_1 + \alpha_2 \in 2E_8$, which implies that $\lambda = (2(\alpha'_1 + \alpha'_2))^2 = 4\lambda' \in 4\mathbf{Z}$ with $\lambda' = (\alpha'_1 + \alpha'_2)^2 < 0$. But then

$$v \cdot I(v) = v^2 - \lambda = -2 - 4\lambda' \geq 2$$

and thus

$$v_I^2 = 4v^2 - 2(\alpha_1 + \alpha_2)^2 = -8 - 2 \cdot (4\lambda') \geq 0,$$

a contradiction.

Case 2: $v^2 = -4$

In this case we have $\text{div}_{T_{dP}}(v) = 2$ and $\alpha_1, \alpha_2, \alpha_1 + \alpha_2 \in 2E_8$. By similar computations as in the first case, we again obtain

$$-\lambda \geq 8 \implies v \cdot I(v) \geq 4 \implies v_I^2 \geq 0,$$

a contradiction. Finally, the converse follows from [Nam85, 2.13 and 2.15], since $T_{\text{En}}[k]/O(T_{\text{En}})$ contains one orbit for $k = -2$ and two orbits for $k = -4$.

□

Thus for every 0-cusp $\tilde{\eta}_i \in T_{dP}$ for $F_{(2,2,0)}$, we have an explicit understanding of how roots α in boundary lattices $\overline{T_{dP\tilde{\eta}_i}}$ ($i = 1, 2$) descend or combine when passing to the I -invariant sublattices, which correspond to $\overline{T_{\text{En}\eta_j}}$ ($j = 1, \dots, 5$). The Coxeter diagrams $G(\Gamma_{dP\tilde{\eta}_i})$ at 0-cusps η_i in $F_{(2,2,0)}$ determine, via their maximal parabolic subdiagrams, the type II curves adjacent to η_i in $\partial \overline{F_{(2,2,0)}}^{\text{BB}}$. Under the folding involution I , only those maximal parabolic subdiagrams that are I -invariant and whose images remain maximal parabolic correspond to maximal parabolic subdiagrams in the folded diagram $G(G(\Gamma_{dP\tilde{\eta}_i}))^I$, which in turn correspond to maximal parabolic subdiagrams in $G(\Gamma_{\text{En},2\eta_j})$ and thus boundary curves in $\partial \overline{F_{\text{En},2}}^{\text{BB}}$. The incidences between 0-cusps and 1-cusps is preserved only for these cases, and thus under this correspondence, we can determine the cusp diagram of $F_{\text{En},2}$ entirely by studying folded diagrams.

CHAPTER 3

ENRIQUES SURFACES AND THEIR K3 COVERS

Building on the foundational lattice theory established in Chapter 1 and Chapter 2, we focus on the concrete geometric constructions, orbit classifications, and boundary correspondences that distinguish Enriques surface moduli theory from general K3 theory. Every degree 2 numerically polarized Enriques surface can be realized via a canonical toric construction, which we describe in this chapter.

3.1 Notational Conventions

We establish standard notation to be used throughout the remainder of this work: for lattice theoretic constructions, we write U for the hyperbolic plane, $\langle a_1, \dots, a_n \rangle$ for the diagonal lattice with Gram matrix $\text{diag}(a_1, \dots, a_n)$, $L(m)$ for rescaling of lattice L by factor m , $L^{\oplus n}$ for the n -fold orthogonal direct sum, and $L^\vee$ for the dual lattice $\text{Hom}_{\mathbf{Z}}(L, \mathbf{Z})$. We write $\text{I}_{p,q}$ and $\text{II}_{p,q}$ for the unique odd (resp. even) nondegenerate unimodular lattice of signature (p, q) over $\mathbf{Z}$. For root lattices, we use A_n, D_n, E_6, E_7, E_8 for the classical root lattices, $\Phi(L)$ for the root system of all roots in L , and $W(L)$ for the Weyl group generated by reflections in all roots. These are typically indexed by vectors of norm $v^2 = -2$ or those of norm $v^2 = -4$ with divisibility 2. We write $A_L := L^\vee/L$ for the discriminant group, $q_L : A_L \rightarrow \mathbf{Q}/2\mathbf{Z}$ for the discriminant quadratic form, and $\ell(L) := \ell(A_L) = \text{rank}_{\mathbf{F}_2} A_L$ for the length of the discriminant group. For group actions, we denote $\text{O}(L)$ for the orthogonal group of lattice L , $\text{O}^*(L) = \ker(\text{O}(L) \rightarrow \text{O}(A_L))$ for the stable orthogonal group, $\text{div}_L(v)$ for the divisibility of vector v

in L , and $L[k] = \{v \in L \mid v^2 = k\}$ for vectors of squared length k . In applications to K3 and Enriques surfaces, L_{K3} denotes the K3 lattice, (r, a, δ) are the classification invariants for 2-elementary lattices, ι_{En} is the Enriques involution, and $F_{\text{En},2}$ is the moduli space of Enriques surfaces with polarization of degree 2.

3.2 The Main Diagram and Geometric Construction

3.2.1 The base surface $Y = \mathbf{P}^1 \times \mathbf{P}^1$

The Basic Geometry of $\mathbf{P}^1 \times \mathbf{P}^1$

Let $Y = \mathbf{P}^1 \times \mathbf{P}^1$ over $\mathbf{C}$, with coordinates (x, y) on standard affine charts. Consider the natural projections onto the two factors,

$$p_1 : Y \rightarrow \mathbf{P}_1^1, \quad p_2 : Y \rightarrow \mathbf{P}_2^1.$$

Let $H_1 \subset \mathbf{P}_1^1$ and $H_2 \subset \mathbf{P}_2^1$ denote hyperplane classes on the first and second factor, respectively, and write their pullbacks as $\ell_1 := p_1^* H_1$ and $\ell_2 := p_2^* H_2$. The group $\text{Pic}(Y) \cong \mathbf{Z}^2$ is freely generated by ℓ_1 and ℓ_2 , so every divisor (and hence every line bundle) corresponds to a vector (a, b) . The intersection numbers of ℓ_1 and ℓ_2 can be readily computed, yielding the following lattice isometry:

$$\ell_1^2 = 0, \quad \ell_2^2 = 0, \quad \ell_1 \cdot \ell_2 = 1 \implies (\text{Pic}(Y), \cdot) \xrightarrow{\sim} \left(\mathbf{Z}^2, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) = U,$$

which follows because fibers do not intersect, while a fiber and a section (or a fiber in the other projection) generically intersect in a single point, and we can readily apply Chow's moving lemma to arrange for this. For any $(a, b) \in \mathbf{Z}^2$, the line bundle $\mathcal{O}_Y(a, b)$ is defined as

$$\mathcal{O}_Y(a, b) = p_1^* \mathcal{O}_{\mathbf{P}^1}(a) \otimes_{\mathcal{O}_Y} p_2^* \mathcal{O}_{\mathbf{P}^1}(b)$$

and its associated divisor class is $a\ell_1 + b\ell_2$. On the standard affine open $U_{00} = \{x_0 \neq 0, y_0 \neq 0\}$ with local coordinates $x = x_1/x_0$, $y = y_1/y_0$,

a section of $\mathcal{O}_Y(a, b)$ has the form

$$f(x, y) = \sum_{\substack{0 \leq i \leq a \\ 0 \leq j \leq b}} a_{ij} x^i y^j, \quad a_{ij} \in \mathbf{C}.$$

Globally, a section of $\mathcal{O}_Y(a, b)$ corresponds to a bihomogeneous polynomial $F(x_0, x_1, y_0, y_1)$ of “bidegree” (a, b) :

$$F(x_0, x_1; y_0, y_1) = \sum_{\substack{i+j=a \\ k+\ell=b}} c_{i,j;k,\ell} x_0^i x_1^j \cdot y_0^k y_1^\ell, \quad c_{i,j;k,\ell} \in \mathbf{C}.$$

Thus $H^0(\mathcal{O}_Y(a, b))$ is generated by the monomials $x^i y^j$, where $0 \leq i \leq a$ and $0 \leq j \leq b$. If f is a section of $\mathcal{O}_Y(a, b)$, we thus say it has **bidegree** (a, b) . Recall that a **del Pezzo surface** is a smooth projective surface Y with ample anticanonical divisor $-K_Y$, and the **degree** of a del Pezzo surface is the intersection number $(-K_Y)^2$. The canonical divisor on Y is $K_Y = -2\ell_1 - 2\ell_2$ and thus corresponds to the vector $(-2, -2)$, making the degree easy to compute:

$$K_Y^2 = (-2, -2)^t \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} (-2, -2) = 8.$$

This of course agrees with the standard calculation,

$$K_Y^2 = (-2\ell_1 - 2\ell_2)^2 = 4(\ell_1^2 + 2(\ell_1 \cdot \ell_2) + \ell_2^2) = 4(0 + 2 \cdot 1 + 0) = 8.$$

We thus recover the well-known fact that Y is a del Pezzo surface of degree 8. It is also well-known that Y is a toric, and the polytope for the projective toric pair $(Y, \mathcal{O}_Y(a, b))$ is the rectangle $Q_{a,b} := [0, a] \times [0, b] \subseteq \mathbf{Z}^2$. In particular, for $\mathcal{O}_Y(4, 4)$, the polytope is the square $Q := Q_{4,4} = [0, 4]^2 \subseteq M_Y := \mathbf{Z}^2$ with sides of lattice length 4.

Involutions in Coordinates

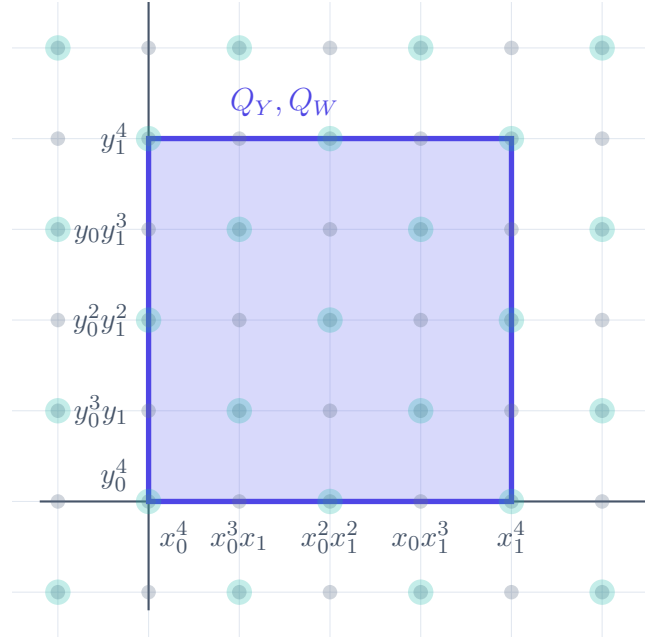
Define the involution $\tau : (x, y) \mapsto (-x, -y)$. In homogeneous coordinates, this is given by $(x_0 : x_1, y_0 : y_1) \mapsto (x_0 : -x_1, y_0 : -y_1)$. The involution τ fixes the four points with coordinates (x, y) equal to either 0 or ∞ , that is, those in $\{0, \infty\} \times \{0, \infty\}$. The action $\tau \curvearrowright H^0(\mathcal{O}_Y(4, 4))$ is given on generators $\tau^*(x^i y^j) = (-1)^{i+j} x^i y^j$. Hence, the τ -invariant monomials are those for which $i+j$ is even. Let $W = Y/\tau$, and denote by $\phi : Y \rightarrow W$ the quotient map. This map is branched precisely at

the four fixed points described above, and the quotient is locally given by $(u, v) \mapsto (u^2, v^2)$, yielding 4 A_1 singularities on W .

Adjunction for finite covers gives $K_Y = \phi^*K_W + R$, where R is the ramification divisor, here the sum of the four fixed points. Since any intersection involving R vanishes (as it is 0-dimensional), it follows that $K_Y^2 = (\phi^*K_W)^2$. The map ϕ has degree two, so $(\phi^*K_W)^2 = 2K_W^2$, and hence $K_W^2 = K_Y^2/2 = 4$. Thus, W is a (singular) del Pezzo surface of degree 4. The ampleness of $-K_W$ follows from the fact that $-K_Y$ is ample and the ramification locus consists only of isolated points. The surface W is again toric, and yields a projective pair (W, Q_W) where $Q_W := Q \cap \mathbf{Z}_{\text{ev}}^2$ where

$$\mathbf{Z}_{\text{ev}}^2 = \{(i, j) \in \mathbf{Z}^2 \mid i + j \in 2\mathbf{Z}\},$$

corresponding to τ -invariant monomials. Thus Q and W share the same polytope, but for the two distinct lattices $M_Y := \mathbf{Z}^2$ and $M_W := \mathbf{Z}_{\text{ev}}^2$. We note that Q_Z has 16 boundary lattice points and 9 interior points, yielding 25 total sections of $\mathcal{O}_Y(4, 4)$, while Q_W has 8 boundary points and 5 interior points, yielding 13 τ -invariant sections:



The K3 Double Cover

We briefly recall the standard construction of a double cover of smooth projective surface Y branched over a reduced divisor B as a method of

producing (families of) Calabi-Yau varieties (see e.g.[Par91]). Select a line bundle L such that $L^{\otimes 2} \cong \mathcal{O}_Y(B)$ – for example, if $B \in |-2K_Y|$, one can take $L = -K_Y$. We can write B as the zero divisor of zeros of a section $f \in H^0(Y, L^{\otimes 2})$, and define a coherent $\mathcal{O}_Y$ -algebra $\mathcal{A} = \mathcal{O}_Y \oplus L^{-1}$, where the algebra structure is locally determined by

$$(o_1 \oplus \ell_1) \cdot (o_2 \oplus \ell_2) = (o_1 o_2 + f|_U \cdot \ell_1 \ell_2) \oplus (o_1 \ell_2 + o_2 \ell_1)z,$$

$$o_i \in \mathcal{O}_Y(U), \ell_i \in L^{-1}(U)$$

where $z \in H^0(L^{-1})$ is a generator of L^{-1} satisfying $z^2 = f$. The double cover $\pi : X \rightarrow Y$ is then constructed as $X = \text{Spec}_Y(\mathcal{A})$; π is a finite morphism, it is etale away from B , and is ramified to order 2 precisely over B . The relation $z^2 + f = 0$ is imposed for an $\mathcal{O}_Y$ -module generator z of L^{-1} , and the algebra structure above ensures that global functions on X over an open $U \subseteq Y$ are of the form $s_1 + z s_2$ with $s_i \in \mathcal{O}_Y(U)$ and $z^2 = f|_U$, and the covering involution is locally modeled by $z \mapsto -z$. By the adjunction formula, one has $K_X = \pi^*(K_Y + L)$, and so for $L = -K_Y$, and $B \in |-2K_Y|$, we have $K_X = \pi^*(K_Y - K_Y) = \mathcal{O}_X$ and X is Calabi-Yau.

The main family of K3 surfaces

We now let $B \subset Y$ be a smooth τ -invariant curve in the linear system $|-2K_Y| = |\mathcal{O}_Y(4, 4)|$. Such a curve is defined by a bihomogeneous polynomial $f(x, y)$ of bidegree $(4, 4)$ that is invariant under τ ; that is, only monomials $x^i y^j$ with $i + j$ even may have nonzero coefficient in f . In order that B does not pass through any of the four fixed points, it suffices to require that the coefficients of the monomials $1, x^4, y^4$, and $x^4 y^4$ are nonzero. We now apply this to B with $L = -K_Y = \mathcal{O}_Y(2, 2)$ as above. Since Y is simply connected and π is a finite cover, it follows that X is simply connected as well. Moreover, one can show by the Leray spectral sequence that $h^1(\mathcal{O}_X) = h^1(\mathcal{O}_Y) = 0$, and thus X is a K3 surface. For a toric description, let P denote the polytope in $\mathbf{R}^3$ formed by forming the pyramid over Q with apex $(2, 2, 2)$. The associated toric threefold V_P then contains X as the affine hypersurface locally defined by the vanishing locus of $z^2 + f$. By varying B , we can thus produce families of K3 surfaces as families of hypersurfaces in the toric threefold V_P . The parameter space for such B corresponds to the space of τ -invariant global sections of $\mathcal{O}_Y(4, 4)$. There are exactly 13 monomials $x^i y^j$ of bidegree $(4, 4)$ with $i + j$ even, which can be easily

seen by counting lattice points in Q , and thus each choice of f defines a point in an open subset $U \subseteq \mathbf{P}^{12}$. We thus obtain a 12-dimensional family of such K3 surfaces. While the full automorphism group of Y is an extensions of PGL_2^2 , the relevant symmetries for this family are the scaling actions of the torus factor $(\mathbf{C}^\times)^2$ on the coordinate axes of Y , as well as the symmetries D_4 of Q . We thus pass to a 10-dimensional family:

$$\mathcal{M}_{\mathrm{En},2} := U / (D_4 \rtimes (\mathbf{C}^\times)^2).$$

Geometric Involutions

Let X be constructed as above; it carries three distinguished involutions:

$$\begin{aligned}\iota_{\mathrm{dP}}(x, y, z) &= (x, y, -z) \\ \iota_{\mathrm{En}}(x, y, z) &= (-x, -y, -z) \\ \iota_{\mathrm{Nik}}(x, y, z) &= (-x, -y, z)\end{aligned}$$

The involution ι_{dP} is the covering involution of the branched double cover $\pi: X \rightarrow Y$. The original involution $\tau: (x, y) \mapsto (-x, -y)$ on Y has two lifts to X , both ι_{En} and ι_{Nik} . By checking directly in coordinates, o

$$\iota_{\mathrm{dP}}^2 = \iota_{\mathrm{En}}^2 = \iota_{\mathrm{Nik}}^2 = \mathrm{id}_X, \quad \iota_{\mathrm{dP}} \circ \iota_{\mathrm{En}} = \iota_{\mathrm{En}} \circ \iota_{\mathrm{dP}} = \iota_{\mathrm{Nik}}$$

and all three commute pairwise, thus generating a faithful representation of the Klein 4-group $G := \langle \iota_{\mathrm{dP}}, \iota_{\mathrm{En}}, \iota_{\mathrm{Nik}} \rangle \cong \mathbf{Z}_2^2 \hookrightarrow \mathrm{Aut}(X)$ acting by algebraic automorphisms on X . Moreover, on affine open sets, a generating holomorphic 2-form for X can be found, where the three involutions act in the following way:

$$\omega_X = \mathrm{Res}_X \left(\frac{dx \wedge dy \wedge dz}{z^2 + f(x, y)} \right) \implies \iota^*(\omega_X): \begin{cases} \iota_{\mathrm{dP}}^*(\omega_X) = -\omega_X, \\ \iota_{\mathrm{En}}^*(\omega_X) = -\omega_X, \\ \iota_{\mathrm{Nik}}^*(\omega_X) = +\omega_X. \end{cases}$$

Thus, ι_{dP} and ι_{En} are **non-symplectic**, i.e. act as $-\mathrm{id}$ on $H^{2,0}(X)$, while ι_{Nik} is symplectic. Let $Z = X/\iota_{\mathrm{En}}$ and $Z' = X/\iota_{\mathrm{Nik}}$, write $\psi: X \rightarrow Z$ and $\psi': X \rightarrow Z'$, and consider the singularities of the

quotient surfaces. Note that Z' has at least 8 A_1 singularities, induced by the 4 fixed points of τ on Y under the 2-to-1 cover $\pi : X \rightarrow Y$. However, provided B does not pass through $\text{Fix}(\tau)$, it is a fixed-point-free involution on X and thus Z is a smooth Enriques surface and X is its universal K3 cover. We summarize the current situation below:

$\iota_\star$	$\iota_\star(x, y, z)$	$\iota_\star^* \omega_X$	$X/\iota_\star$	Type
ι_{dP}	$(x, y, -z)$	$-\omega_X$	$\pi : X \rightarrow Y$	K3, smooth
ι_{En}	$(-x, -y, -z)$	$-\omega_X$	$\psi : X \rightarrow Z$	Enriques, smooth
ι_{Nik}	$(-x, -y, z)$	$+\omega_X$	$\psi' : X \rightarrow Z'$	$\widehat{\text{K3}}, \geq 8A_1$

Branch and Ramification Divisors

Let τ_{dP} be the involution on Z induced by the deck transformation ι_{dP} for the original cover $\pi : X \rightarrow Y$, and write $\rho : Z \rightarrow Z/\tau_{\text{dP}} = W$ for the corresponding branched cover. Let B and R denote the branch and ramification divisors of $\pi : X \rightarrow Y$, and B_W and R_Z the branch and ramification divisors of $\rho : Z \rightarrow W$. We record the classical formulas $\pi^*(B) = dR$ and $R = \frac{1}{d}\pi^*(B)$ for ramified d -to-1 covers, and we thus have

$$R = \psi^* R_Z = \frac{1}{2} \pi^*(B), \quad R_Z = \frac{1}{2} \psi_* R.$$

Since B is ample, R is ample as the pullback of an ample divisor under a finite surjective morphism. Since ψ is a finite quotient by a group acting freely in codimension 1, the pushforward R_Z of R is ample as well. The associated line bundle on Z satisfies $\mathcal{M} := \mathcal{O}_Z(R_Z) = \mathcal{L}_Z^{\otimes 2}$ for a suitable element $\mathcal{L}_Z \in \text{Pic}(Z)$, and thus $[\mathcal{L}_Z] \in \text{Pic}(Z)/C_2$ is a well-defined numerical polarization on the Enriques surface Z . We can check that its degree is 2, and the degree of $\mathcal{M}$ is 8. Conversely, if $\mathcal{M}$ is any polarization on an Enriques surface Z with at worst ADE singularities which is 2-divisible, so $\mathcal{M} = \mathcal{L}_Z^{\otimes 2}$ for some $\mathcal{L}_Z \in \text{Pic}(Z)$, then $|\mathcal{M}|$ is big and nef and thus by [Cos83; CD89; CDL24] defines a 2-to-1 branched cover $\rho : Z \rightarrow W$ where W is a quartic del Pezzo with $4A_1$ or $A_3 + 2A_1$ singularities with ample ramification divisor R_Z . In this situation, $(Z, \varepsilon R_Z)$ is a log canonical pair for small enough ε , and thus there is a well-defined KSBA moduli space of such pairs which we will refer to as $F_{\text{En}, 2}$. We summarize the overall situation in the following diagram:

Here,

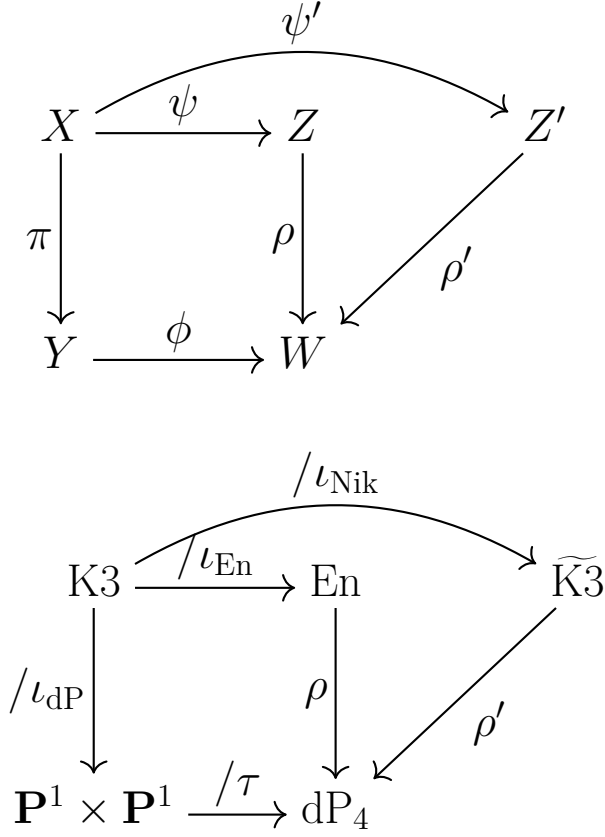


Figure 3.2.1: The main diagram.

- $\pi : X \rightarrow Y$ is the double cover branched over $B \subset Y$, recalling that $Y = \mathbf{P}^1 \times \mathbf{P}^1$ and X is a K3 surface,
- $\phi : Y \rightarrow W$ is the quotient by τ ,
- $\psi : X \rightarrow Z := X/\iota_{\text{En}}$ is the quotient by the Enriques involution ι_{En} and Z is a smooth Enriques surface,
- $\rho : Z \rightarrow W$ is the further quotient by the involution induced by ι_{dP} , where W is a singular quartic del Pezzo with $4A_1$ or $A_3 + 2A_1$ singularities,
- $\psi' : X \rightarrow Z' := X/\iota_{\text{Nik}}$ is the quotient by the Nikulin involution ι_{Nik} , where Z' is a singular K3 surface with (at least) $8A_1$ singularities.

This diagram has been well-studied in the literature, see e.g. [Cos83; CD89; Hor78; Enr06]. The GIT quotient

$$\overline{\mathcal{M}}_{\text{En},2}^{\text{GIT}} := \mathbf{P}^{12} // D_4 \ltimes (\mathbf{C}^\times)^2,$$

was studied in [Sha81], where it is determined exactly which forms of f result in stable and unstable quartic K3 surfaces [Hor78] analyzed the period map and its extension. In particular, if $f(x, y)$ vanishes at a torus-fixed point, the K3 cover X acquires a node and the quotient by the Enriques involution is a **Coble surface**: a smooth rational projective surface with empty anti-canonical linear system $|-K| = \emptyset$ but non-empty anti-bicanonical system $|-2K| \neq \emptyset$.

3.3 Enumerating Cusps

As a starting point to any compactification procedure of lattice polarized K3 surfaces F_S , we must first find the cusps of $\overline{F}_S^{\text{BB}}$. We give an overview here of various methods in the literature for similar moduli spaces, and how their cusps can be found and studied.

3.3.1 F_{2d} : Degree 2d Polarized K3 Surfaces (Scattone's Description)

Let L_{K3} denote the K3 lattice, and let $h \in L_{K3}$ be a primitive vector of square $2d > 0$. The lattice orthogonal to the polarization is defined as

Separate the Hodge-theoretic work

$$T_{2d} := h_{L_{K3}}^\perp \cong \langle -2d \rangle \oplus U^{\oplus 2} \oplus E_8^{\oplus 2},$$

which is even and of signature $(2, 19)$. The Type IV Hermitian symmetric period domain for T_{2d} ,

$$D_{T_{2d}} = \left\{ [\omega] \in \mathbf{P}(T_{2d, \mathbf{C}}) \mid (\omega, \omega) = 0, (\omega, \overline{\omega}) > 0 \right\},$$

admits a natural right action by the arithmetic group $\Gamma_{2d} \subset \text{O}^+(T_{2d})$, the intersection of the original orthogonal group with the subgroup stabilizing h (and, if necessary, a choice of connected component). The arithmetic quotient

$$F_{2d} := D_{T_{2d}} / \Gamma_{2d}$$

is the coarse moduli space parametrizing degree $2d$ polarized K3 surfaces.

The Baily–Borel compactification $\overline{F}_{2d}^{\text{BB}}$ is projective, and its boundary strata correspond bijectively to Γ_{2d} -orbits of primitive isotropic sublattices of T_{2d} of ranks 1 and 2. More precisely, 0-cusps are in correspondence with orbits of primitive isotropic planes up to the action of Γ_{2d} . The incidence relations between cusps are set by lattice inclusions: any 0-cusp (an isotropic line) is contained in the closure of all 1-cusps (isotropic planes) in which it lies.

As a concrete and illustrative case, set $d = 1$. Then $h^2 = 2$ and

$$T_2 = \langle -2 \rangle \oplus U^{\oplus 2} \oplus E_8^{\oplus 2}.$$

Scattone shows that for squarefree d (in particular, $d = 1$), there is exactly one Γ_{2d} -orbit of primitive isotropic lines in T_{2d} , so the Baily–Borel boundary of F_2 has a unique 0-cusp.

The 1-cusps are determined by the negative-definite lattices $\overline{T}_{2d,I} := I^{\perp T}/I$, for $I \subset T$ a primitive isotropic plane. Scattone showed that, up to isomorphism and the action of the arithmetic group, there are exactly four possible such lattices, each of rank 18 and discriminant 2, characterized by their root sublattices:

$$A_1 \oplus E_8^{\oplus 2}, \quad E_7 \oplus D_{10}, \quad A_1 \oplus D_{16}, \quad A_{17}. \quad (3.1)$$

These emerge as orthogonal complements to embeddings of E_7 into the four Niemeier lattices U of rank 24 that admit such sublattices. Each of these possibilities labels a modular curve in the boundary of $\overline{F}_2^{\text{BB}}$, and the closure of each of these modular curves contains the unique 0-cusp as every isotropic line is contained in some isotropic plane.

The structure of $\partial \overline{F}_{2d}^{\text{BB}}$ can be studied through the asymptotic behavior of the period map, governed by limiting mixed Hodge structures and their associated monodromy operators, following the work of [FS86]. For a one-parameter degeneration $\mathcal{X} \rightarrow \Delta$ of polarized degree $2d$ K3 surfaces over a punctured disk Δ^* , the unipotent monodromy operator $T \in O(T_{2d})$ determines the degeneration structure. Its nilpotent logarithm $N = \log T \in \text{End}(T_{2d, \mathbf{Q}})$ induces the canonical monodromy weight filtration $W^\bullet$ on $T_{2d, \mathbf{Q}}$, uniquely characterized by the properties $N(W_k) \subseteq W_{k-2}$ for all k , and that for each $j > 0$, the maps $N^j : \text{Gr}_{k+j}^W \rightarrow \text{Gr}_{k-j}^W$ are isomorphisms. Schmid’s Nilpotent

Orbit Theorem establishes that the period map asymptotically approaches a nilpotent orbit, defining a limiting Hodge filtration F^{lim} on $T_{2d, \mathbb{C}}$ which, together with $W^\bullet$, constitutes the limiting mixed Hodge structure (LMHS). This framework provides the connection between geometric degenerations and arithmetic lattice structures.

The boundary components of $\overline{F}_{2d}^{\text{BB}}$ are classified by the nilpotency index of N . The Type II boundary components (1-cusps) correspond to degenerations where $N \neq 0$ but $N^2 = 0$. For such degenerations, associated with a primitive isotropic plane $I \subset T_{2d}$, is a three-step monodromy weight filtration:

$$0 = W_0 \subset W_1 = I_{\mathbf{Q}} \subset W_2 = I_{\mathbf{Q}}^{\perp T_{2d}} \subset W_3 = T_{2d, \mathbf{Q}}.$$

The LMHS induces a pure polarized Hodge structure of weight 2 on the graded piece $\text{Gr}_2^W = I^\perp/I$, which is precisely the boundary lattice $\overline{T}_I$ arising in Scattone's combinatorial classification. This identification reveals that $\overline{T}_I$ is not merely a lattice-theoretic invariant, but rather the natural target of the weight-2 component of the limiting mixed Hodge structure.

For Type III degenerations (0-cusps), where $N^2 \neq 0$ but $N^3 = 0$, the weight filtration is of maximal length. The classifying space of such Hodge structures on $\overline{T}_I$ is itself a Type IV Hermitian symmetric domain $D_{\overline{T}_I}$ and the boundary component itself is a modular variety F_{Γ_I} for an appropriate arithmetic subgroup $\Gamma_I \leq \text{O}^+(\overline{T}_I)$. These degenerations correspond to normal crossing varieties whose dual complex is a triangulation of S^2 . The computational accessibility of these mixed Hodge structures relies on several key tools developed in the work starting in [FS86]: the Clemens-Schmid exact sequence relating the cohomology of central and $\mathcal{X}_t$ s, and the Steenbrink weight spectral sequence, which for K3 surfaces degenerates integrally at the E_2 -page. This integral degeneration is a special property of K3 surface degenerations that enables explicit computation of the limiting mixed Hodge structure components. The vanishing cycle analysis developed by Friedman and Scattone provides detailed control over how cohomology classes behave under degeneration. Through Mayer-Vietoris techniques and careful analysis of the dual complex structure, they established the precise relationship between the geometric combinatorics of singular fibers and the arithmetic invariants encoded in LMHS. The stratification $\overline{F}_{2d}^{\text{BB}}$ is thus realized by the asymptotic behavior of the period map at the various boundary cusps.

3.3.2 F_2 : Scattone's Description

For the specific case $d = 1$, [Sca87] finds exactly four possible isometry classes for $\overline{T}_I$ determined by a 1-cusp I adjacent to the unique 0-cusp η , distinguished by their root sublattices as defined above. These correspond to the four distinct Type II modular curves in the boundary of $\overline{F}_2^{\text{BB}}$. The unique 0-cusp η correspond to a Type III component, and thus a degeneration where $N^2 \neq 0$ but $N^3 = 0$, yielding a weight filtration of maximal length. The geometric inclusion of the 0-cusp in the closure of each 1-cusp reflects the lattice-theoretic fact that any primitive isotropic plane I contains the primitive isotropic line η , all up to $\Gamma_{T_{2d}}$ -invariance. This incidence structure is encoded combinatorially in the Coxeter diagram of the lattice $\overline{T}_{2d,\eta}$ associated to the 0-cusp, whose maximal parabolic subdiagrams are in bijection with the possible isomorphism classes of the lattices $\overline{T}_I$ for the adjacent 1-cusps.

The cusps in $\overline{F}_2^{\text{BB}}$ are obtained by classifying primitive isotropic sublattices of the lattice

$$T_2 := \langle -2 \rangle \oplus U^{\oplus 2} \oplus E_8^{\oplus 2}$$

using discriminant–form methods, from which Scattone shows:

- There is a single Γ_2 -orbit of primitive isotropic lines η in T_2 , giving one 0-cusp (Type III);
- There are four Γ_2 -orbits of primitive isotropic planes I in T_2 , whose corresponding boundary lattices $\overline{T}_I$ are the rank-18, negative-definite lattices given in Equation (3.1), yielding four Type II boundary curves;
- These four curves meet transversely at the unique Type III point.

We assemble this data into the following **cusp diagram**:

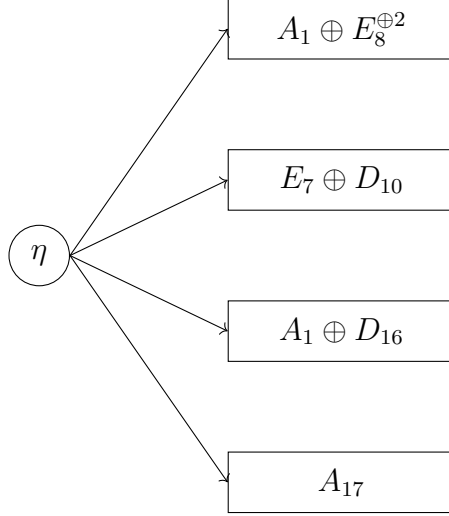


Figure 3.3.1: The cusp diagram of F_2 , the moduli space of degree 2 polarized K3 surfaces, which contains one 0-cusp η adjacent to four 1-cusps.

The general enumeration of $\partial \overline{F}_{2d}^{\text{BB}}$ reduces to finite problems in the discriminant group $A_{T_{2d}}$: cusps can be classified by studying isotropic subgroups of the finite discriminant group $A_{T_{2d}} := (T_{2d})^*/T_{2d}$, and applying Nikulin's theorem that the genus of an even lattice is determined by its signature and the isomorphism class of its discriminant form [Nikulin 1980]. The classification proceeds by associating to a primitive isotropic sublattice $I \subset T_{2d}$ an isotropic subgroup of $A_{T_{2d}}$. The problem of classifying orbits of such sublattices under the infinite group Γ_{2d} is thereby reduced to classifying orbits of isotropic subgroups of the finite group $A_{T_{2d}}$ under the action of a subgroup of $O^*(A_{T_{2d}})$. For the case $d = 1$, the four isomorphism classes of the rank-18 lattice $\overline{T}_I$ are constructed by leveraging the classification of the 24 Niemeier lattices: each isometry class of $\overline{T}_I$ is realized as the orthogonal complement $E_7^\perp \subset M$, where M is one of the Niemeier lattices that admit a primitive embedding of the E_7 root lattice [Scatone 1987]. This reduces the classification to a relatively well-known, finite set of possibilities. For general d , cusp enumeration becomes a number-theoretic problem, since structure of $A_{T_{2d}}$ is highly dependent on the arithmetic properties of d itself, including various the numbers of solutions to various congruences, as well as the prime factorization of d . [Sca87] uses these techniques to explicitly describe cusp diagrams for certain (sparse) families of values of d . For further details and the explicit Coxeter diagrams, see [Sca87, §6.2] and [AET23b, Fig. 2].

3.3.3 F_{ell} : Elliptic Surfaces

The moduli space F_{ell} parametrizes elliptic K3 surfaces with a chosen section, a condition that fixes a primitive embedding of a hyperbolic plane $U_1 \subset L_{K3}$. The relevant period map is thus defined on the orthogonal complement $T_{\text{ell}} := U_1^{\perp L_{K3}} \cong U^2 \oplus E_8^2$, and the moduli space is the 18-dimensional arithmetic quotient F_{ell} . A geometric description of the compactifications of F_{ell} is given in [ABE22], who construct KSBA compactifications and prove their isomorphism to specific semitoroidal compactifications. There a unique F_{ell} -orbit of 0-cusps in T_{ell} , represented by $\eta = e$, and the analysis falls on $(\overline{T_{\text{ell}}})_{\eta} \cong \text{II}_{1,17}$. Semitoroidal compactifications are defined by fans constructed in the rational closure $\mathfrak{C}_{\eta, \mathbf{Q}}$ of the positive cone in $\text{II}_{1,17}$. Two separate KSBA compactifications are constructed:

1. **The Ramification Divisor Compactification ($\overline{F}^{\text{ram}}$):** The polarization is given by the class of the ramification divisor R from the representation of the K3 surface as a double cover of $\mathbf{P}(1, 1, 4)$, so that $[R] = 3(s + 2f)$, where s is the section class and f is the fiber class. This defines a KSBA compactification parametrizing pairs $(X, \epsilon R)$ with X an slc K3 surface.
2. **The Rational Curve Divisor Compactification ($\overline{F}^{\text{rc}}$):** The polarization is taken to be $R = s + m \sum_{i=1}^{24} f_i$, where the f_i are the 24 singular fibers of the elliptic fibration for a generic elliptic K3 surface.

The core result [ABE22] is the identification of these KSBA moduli spaces with semitoroidal compactifications defined by specific fans in $\mathfrak{C}_{\mathbf{Q}}$. The fundamental fan is the **Coxeter fan** F^{Cox} , whose cones are the chambers of the reflection group $W(\text{II}_{1,17})$. The **ramification fan** F^{ram} is a coarsening of F^{Cox} whose fundamental chamber is a union of four Coxeter chambers. The **rational curve fan** F^{rc} is a refinement of F^{Cox} obtained by subdividing its fundamental chamber into nine sub-chambers. [ABE22] prove that the normalizations of $\overline{F}^{\text{ram}}$ and $\overline{F}^{\text{rc}}$ are isomorphic to the semitoroidal compactifications defined by the fans F^{ram} and F^{rc} , respectively, laying the groundwork for our main result on $F_{\text{En},2}$.

The geometric models for the boundary strata are constructed using the theory of **integral-affine spheres with 24 singularities (IAS²)**. A Type III Kulikov degeneration of an elliptic K3 surface corresponds

bijectively to a triangulated IAS², and the monodromy of a one-parameter degeneration determines a vector $\lambda \in \mathfrak{C}_{\mathbf{Q}}$, the **monodromy invariant**, which determines the combinatorial type of the stable limit $(X_0, \epsilon R)$ and is constant for all λ within the interior of a cone of the relevant fan. This provides a description of the boundary strata as unions of rational surfaces with prescribed singularities determined by the IAS².

Its Baily–Borel boundary contains a unique 0-cusp and two 1-cusps. The latter correspond to the two Γ^{ell} -orbits of primitive isotropic planes $I \subset T_{\text{ell}}$, distinguished by the isomorphism class of the negative-definite rank-16 lattice $\overline{T}_I = I^\perp/I$. These two classes are isometric to the root lattices $E_8 \oplus E_8$ and D_{16} , respectively, and the cusp diagram is as follows:

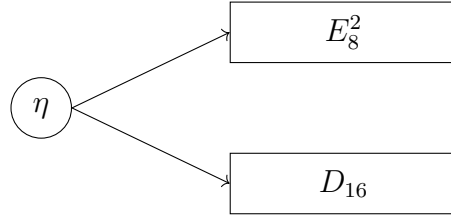


Figure 3.3.2: The cusp diagram of F_{ell} , the moduli space of elliptic K3 surfaces, which contains one 0-cusp η adjacent to two 1-cusps.

3.3.4 F_{En} : Unpolarized Enriques Surfaces

The moduli space of *unpolarized* Enriques surfaces corresponds to the lattice T_{En} and $\Gamma_{\text{En}} := \text{O}^+(T_{\text{En}})$, yielding the orthogonal modular variety F_{En} . To enumerate the 0-cusps of $\overline{F}_{\text{En}}^{\text{BB}}$, one can replace T_{En} by an auxiliary lattice $K = U \oplus E_8 \oplus \text{I}_{1,1}$ and utilize a bijection

$$T_{\text{En}}/\text{O}(T_{\text{En}}) \cong K/\text{O}(K),$$

allowing for a classification in terms of simpler lattices. A primitive isotropic vector $\eta \in \text{I}_{1,1} \subset K$ yields a unimodular lattice $\overline{T}_\eta$ of signature $(1, 9)$. There are precisely two such lattices up to isometry, $\text{I}_{1,9}$ and $\text{II}_{1,9}$. Pulling back, some slightly finer analysis shows there are exactly two Γ_{En} -orbits of primitive isotropic lines in T_{En} , corresponding to two 0-cusps η_1, η_2 . A direct approach for **1-cusps** is analytic: any isotropic plane $P \subset K$ must contain an odd primitive isotropic vector η , due to the indefinite form $\text{I}_{1,1}$. There is only one orbit of such under the full orthogonal group, so P can always be assumed to contain η .

The remaining problem is to classify the possible isometry classes of primitive isotropic lines in $\overline{T}_\eta \cong I_{1,9}$. By examining the parities of a basis $\{w, w'\}$ for P , one finds two inequivalent types: planes where both generators have the same parity (even/even or odd/odd), and planes where the two have different parity (even/odd). Hence, there are exactly two distinct $O(T_{\text{En}})$ -orbits of primitive isotropic planes, corresponding to two 1-cusps.

Explicit representatives can be given as follows. The 0-cusps correspond to the isotropic lines $\langle e \rangle$ and $\langle e' \rangle$, where $U = \langle e, f \rangle$ and $U(2) = \langle e', f' \rangle$ as sublattices of T_{En} . The 1-cusps can be represented by the planes $\langle e, e' \rangle$, and $\langle e', f' + \alpha \rangle$ where $\alpha \in E_8$ is a fixed primitive vector with $\alpha^2 = -4$. We thus obtain the following cusp diagram:

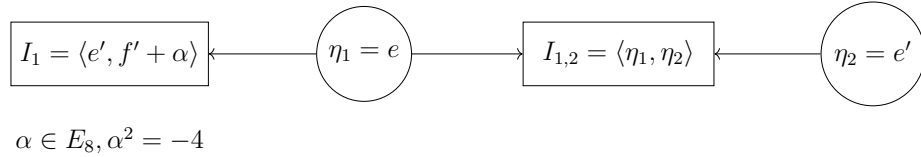


Figure 3.3.3: The cusp diagram of F_{En} , the moduli space of unpolarized Enriques surfaces.

Mirror Moves

All of the above situations involved somewhat ad-hoc analyses, which can be verified by computing the Coxeter-Vinberg diagrams $G(\Gamma_\eta)$ at the 0-cusps η associated to the arithmetic subgroup Γ and classifying their maximal elliptic and parabolic subdiagrams. This can be computationally difficult, so we note an algorithm that can unify most of the above situations when Γ is the full (stable) orthogonal group. Let T be a 2-elementary, nondegenerate, even unimodular lattice of signature $(2, n)$ embedding into the K3 lattice L_{K3} , let $\Gamma = O^*(T)$, and consider the corresponding arithmetic quotient $F_\Gamma = D_T/\Gamma$. All such lattices are encoded in Nikulin's pyramid diagram (see [AE23b]) of 2-elementary lattices. The **cusp diagram** of $\overline{F}_\Gamma$ is a labeled, directed graph whose vertices are boundary points and curves in $\overline{F}_\Gamma$, corresponding (up to Γ -equivalence) with primitive isotropic vectors η, I in T and their associated boundary lattices $\overline{T}_\eta, \overline{T}_I$, with an edge $v_1 \rightarrow v_2$ between vertices if the corresponding boundary stratum F_1 is contained in the closure of F_2 . The cusp diagram is computed as follows.

A **mirror move** is a lattice-theoretic operation governed by the existence of a primitive isotropic vector $\eta \in T$ of a specified type and splitting as proved in [AE23b]:

- **Odd/simple:** $\text{div}_T(\eta) = 1$, and $T \cong U \oplus K$
- **Even, ordinary:** $\text{div}_T(\eta) = 2$ and η^* is ordinary in A_T , and $T \cong U(2) \oplus K$
- **Even, characteristic:** $\text{div}_T(\eta) = 2$ and η^* is characteristic, and $T \cong I_{1,1}(2) \oplus K$

The move replaces T with $\overline{T}_\eta$, computing the new invariants in each case. A mirror move is only possible if the resulting lattice is realized (i.e., if the corresponding node exists in Nikulin's pyramid, see [AE23b, Fig. 1]) and admits the required splitting.

Step 1: Find All Orbits of 0-Cusps η

1. **Start at the node** (r_0, a_0, δ_0) in Nikulin's pyramid associated to T .
2. **For every admissible mirror move**, check for outgoing arrows (*mirror moves*) from this node, corresponding to possible splittings of boundary lattices T_{η_a} for isotropic vectors η_a . Each outgoing arrow from (r_0, a_0, δ_0) will be of one of the following types:

Type of $\eta_a \in T$	Destination (r_1, a_1, δ_1)	Splitting of $\overline{T}_{\eta_a}$
Odd/simple	$(r_0 - 2, a_0, 1)$	$U \oplus K$
Even, ordinary	$(r_0 - 2, a_0 - 2, 1)$	$U(2) \oplus K$
Even, characteristic	$(r_0 - 2, a_0 - 2, 0)$	$I_{1,1}(2) \oplus K$

A **0-cusp** η_a **exists** if the arrow exists: the destination represents a realizable, hyperbolic, 2-elementary lattice $\overline{T}_{\eta_a}$ in Nikulin's table, and the node at the head of the arrow gives the invariants (r_1, a_1, δ_1) of the boundary lattice $\overline{T}_{\eta_a}$.

Step 2: All 1-Cusps and Incidence Structure

1. For each 0-cusp η_a determined in Step 1, recursively use $\overline{T}_{\eta_a}$ with invariants (r_1, a_1, δ_1) as a new starting point and consider all outgoing arrows from (r_1, a_1, δ_1) to construct (r_2, a_2, δ_2) . Each move corresponds to a splitting of $\overline{T}_I := \overline{(T_{\eta_a})_{\eta_b}}$ for a primitive isotropic vector $\eta_b \in \overline{T}_{\eta_a}$:

Type of $\eta_b \in \overline{T}_{\eta_a}$	Destination (r_2, a_2, δ_2)	Splitting of $\overline{T}_I = \overline{(T_{\eta_a})_{\eta_b}}$
Odd/simple	$(r_1 - 2, a_1, \delta_1)$	$U \oplus K'$
Even, ordinary	$(r_1 - 2, a_1 - 2, \delta_1)$	$U(2) \oplus K'$
Even, characteristic	$(r_1 - 2, a_1 - 2, 0)$	$I_{1,1}(2) \oplus K'$

2. **A 1-cusp I of type (η_a, η_b) exists** if and only if the move $(r_1, a_1, \delta_1) \rightarrow (r_2, a_2, \delta_2)$ exists, corresponding to a 2-elementary lattice $\overline{T}_I$ with the specified invariants. The 1-cusp I is incident to η_a if and only if it arises from such a two-step sequence.

Step 3: Diagram Extraction The **nodes for 0-cusps** are given by a single admissible mirror move from (r_0, a_0, δ_0) , labeled by (r_1, a_1, δ_1) , the corresponding lattice splitting, and the root system $\Phi(\overline{T}_\eta)$. The **nodes for 1-cusps** are determined by two-step sequences $(r_0, a_0, \delta_0) \rightarrow (r_1, a_1, \delta_1) \rightarrow (r_2, a_2, \delta_2)$, with associated lattice splitting and root system $\Phi(\overline{T}_I)$. Each 1-cusp is incident to every 0-cusp (r_1, a_1, δ_1) that occurs as an intermediate node in such a sequence. While the boundary lattice $\overline{T}_I$ for a 1-cusp is determined up to isometry by invariants (r_2, a_2, δ_2) , distinct Γ -orbits of 1-cusps are distinguished by their root lattices $\Phi(\overline{T}_I)$, either tabulated in [AE23b] or computed via Vinberg's algorithm. We conclude by tabulating several useful references to use when carrying out this algorithm:

Cusp	Invariants	Existence Condition
0-cusp η	(r_1, a_1, δ_1)	$\exists (r_0, a_0, \delta_0) \rightarrow (r_1, a_1, \delta_1)$

Cusp	Invariants	Existence Condition
1-cusp I	$(r_2, a_2, \delta_2), \Phi(\overline{T}_I)$	$\exists (r_0, a_0, \delta_0) \rightarrow (r_1, a_1, \delta_1) \rightarrow (r_2, a_2, \delta_2)$

Step 1 Type, η_a	Step 2 Type, η_b	(r_1, a_1, δ_1)	(r_2, a_2, δ_2)
Odd/simple	Odd/simple	$(r_0 - 2, a_0, 1)$	$(r_0 - 4, a_0, 1)$
Odd/simple	Even, ordinary	$(r_0 - 2, a_0, 1)$	$(r_0 - 4, a_0 - 2, 1)$
Odd/simple	Even, characteristic	$(r_0 - 2, a_0, 1)$	$(r_0 - 4, a_0 - 2, 0)$
Even, ordinary	Odd/simple	$(r_0 - 2, a_0 - 2, 1)$	$(r_0 - 4, a_0 - 2, 1)$
Even, ordinary	Even, ordinary	$(r_0 - 2, a_0 - 2, 1)$	$(r_0 - 4, a_0 - 4, 1)$
Even, ordinary	Even, characteristic	$(r_0 - 2, a_0 - 2, 1)$	$(r_0 - 4, a_0 - 4, 0)$
Even, characteristic	Odd/simple	$(r_0 - 2, a_0 - 2, 0)$	$(r_0 - 4, a_0 - 2, 0)$
Even, characteristic	Even, ordinary	$(r_0 - 2, a_0 - 2, 0)$	$(r_0 - 4, a_0 - 4, 0)$
Even, characteristic	Even, characteristic	$(r_0 - 2, a_0 - 2, 0)$	$(r_0 - 4, a_0 - 4, 0)$

Table 3.3.4: Change of invariants under 2-step mirror moves.

Carrying out this algorithm for F_{En} shows that there are exactly two 1-cusps and two 0-cusps. The following diagram encodes the mirror-move procedure, recording a sequence of moves starting from a primitive sublattice $S \hookrightarrow L_{K3}$, computing $T := S^{\perp L_{K3}}$, finding $\overline{T}$, the first type of boundary lattice corresponding to a 1-step mirror move (corresponding to 0-cusps) and finally finding $\overline{\overline{T}}$, the target of a 2-step mirror move. We find that there are two possibilities for $\overline{T}$, indicated in the $\overline{T}$ column as $(10, 10, 0)_1$ and $(10, 8, 0)_1$, and two possibilities present in the $\overline{\overline{T}}$ column. There are three 2-step paths through the diagram, but only two possibilities for $\overline{\overline{T}}$, yielding two 1-cusps and two 0-cusps. We record the resulting cusp diagram below as well.

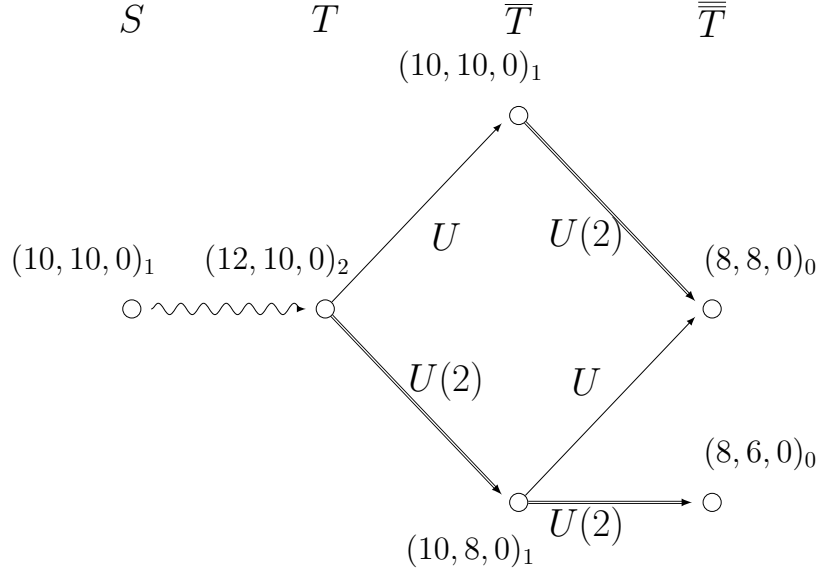


Figure 3.3.4: A concise summary of the mirror move algorithm applied to $(10, 10, 0)_1$, corresponding to F_{En} .

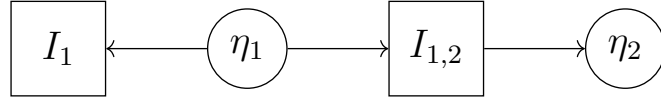


Figure 3.3.5: The cusp diagram of F_{En} , indicating two 0-cusps η_1, η_2 and two 1-cusps $I_1, I_{1,2}$. We note that this recovers the known cusp diagram shown in Figure 3.3.3.

3.4 Enumeration of Baily–Borel 0-cusps via Diagram Folding

The boundary lattices at 0-cusps η have the form $\overline{T}_\eta = \eta^\perp / \langle \eta \rangle$ for each primitive isotropic vector η . The typical situation is two have some number n of possible isometry classes for boundary lattices, and some number $m \geq n$ of *actual* boundary lattices, where m depends on Γ , reflecting isometry classes splitting into further possibilities. For $F_{(2,2,0)}$, there are two isometry classes, each containing one sub-type of lattice, so the two cusps can be labeled

$$\tilde{\eta}_1 := (18, 2, 0)_1, \quad \tilde{\eta}_2 := (18, 0, 0)_1.$$

For $F_{\text{En},2}$, there are two isometry classes, one which does not split into further subclass corresponding to orbits of divisibility one vectors in T_{En} , the other splitting into four sub-classes, reflecting an $O(T_{\text{En}})$ orbit of divisibility two vectors that splits into 4 separate orbits. We label the cusps $\eta_1, \dots, \eta_5$, and note that the two isometry classes are given by

$$\text{div}_{T_{\text{En}}}(\eta) = 1: (10, 10, 0)_1, \quad \text{div}_{T_{\text{En}}}(\eta) = 2: (10, 8, 0)_1$$

$F_{\text{En},2}$ Cusp	$F_{(2,2,0)}$ Cusp	Involution	Folded Lattice
η_1 ($\text{div}(\eta_1) = 1$)	$\tilde{\eta}_1 := (18, 2, 0)_1$	180° rotation	$(10, 10, 0)_1$
η_2 ($\text{div}(\eta_2) = 2$)	$\tilde{\eta}_2 := (18, 0, 0)_1$	Vertical reflection	$(10, 8, 0)_1$
η_3 ($\text{div}(\eta_3) = 2$)	$\tilde{\eta}_1 := (18, 2, 0)_1$	Diagonal + reflection	$(10, 8, 0)_1$
η_4 ($\text{div}(\eta_4) = 2$)	$\tilde{\eta}_1 := (18, 2, 0)_1$	Horizontal reflection	$(10, 8, 0)_1$
η_5 ($\text{div}(\eta_5) = 2$)	$\tilde{\eta}_1 := (18, 2, 0)_1$	8 commuting reflections	$(10, 8, 0)_1$

3.5 Cusps by Folding

Following [Ale+23, Lemmas 3.10, 3.19], the five 0-cusps of the Bailly–Borel compactification $\overline{F}_{\text{En},2}^{\text{BB}}$ correspond to five different involutions on the K3 lattice T_{dP} . Let T be an even indefinite lattice and let $\eta \in T$ be a primitive isotropic vector. Define the boundary lattice $\overline{T}_\eta := \eta^\perp T / \text{gen } \eta$, which we will often simply write as $\overline{T} := \eta^\perp / \eta$ when working with a fixed vector η . Write $\Phi(\overline{T})$ for the root system of $\overline{T}_\eta$. Let $\Gamma \leq O(T)$ be a fixed arithmetic subgroup acting on T . For each such η , we write U_η for the maximal unipotent subgroup of the stabilizer of η in Γ which acts trivially on $\overline{T}_\eta$. This group thus fits into an exact sequence

$$0 \rightarrow U_\eta \rightarrow \text{Stab}_\Gamma(\eta) \rightarrow \Gamma_\eta \rightarrow 0.$$

We refer to Γ_η as the **stable boundary group at η** and its corresponding reflection subgroup $W(\Gamma_\eta)$ as the **stable reflection group at η** . When

$\Gamma = \mathrm{O}(T)$, we will write this as $W(\overline{T})$. We write $\mathfrak{C}(\Gamma_\eta)$ for the corresponding fundamental chamber of $W(\Gamma_\eta)$, and $G(\Gamma_\eta)$ for its Coxeter diagram. When $\overline{T}_\eta$ is hyperbolic, it is known that $\mathrm{O}(\overline{T}_\eta)$ is an extension of $W(\overline{T}_\eta)$ by a subgroup of chamber symmetries $\mathrm{Aut}(\mathfrak{C}(\Gamma_\eta))$, and thus any diagram automorphism of $G(\Gamma_\eta)$ induces a chamber symmetry and thus an isometry of $\overline{T}_\eta$. Let J be an involution of $\overline{T}_\eta$, for example induced by a diagram involution on $G(\Gamma_\eta)$. We then write $G(\Gamma_\eta)^J$ for the corresponding folded diagram, $\mathfrak{C}(\Gamma_\eta)^J$ its folded fundamental chamber, and so on.

We recall the definitions

$$T_{\mathrm{En}} = U \oplus U(2) \oplus E_8(2) = (12, 10, 0)_2 \hookrightarrow T_{\mathrm{dP}} = U \oplus U(2) \oplus E_8^2 = (20, 2, 0)_2,$$

and write I_{En} , I_{dP} , and I_{Nik} for the involutions on L_{K3} induced by the geometric involutions ι_{En} , ι_{dP} , and ι_{Nik} respectively. In the decomposition $L_{\mathrm{K3}} = U^3 \oplus E_8^2$, these involutions can be written in block form as follows:

$$I_{\mathrm{dP}} : \begin{pmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}, \quad I_{\mathrm{En}} : \begin{pmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad I_{\mathrm{Nik}} : \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \end{pmatrix},$$

or as in Equation (2.1),

$$\begin{aligned} I_{\mathrm{dP}} : (u_1, u_2, u_3, \alpha_1, \alpha_2) &\rightarrow (-u_1, u_3, u_2, -\alpha_1, -\alpha_2) \\ I_{\mathrm{En}} : (u_1, u_2, u_3, \alpha_1, \alpha_2) &\rightarrow (-u_1, u_3, u_2, \alpha_2, \alpha_1) \\ I_{\mathrm{Nik}} : (u_1, u_2, u_3, \alpha_1, \alpha_2) &\rightarrow (u_1, u_2, u_3, -\alpha_2, -\alpha_1) \end{aligned}$$

By [Ale+23, Lem. 3.4], the coinvariant sublattice $T_{\mathrm{dP}}^{I_{\mathrm{En}}=-1} \xrightarrow{\sim} T_{\mathrm{En}}$, and so to simplify matters we write $I := -I_{\mathrm{En}} = I_{\mathrm{Nik}}$ and thus $T_{\mathrm{dP}}^I := T_{\mathrm{dP}}^{I=-1} = T_{\mathrm{En}}$. Fixing a primitive isotropic vector $\eta \in T_{\mathrm{En}}$, we write $\overline{T_{\mathrm{En}}\eta} := \eta^\perp_{T_{\mathrm{En}}} / \langle \eta \rangle_{T_{\mathrm{En}}}$ and $\overline{T_{\mathrm{dP}}\eta} := \eta^\perp_{T_{\mathrm{dP}}} / \langle \eta \rangle_{T_{\mathrm{dP}}}$, and so on. Under the primitive embedding $T_{\mathrm{En}} \hookrightarrow T_{\mathrm{dP}}$, we can construct two distinct boundary lattices associated to η ,

$$\overline{T_{\mathrm{En}}\eta} := \eta^\perp_{T_{\mathrm{En}}} / \langle \eta \rangle, \quad \overline{T_{\mathrm{dP}}\eta} := \eta^\perp_{T_{\mathrm{dP}}} / \langle \eta \rangle,$$

where we carefully identify η with its image in T_{dP} under the embedding. There are two possible isometry classes possible for $\overline{T_{\mathrm{dP}}\eta}$:

$$\overline{T_{\mathrm{dP}}\eta} : (18, 2, 0)_1 = U(2) \oplus E_8^2, \quad \text{or} \quad (18, 0, 0)_1 = U \oplus E_8^2,$$

corresponding to the two 0-cusps of the Baily-Borel compactification $\overline{F}_{(2,2,0)}^{\text{BB}}$, and distinguished by the divisibility $\text{div}_{T_{\text{dP}}}(\eta)$ of η in T_{dP} . From a similar analysis of $\overline{F}_{\text{En}}^{\text{BB}}$, the moduli space of *unpolarized* Enriques surfaces, one finds that there are similarly two possibilities

$$\overline{T}_{\text{En}\eta} : \quad (10, 10, 0)_1 = U(2) \oplus E_8(2), \quad \text{or} \quad (10, 8, 0)_1 = U \oplus E_8(2).$$

Both possibilities for $\overline{T}_{\text{dP}\eta}$ are hyperbolic, 2-elementary lattices with induced involutions $\overline{I}_{\text{En}}$ and $\overline{I}_{\text{dP}}$, and if we write $J := -\overline{I}_{\text{En}}$ then we recover $\overline{T}_{\text{dP}\eta}^J = \overline{T}_{\text{En}\eta}$ as the invariant sublattice. We immediately note that the involution J is highly sensitive to the choice of η , as are the isometry classes of $\overline{T}_{\text{dP}\eta}$ and thus $\overline{T}_{\text{En}\eta}$ – in our situation, this highly depends on the geometry of the moduli spaces $F_{\text{En},2}$ and $F_{(2,2,0)}$. To make this dependence explicit, we define the arithmetic groups used to construct period domains for these spaces. We define a distinguished polarization

$$h = e + f \in S_{\text{dP}} \xrightarrow{\sim} U(2) \hookrightarrow S_{\text{En}} \xrightarrow{\sim} U(2) \oplus E_8(2) = (10, 10, 0)_1,$$

and writing $\Psi : \text{O}(L_{\text{K3}}) \rightarrow \text{O}(T_{\text{En}})$ for the restriction map,

$$\Gamma_{\text{En},2} := \Psi \left(\left\{ g \in \text{O}(L_{\text{K3}}) \mid g \circ I_{\text{En}} = I_{\text{En}} \circ g, g(h) = h \right\} \right), \quad \Gamma_{\text{dP}} := \text{O}(T_{\text{dP}}),$$

where we form $\Gamma_{\text{En},2}$ by taking the intersection of the commutator of I_{En} in $\text{O}(L_{\text{K3}})$, intersecting it with the stabilizer of the polarization, and taking the image in $\text{O}(T_{\text{En}})$. This is the correct monodromy group for $F_{\text{En},2}$; the full details can be found e.g. in [Ste91]. For any lattice T of signature $(2, n)$, we define its period domain as

$$D_T := \left\{ [v] \in \mathbf{P}(T_{\mathbf{C}}) \mid v^2 = 0, v\bar{v} > 0 \right\}^{\circ},$$

where $(-)^{\circ}$ denotes taking one connected component. We then define the two moduli spaces

$$F_{\text{En},2} := D_T / \Gamma_{\text{En},2}, \quad F_{(2,2,0)} := D_T / \Gamma_{\text{dP}} = D_T / \text{O}(T_{\text{dP}}).$$

corresponding to degree 2 numerically polarized Enriques surfaces and quartic hyperelliptic K3 surfaces respectively. To study the Baily-Borel compactification $\overline{F}_{\text{En},2}^{\text{BB}}$, we will be interested in $\Gamma_{\text{En},2}$ -orbits of primitive isotropic vectors η_i , which correspond to 0-cusps, and the

Coxeter diagrams for the stable reflection groups $\Gamma_{\text{En},2,\eta_i}$ at each η_i , which can be used to determine the 1-cusps and their adjacencies. We note that if $\alpha \in \Phi(T_{\text{dP}})$ is a (negative) root of T_{dP} , then the folded root α_I is in $\Phi(T_{\text{dP}}^I) = \Phi(T_{\text{En}})$ and is thus a root of T_{En} . We can thus use the known stable reflection groups of T_{dP} at its known 0-cusps to study those of T_{En} and $\overline{T_{\text{En}\eta_i}}$ at each cusp η_i . We analyze the structure of reflection groups and invariant lattices associated to each 0-cusp η_i on the moduli space, using the correspondence between the (negative) roots of the covering domain and its foldings. The results are organized by cusp.

Let $\eta_1, \dots, \eta_5$ denote the five distinguished 0-cusps with the invariants, lattices, and automorphism relations described as follows:

3.5.1 Cusp $\eta_1 = e$ (divisor 1)

$$\begin{array}{ll}
\text{Transcendental lattice:} & (18, 2, 0)_1, \quad U(2) \oplus E_8^2 \\
\text{Automorphism: rotation by } 180^\circ : & \begin{cases} \alpha_i \leftrightarrow \alpha_{8+i}, & i = 0, \dots, 7 \\ \alpha_{16} \leftrightarrow \alpha_{18} \\ \alpha_{17} \leftrightarrow \alpha_{19} \end{cases} \\
\text{Invariants:} & (10, 10, 0)_1, \quad U(2) \oplus E_8(2)
\end{array}$$

3.5.2 Cusp $\eta_2 = e'$ (divisibility 2)

$$\begin{array}{ll}
\text{Transcendental lattice:} & (18, 0, 0)_1, \quad U \oplus E_8^2 \\
\text{Automorphism: vertical reflection:} & \alpha_i \leftrightarrow \alpha_{20-i}, \quad i = 1, \dots, 8 \\
\text{Invariants:} & (10, 8, 0)_1, \quad U \oplus E_8(2)
\end{array}$$

3.5.3 Cusp $\eta_3 = e' + f' + \omega$, $\omega^2 = -4$ (divisibility 2)

$$\begin{array}{ll}
\text{Transcendental lattice:} & (18, 2, 0)_1, \quad U(2) \oplus E_8^2 \\
\text{Automorphism: diagonal reflection:} & \begin{cases} \alpha_i \leftrightarrow \alpha_{16-i}, & i = 1, \dots, 7 \\ \alpha_{17} \leftrightarrow \alpha_{19} \\ s_{\alpha_{20}} \text{ fixes} \end{cases} \\
\text{Invariants:} & (10, 8, 0)_1, \quad U \oplus E_8(2)
\end{array}$$

3.5.4 Cusp $\eta_4 = e' + 2f' + \alpha$, $\alpha^2 = -8$ (divisibility 2)

Transcendental lattice:	$(18, 2, 0)_1, \quad U(2) \oplus E_8^2$
Automorphism: horizontal reflection:	$\begin{cases} \alpha_{14+i} \leftrightarrow \alpha_{14-i} \bmod 16 \\ \alpha_{19} \leftrightarrow \alpha_{16} \\ \alpha_{17} \leftrightarrow \alpha_{18} \\ \alpha_{20} \leftrightarrow \alpha_{21} \end{cases}$
Invariants:	$(10, 8, 0)_1, \quad U \oplus E_8(2)$

3.5.5 Cusp $\eta_5 = 2e + 2f + \alpha$, $\alpha^2 = -8$ (divisibility 2)

Transcendental lattice:	$(18, 2, 0)_1, \quad U(2) \oplus E_8^2$
Automorphism: 8 reflections:	$s_{\alpha_{2i+1}}, \quad i = 0, \dots, 7$
Invariants:	$(10, 8, 0)_1, \quad U \oplus E_8(2)$

3.5.6 The main cusp correspondence

In this section, we record the cusp diagrams of the main moduli spaces of interest: $F_{(2,2,0)}$ and $F_{\text{En},2}$. The cusp diagram for $F_{(2,2,0)}$ is shown below, which can be found in [AE22] or reconstructed using the mirror move algorithm. The boundary lattices for $F_{(2,2,0)}$ at its two 0-cusps are $(18, 0, 0)_1$ and $(18, 2, 0)_1$. The cusps, their Coxeter diagrams, and the KSBA compactification $\overline{F_{(2,2,0)}}$ were analyzed in detail in [AE22, §10]. Consider the Enriques transcendental lattice

$$T_{\text{En}} := U \oplus U(2) \oplus E_8(2).$$

The classification of $\Gamma_{\text{En},2}$ -orbits of primitive isotropic vectors in T_{En} provides an enumeration of the 0-cusps in the Baily–Borel compactification of $F_{\text{En},2}$. We know by [Ste91] what the five 0-cusps $\eta_1, \dots, \eta_5$ of $\overline{F_{\text{En},2}}^{\text{BB}}$ are, as well as their stable reflection groups, Coxeter diagrams, and the associated 1-cusps. We collect below some of the lattice-theoretic calculations that will be relevant to showing that folding methods can be used to recover this data. The following shows the cusp diagram for $F_{\text{En},2}$, where we note that the mirror move algorithm does *not* apply, since it only determines cusps when Γ is the full stable orthogonal group. We can instead appeal to [Ste91],

who computed these cusps and their incidences in their entirety. In the diagram below, we recapitulate these incidences, adding new information: recalling if D is a G -space and $H \leq G$ is a subgroup, the chain of subgroups $1 \hookrightarrow H \hookrightarrow G$ induces a chain of surjective morphisms $D \twoheadrightarrow D/H \twoheadrightarrow D/G$. Thus there is a chain of maps $D_{T_{\text{En}}} \rightarrow F_{\text{En},2} \rightarrow F_{\text{En}}$, and we can consider the images of the cusps of $F_{\text{En},2}$ in F_{En} as well as their images in $F_{(2,2,0)}$.

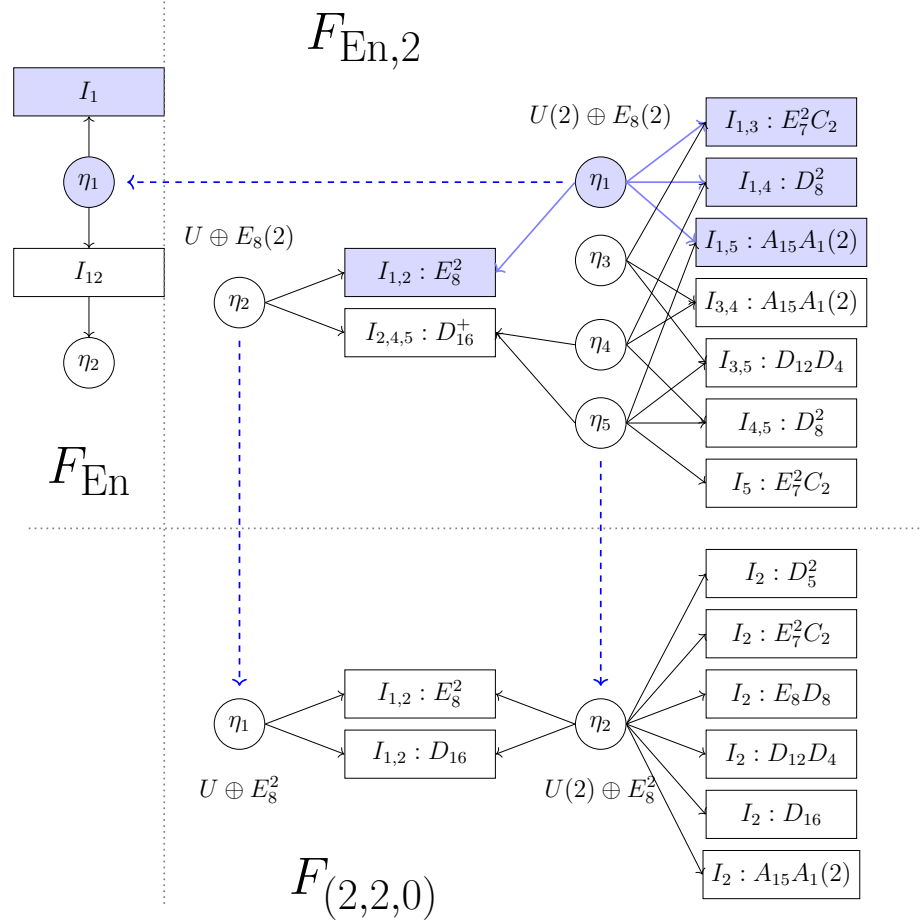


Figure 3.5.1: Mappings of boundary cusps under $F_{\text{En}} \leftarrow F_{\text{En},2} \rightarrow F_{(2,2,0)}$.

We briefly describe the methods that go into finding these cusps, due to [Ste91]. The following result generalizes the Eichler transvection method:

Theorem 3.5.6.1 (Orbit Classification in $T = U \oplus \bar{T}_\eta$ ([Ste91, Cor. 3.3])). *Fix an even lattice T with a splitting $T = U \oplus \bar{T}_\eta$ with $\bar{T}$ negative-*

definite and even. Let $\eta_1, \eta_2 \in T$ be primitive isotropic vectors satisfying

1. $\eta_1^2 = \eta_2^2 = k$;
2. $\text{div}_T(\eta_1) = \text{div}_T(\eta_2) = p > 0$ (so $(\eta_i, T) = p\mathbb{Z}$);
3. $\eta_1 \equiv \eta_2 \pmod{pT}$.

Then there exists an isometry $\phi \in O^*(T)$ such that $\phi(\eta_1) = \eta_2$. In particular, η_1 and η_2 lie in the same $O^*(T)$ -orbit.

Sketch. The idea is to let $\eta_i = a_i e + b_i e + c_i$ where $c_i \in \overline{T}_\eta$, and then put both η_i into normal form. Sterk shows that you can arrange for $\eta_1 \cdot e = p$, so $a_1 = 0, b_1 = p$ by some $g \in O^*(T)$, and that you can arrange for $\eta_2 \cdot e = p$ simultaneously, so $a_2 = 0, b_2 = p$. Thus

$$\eta_1 = pe + c_1, \eta_2 = pe + c_2, \quad \eta_1 f = \eta_2 f = p.$$

One then checks that

$$\eta_1 \equiv_{pT} \eta_2 \implies \eta_1 - \eta_2 \in pT \implies c_1 - c_2 = pc \in p\overline{T}_\eta$$

by canceling the now-identical pf components, and writes

$$\eta_1 - \eta_2 = pc \implies \eta_1 = \eta_2 + pc, \quad c \in \overline{T}_\eta.$$

Constructing the Siegel-Eichler transvection

$$E_{f,c}(x) = x - (x, f)c + \left((x, c) - \frac{1}{2}c^2(x, f) \right) f,$$

one finds that

$$\begin{aligned} E_{f,c}(\eta_1) &\equiv_{\mathbf{Z}_f} \eta_1 - (\eta_1, f)c \\ &\equiv_{\mathbf{Z}_f} (pe + c_1) - pc \\ &\equiv_{\mathbf{Z}_f} (pe + c_1) - (c_1 - c_2) \\ &\equiv_{\mathbf{Z}_f} pe + c_2 \\ &\equiv_{\mathbf{Z}_f} \eta_2 \end{aligned}$$

Writing $\eta_1 = \eta_2 + \lambda f$, one concludes by checking that

$$k = \eta_1^2 = E_{f,c}(\eta_1)^2 = (\eta_2 + \lambda f)^2 = \eta_2^2 + 2\lambda(\eta_2, f) + \lambda^2 f^2 = k + 2\lambda p,$$

forcing $\lambda = 0$ and $\eta_1 = \eta_2$. □

We then find that all divisibility one vectors are in the same orbit:

Proposition 3.5.6.2 ([Ste91, Lem. 4.2.1]). *If $\text{div}_{T_{\text{En}}}(v) = 1$, then $v \sim_{\Gamma_{\text{En},2}} v_1 := e \in U$.*

Proof. We apply Theorem 3.5.6.1: since

- $v^2 = e^2 = 0$,
- $\text{div}_{T_{\text{En}}}(v) = \text{div}_{T_{\text{En}}}(e) = 1$, and
- $v \equiv e \pmod{T_{\text{En}}}$,

we have $v \sim_{O^*(T_{\text{En}})} e$ and thus $v \sim_{\Gamma_{\text{En},2}} e$. □

The divisibility 2 vectors require a slightly finer analysis, but this quickly reduces to studying the large (but finite) discriminant group $A_{T_{\text{En}}}$: [Ste91, §. 4.2.2] first uses the fact that there is a decomposition

$$A_{T_{\text{En}}} = A \oplus B := \frac{1}{2}U/U \oplus \frac{1}{2}E_8/E_8$$

and if $\text{div}_{T_{\text{En}}}(\eta) = k \geq 2$ then $\eta/k \in T_{\text{En}}^\vee$ induces a nontrivial class in $A_{T_{\text{En}}}$. Any such class can be written as $g = a + b$ with $a \in A, b \in B$ and $q(g) = q_A(a) + q_B(b)$. If $q(g) = 0 \pmod{2\mathbf{Z}}$, there are exactly two possibilities:

1. $q_A(a) = q_B(b) = 0$, or
2. $q_A(a) = q_B(b) = 1$.

In the first case, one checks that the possibilities are

$$a = 0, \quad \frac{1}{2}e', \quad \frac{1}{2}f', \quad b = 0, \quad \frac{1}{2}\alpha, \quad \alpha \in E_8(2), \quad \alpha^2 = -8,$$

using explicit computations for $U(2)$ and known facts about $E_8(2)$, and one can immediately rule out $\frac{1}{2}f'$ up to stable isometries. In the second case, one has

$$a = \frac{1}{2}(e' + f'), \quad b = \frac{1}{2}\omega, \quad \omega \in E_8(2), \quad \omega^2 = -4.$$

This gives five possibilities, only four of which are new, which admit explicit lifts that are shown to be inequivalent under the action of $\Gamma_{\text{En},2}$:

$$A_T : \begin{cases} 0 \\ \frac{1}{2}e' \\ \frac{1}{2}\alpha & (\alpha^2 = -8) \\ \frac{1}{2}e' + \frac{1}{2}\alpha & (\alpha^2 = -8) \\ \frac{1}{2}e' + \frac{1}{2}f' + \frac{1}{2}\omega & (\omega^2 = -4) \end{cases} \rightsquigarrow T : \begin{cases} e \\ e' \\ 2e + 2f + \alpha & (\alpha^2 = -8) \\ e' + 2f' + \alpha & (\alpha^2 = -8) \\ e' + f' + \omega & (\omega^2 = -4) \end{cases}$$

Sterk then shows that under the full isometry group $O(T_{\text{En}})$, the four divisibility-two orbits collapse to a single orbit, and the divisibility-one orbit remains unique. Therefore, up to $O(T_{\text{En}})$, there are exactly two orbits of primitive isotropic vectors, partially affirming the known cusp diagram of F_{En} shown in the previous section.

3.5.7 Folded Coxeter Diagrams

We now describe—in precise terms following [Ale+23]—how the Coxeter diagrams for the 0-cusps of $F_{\text{En},2}$ are obtained by folding the Coxeter diagrams for the 0-cusps of the related quartic hyperelliptic K3 moduli $F_{(2,2,0)}$ under the involution $I = -I_{\text{En}}$. Recall there are two $O(T_{\text{dP}})$ -orbits of primitive isotropic vectors in T_{dP} , associated to the boundary lattices $(18, 2, 0)_1 = U(2) \oplus E_8^2$ and $(18, 0, 0)_1 = U \oplus E_8^2$. Each determines a *Coxeter diagram* encoding the walls of the fundamental chamber for the stable reflection group. The five 0-cusps of $F_{\text{En},2}$ correspond to five distinct orbits of primitive isotropic vectors in T_{En} (Sterk), and each is *realized as a folded image* of one of the K3 diagrams under the involution.

The fundamental fact is that the set of simple roots defining the faces of the Coxeter chamber are determined by Lemma 2.8.0.1 The *folded chamber* for the reflection group in T_{En} is the intersection

$$\mathfrak{C}^I = \mathfrak{C} \cap \overline{T}_{\eta, \mathbf{R}}^{I=1}$$

where $\mathfrak{C}$ is the Coxeter chamber for T_{dP} , yielding a fundamental chamber for the reflection group in T_{En} whose faces correspond to the roots described above. Simple roots fixed by I correspond to cases 1 and 2 above, and pass directly to the folded diagram as roots of the same

norm. Pairs of simple (-2) -roots swapped by the involution I and orthogonal to their images, i.e. $I(\alpha) \in \alpha^\perp$, are “averaged” into a new (-4) -root wall of the folded diagram. Moreover, every wall of the folded chamber, and hence every boundary divisor at each 0-cusp of $F_{\text{En},2}$, arises by this procedure. Maximal *parabolic subdiagrams* of the K3 diagrams invariant under I (i.e., unions of nodes fixed or swapped under I as above) yield, upon folding, the parabolic subdiagrams of the Enriques diagram, which can thus be used to find the 1-cusps and complete the full cusp diagram.

To summarize, the five Coxeter diagrams for the five 0-cusps of $F_{\text{En},2}$ are obtained by explicit folding of the diagrams $G(18, 2, 0)$ and $G(18, 0, 0)$ of $F_{(2,2,0)}$ under I . This process precisely matches Sterk’s list: each 0-cusp boundary lattice in T_{En} is the fixed lattice under I of the appropriate K3 boundary lattice for T_{dP} , and the combinatorics of simple roots and walls are gotten by descending them from T_{dP} under the involution. Moreover, each Coxeter diagram for each 0-cusp of $F_{\text{En},2}$ arises from an explicit involution on a Coxeter diagram for $F_{(2,2,0)}$, which we list below. In each diagram, a folding symmetry is defined by a combination of blue arrows and crossed out red nodes. Each folding involution is strictly speaking an element of $O(\Phi(T))$ for an appropriate lattice T , which decomposes as the semidirect product of a reflection group and a subgroup of diagram symmetries, as described above. We can thus specify *some* isometries of $\Phi(T)$ by combining elements from each factor. In these diagrams, the blue decorations indicate isometries taken from the group of diagram symmetries, while red crossed-out nodes indicate elements taken from the Weyl group. Explicitly, in each case we have:

1. A counter-clockwise rotation by π ,
2. A left-to-right reflection about the center,
3. A reflection about the line $y = x$, supposing the bottom-left node is at $(0, 0)$ in the plane, composed with a reflection in a single root,
4. A top-to-bottom reflection about the center, and
5. A composition of commuting reflections in 8 simple roots.

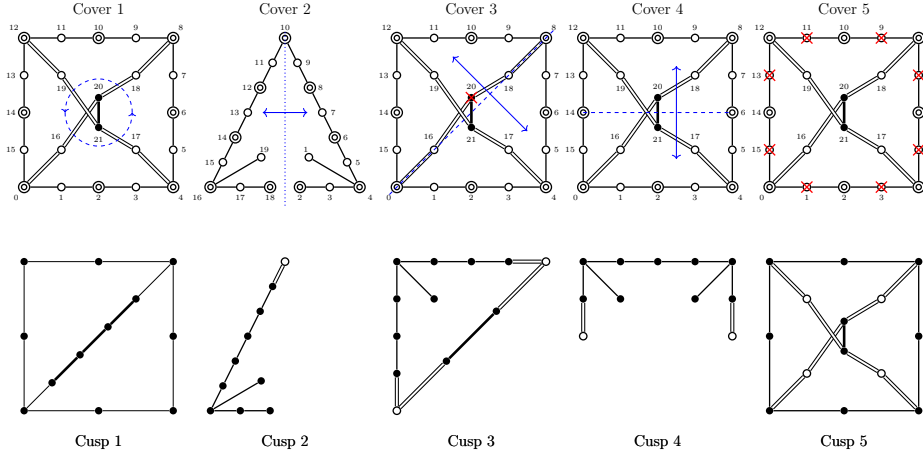


Figure 3.5.2: The five 0-cusps η_i in $F_{\text{En},2}$, along with the five "covering" relations: each corresponds to one of the two 0-cusps of $F_{(2,2,0)}$, along with an involution specific to each η_i .

Part II

**Compactifications and Integral
Affine Structures**

Let D be a Hermitian symmetric domain of type IV, such as the domains D_T arising from indefinite lattices of signature $(2, n)$, and Γ be an arithmetic subgroup of $\text{Aut}(D)$, e.g. an arithmetic subgroup of $\text{O}(T)$. The typical compactifications of $F_\Gamma = \Gamma \backslash D$ include the Baily–Borel compactification $\overline{F}_\Gamma^{\text{BB}}$, toroidal compactifications $\overline{F}_\Gamma^{\Sigma^\bullet}$, semitoroidal compactifications $\overline{F}_\Gamma^{\mathcal{F}^\bullet}$, and the KSBA compactification $\overline{F}_\Gamma$. Discussions of the following material can be found in [Sca87], [Ash+75], [BB66], [FM13], [KM98b].

3.5.8 Baily–Borel Compactification

The Baily Borel compactification $\overline{F}_\Gamma^{\text{BB}}$ is defined as $\text{Proj } A(\Gamma)$, where $A(\Gamma)$ is the graded $\mathbf{C}$ -algebra of automorphic forms for Γ . It is a projective, normal variety, and is canonical in the sense that it does not depend on any auxiliary choices. It is the unique *minimal* geometric compactification in the following sense: for any other normal, projective compactification X of F_Γ admitting an extension of the period map, there exists a unique, regular, projective morphism $X \rightarrow \overline{F}_\Gamma^{\text{BB}}$ restricting to the identity on F_Γ . The boundary $\partial \overline{F}_\Gamma^{\text{BB}}$ consists of finitely many strata, each itself a quotient of a Hermitian symmetric domain by an arithmetic group, and is described via rational boundary components indexed by maximal rational parabolics subgroups of $\text{Aut}(D)$. However, the boundary is generally of high codimension and almost never divisorial for Type IV domains, for which it is generically comprised of only points and modular curves. The analytic structure at the boundary is highly singular and generally not even $\mathbf{Q}$ -factorial. Moreover, boundary points typically do not correspond to moduli of degenerate surfaces. However, $\overline{F}_\Gamma^{\text{BB}}$ is functorial and canonical, and the representation-theoretic nature of the boundary components often reduces their study to combinatorics and lattice theory.

3.5.9 Toroidal Compactifications

Toroidal compactifications $\overline{F}_\Gamma^{\Sigma^\bullet}$ are defined by refining the structure near the cusps of $\overline{F}_\Gamma^{\text{BB}}$. For each cusp, one chooses an admissible rational polyhedral fan in a particular cone at that cusp. The result is a proper algebraic variety with a divisorial boundary which is locally toric. The construction is highly non-canonical, and depends on the choice of fans, of which there are typically infinitely many. Different

choices give different, though birationally equivalent, compactifications, all mapping to $\overline{F}_\Gamma^{\text{BB}}$. The boundary $\partial\overline{F}_\Gamma^{\Sigma^\bullet}$, after potentially a refinement of fans and a finite cover, is a snc divisor. Each boundary component is constructed from tori associated to unipotent radicals U_P of rational parabolic subgroups P of $\Gamma \leq \text{Aut}(D)$. The fibers of the canonical morphism $\overline{F}_\Gamma^{\Sigma^\bullet} \rightarrow \overline{F}_\Gamma^{\text{BB}}$ over boundary strata are toric. The boundary singularities are typically milder than in $\overline{F}_\Gamma^{\text{BB}}$, and for well-chosen admissible decompositions, it is possible to construct smooth compactifications. Moreover, toroidal compactifications often allow the extension of period maps and universal families over boundary strata, yielding a much stronger modular interpretation than $\overline{F}_\Gamma^{\text{BB}}$.

3.5.10 Semitoroidal Compactifications

Semitoroidal compactifications, introduced by [Loo85], are useful when $\overline{F}_\Gamma^{\text{BB}}$ is insufficient, but a full toroidal compactification $\overline{F}_\Gamma^{\Sigma^\bullet}$ is unnecessary or obstructed, or there are no particularly canonical choice of the necessary toric fan data. While $\overline{F}_\Gamma^{\text{BB}}$ has boundary components of codimension typically at least 2 and toroidal compactifications $\overline{F}_\Gamma^{\Sigma^\bullet}$ *fully* refine all cusps to divisorial components, semitoroidal compactifications only partially “toroidalize” along a subset of boundary strata. They thus allow individual cusps to be differentiated, some being fully toroidal, and others being only partially toroidal, hence the name *semitoroidal*. Formally, choose a subset $\mathcal{S}$ of boundary strata in $\overline{F}_\Gamma^{\text{BB}}$ and assign, for each $F \in \mathcal{S}$, an admissible rational polyhedral decomposition (i.e. a *semifan*) in the relevant cone at that cusp. One then performs a blowup up only along these loci, glues in toroidal models over $\mathcal{S}$, and leave other strata untouched. The boundary is divisorial along those strata $F \in \mathcal{S}$, but retains higher codimension strata elsewhere. The construction is not canonical, since there are still infinitely many choices of both semitoric data at each cusp. There are still natural projective morphisms

$$\overline{F}_\Gamma^{\Sigma^\bullet} \longrightarrow \overline{F}_\Gamma^{\mathcal{F}^\bullet} \longrightarrow \overline{F}_\Gamma^{\text{BB}}$$

restricting to the identity on the open locus F_Γ , yielding a compactification birationally equivalent to all toroidal compactifications. By [AE21, Thm. 5.14] the semitoroidal compactifications are *precisely* the compactifications fitting into such a tower. Such compactifications are particularly effective when compactifying Type IV domains, and

often admit extensions of period maps and (partial) universal families over the cusps.

3.5.11 KSBA Compactification

The KSBA compactification $\overline{F}_\Gamma$ arises from the (log) minimal model program (MMP), for varieties of log general type. For a moduli problem, e.g., polarized varieties or stable pairs, one seeks a modular compactification where points correspond to geometric degenerations: namely, stable slc pairs (X, D) with prescribed numerical invariants with $K_X + D$ ample. Forming the moduli stack of such pairs, by properness of the moduli functor (see [KS88; Ale96b]), one obtains a proper algebraic space that compactifies the stack of smooth pairs. The resulting **KSBA moduli space** $\overline{F}_\Gamma$ includes boundary divisors corresponding to *all* possible stable degenerations, including those not accessible via general methods for locally symmetric spaces. The boundary is divisorial, though possibly reducible and singular, and every 1-parameter family over a DVR of smooth surfaces admits, after base-change, a unique stable limit. Universal families exist over open subsets, and a coarse moduli interpretation is valid everywhere. When a period map exists (e.g. by a Torelli theorem), there is a regular but not generally finite morphism $\overline{F}_\Gamma \rightarrow \overline{F}_\Gamma^{\text{BB}}$ and $\overline{F}_\Gamma$ is generally the finest among all of the compactifications discussed. In general, these maps contract or identify boundary components, and the typical situation is shown in Figure 3.5.3. However, in situations such as $F_{\text{En},2}$, the morphism $\overline{F}_\Gamma \rightarrow \overline{F}_\Gamma^{\text{BB}}$ lifts to an isomorphism $\overline{F}_\Gamma \xrightarrow{\sim} \overline{F}_\Gamma^{\mathcal{F}\bullet}$ for a proper choice of semifans $\mathcal{F}$.

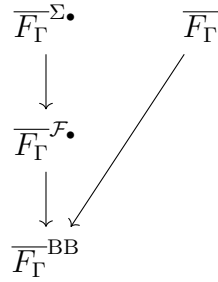


Figure 3.5.3: The typical relationship between toroidal, semitoroidal, Baily-Borel, and KSBA compactifications.

CHAPTER 4

COMPACTIFICATIONS

4.1 Comparing Compactifications

Given a smooth projective family of varieties $\mathcal{X} \rightarrow \Delta$ over the unit disk with $\mathcal{X}_0$ a simple normal crossings (SNC) divisor, the Clemens–Schmid exact sequence provides a powerful tool to analyze the limit mixed Hodge structure on the cohomology of the general fiber. This sequence, however, depends on the presence of semistable (SNC) reduction, and in general, no analogous sequence or description is known when the total space $\mathcal{X}$ is itself singular or the $\mathcal{X}_0$ has worse singularities. To systematically handle degenerating families of higher-dimensional varieties – where semistable reduction is either inadequate or inapplicable – [KS88] introduced the theory of stable slc pairs. A **stable pair** consists of a projective variety X (possibly reducible and with mild singularities) together with a $\mathbf{Q}$ -divisor D such that $K_X + D$ is ample and $\mathbf{Q}$ -Cartier, and the pair (X, D) has semi log canonical (slc) singularities. This generalizes the definition of stable curves from the classical Deligne–Mumford compactification $\overline{\mathcal{M}}_g$ to all dimensions, and stable slc pairs provide a modular, proper, algebraic compactification of moduli spaces of varieties and pairs. For moduli of K3 surfaces, this leads to the **KSBA compactification**: for a moduli space F_{2d} of degree $2d$ polarized K3 surfaces, and for a suitable **recognizable** divisor R on a generic such surface, one studies the closure of the locus of pairs $(X, \varepsilon R)$ (for sufficiently small $\varepsilon > 0$) in the moduli space of all KSBA stable pairs, sometimes referred to as *slc pairs* in the relevant literature. For example, for a degree two polarized K3 surface (X, L) , the linear system $|L|$ gives a double cover $X \rightarrow \mathbf{P}^2$ branched along a sextic, and R can be chosen as the ramification divisor for the covering

involution. When R is recognizable, there exists a semifan $\mathcal{F}_R$ so that the normalization of the KSBA compactification $\overline{F}_{2d}$ is isomorphic to the corresponding **semitoroidal compactification** $\overline{F}_{2d}^{\mathcal{F}_R}$. The precise construction of such semifans is highly nontrivial and depends on the geometry of the polarization.

Drawing an analogy with the Deligne–Mumford compactification of $\mathcal{M}_g$, one obtains the following procedure for families of surfaces: after base change and birational modifications, any family $\mathcal{X} \rightarrow \Delta$ of surfaces can be replaced by a semistable model, that is, with $\mathcal{X}$ smooth and the $\mathcal{X}_0$ a reduced snc divisor. One then contracts (-1) -curves in the fibers to obtain a relatively minimal model with smooth total space. To construct a stable model, one further contracts all (-2) -curves in the fibers; this produces a family whose total space is typically singular, but whose $\mathcal{X}_0$ is uniquely determined and has at worst nodal singularities. This uniqueness (up to isomorphism), together with the finiteness of the automorphism group and the ampleness of the dualizing sheaf, are the properties which extend to higher dimensional varieties. Recall that a divisor D on a normal variety X has simple normal crossings (snc) if, in a Zariski neighborhood of every point, its support is defined by the vanishing of a product of coordinate functions. A log pair (X, D) is said to be **divisorial log terminal** (dlt) if there exists an open subset $U \subseteq X$ such that $D|_U$ has snc support, and for any divisorial valuation E centered in $X \setminus U$, the log discrepancy $a(E, X, D) > -1$. A **KSBA-stable pair** is a pair $(X, D = \sum b_j D_j)$ where X is a projective demi-normal variety (that is, X satisfies Serre’s condition S_2 and has normal crossing singularities in codimension 1), the divisorial components D_j are effective Weil divisors of X not contained in the singular locus of X , and the coefficients b_j are rational numbers with $0 < b_j < 1$, such that:

1. The divisor $K_X + D$ is ample and $\mathbf{Q}$ -Cartier.
2. The pair (X, D) has semi log canonical (slc) singularities

This notion generalizes the stability condition for curves: for X a nodal curve and $D = \sum b_j p_j$ a divisor with $0 < b_j < 1$, the pair (X, D) is KSBA stable if and only if X is stable in the sense of Deligne–Mumford. The KSBA compactification provides a canonical and proper compactification of the moduli space of varieties and pairs of log general type by allowing stable pairs (X, D) with semi log canonical singularities and ample $K_X + D$. As a consequence, these

moduli spaces are proper and projective, and their construction is compatible with the minimal model program.

4.2 Recognizable Divisors

Let F_S be the moduli space of S -polarized K3 surfaces, and let R be a canonical ample divisor on the generic surface in F_S . The divisor R is called **recognizable** (for F_S) if for any S -quasipolarized Kulikov degeneration $\mathcal{X} \rightarrow \Delta$, there exists a divisor $R_0 \subset \mathcal{X}_0$ such that, for any other smoothing $\tilde{\mathcal{X}} \rightarrow \Delta$ with $\mathcal{X}_0 \xrightarrow{\sim} \tilde{\mathcal{X}}_0$, the divisor R_0 is, up to the connected component of the identity $\text{Aut}^0(\tilde{\mathcal{X}}_0)$, the unique flat limit of the divisors $R_t \subset \mathcal{X}_t$ as $t \rightarrow 0$. This yields a form of uniqueness and “path-independence” for Kulikov models, namely that R extends unambiguously to all boundary components of F_S . For any choice of recognizable divisor, the normalization of the KSBA stable pair compactification $\overline{F}^R$ is a semitoroidal compactification of the period domain, with the semifan determined by R . A canonical example of a recognizable divisor comes from [AE23a], who compactify the moduli spaces F_{2d} of polarized K3 surfaces (X, L) of degree $2d$. The **rational curve divisor** is the formal sum R_{rc} of all smooth genus zero curves in the linear system $|L|$:

$$R_{\text{rc}} := \sum_{i=1}^{n_d} R_i \in |n_d L|$$

where each R_i is an irreducible rational curve and n_d is given by the *Yau–Zaslow* formula, see e.g. [AE23a, Thm. 10.2]. In loc.cit., this divisor is shown to be recognizable for all d , and thus there are semifans such that $\overline{F}_{2d}^{\mathcal{F}^\bullet} \cong \overline{F}_{2d}$. The proof uses the Kontsevich moduli space of stable maps, Gromov–Witten invariants, and properties of rational stable maps to surfaces with K-trivial degenerations.

This follows a similar line of work: for F_2 , see [AET19a], and for degree 2 elliptic K3 surfaces F_{ell} , see [ABE22]. These are S -polarized K3 surfaces for S a 2-elementary lattice that primitively embeds in L_{K3} , of which there are exactly 75 by [Nik79b], and [AE23b, Thm. 9.10] handles the remaining cases, including $F_{(2,2,0)}$ corresponding to $S = U(2) = (2, 2, 0)_1$, using the fact that such surfaces carry a nonsymplectic involution ι whose ramification divisor R_ι is recognizable. For 50 of these cases, there is result similar to Theorem 1.0.0.1: $\overline{F}_S^{\mathcal{F}^\bullet} \xrightarrow{\sim} \overline{F}_S$ for a semifan associated to R_ι – they are precisely the closures of ir-

reducible components on which monodromy invariants λ maintain a fixed *combinatorial type*, as defined in [AE21]. By passing to Kulikov models with nonsymplectic involutions, they show that stable limits can be constructed using the fact that the central fibers $\mathcal{X}_0 = \cup_i V_i$ of Type II Kulikov models $\mathcal{X}$ of K3 surfaces admit dual complexes $\Gamma(\mathcal{X}_0)$ homeomorphic to S^2 , which can naturally be equipped with integral affine structures, yielding an ι -symmetric IAS², where ι typically acts on S^2 by $(x, y, z) \mapsto (x, y, -z)$ in the standard coordinates on $\mathbf{R}^3$. Thus one can reverse-engineer this procedure, starting with a model for a hemisphere of S^2 which is homeomorphic to $\mathbf{D}^2$, and realizing $S^2 \cong \mathbf{D}^2 \sqcup_{\partial \mathbf{D}^2} \mathbf{D}^2$ as the pushout of two discs along their boundaries, naturally enforcing ι -invariance. The process starts from a *monodromy invariant* λ encoding a degeneration, then constructs an integral-affine polygon $P(\lambda)$ in $\mathbf{R}^2$ with singularities, performing Symington surgeries on $P(\lambda)$ that encode the degeneration. Passing to a complete triangulation and taking the pushout

$$B(\lambda) := P(\lambda) \bigsqcup_{\partial P(\lambda)} (-P(\lambda)),$$

where $-P(\lambda)$ denotes reversing the orientation, yields an IAS².

From this, one can construct a *d-semistable Kulikov surface* such that $B(\lambda) \cong \Gamma(\mathcal{X}_0(\lambda))$, which by [Fri83b] smooths to K3 surface $\mathcal{X}_t := \mathcal{X}_t(\lambda)$ and thus specifies a family $\mathcal{X} := \mathcal{X}(\lambda)$. One extends the induced involution ι on $m\mathcal{X}_0$ to $\mathcal{X}$, passes to a carefully chosen divisorial component of the ramification divisor R_t on $\mathcal{X}$ and performs modifications to obtain a pair $(X, \varepsilon R) := (X(\lambda), \varepsilon R(\lambda))$. One then shows that the limit of R_t is big and nef and thus defines a contraction $\pi : (X, \varepsilon R) \rightarrow (\overline{X}, \varepsilon \overline{R})$ to a KSBA stable pair. There is a decomposition $X = \cup_i (V_i, D_i)$ into irreducible components indexed by the lattice points in $B(\lambda)$, each of which forms an anticanonical pair, and so we can write $(X, \varepsilon R) = \cup_i ((V_i, D_i), \varepsilon R_i)$, which contracts under π to a decomposition $(\overline{X}, \varepsilon \overline{R}) = \cup_i ((\overline{V}_i, \overline{D}_i), \varepsilon \overline{R}_i)$ of the stable model. Tracing this construction backwards, we thus access the irreducible components of the stable model by understanding how the contraction π combinatorially acts at the level of $B(\lambda) = \Gamma(\mathcal{X}_0)$, and in particular how the contraction acts on individual anticanonical pairs (V_i, D_i) . By analogy, we seek a similar method of studying Enriques surfaces.

4.3 The Baily-Borel Compactification

4.3.1 Hermitian Symmetric Domains and Arithmetic Quotients

Let G be a connected, simple linear algebraic group defined over $\mathbf{Q}$ and fix a maximal compact subgroup $K \subset G(\mathbf{R})$. The quotient $D = G(\mathbf{R})/K$ is an irreducible symmetric space. This data defines a **Riemannian symmetric space** (G, K) with a decomposition of the Lie algebra of $G(\mathbf{R})$:

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p},$$

where $\mathfrak{p}$ is the orthogonal complement to $\mathfrak{k}$ for an invariant symmetric bilinear form β on $\mathfrak{g}$. These spaces frequently arise in moduli problems: for an arithmetic subgroup $\Gamma \subset G(\mathbf{Q})$ acting properly discontinuously on $D := G/K$, the quotient D/Γ is a locally symmetric space. It can be expressed as the double coset space

$$F_\Gamma := D/\Gamma \cong \Gamma \backslash G/K = \left\{ \Gamma g K \mid g \in G \right\},$$

which is typically a non-proper, quasi-projective complex variety parameterizing representation-theoretic data such as Hodge structures of a fixed type. Such symmetric spaces (G, K) are classified by the restriction of β :

- **Euclidean type:** $\beta|_{\mathfrak{p}} = 0$,
- **Compact type:** $\beta|_{\mathfrak{p}} < 0$ (negative definite),
- **Non-compact type:** $\beta|_{\mathfrak{p}} > 0$ (positive definite).

For non-compact types, D is a **Hermitian symmetric domain** (HSD): a non-compact, irreducible symmetric space with a $G(\mathbf{R})$ -invariant complex structure and Hermitian metric. All Hermitian symmetric domains can be realized as bounded domains in $\mathbf{C}^n$ via the **Harish-Chandra embedding**. They admit a classification due to Cartan; the non-exceptional types are:

Type	Standard Description	Realization as Homogeneous Space
$I_{p,q}$	Complex Grassmannian of p -dimensional subspaces	$SU_{p,q}/S(U_p \times U_q)$
II_n	Lagrangian Grassmannian (orthogonal type)	$SO_{2n}^*(\mathbf{R})/U_n$
III_n	Siegel upper half space (symplectic type)	$Sp_n(\mathbf{R})/U_n$
IV_n	Period domains	$O_{2,n}(\mathbf{R})/(O_2(\mathbf{R}) \times O_n(\mathbf{R}))$

Type I

Type I domains arise as quotients $D_{p,q}^I = SU_{p,q}/S(U_p \times U_q)$ where $p \leq q$, and consist of p -dimensional positive subspaces in $\mathbf{C}^{p+q}$ with respect to the Hermitian form

$$h(z, w) = z_1 \bar{w}_1 + \cdots + z_p \bar{w}_p - z_{p+1} \bar{w}_{p+1} - \cdots - z_{p+q} \bar{w}_{p+q}.$$

A fundamental example is the **complex upper half-plane**,

$$\mathbb{H} = \left\{ \tau \in \mathbf{C} \mid \Im(\tau) > 0 \right\} \cong SL_2(\mathbf{R})/SO_2(\mathbf{R}),$$

where $\Im(\tau)$ is the imaginary part of τ . The action of $SL_2(\mathbf{Z})$ (or congruence subgroups $\Gamma \leq SL_2(\mathbf{Z})$ thereof) on $\mathbb{H}$ via Möbius transformations

$$\gamma \cdot \tau = \frac{a\tau + b}{c\tau + d}, \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$$

gives rise to the **modular curves** $Y_\Gamma = \Gamma \backslash \mathbb{H}$, which are coarse moduli spaces for elliptic curves with level structure.

Type III

Type III domains take the form $D_g^{III} = Sp_g(\mathbf{C})/U_g$ and consist of $g \times g$ symmetric complex matrices Z whose imaginary part is positive definite – otherwise known as the **Siegel upper half space**,

$$\mathbb{H}_g = \left\{ Z \in M_{g \times g}(\mathbf{C}) \mid Z^t = Z, \Im(Z) > 0 \right\},$$

where $\Im(Z)$ denotes the imaginary part of the matrix Z . The group $\mathrm{Sp}_{2g}(\mathbf{Z})$ acts by

$$\gamma \cdot Z = (AZ + B)(CZ + D)^{-1}, \quad \gamma = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathrm{Sp}_{2g}(\mathbf{Z}),$$

producing a coarse moduli space for complex principally polarized abelian varieties: $\mathcal{A}_g = \mathrm{Sp}_{2g}(\mathbf{Z}) \backslash \mathbb{H}_g$. For genus $g > 1$, via the Torelli map, the moduli space of curves $\mathcal{M}_g$ embeds as an open subset of a suitable arithmetic quotient of a Siegel space.

Type IV

Type IV domains are open subsets of quadric surfaces defined as

$$D_T := \left\{ [v] \in \mathbf{P}(T_{\mathbf{C}}) \mid v^2 = 0, v\bar{v} > 0 \right\}^{\circ}$$

where L is a lattice of signature $(2, n)$ and $(\cdot, \cdot)$ denotes the complex *bilinear* extension of the intersection pairing to $L_{\mathbf{C}}$. These domains can be realized as open subsets of quadric hypersurfaces in projective space, $Q = \left\{ [\omega] \in \mathbf{P}^{n+1} \mid (\omega, \omega) = 0 \right\}$, and the boundary components of their Baily-Borel compactifications encode possible limiting mixed Hodge structures of varieties.

4.3.2 Cusps Parabolic Subgroups

To compactify F_{Γ} , one must analyze the boundary of D . This is achieved via the **Borel embedding**, a $G(\mathbf{R})$ -equivariant holomorphic open immersion $D \hookrightarrow D^{\vee}$, where $D^{\vee}$ is the **compact dual**. $D^{\vee}$ can be realized as $\tilde{G}/K$, where $\tilde{G}$ is the simply connected complex Lie group with Lie algebra $\tilde{\mathfrak{g}} := \mathfrak{k} + i\mathfrak{p} \subset \mathfrak{g} \otimes_{\mathbf{R}} \mathbf{C}$. The boundary of the closure of D in $D^{\vee}$, denoted ∂D , decomposes as a disjoint union of maximal connected complex-analytic subsets, known as **boundary components** or **cusps**, $\partial D = \bigsqcup F_i$. For each such component F , its stabilizer in $G(\mathbf{R})$ is the parabolic subgroup

$$N_F := \left\{ g \in G(\mathbf{R}) \mid gF = F \right\}.$$

The maximal parabolic subgroups of $G(\mathbf{R})$ are precisely those arising as stabilizers N_F of boundary components F , and thus there is a

bijective correspondence

$$\{\text{Boundary components } F \subseteq \partial D\} \longleftrightarrow \{\text{Max. parabolic subgroups } P \leq G\}.$$

A boundary component F is **rational** if its stabilizer N_F is defined over $\mathbf{Q}$. Let $B(D)$ be the collection of proper rational boundary components. The set $B(D)$ is in a natural bijection with the set of proper maximal parabolic $\mathbf{Q}$ -subgroups of G . The boundary is thus stratified by its rational components:

$$\partial D = \bigsqcup_{F \in B(D)} F \subseteq D^\vee$$

4.3.3 The Rational Closure

The Baily–Borel compactification of $F_\Gamma := D/\Gamma$, denoted $\overline{F}_\Gamma^{\text{BB}}$, is constructed from the **rational closure** of D :

$$D^* := D \cup \partial D := D \cup \bigsqcup_{F \in B(D)} F$$

The arithmetic group Γ acts naturally on D^* with only finitely many orbits of boundary strata. The compactification is the quotient space, which can also be expressed as the quotient of D with its boundary adjoined:

$$\overline{F}_\Gamma^{\text{BB}} := D^*/\Gamma = (D \cup \partial D)/\Gamma.$$

By the main theorem of Baily and Borel, $\overline{F}_\Gamma^{\text{BB}}$ is a compact and Hausdorff space that contains F_Γ as a dense open subset whose boundary $\partial \overline{F}_\Gamma^{\text{BB}} := \overline{F}_\Gamma^{\text{BB}} \setminus F_\Gamma$ is a finite disjoint union of closed, locally symmetric varieties: $\overline{F}_\Gamma^{\text{BB}} \setminus F_\Gamma = \bigsqcup [F] \in B(D)/\Gamma V_F$ where the indices $[F]$ run over Γ -orbits rational boundary components. Each component F is itself a Hermitian symmetric domain (or possibly a point), and the stratum V_F is its arithmetic quotient, and we can write

$$V_F = F/N_\Gamma(F) \text{ where } N_\Gamma(F) := \text{Stab}_\Gamma(F)/\text{Fix}_\Gamma(F), \\ \text{Aut}(V_F) = G_F := \text{Stab}_{G(\mathbf{R})}(F)/\text{Fix}_{G(\mathbf{R})}(F).$$

Since F is rational, $N_\Gamma(F)$ is a discrete arithmetic subgroup of the Lie group G_F . As F is an HSD, the quotient V_F inherits the structure of a locally symmetric variety.

4.3.4 Projectivity via Automorphic Forms

A foundational result of [BB66] is that $\overline{F}_\Gamma^{\text{BB}}$ can alternatively be constructed from the space of automorphic forms for Γ and a canonical automorphic line bundle, showing it is a normal projective variety. Letting $D = G(\mathbf{R})/K$ be a Hermitian symmetric domain associated to a symmetric pair (G, K) , and $\Gamma \subset G(\mathbf{Q})$ be an arithmetic subgroup acting properly discontinuously on D as above, there is a distinguished $G(\mathbf{R})$ -equivariant ample line bundle $\mathcal{L} = \mathcal{L}_\chi$ on D defined by a particular character $\chi : K \rightarrow \mathbf{C}^\times$. In cases of interest, such as the Siegel and Type IV cases $D = \text{Sp}_g(\mathbf{R})/\text{U}_g$ or $D = \text{O}_{2,n}(\mathbf{R})/(\text{O}_2(\mathbf{R}) \times \text{O}_n(\mathbf{R}))$, the bundle $\mathcal{L}$ is the determinant of the Hodge bundle or the inverse tautological bundle, respectively, where for a smooth, proper family of n -dimensional varieties $\pi : X \rightarrow S$, the **Hodge bundle** is the vector bundle $\mathbb{E} := \pi_* \Omega_{X/S}^n$ whose fiber over $s \in S$ is $H^0(\Omega_{X_s}^n)$, the space of global holomorphic n -forms on the fiber X_s . For $\mathcal{M}_g$, this reduces to the pushforward $\pi_* \omega_{\mathcal{C}/\mathcal{M}_g}$ of the relative dualizing sheaf of the universal curve $\pi : \mathcal{C} \rightarrow \mathcal{M}_g$, and for $\mathcal{A}_g$ one often passes to its determinant, the **Hodge line bundle**.

For arithmetic groups Γ as above, an **automorphic form of weight k for Γ** (with *factor of automorphy* j) is a holomorphic section $f \in H^0(\mathcal{L}^{\otimes k})$ of the k th tensor power of $\mathcal{L}$ such that the $\gamma^* f = f$ for all $\gamma \in \Gamma$. Locally, this recovers the familiar automorphy condition

$$f(\gamma \cdot z) = j(\gamma, z)^k \cdot f(z) \quad \text{for all } \gamma \in \Gamma, z \in D.$$

We define the space of weight k Γ -automorphic forms as the invariants sections of $\mathcal{L}^{\otimes k}$, which assemble to a graded ring:

$$A_k(\Gamma) := H^0(D, \mathcal{L}^{\otimes k})^\Gamma, \quad R_\Gamma := \bigoplus_{k \geq 0} A_k(\Gamma).$$

[BB66] shows that that R_Γ is a finitely generated $\mathbf{C}$ -algebra, and there is an identification $\overline{F}_\Gamma^{\text{BB}} \cong \text{Proj}(R_\Gamma)$. Moreover, $\mathcal{L}$ descends to an ample line bundle on $\overline{F}_\Gamma^{\text{BB}}$ and the sections of $\mathcal{L}^{\otimes k}$ satisfy the analytic growth conditions at cusps of F_Γ in analogy to classical modular forms for $\text{SL}_2(\mathbf{Z})$. This construction is canonical and functorial: any (Γ_1, Γ_2) -

equivariant morphism $F_{\Gamma_1} \rightarrow F_{\Gamma_2}$ compatible with $\mathcal{L}_1$ and $\mathcal{L}_2$ extends uniquely to a morphism between their Baily–Borel compactifications. However, this typically introduces singularities on $\partial \overline{F}_{\Gamma}^{\text{BB}}$, motivating further refinements, e.g., semitoroidal or KSBA compactifications.

4.3.5 Cusps and Boundary Strata

Let T be an even lattice of signature $(2, n)$, let $\Gamma \leq \text{O}(L)$ be a (neat) arithmetic subgroup, and let $D := D_T$ be the corresponding period domain with arithmetic quotient $F_{\Gamma} := D_T/\Gamma$. The rational boundary components of D have dimensions 0 (“Type III”) or 1 (“Type II”). There are canonical bijections between Γ -orbits of the following sets:

- Rational boundary components of D ,
- Rational parabolic subgroups $P \leq \text{O}(T_{\mathbf{Q}})$, and
- Primitive isotropic subspaces $I \subset T_{\mathbf{Q}}$.

This correspondence is explicitly as follows: to each primitive isotropic subspace $I \subset T_{\mathbf{Q}}$ of rank k (with $1 \leq k \leq 2$), associate the boundary component F_I of dimension $k - 1$ described as follows: writing D as $G(\mathbf{R})/K$, a boundary component F_I corresponds to an isotropic sublattice I if and only if $\text{Stab}_G(I) = \text{Stab}_G(F_I)$. If $k = 1$, then I is a line and F_I is a point, and if $k = 2$, the component F_I is a modular curve with level structure. These bijections are compatible with the natural poset structures: inclusion $I_1 \subset I_2$ induces F_{I_2} contained in the closure of F_{I_1} , and parabolic subgroups are partially ordered by inclusion. We grade isotropic sublattices by rank, boundary components by dimension, and parabolic subgroups by the dimension of their unipotent radical. The above bijection then extends to an isomorphism of $\mathbf{Z}$ -graded posets, up to a shift: rank k isotropic sublattices correspond to boundary components of dimension $k - 1$. This is the fundamental bijection which makes $\overline{F}_{\Gamma}^{\text{BB}}$ accessible by lattice-theoretic methods.

To conclude, let $\eta \in T$ be a primitive isotropic vector and set $\overline{T}_{\eta} := \eta^{\perp T} / \langle \eta \rangle$, which is an even lattice of signature $(1, n - 1)$. Recall that the **stable boundary group** is defined as $\Gamma_{\eta} := \text{Stab}_{\Gamma}(\eta)/U_{\eta}$, where $\text{Stab}_{\Gamma}(\eta)$ is the stabilizer of η in Γ and U_{η} is its unipotent radical. Associated to Γ_{η} is its **stable reflection group** $W(\Gamma_{\eta}) \subseteq \text{O}(\overline{T}_{\eta})$, generated by reflections in roots of $\overline{T}_{\eta}$ contained in Γ_{η} . Denote

by $\mathfrak{C}(\Gamma_\eta)$ the *fundamental chamber*, which is the convex polyhedral cone in $\overline{T}_{\eta, \mathbf{R}}$ consisting of those vectors v satisfying $(\alpha, v) \geq 0$ for each simple root α corresponding to the chosen system of reflections. The local data for a toroidal compactification at the 0-cusp F_η comes from attaching a toric variety constructed from the positive cone and its rational closure,

$$\mathfrak{C}_\eta := \left\{ v \in \overline{T}_{\eta, \mathbf{R}} \mid v^2 > 0 \right\} \quad \mathfrak{C}_{\eta, \mathbf{Q}} := \mathfrak{C}_{\eta, \mathbf{R}} \cup \left\{ v \in \overline{T}_{\eta, \mathbf{Q}} \mid v^2 = 0 \right\},$$

where we implicitly choose a component of the full cone satisfying $x \cdot h \geq 0$ for a fixed vector h . After projectivization, $\mathbf{P}(\mathfrak{C}_\eta) \cong \mathfrak{C}_\eta / \mathbf{R}_{>0}$ is identified with $\mathbb{H}^{n-1}$. The stable reflection group $W(\Gamma_\eta)$ acts discretely on this space by isometries, with fundamental chamber $\mathfrak{C}(\Gamma_\eta)$ yielding a convex (possibly non-compact) hyperbolic polytope bounded by the reflection hyperplanes in the roots of $\overline{T}_\eta$.

One must choose a Γ_η -invariant rational polyhedral fan $\Sigma(\eta)$ supported on $\mathfrak{C}_{\eta, \mathbf{Q}}$. A particularly canonical choice, when $W(\Gamma_\eta)$ is sufficiently large enough and defines a locally finite arrangement, is the **Coxeter fan**: the rational polyhedral decomposition of $\mathfrak{C}_{\eta, \mathbf{Q}}$ whose cones are in bijection with the $W(\Gamma_\eta)$ -translates of $\mathfrak{C}(\Gamma_\eta)$. This yields a tiling of $\mathbb{H}^{n-1}$ by Coxeter polytopes, where each chamber is in bijection with a fan in $\mathfrak{C}_{\eta, \mathbf{Q}}$ and their gluing data is determined by the combinatorics of $W(\Gamma_\eta)$. In general, other Γ_η -invariant fans $\Sigma(\eta)$ may be chosen; however, when the Coxeter fan is well-defined and locally finite, it provides a natural, symmetric choice. Globally, the collection of all such fans $\{\Sigma(\eta)\}$ ranging over all k -cusps η produces a **toroidal compactification** $\overline{F}_\Gamma^{\Sigma^\bullet}$. Locally, analytic neighborhoods of cusps η are described via open subsets in the respective toric varieties $X_{\Sigma(\eta)}$, which assemble to form a normal, complex algebraic space $\overline{F}_\Gamma^{\Sigma^\bullet}$ that naturally maps to $\overline{F}_\Gamma^{\text{BB}}$.

4.4 Toroidal Compactifications

Let $F_\Gamma := D/\Gamma$ be a Hermitian symmetric domain of Type IV associated to an indefinite lattice T of signature $(2, n)$, as in the previous section, and let $\overline{F}_\Gamma^{\text{BB}}$ be its Baily-Borel compactification. We make an additional technical assumption that Γ is *neat*, which ensures that F_Γ is smooth. The Baily-Borel compactification $\overline{F}_\Gamma^{\text{BB}}$ is projective but is typically highly singular along its cusps. To obtain a compactification

with better local and algebro-geometric properties, one constructs a **toroidal compactification**. This requires, for each cusp η of $\overline{F}_\Gamma^{\text{BB}}$, the choice of a Γ -admissible rational polyhedral decomposition Σ_η of the rational closure of the positive cone, i.e. a fan. Locally, a neighborhood of η is described by a toric variety $X_{\Sigma(\eta)}$ associated to these fans. The global compactification $\overline{F}_\Gamma^{\Sigma_\bullet}$ is patched from these local models and contains F_Γ as a dense open subset, with a proper, Γ -equivariant morphism $\overline{F}_\Gamma^{\Sigma_\bullet} \rightarrow \overline{F}_\Gamma^{\text{BB}}$. Unlike the Baily–Borel compactification, $\partial\overline{F}_\Gamma^{\Sigma_\bullet}$ can be made smooth for suitable choices of fans. We now describe the construction in more detail.

4.4.1 Admissible Fans

Let T be an even indefinite lattice of signature $(2, n)$ and let $\Gamma \leq O(T)$ be an arithmetic subgroup. Fix a primitive isotropic sublattice $I \subset T$ of rank k and set the boundary lattice $\overline{T}_I := I^{\perp T}/I$, which is an even lattice with $\text{sig}(I) = (2 - k, n - k)$ and thus $\text{rank}(I) = \text{rank}(T) - 2k$. Denote by $\mathfrak{C}_I \subset \overline{T}_{I, \mathbf{R}}$ the positive cone, with $\mathfrak{C}_{I, \mathbf{Q}} \subset \overline{T}_{I, \mathbf{R}}$ its rational closure: A **rational polyhedral fan** Σ_I in $\overline{T}_{I, \mathbf{R}}$ is a collection of strongly convex, rational polyhedral cones $\sigma \subset V$ with the following properties:

- For every face τ of every $\sigma \in \Sigma_I$, we have $\tau \in \Sigma_I$.
- For all $\sigma_1, \sigma_2 \in \Sigma_I$, the intersection $\sigma_1 \cap \sigma_2$ is a common face of both.
- Σ_I is **locally finite**: every compact subset of $\overline{T}_{I, \mathbf{R}}$ meets only finitely many cones of Σ_I .
- Each cone σ is **rational**: it is generated by finitely many elements of $\overline{T} \cap (T \otimes \mathbf{Q})$.
- Each cone σ is **strongly convex**: it contains no nontrivial linear subspace.

4.5 Admissible Fans at Boundary Cusps

Let T be an integral lattice and $\Gamma \leq O(T)$ an arithmetic subgroup. For a primitive isotropic sublattice $I \subset T$, consider a fan Σ_I in the

associated real vector space $\overline{T}_{I,\mathbf{R}}$. The **support** of Σ_I is defined as

$$|\Sigma_I| := \bigcup_{\sigma \in \Sigma_I} \sigma.$$

Let $\text{Stab}_\Gamma(I) \leq \Gamma$ denote the stabilizer of I in Γ . The fan Σ_I is said to be **Γ -admissible at the cusp determined by I** if the following conditions are satisfied:

- **(i) Invariance:** Σ_I is invariant under the action of $\text{Stab}_\Gamma(I)$;
- **(ii) Local finiteness:** Σ_I is locally finite in $\overline{T}_{I,\mathbf{R}}$;
- **(iii) Finiteness of orbits:** There are only finitely many $\text{Stab}_\Gamma(I)$ -orbits of cones in Σ_I ;
- **(iv) Full support:** $|\Sigma_I| = \mathfrak{C}_{I,\mathbf{Q}}$, where $\mathfrak{C}_{I,\mathbf{Q}}$ is the rational closure of the relevant positive cone associated to I ;
- **(v) Compatibility under projection:** For every inclusion of primitive isotropic sublattices $I \subset J \subset T$, let $W := (J^\perp/J) \otimes_{\mathbf{Z}} \mathbf{R}$ and denote by $\pi_{IJ}: T_{\mathbf{R}} \rightarrow W$ the canonical projection. Then the fan at J is given by

$$\Sigma_J = \left\{ \pi_{IJ}(\sigma) \mid \sigma \in \Sigma_I, \pi_{IJ}(\sigma) \text{ is a cone of positive dimension} \right\}$$

that is, Σ_J is the image of Σ_I under the projection π_{IJ} , omitting cones of dimension zero.

A **compatible system of Γ -admissible fans** is a family $\{\Sigma_I\}$, indexed by all primitive isotropic sublattices $I \subset T$, such that for every inclusion $I \subset J$, the compatibility condition above (v) is satisfied.

Given such a system, the local model of the toroidal compactification at the cusp associated to I is the toric variety X_{Σ_I} determined by Σ_I , and $\overline{F}_\Gamma^{\Sigma_\bullet}$ is then constructed by gluing these toric varieties to neighborhoods of the cusps in $\overline{F}_\Gamma^{\text{BB}}$ according to the configuration of cones $\Sigma_\bullet := \{\Sigma_I\}$. If $\{\Sigma_I\}$ is sufficiently fine and regular, $\partial \overline{F}_\Gamma^{\Sigma_\bullet}$ is divisorial and locally a union of snc toric varieties. It is a normal, projective algebraic variety containing F_Γ as a dense open subset and is equipped with a proper morphism $\overline{F}_\Gamma^{\Sigma_\bullet} \rightarrow \overline{F}_\Gamma^{\text{BB}}$ extending the identity on F_Γ and resolving the singularities at the boundary of the Baily–Borel compactification.

The compactification $\overline{F}_\Gamma^{\Sigma^\bullet}$ is canonical up to the choice of admissible fan. For each 0-cusp η of the Baily–Borel compactification, corresponding to a primitive isotropic sublattice $I \subset T$, the local structure of the toroidal compactification near η is modeled on a toroidal embedding; that is, a formal (or analytic) neighborhood is locally isomorphic to an open subset of a finite quotient of a toric variety $X(\Sigma_\eta)$, where Σ_η is a Γ_η -invariant rational polyhedral fan as above. For higher-dimensional boundary strata, the local model is a finite quotient of a toric fibration over the corresponding stratum of $\overline{F}_\Gamma^{\text{BB}}$.

Globally, there is a proper, Γ -equivariant morphism $\sigma: \overline{F}_\Gamma^{\Sigma^\bullet} \rightarrow \overline{F}_\Gamma^{\text{BB}}$ which is an isomorphism over F_Γ . Over a 0-cusp η , the preimage $\sigma^{-1}(\eta)$ is a finite quotient of the union of toric boundary strata in $X(\Sigma_\eta)$. For higher-dimensional boundary strata S , the preimage $\sigma^{-1}(S)$ is locally a finite quotient of a toric fibration over S , whose fibers are unions of toric strata as above. The contraction induced by σ contracts each such fiber to the corresponding cusp, and is an isomorphism on the interior F_Γ . Since the entire construction is compatible with Γ , the resulting space is generally a normal, proper algebraic variety when Γ is neat.

4.5.1 Example: $\mathcal{A}_g$ and Voronoi Fans

The Siegel modular variety $\mathcal{A}_g = \text{Sp}_{2g}(\mathbf{Z}) \backslash \mathbb{H}_g$ parameterizes principally polarized abelian varieties (PPAVs) of dimension g , where $\mathbb{H}_g$ is the Siegel upper half-space of $g \times g$ complex symmetric matrices with $\Im(\tau) > 0$. A toroidal compactification $\overline{\mathcal{A}}_g^{\Sigma^\bullet}$ is specified by a collection of admissible, rational polyhedral fans Σ_I in the cone C_g of real, symmetric, positive-definite $g \times g$ matrices at each cusp, invariant under a finite index subgroup of $\text{Stab}_\Gamma(I)$. Each cone $\sigma \in \Sigma$ determines a boundary stratum D_σ , and locally the boundary is modeled by toric varieties U_σ/Γ_σ glued along shared faces, as described above. For $Q \in C_g$ and

$$M(Q) := \left\{ v \in \mathbf{Z}^g \setminus \{0\} \mid Q[v] = \min_{w \neq 0} Q[w] \right\},$$

define the **perfect cone** generated by Q as

$$\sigma(Q) = \left\langle vv^t \mid v \in M(Q) \right\rangle_{\mathbf{R}_{\geq 0}}.$$

A form Q is **perfect** if it is determined by $M(Q)$ up to scaling. The **first Voronoi fan** Σ^{perf} , is the union of all such cones as Q varies over all such forms, and determines a compactification $\overline{\mathcal{A}}_g^{\text{perf}}$. Each form $Q \in C_g$ induces a Delaunay decomposition: a tiling of $\mathbf{R}^g$ into convex polytopes determined by minima of Q , and two forms are Delaunay-equivalent if their tilings are $\text{GL}_g(\mathbf{Z})$ -equivalent. The **second Voronoi fan** Σ^{Vor} consists of cones dual to faces of the secondary polytope parameterizing all possible Delaunay decompositions, which determines a dual compactification $\overline{\mathcal{A}}_g^{\text{Vor}}$. Each boundary stratum in $\overline{\mathcal{A}}_g^{\text{Vor}}$ corresponds to a stable *semiabelic pair* (G, Θ) , where G is a semiabelian scheme, i.e. an extension $0 \rightarrow \mathbf{G}_m^r \rightarrow G \rightarrow A' \rightarrow 0$, where Θ is a stable limit of a theta divisor. The face poset of Σ^{Vor} encodes the combinatorics of Θ , and there is an identification $\overline{\mathcal{A}}_g^{\text{Vor}} \cong \overline{\mathcal{A}}_g$ where $\overline{\mathcal{A}}_g$ is the KSBA compactification by stable semiabelic pairs due to [Ale03]. This mirrors $\overline{\mathcal{M}}_g$, where dual graphs index boundary points of the Deligne-Mumford compactification by stable curves [DM69]; here, cones in Σ^{Vor} play an analogous role. For each $\sigma \in \Sigma$, neighborhoods of D_σ are modeled analytically on toric varieties U_σ/Γ_σ , where face inclusions determine the gluing data.

4.5.2 Toward semitoroidal compactifications

The theory of toroidal compactifications guarantees that local models around boundary cusps have at worst finite quotient singularities if Γ is not neat, and that the resulting compactification $\overline{F}_\Gamma^{\Sigma^\bullet}$ is a normal, projective algebraic space. However, in many cases, the requirement that Σ_I is globally polyhedral and locally finite at every cusp is too restrictive, making $\overline{F}_\Gamma^{\text{BB}}$ too coarse and singular for many applications. To address this, semitoroidal compactifications allow, at each cusp, the use of a semifan: a collection of convex rational cones which need not be polyhedral or locally finite, which may thus contain infinitely generated cones or infinitely many cones accumulating at the boundary. The combinatorial data $\mathcal{F}_\bullet = \mathcal{F}_I$ indexed by cusps I is still required to be Γ -admissible, i.e. invariant with only finitely many orbits under the stabilizer, and satisfy similar compatibility conditions. While the local boundary models in semitoroidal compactifications $\overline{F}_\Gamma^{\mathcal{F}_\bullet}$ are no longer strictly toric varieties, the resulting compactification still admits a proper, Γ -equivariant morphism $\overline{F}_\Gamma^{\mathcal{F}_\bullet} \rightarrow \overline{F}_\Gamma^{\text{BB}}$ and contains F_Γ as a dense open subset. In the next section, we review

the formal definitions of semifans and the construction of semitoroidal compactifications, following [Loo85] and related work.

4.6 Semitoroidal Compactifications

Throughout this section, let T be an even lattice of signature $(2, n)$, $\overline{T}_\eta := \overline{T}_\eta = \eta^\perp T / \langle \eta \rangle$ be the boundary lattice of signature $(1, n-1)$ at a 0-cusp $\eta \in T$, and let $\Gamma \leq \mathrm{O}(T)$ be a neat arithmetic subgroup, and let $\eta \in T$ denote a primitive isotropic line in T and $I \subseteq T$ a primitive isotropic plane, corresponding to a 0-cusp of $\overline{F}_\Gamma^{\mathrm{BB}}$ and a 1-cusp respectively. Let $W := W(\Gamma_\eta)$ denote the stable reflection group acting on $\overline{T}_\eta$. Let $\mathfrak{C} = \mathfrak{C}(\Gamma_\eta)$ be a fixed fundamental chamber for W defined by the inequalities $(v, \alpha) \geq 0$ for all $\alpha \in \Phi(\Gamma_\eta)$. Let $G(\Gamma_\eta)$ be the associated Coxeter diagram and $\Phi(\Gamma_\eta)$ be the set of simple roots for $W(\Gamma_\eta)$. **Semitoroidal compactifications** ([Loo85; Loo03]) $\overline{F}_\Gamma^{\mathcal{F}\bullet}$ generalize the construction of toroidal compactifications $\overline{F}_\Gamma^{\Sigma\bullet}$ by replacing each polyhedral fan Σ_I with a Γ -admissible **semifan**. The local models are toroidal embeddings attached to semifans, compatibly glued over all cusps. This section formalizes semifans and semitoroidal compactifications, describes admissibility and compatibility, and relates semitoroidal, toroidal, and Baily–Borel compactifications via a tower of birational morphisms:

$$\overline{F}_\Gamma^{\mathrm{BB}} \longleftarrow \overline{F}_\Gamma^{\mathcal{F}\bullet} \longleftarrow \overline{F}_\Gamma^{\Sigma\bullet}.$$

More precisely, let V be a real finite-dimensional vector space, and let $C \subset V$ be an open, nondegenerate convex cone. A **semifan** $\mathcal{F}$ in C is a collection of closed, convex, rational polyhedral cones $\sigma \subset \mathfrak{C}_\mathbf{Q}$ such that:

- every face of every $\sigma \in \mathcal{F}$ belongs to $\mathcal{F}$;
- for any $\sigma, \tau \in \mathcal{F}$, the intersection $\sigma \cap \tau$ is a face of both σ and τ ;
- the interiors of the cones in $\mathcal{F}$ are pairwise disjoint.

Unlike a fan, $\mathcal{F}$ is not required to be locally finite, nor to cover all of C . In particular, $\mathcal{F}$ may be infinite and its support $\bigcup_{\sigma \in \mathcal{F}} \sigma$ may be a proper subset of $\mathfrak{C}_\mathbf{Q}$. This generalization allows, for example, arrangements arising from infinite or non-polyhedral wall structures,

as well as dual complexes of degenerations; it thus extends the toroidal theory to situations where the properties of being globally polyhedral or locally finite fail.

Remark 4.6.0.1. Recall that the positive cone associated to a primitive isotropic sublattice and its rational closure define a real hyperbolic space by projectivization:

$$\mathbb{H}_I := \mathbf{P}(\mathfrak{C}_I) := \left\{ \langle v \rangle_{\mathbf{R}_{>0}} \subset \overline{T}_{I,\mathbf{R}} \mid v^2 > 0 \right\}.$$

A **tiling** $\mathcal{T}_I$ of $\mathbb{H}_I$ is a locally finite collection of convex polyhedral subsets $\{\tau_\alpha\}_{\alpha \in A}$ such that $\mathbb{H}_I = \bigcup_\alpha \tau_\alpha^\circ$ is disjoint union of the relative interiors of the tiles, and for each α , there is a cone $\sigma \in \mathcal{F}_I$ with $\tau_\alpha = \mathbf{P}(\sigma)$ where $\mathcal{F}_I$ is a semifan in $\mathfrak{C}_{I,\mathbf{Q}}$. If for every cone $\sigma \in \mathcal{F}_I$ there is a decomposition $\sigma = P \times H_{I,\mathbf{R}}$ where $P \subset \overline{T}_{I,\mathbf{R}}$ is a convex polytope and $H_{I,\mathbf{R}} \subset \overline{T}_{I,\mathbf{R}}$ is a fixed real linear subspace, then we say the lattice $H_I := H_{I,\mathbf{R}} \cap \overline{T}_I$ is the **coning direction**, and the semifan (or equivalently the tiling) is **coned in the direction of H_I** .

Definition 4.6.0.2 (Γ -admissible semifans). Let T be an even lattice of signature $(2, n)$ and $\Gamma \leq \mathrm{O}(T)$ a neat arithmetic group. Let F_Γ be the arithmetic quotient, and $\overline{F}_\Gamma^{\mathrm{BB}}$ its Baily–Borel compactification. Each cusp of $\overline{F}_\Gamma^{\mathrm{BB}}$ corresponds to a primitive isotropic subspace $I \subset T$. Let $\overline{T}_I = I^\perp / I$ be the associated boundary lattice and $\mathfrak{C}_{I,\mathbf{Q}}$ the rational closure of the positive cone in $\overline{T}_{I,\mathbf{R}}$. A **Γ -admissible semifan at the cusp determined by I** is a semifan $\mathcal{F}_I$ in $\mathfrak{C}_{I,\mathbf{Q}}$, invariant under $\mathrm{Stab}_\Gamma(I)$ with finitely many $\mathrm{Stab}_\Gamma(I)$ -orbits of cones. The support of $\mathcal{F}_I$ need not cover all of $\mathfrak{C}_{I,\mathbf{Q}}$ nor be locally finite. A **compatible system of Γ -admissible semifans** is a collection $\mathcal{F}_\bullet$, one for each cusp I , such that for every inclusion of cusps $I \subset J$ (that is, for inclusions of the underlying isotropic subspaces), the natural projection $\pi_{IJ}: \overline{T}_{I,\mathbf{R}} \rightarrow \overline{T}_{J,\mathbf{R}}$ satisfies

$$\mathcal{F}_J = \left\{ \pi_{IJ}(\sigma) \mid \sigma \in \mathcal{F}_I, \pi_{IJ}(\sigma) \text{ a cone of positive dimension} \right\}.$$

As an example, any fan is a semifan, and if a semifan is not a fan, we say it is a **strict semifan**. In particular, the classical toroidal case is recovered by choosing for each I a rational polyhedral fan Σ_I supported on $\mathfrak{C}_{I,\mathbf{Q}}$ and requiring Σ_I to be invariant under $\mathrm{Stab}_\Gamma(I)$ with finitely many orbits. The corresponding compactification is the toroidal compactification $\overline{F}_\Gamma^{\Sigma_\bullet}$. The “trivial” semifan consists of, for each I , the full positive cone $\mathfrak{C}_{I,\mathbf{Q}}$ in $\overline{T}_{I,\mathbf{R}}$, which yields $\overline{F}_\Gamma^{\mathrm{BB}}$, and

thus we will refer to this as the **Baily-Borel semifan** $\mathcal{F}^{\text{BB}}$. In this case each local model is essentially the “one-point” (or “one-orbit”) compactification, and $\mathcal{F}^{\text{BB}}$ is a fan in the toroidal sense. Generalized Coxeter semifans, defined below, provide intermediate examples of strict semifans. For instance, partial wall data associated to a set of relevant roots yields a semifan that is a coarsening of the Coxeter fan. These may fail to be locally finite – the boundary decomposition then glues together chambers along the omitted (irrelevant) walls.

4.6.1 Construction

Theorem 4.6.1.1 (Existence of semitoroidal compactifications [Loo03]). *Let T be an even lattice of signature $(2, n)$, $\Gamma \leq \text{O}^*(T)$ a neat arithmetic group, and F_Γ the associated locally symmetric modular variety. Let $\mathcal{F}_\bullet$ be a compatible system of Γ -admissible semifans ranging over Baily-Borel cusps I of $\partial \overline{F}_\Gamma^{\text{BB}}$, as defined above. Then there exists a normal compactification $\overline{F}_\Gamma^{\mathcal{F}_\bullet}$ containing F_Γ as an open dense subset, called the **semitoroidal compactification** associated to $\mathcal{F}_\bullet$, with the following properties:*

1. *The boundary strata are in bijection with the set of Γ -orbits of pairs (I, σ) , where I is a cusp and σ is the class of a cone in $\mathcal{F}_I$, modulo the stabilizer in Γ .*
2. *If $\sigma \in \mathcal{F}_\eta$ is a cone with η an isotropic line corresponding to a Type III cusp, let $L_{\eta, \sigma} = \eta^{\perp T} / \langle \eta, \sigma \rangle$. Then the corresponding stratum is a finite quotient of $L_{\eta, \sigma, \mathbb{C}^\times}$.*
3. *If $\sigma \in \mathcal{F}_I$ with I an isotropic plane corresponding to a Type II cusp, there is a subspace H_I depending on σ and an algebraic group $\mathcal{E}$ defined in [Loo03]. Let $L_{I, \sigma} := I^\perp / \langle I, H_I \rangle$, then the corresponding stratum is a finite quotient of $L_{I, \sigma, \mathcal{E}}$.*
4. *For any compatible system of semifans $\mathcal{G}_\bullet$ refining $\mathcal{F}_\bullet$, there is a natural morphism $\overline{F}_\Gamma^{\mathcal{G}_\bullet} \rightarrow \overline{F}_\Gamma^{\mathcal{F}_\bullet}$ which maps strata to strata according to inclusion of cones: for $\tau \subset \sigma$, the stratum indexed by (I, τ) maps to the stratum indexed by (I, σ) .*

Note that unlike for fans, strata corresponding to Type III cones of a semifan may have infinite stabilizer in $\text{Stab}_\Gamma(\eta)$ – however, the stratum is still a finite quotient of $L_{\eta, \sigma, \mathbb{C}^\times}$.

Definition 4.6.1.2 (Semitoroidal compactifications). Given a compatible system $\mathcal{F}_\bullet$ of Γ -admissible semifans, the local model of the semitoroidal compactification near the cusp associated to I is a toroidal embedding constructed from $\mathcal{F}_I$, typically a finite quotient of a toric embedding $X(\mathcal{F}_\bullet)$ associated with the semifan. For 0-cusps, the local neighborhood is modeled on such a quotient of a torus fibration, and the boundary strata correspond to cones of $\mathcal{F}_\bullet$. For higher rank cusps ($\text{rank}(I) > 1$), the local model is, locally in the analytic or formal topology, a toric fibration over the boundary stratum corresponding to I , whose fibers are toroidal embeddings associated to semifans. The boundary of $\overline{F}_\Gamma^{\mathcal{F}_\bullet}$ is stratified by the cones and faces of all semifans $\mathcal{F}_\bullet$, with strata glued compatibly respecting the combinatorial morphisms given by face projections. This local-to-global structure enables interpolation between the $\overline{F}_\Gamma^{\text{BB}}$ and the maximal $\overline{F}_\Gamma^{\Sigma_\bullet}$ by allowing the semifans to vary from trivial to polyhedral and locally finite.

Theorem 4.6.1.3 (Birational tower characterization [AE23a, Theorem 5.14]). *Let F_Γ be a Type IV arithmetic quotient, and let $\overline{F}_\Gamma$ be any normal compactification of F_Γ . The following are equivalent:*

1. *There exist proper morphisms of normal compactifications*

$$\overline{F}_\Gamma^{\Sigma_\bullet} \longrightarrow \overline{F}_\Gamma \longrightarrow \overline{F}_\Gamma^{\text{BB}}$$

where $\overline{F}_\Gamma^{\Sigma_\bullet}$ is a toroidal compactification and $\overline{F}_\Gamma^{\text{BB}}$ is the Baily–Borel compactification.

2. *There exists a collection of Γ -admissible semifans $\mathcal{F}_\bullet$ such that $\overline{F}_\Gamma^{\mathcal{F}_\bullet} \xrightarrow{\sim} \overline{F}_\Gamma$.*

Theorem 4.6.1.4 (Recognition theorem for KSBA compactifications [AE23a, Theorem 9.1]). *Let R be a recognizable divisor in the moduli space of S -polarized K3 surfaces (in the sense of [AE23a]) corresponding to the moduli space F_Γ , where Γ is the appropriate arithmetic subgroup of $\text{O}(T)$ and $T := S^{\perp L_{\text{K3}}}$. Then there exists a unique semifan $\mathcal{F}_R$ such that the normalization morphism $\overline{F}_\Gamma^{\mathcal{F}_R} \rightarrow \overline{F}_\Gamma^R$ identifies $\overline{F}_M^{\mathcal{F}_R}$ as the normalization of the KSBA compactification $\overline{F}_\Gamma^R$ associated to R . The cones of $\mathcal{F}_R$ are precisely the maximal subsets in which the combinatorial type of slc stable pairs is constant as a function of the monodromy invariant λ .*

Remark 4.6.1.5. For any recognizable divisor R on F_S , the normalization of the KSBA compactification $\overline{F}^R$ is a semitoroidal compactification, with the associated semifan $\mathcal{F}_R$ determined by R . A fundamental example of a recognizable divisor is the **rational curves divisor** R_{rc} for moduli of degree $2d$ K3 surfaces:

$$R_{\text{rc}} := \sum_{i=1}^{n_d} R_i \in |n_d L|$$

where R_i runs over all irreducible rational curves in the linear system $|L|$ of the polarization and n_d is given by the Yau–Zaslow formula ([AE23a, Thm. 10.2]). In [AE23a], R_{rc} is shown to be recognizable for all d , so there exists a canonical semifan $\mathcal{F}^{\text{rc}}$ with

$$\overline{F_{2d}^{\mathcal{F}^\bullet}} \cong \overline{F_{2d}}.$$

In degree 2, $\mathcal{F}_{\text{rc}}$ is a semifan but not a fan, and coarsens the Coxeter fan. For elliptic K3 surfaces with divisor $s + \sum_{i=1}^{24} f_i$, the semifan is in fact a fan, which further refines the maximal cone of the Coxeter fan. It is currently a widely open problem to describe the semifan $\mathcal{F}_R$ for a recognizable divisor in full generality, and is still open in the case of $R = R^{\text{rc}}$ – these are currently only characterized only as the coarsest subdivision such that the combinatorial type of stable pairs is constant on its cones. Our work in [Ale+23] relies on the fact that recognizability was proved for ramification loci of non-symplectic automorphisms in [AE22], and explicit constructions of the corresponding semifans are provided.

4.6.2 Generalized Coxeter Semifans and Relevant Roots

Definition 4.6.2.1 (Relevant and irrelevant roots). Let $\overline{T}_\eta$ be the boundary lattice of signature $(1, n-1)$ for a 0-cusp η , with stable reflection group $W = W(\Gamma_\eta)$, simple root system $\Phi = \Phi(\Gamma_\eta)$, and fundamental chamber $\mathfrak{C} = \mathfrak{C}_\eta$. The Coxeter diagram is $G = G(\Gamma_\eta)$. A partition $\Phi = \Phi^{\text{rel}} \sqcup \Phi^{\text{irr}}$ into **relevant** and **irrelevant** simple roots is defined as follows:

- The roots in Φ^{irr} (the **irrelevant roots**) are those for which, in every Kulikov model over the cusp η , the faces of $\mathfrak{C}$ defined using only mirrors from Φ^{irr} correspond to strata that are contracted in the KSBA stable model over η .

- The **relevant roots** are those in $\Phi^{\text{rel}} = \Phi \setminus \Phi^{\text{irr}}$.

Definition 4.6.2.2 (Generalized Coxeter Semifan [AET19b, Def. 4.16]).

Fix a partition $\Phi = \Phi^{\text{irr}} \sqcup \Phi^{\text{rel}}$. Let $W^{\text{irr}} = \langle w_\alpha \mid \alpha \in \Phi^{\text{irr}} \rangle \subset W$ denote the reflection subgroup generated by reflections in the irrelevant roots. The corresponding generalized chamber is

$$\mathcal{L} = \bigcup_{h \in W^{\text{irr}}} h(\mathfrak{C})$$

The corresponding **generalized Coxeter semifan** $\mathcal{F}_{\text{gen}}$ is the semifan whose maximal cones are $g(\mathfrak{C}_{\text{gen}})$ for $g \in W$, with faces given by all intersections of maximal cones not contained in any wall $\alpha^\perp$ for $\alpha \in \Phi^{\text{rel}}$.

Remark 4.6.2.3. Passing from the Coxeter fan to the generalized Coxeter semifan corresponds to deleting nodes of $G(\Gamma_\eta)$ representing the irrelevant roots: walls $\alpha^\perp$ for $\alpha \in \Phi^{\text{irr}}$ are omitted, so maximal cones become unions of Weyl chambers glued along these “inactive mirrors”. If $\Phi^{\text{irr}} = \emptyset$, this recovers the Coxeter fan, while if $\Phi^{\text{rel}} = \emptyset$, there is a single chamber and this recovers the Baily–Borel fan. Moreover, if $|W^{\text{irr}}| = \infty$, the resulting semifan is not locally finite, and the compactification is strictly semitoroidal. Crossing a relevant wall $\alpha^\perp$ corresponds to a birational transformation – such as a flip, flop, or divisorial contraction – between KSBA stable models. The chambers cut out by relevant walls in $\mathfrak{C}$ correspond to regions where the combinatorial type of Kulikov (and hence stable) models remains constant.

4.6.3 Generalized Coxeter compactifications

We thus obtain a natural, purely combinatorial way to interpolate between the maximal and minimal compactifications of F_Γ , which can be encoded in a single combinatorial object that we now describe.

Definition 4.6.3.1 (The semifan poset of a cone). Let V be a real finite-dimensional vector space and $\mathfrak{C} \subset V$ an open convex cone. The **semifan poset** $\text{SFan}(\mathfrak{C})$ is the set of all semifans in $\mathfrak{C}$, partially ordered by refinement: $\mathcal{F} \leq \mathcal{F}'$ if every cone of $\mathcal{F}$ is contained in a cone of $\mathcal{F}'$. The maximal (finest) and minimal (coarsest) elements correspond to the finest locally polyhedral subdivision and the single full cone, respectively.

Definition 4.6.3.2 (The semitoroidal compactification poset). The **semitoroidal compactification poset** $\mathcal{S}_\Gamma$ of F_Γ is the set of isomorphism classes of semitoroidal compactifications $\overline{F}_\Gamma, \mathcal{F}_\bullet^{\mathcal{F}}$ constructed from compatible systems of Γ -admissible semifans $\mathcal{F}_\bullet$ ranging over the cusps I of $\overline{F}_\Gamma^{\text{BB}}$. $\mathcal{S}_\Gamma$ is partially ordered: $\overline{F}_\Gamma^{\mathcal{F}} \leq \overline{F}_\Gamma^{\mathcal{G}}$ if $\mathcal{F}_I$ refines $\mathcal{G}_I$ for all I . This order is reversed under induced morphisms, i.e., coarser semifans yield “smaller” compactifications. The maximal element corresponds to the maximal toroidal compactification; the minimal to the Baily–Borel compactification.

Definition 4.6.3.3 (The Coxeter semitoroidal compactification poset). For each 0-cusp η of $\overline{F}_\Gamma^{\text{BB}}$, let $G(\Gamma_\eta)$ be the stable Coxeter diagram, and $\mathcal{P}_{G(\Gamma_\eta)}$ the poset of subdiagrams under inclusion. Define the **Coxeter semitoroidal compactification poset** of F_Γ as the coproduct poset

$$\mathcal{P}_\Gamma := \coprod_{\eta \in \partial \overline{F}_\Gamma^{\text{BB}}} \mathcal{P}_{G(\Gamma_\eta)}.$$

An element $(D_\eta) \in \mathcal{P}_\Gamma$ specifies, for each 0-cusp, a subdiagram $D_\eta \subset G(\Gamma_\eta)$ and thus a set of irrelevant roots at each cusp.

Proposition 4.6.3.4 (Coxeter semifan system and main identification). *There is a canonical poset morphism $\Psi: \mathcal{P}_\Gamma \rightarrow \mathcal{S}_\Gamma$ which sends a tuple (D_η) of subdiagrams, one for each 0-cusp η , to the induced semitoroidal compactification, constructed by gluing together maximal cones of generalized Coxeter semifans (as in Definition 4.6.3.3) along all omitted walls, corresponding to the nodes omitted in D_η . The maximal element of $\mathcal{P}_\Gamma$, where no nodes are omitted, corresponds to the Coxeter fan and thus the toroidal compactification. The minimal element, where all nodes are omitted at each cusp, corresponds to the Baily–Borel fan Σ^{BB} and thus $\overline{F}_\Gamma^{\text{BB}}$. Morphisms in $\mathcal{P}_\Gamma$ corresponding to deleting more nodes correspond to coarsenings of the semifans and thus to proper birational morphisms between the corresponding semitoroidal compactifications.*

Remark 4.6.3.5. The poset $\mathcal{S}_\Gamma$ of semitoroidal compactifications organizes all normal compactifications arising from systems of Γ -admissible semifans over the boundary strata of F_Γ . The Coxeter semitoroidal subposet $\mathcal{P}_\Gamma$ parameterizes those compactifications obtained as Coxeter-type (generalized Coxeter semifan) coarsenings of local reflection decompositions, and is canonically identified with the product of sub-

diagram posets at all 0-cusps. The product $\mathcal{P}_\Gamma$ thus parameterizes exactly the semitoroidal compactifications of F_Γ arising from generalized Coxeter semifans, distinguished in the full semitoroidal poset $\mathcal{S}_\Gamma$. Note that the latter may contain other compactifications, and the author is not aware of any choices of F_Γ for which Ψ is known to be surjective or bijective.

4.7 KSBA Compactifications

4.7.1 Introduction

For higher-dimensional varieties, GIT, toroidal, and semitoroidal compactifications are often inadequate, producing boundary points that lack modular interpretations and possibly corresponding to degenerations with excessively severe singularities. For instance, GIT compactification F_2 result in boundary strata containing highly singular, non-separated, or even non-reduced curves. Such limiting surfaces may not be uniquely determined by one-parameter degenerations, violating the valuative criterion for properness and separatedness. Moreover, GIT boundaries often allow non-slc singularities, which are more severe than those typically permitted by the MMP (see e.g., [Sha81], [Loo86]). In contrast, for curves of genus $g \geq 2$, Deligne and Mumford compactify $\mathcal{M}_g$ using **stable curves**: a (connected, reduced, projective) curve C over an algebraically closed field is **stable** if all singularities of C are nodes and every rational component of C meets the rest of C (including the marked points) in at least 3 points. The moduli functor $\mathcal{M}_g^{\text{DM}}$ assigning to each connected base S the groupoid of flat, proper families $\mathcal{C} \rightarrow S$ whose fibers are stable curves, is represented by a proper Deligne–Mumford stack. The boundary $\partial\mathcal{M}_g$ parameterizes connected, nodal curves with finite automorphism groups, and every family C° over $S^\circ = S \setminus \{0\}$ admits, after a ramified base change, a unique stable limit by semistable reduction and relative abundance of $\omega_{C/S}$ (see [DM69], [KM98a]). The **boundary strata** of $\partial\mathcal{M}_g$ are indexed by **dual graphs** recording the incidence data of components and their intersections – each stable degeneration’s combinatorial “type” corresponds to the dual graph of its $\mathcal{X}_0$, and each such graph describes a distinct boundary stratum.

To generalize this to higher dimension, one introduces **stable pairs** (X, B) , following [KS88], [Ale96a]. Such a pair is **KSBA stable** if:

- X is a projective, reduced, equidimensional, *demi-normal* variety of dimension $d \geq 2$,
- $B = \sum b_i B_i$ is an effective $\mathbf{Q}$ -divisor, $0 < b_i \leq 1$,
- The pair (X, B) has **semi-log-canonical (slc) singularities**,
- $K_X + B$ is ample.

KSBA stability ensures:

- (**Properness**) Any flat family $(\mathcal{X}^*, \mathcal{B}^*) \rightarrow \Delta^*$ over the punctured disk extends, after finite base change, to a family $(\mathcal{X}, \mathcal{B}) \rightarrow \Delta$ of stable pairs with $\mathcal{X}_0 (X_0, B_0)$ slc and $K_{X_0} + B_0$ ample.
- (**Separatedness**) Any isomorphism over Δ^* extends uniquely over Δ .
- (**Modularity**) Every point of the boundary parameterizes a unique, geometrically meaningful, slc limit.

For curves, setting $d = 1$ and B be the sum of markings, this recovers the notion of stable curves as $\omega_C(\sum_i p_i)$ ample and only nodes allowed. However, unlike the case of curves, where all stable curves are smoothable, not every stable variety is a limit of smooth ones. Thus, boundary components can include non-smoothable varieties, yielding compactifications that potentially have multiple irreducible components. Fixing discrete invariants – the dimension d , a Hilbert polynomial h , boundary coefficients $\{b_i\}$, and the **volume** $(K_X + B)^d$ –, the **KSBA moduli functor** $\overline{\mathcal{M}}_{d, \vec{b}, v} : \text{Sch}^{\text{op}} \rightarrow \mathbf{Sets}$ assigns to S the set of isomorphism classes of flat families $(\mathcal{X}, \mathcal{B}) \rightarrow S$ of KSBA stable pairs with these invariants. This functor is represented by a proper, separated Deligne–Mumford stack $\mathcal{M}$, whose coarse moduli space M is a projective scheme [KSB88]. The closure of the locus of smooth pairs in $\mathcal{M}$ provides a geometrically meaningful compactification by stable pairs.

For K3 or Enriques surfaces, $K_X \equiv 0$ is numerically trivial, so to ensure ampleness one considers **pairs** $(X, \varepsilon R)$ for $0 < \varepsilon \ll 1$, R ample divisor, and studies the stable pair locus for these data. The divisor R is typically chosen to be the ramification divisor of an automorphism, and the compactification is independent of ε for ε sufficiently small ([KSB88], [Alexeev96], [Kol23, Lemma VI.1.1], [AET19a]). For $K3$

surfaces with polarization of degree $2d$, a **divisor model** refers to the representation of a K3 surface X together with an ample Cartier divisor L of degree $2d$ – concretely, for $d = 1$, this is a double cover of $\mathbf{P}^2$ branched along a sextic. The stability condition requires (X, L) to be log canonical with $K_X + L$ ample. The KSBA compactification $\overline{F}_{2d}$ for such pairs compactifies the moduli space of smooth pairs (X, L) by adding K3 surface pairs with at worst ADE/slc singularities. These are parameterized by **integral affine spheres** IAS^2 with 24 singularities. For Enriques surfaces the boundary of the KSBA compactification naturally includes **half-divisor models**: pairs (Z, Δ) , where Z is an Enriques surface and Δ is an effective Weil divisor determined by the universal K3 cover $X \rightarrow Z$, often defined only up to numerical equivalence or descent X . These are in turn parametrized by IAS^2 with involutions.

4.7.2 Singularities

Definition 4.7.2.1 (Demi-normal Variety). A variety X is **demi-normal** if X is reduced, X is S_2 and normal crossing in codimension one (every codimension one singularity is analytically isomorphic to a normal crossing or ordinary node).

Definition 4.7.2.2 (Serre’s S_2 Condition). A Noetherian scheme X satisfies **Serre’s S_2 condition** if, for every $x \in X$, $\text{depth } \mathcal{O}_{X,x} \geq \min\{2, \dim \mathcal{O}_{X,x}\}$. For surfaces, S_2 implies X is Cohen–Macaulay in codimension one.

Definition 4.7.2.3 (Simple Normal Crossings (snc)). A variety X has **simple normal crossings (snc)** singularities if, at each point $x \in X$, there exists an analytic or étale neighborhood isomorphic to

$$(x_1 x_2 \cdots x_k = 0) \subset \mathbf{A}^n$$

for some $k \leq n$, i.e., X locally looks like k coordinate hyperplanes.

Definition 4.7.2.4 (Gorenstein and $\mathbf{Q}$ -Gorenstein Varieties). A scheme X of pure dimension is **Gorenstein** if it is Cohen–Macaulay and its dualizing sheaf ω_X is invertible. X is **$\mathbf{Q}$ -Gorenstein** if ω_X is $\mathbf{Q}$ -Cartier, i.e., some positive tensor power $\omega_X^{\otimes m}$ is invertible for $m > 0$.

Definition 4.7.2.5 (ADE Singularities (Du Val, Rational Double Points)). An **ADE singularity** (Du Val or rational double point) is a normal

surface singularity whose minimal resolution has exceptional curves intersecting according to an ADE Dynkin diagram with analytic forms:

$$\begin{aligned} A_n: x^2 + y^2 + z^{n+1} &= 0 \\ D_n: x^2 + y^{n-1} + yz^2 &= 0 \\ E_6: x^2 + y^3 + z^4 &= 0 \\ E_7: x^2 + y^3 + yz^3 &= 0 \\ E_8: x^2 + y^3 + z^5 &= 0 \end{aligned}$$

Definition 4.7.2.6 (Normalization and Double Locus). Let X be a reduced, demi-normal scheme, as defined above. The **normalization** of X is a finite birational morphism

$$\nu: \bar{X} \rightarrow X$$

with X^ν normal, universal among morphisms from normal schemes to X .

Definition 4.7.2.7 (The Conductor Subscheme and Double Locus). Let X be a reduced scheme and let $\nu: \tilde{X} \rightarrow X$ denote its normalization. The **conductor ideal sheaf** is defined as

$$\mathcal{C}_X := \text{Ann}_{\mathcal{O}_X} \left(\nu_* \mathcal{O}_{\tilde{X}} / \mathcal{O}_X \right).$$

This ideal sheaf gives rise to two closed subschemes:

- The **conductor subscheme** (or **conductor locus**) on X , defined by $\mathcal{C}_X \subseteq \mathcal{O}_X$. This locus coincides with the points where ν is not an isomorphism, i.e., where X is not normal.
- The **conductor divisor** on $\tilde{X}$, which is the closed subscheme cut out by pulling back the conductor ideal via ν .

The **double locus** of X is the support of the conductor subscheme on X . Equivalently, it is the locus where X fails to be normal—typically, the set of points where two or more local analytic branches of the normalization are identified in X .

For algebraic surfaces, the double locus is a (possibly reducible) disjoint union of curves (i.e., it is of pure codimension one). In higher dimensions, the conductor always has pure codimension one in X .

The preimage of the conductor divisor in $\widetilde{X}$ records the precise locations where the gluing occurs in the normalization, thus encoding the identification data required to reconstruct X from its normalization. Normalization replaces a reduced scheme X with a normal scheme X^ν up to birational equivalence, and the fibers of $X^\nu \rightarrow X$ encode the branching behavior of the singularity at that point. Its practical implications by dimension are as follows:

- $\dim(X) = 1$: The normalization of a reduced curve C is a smooth curve $\widetilde{C}$.
- $\dim(X) = 2$: Normalization resolves all non-normal singularities, such as double curves and cusps, and more generally all 1-dimensional singularities, leaving only singularities at isolated points (which are typically ADE or quotient singularities).
- $\dim(X) \geq 3$: The singular locus of X^ν is of codimension at least 2, and consists of *normal singularities* – these can generally be complicated.

Theorem 4.7.2.8 (Zariski’s Main Theorem (Recognition Theorem for Normalizations)). *Let X be a reduced, separated, Noetherian scheme, and let $f: Y \rightarrow X$ be a morphism. Then f is (up to unique isomorphism) the normalization of X if and only if:*

1. Y is normal,
2. f is finite and birational,
3. f restricts to an isomorphism over the open subset of X where X is normal (i.e., over the normal locus of X).

In other words, any morphism with these three properties realizes Y as the normalization of X . In particular, if X is an irreducible, reduced, separated variety over $\mathbf{C}$ such that

1. Y is normal and irreducible, and
2. f is finite and birational,

then Y is the normalization of X and f is the normalization morphism. In this case, f is an isomorphism over the smooth locus of X .

4.7.3 Pairs

Definition 4.7.3.1 (Log Pair (X, D)). A **log pair** is a pair (X, D) , where X is a normal variety over $\mathbf{C}$ and $D = \sum d_i D_i$ is an effective $\mathbf{Q}$ -divisor on X with coefficients $0 \leq d_i \leq 1$.

Definition 4.7.3.2 (Log Canonical (lc) Singularities). Let X be a normal variety over $\mathbf{C}$, and let $R = \sum r_i R_i$ be an effective $\mathbf{Q}$ -divisor with $0 \leq r_i \leq 1$. For any log resolution $f: Y \rightarrow X$, and for each prime divisor E on Y , the **discrepancy** $a(E, X, R)$ is defined by

$$K_Y = f^*(K_X + R) + \sum_E a(E, X, R) E.$$

Let (X, D) be a log pair, where X is a normal variety and D is a $\mathbf{Q}$ -divisor. The pair is said to be **log canonical** (lc) if $K_X + D$ is $\mathbf{Q}$ -Cartier and, for every log resolution $f: Y \rightarrow X$ and every prime divisor E on Y , the discrepancy satisfies $a(E, X, D) \geq -1$. A **log canonical center** of (X, D) is the image $f(E) \subseteq X$ of a prime divisor E on some log resolution $f: Y \rightarrow X$ with discrepancy $a(E, X, D) = -1$; this locus is precisely where the singularities of the pair are exactly log canonical.

4.8 Classes of Singularities for Pairs (X, D)

The minimal model program (MMP) distinguishes four main classes of singularities for pairs, ordered by the values their discrepancies may attain. Let $f: Y \rightarrow X$ be any log resolution and E a prime divisor on Y . Let (X, D) be a normal pair with $K_X + D$ $\mathbf{Q}$ -Cartier. For a birational morphism $f: Y \rightarrow X$, and a prime divisor $E \subset Y$, the discrepancy $a(E, X, D)$ is defined via the relation

$$K_Y + D_Y = f^*(K_X + D) + \sum_E a(E, X, D) \cdot E,$$

where D_Y is the strict transform of D . The classification of singularities according to the discrepancy function is given below.

Class	Discrepancy	
	$a = a(E, X, D)$	Typical Applications
Terminal	$a > 0$	Minimal models in $\dim \geq 3$
Canonical	$a \geq 0$	Canonical models; moduli of varieties of general type

Class	Discrepancy $a = a(E, X, D)$	Typical Applications
Kawamata log terminal (klt)	$a > -1$	Singularities allowed in the MMP
Log canonical (lc)	$a \geq -1$	Stable pairs; compactifications

- **Terminal:** All discrepancies are strictly positive. Such singularities are the mildest allowed in the context of minimal models in dimension at least three, ensuring $\mathbf{Q}$ -factoriality and smoothness in codimension two.
- **Canonical:** Discrepancies are nonnegative. Canonical singularities permit discrepancies to vanish but exclude boundary contributions with coefficient one. They are characteristic of canonical models and appear naturally in the classification of varieties of general type.
- **Kawamata log terminal (klt):** All discrepancies satisfy $a(E, X, D) > -1$. These include quotient singularities and allow boundary divisors with coefficients in $(0, 1)$. klt pairs form the primary class of singularities admissible in the Minimal Model Program.
- **Log canonical (lc):** Discrepancies satisfy $a(E, X, D) \geq -1$. This is the broadest class considered in the birational classification of pairs, accommodating boundary components with coefficient one. lc singularities are essential in the theory of stable pairs and in the construction of compactified moduli spaces.

When the condition $a(E, X, D) \geq -1$ is enforced, the essential steps of the MMP – flips, divisorial contractions, and the extraction of minimal and canonical models – can be performed, and the canonical ring remains finitely generated. If one allows singularities with discrepancies less than -1 , these procedures can fail, and fundamental theorems such as the existence of minimal models or the finiteness of the canonical ring may break down. Thus, log canonical singularities constitute the maximal class for which the program is expected to be valid. Canonical and log canonical singularities appear naturally on canonical models of varieties of general type, and this inclusion ensures that the canonical ring has the necessary finiteness properties. The

MMP is specifically designed so that, under finite generation, canonical models admit at worst lc singularities, which are thus the natural endpoint for varieties of general type constructed via the MMP.

Definition 4.8.0.1 (Divisorial Log Terminal (dlt) Singularities). A pair (X, D) , where X is normal and D is an effective $\mathbf{Q}$ -divisor, is said to be **divisorial log terminal (dlt)** if $K_X + D$ is $\mathbf{Q}$ -Cartier, and there exists a log resolution $f: Y \rightarrow X$ such that:

- for every f -exceptional divisor E , the discrepancy $a(E, X, D) > -1$,
- the union of the strict transform of D with the f -exceptional divisors is a simple normal crossings (snc) divisor,
- all log canonical centers are contained in the support of the strict transform of D .

Definition 4.8.0.2 (Semi-Log Canonical (slc) Pairs). Let X be a demi-normal scheme and $R = \sum r_i R_i$ an effective $\mathbf{Q}$ -divisor on X with $0 \leq r_i \leq 1$. The pair (X, R) is called **semi-log canonical (slc)** if:

1. $K_X + R$ is $\mathbf{Q}$ -Cartier.
2. (X^ν, R^ν) is log canonical, where $\nu: X^\nu \rightarrow X$ is the normalization of X and

$$R^\nu := D + \sum r_i \nu^*(R_i).$$

where D be the conductor divisor on X^ν .

Semi-log canonical singularities generalize lc singularities to possibly non-normal varieties. In this setting, X is allowed to have certain mild singularities, notably double crossings in codimension one. The normalization $\nu: \widetilde{X} \rightarrow X$ separates these non-normal loci, and the conductor divisor $D \subset \widetilde{X}$ records the preimage of the non-normal (double) locus of X . The pair $(\widetilde{X}, R^\nu)$ combines the pullback of R and the conductor divisor, and being lc in this setting captures the requirement that singularities of the normalization and the identifications along the conductor divisor are at worst log canonical; that is, all discrepancies for the pair $(\widetilde{X}, R^\nu)$ are at least -1 , both on the components of $\widetilde{X}$ and along the loci where the components are glued

together via the conductor. This allows the extension of the MMP and moduli of pairs to schemes that are not necessarily normal or irreducible. In particular, slc pairs arise naturally as stable limits of pairs in families where the total space acquires non-normal singularities, making them central objects in the compactification of moduli spaces of pairs.

Definition 4.8.0.3 (Quasi-polarized Minimal Resolutions and Deformation Types). Given a pair (X, R) , where X is a surface with at worst ADE or quotient singularities and R is a nef (or ample) divisor, the **quasi-polarized minimal resolution** is the pair $(\tilde{X}, f^*R)$, where $f: \tilde{X} \rightarrow X$ is the minimal crepant resolution of X . The **deformation type** of a (possibly singular) surface with an ample line bundle (or quasi-polarization) is the isomorphism class (up to analytic or algebraic equivalence) of its quasi-polarized minimal resolution.

4.8.1 Numerical Conditions and Models

Definition 4.8.1.1 (Big and Nef Divisors). Let D be a $\mathbf{Q}$ -Cartier divisor on a projective surface X :

- D is **nef** (numerically effective) if $D \cdot C \geq 0$ for every irreducible curve $C \subset X$.
- D is **big** if its volume is positive,

$$\mathrm{vol}_X(D) := \limsup_{m \rightarrow \infty} \frac{h^0(X, \mathcal{O}_X(mD))}{m^{\dim X} / \dim X!} > 0.$$

Definition 4.8.1.2 (Polarizing, Primitive, and Quasi-Polarizing Divisors). Let X be a projective variety. A **polarizing divisor** is an ample Cartier divisor R on X . We say R is a **quasi-polarizing divisor** if it is only big and nef, but not necessarily ample. It is **primitive** if its class in $\mathrm{Pic}(X)$ cannot be written as kL' for any integer $k > 1$ and any divisor L' .

Ample divisors guarantee that X is projective. making polarizations especially common in moduli problems, where it ensures that the moduli stack is proper and separated. Nefness provides a numerical positivity condition that yields well-defined numerical invariants, and since it is preserved under birational equivalence, it ensures that the singularities of degenerations are mild enough to be controlled by the

MMP. Bigness is a maximality condition, selecting divisors whose sections grow like ample divisors. Quasi-polarizations arise naturally as limits of polarizations: degenerations of polarized KSBA pairs can induce a loss of ampleness, while bigness and nefness persist. Finally, primitivity imposes a minimality and uniqueness condition (up to isomorphism) on the divisor class, ensuring that each isomorphism class is represented exactly once in the corresponding moduli problem, avoiding redundancies due to rescaling.

Definition 4.8.1.3 (Nef Model). Let $\mathcal{X}^* \rightarrow \Delta^*$ be a flat family over a punctured disk, and let $\mathcal{L}^*$ be a relatively big and nef line bundle on $\mathcal{X}^*$. A **nef model** for $(\mathcal{X}^*, \mathcal{L}^*)$ is a pair $(\mathcal{X}, \mathcal{L})$ where:

- $\mathcal{X} \rightarrow \Delta$ is a flat extension of $\mathcal{X}^* \rightarrow \Delta^*$ (often a Kulikov or semistable model),
- $\mathcal{L}$ is a line bundle on $\mathcal{X}$ extending $\mathcal{L}^*$,
- $\mathcal{L}$ is relatively nef and big over Δ .

Given a family $\mathcal{X}^* \rightarrow \Delta^*$ of smooth varieties over a punctured disk and a relatively big and nef line bundle $\mathcal{L}^*$ or an effective divisor $\mathcal{R}^* \subset \mathcal{X}^*$, the problem is to construct canonical extensions of this data over the whole disk Δ , especially over the $\mathcal{X}_0$.

Such extensions must satisfy several requirements dictated by the geometry of degenerations and the construction of compactified moduli spaces:

- The extension $(\mathcal{X}, \mathcal{L})$ or $(\mathcal{X}, \mathcal{R})$ must be flat over Δ , so that the fibers capture the correct limit structure as the family degenerates. Note that flatness of a family ensures that the fibers vary in a “continuous” manner from a scheme-theoretic perspective, yielding equidimensionality and consistent Hilbert polynomial across the family.
- The positivity properties of the line bundle or divisor—such as nefness and ampleness—must be preserved on the total space to ensure that the relevant moduli functor remains separated and proper.
- In the case of divisors, compatibility with the stratification of the $\mathcal{X}_0$ is essential: the $\mathcal{X}_0$ of the divisor should avoid all strata of the possibly singular $\mathcal{X}_0$, so as not to introduce unwanted components or increase the complexity of the limit.

Precisely formulating these extensions is essential for defining and constructing stable limits. They form the starting point for constructing stable limits, which are stable pairs arising as canonical limits of smooth pairs in families. The following definition captures the precise requirements for extending divisors across degenerations:

Definition 4.8.1.4 (Divisor Model). Let $\mathcal{X}^* \rightarrow \Delta^*$ be a family of varieties over a punctured disk and let $\mathcal{R}^* \subset \mathcal{X}^*$ be an effective divisor (usually a section of a relatively nef and big line bundle). A **divisor model** for $(\mathcal{X}^*, \mathcal{R}^*)$ is a pair $(\mathcal{X}, \mathcal{R})$, where:

- $\mathcal{X} \rightarrow \Delta$ is a flat family extending $\mathcal{X}^* \rightarrow \Delta^*$,
- $\mathcal{R}$ is an effective divisor on $\mathcal{X}$ restricting to $\mathcal{R}^*$ on $\mathcal{X}^*$,
- $\mathcal{R}$ is relatively nef over Δ ,
- The $\mathcal{X}_0 \setminus \mathcal{R}_0$ does **not contain any stratum** of the $\mathcal{X}_0 \setminus \mathcal{R}_0$; that is, $\mathcal{R}_0$ does not contain any irreducible component or singular locus (double curves, triple points, etc.) of $\mathcal{X}_0$.

Proposition 4.8.1.5 (Existence and Uniqueness of Stable Limits via Divisor Models). *Let $(\mathcal{X}^*, \mathcal{R}^*)$ be a flat family of smooth pairs over Δ^* . After finite base change, there exists a divisor model $(\mathcal{X}, \mathcal{R})$ as above. The associated **stable model** is the pair $(\overline{\mathcal{X}}, \overline{\mathcal{R}})$, where*

$$(\overline{\mathcal{X}}, \overline{\mathcal{R}}) := \text{Proj}_\Delta \left(\bigoplus_{n \geq 0} H^0(\mathcal{X}, \mathcal{O}_{\mathcal{X}}(n\mathcal{R})) \right).$$

For $0 < \varepsilon \ll 1$, the pair $(\overline{\mathcal{X}}, \varepsilon \overline{\mathcal{R}})$ is KSBA-stable: $\overline{\mathcal{X}}$ has slc singularities and $K_{\overline{\mathcal{X}}} + \varepsilon \overline{\mathcal{R}}$ is ample.

This stable limit is unique up to isomorphism (after base change), and the construction is functorial in families. It provides a canonical procedure for extending any family of smooth pairs to a stable pair in the boundary of the KSBA moduli space, ensuring the properness of the compactification.

Definition 4.8.1.6 (Half-Divisor Model). A **half-divisor model** is a pair $(\mathcal{Z}, \mathcal{R}_{\mathcal{Z}})$ consisting of a flat family $\mathcal{Z} \rightarrow C$ over a base curve C , together with a divisor $\mathcal{R}_{\mathcal{Z}} \subset \mathcal{Z}$, such that the pair arises as the quotient of a divisor model $(\mathcal{X}, \mathcal{R}) \rightarrow C$ by a fixed-point-free involution τ with $\mathcal{R}$ anti-invariant (i.e., $\mathcal{R}$ does not descend as a Cartier divisor, but $2\mathcal{R}$

does). Equivalently, $(\mathcal{Z}, \mathcal{R}_{\mathcal{Z}})$ is locally modeled as $(\mathcal{X}/\tau, \mathcal{R}/\tau)$ where $\mathcal{R}$ is a “half” of a Weil divisor that becomes Cartier only after passing to the double cover.

Let $\mathcal{X} \rightarrow C$ be a flat family of K3 surfaces over a smooth curve, equipped with a fixed-point-free involution τ , and let $\mathcal{R} \subset \mathcal{X}$ be an effective Cartier divisor that is anti-invariant under τ (i.e., $\tau^*\mathcal{R} = -\mathcal{R}$). The quotient family

$$\pi: (\mathcal{X}, \mathcal{R}) \rightarrow (\mathcal{Z}, \mathcal{R}_{\mathcal{Z}})$$

with $\mathcal{Z} = \mathcal{X}/\langle\tau\rangle$, yields a family of Enriques surfaces together with a divisor $\mathcal{R}_{\mathcal{Z}}$ defined as the scheme-theoretic image of $\mathcal{R}$. In general, $\mathcal{R}_{\mathcal{Z}}$ is a Weil divisor on $\mathcal{Z}$ that is not Cartier, but its double $2\mathcal{R}_{\mathcal{Z}}$ is always Cartier. This reflects a fundamental feature of Enriques surfaces: the divisor defining the marking or polarization typically does not descend to a Cartier divisor through a degree two étale cover with empty branch locus. Instead, the presence of the involution ensures there is global 2-torsion in the divisor class group, leading to the condition $2\mathcal{R}_{\mathcal{Z}} \in \text{CaDiv}(\mathcal{Z})$ but $\mathcal{R}_{\mathcal{Z}} \notin \text{CaDiv}(\mathcal{Z})$. This half-divisibility characterizes polarized Enriques surfaces and persists in their degenerations. When the $\mathcal{X}_0$ degenerates $((\mathcal{X}_0, \mathcal{R}_0))$, it may become reducible, and the involution specializes to τ_0 on $\mathcal{X}_0$. The quotient $\mathcal{Z}_0 = \mathcal{X}_0/\langle\tau_0\rangle$ is then a demi-normal surface, and the induced divisor $\mathcal{R}_{\mathcal{Z}_0}$ remains a Weil divisor with $2\mathcal{R}_{\mathcal{Z}_0}$ Cartier. In the context of the KSBA compactification, every boundary stratum corresponding to a stable limit of Enriques surfaces is thus naturally modeled by half-divisor pairs $(\mathcal{Z}_0, \frac{1}{2}\mathcal{R}_{\mathcal{Z}_0})$, with log-canonical polarization $K_{\mathcal{Z}_0} + \frac{1}{2}\mathcal{R}_{\mathcal{Z}_0}$ ample, and with semi-log-canonical singularities that may arise both from the quotient construction and from singularities already present in the K3 $\mathcal{X}_0$. As discussed in [Ale+23], the structure of degenerations of Enriques surfaces are thus governed by half-divisor models in this way.

4.8.2 The KSBA Moduli Stack

Definition 4.8.2.1 (KSBA Stable Pair). Let X be a projective, demi-normal (in particular, S_2 and normal crossing in codimension one) variety over an algebraically closed field of characteristic 0, and let $D = \sum_j a_j D_j$ be an effective $\mathbf{Q}$ -divisor with $0 < a_j < 1$ and each component D_j a Weil divisor whose support does not contain any

component of the double locus of X . The pair (X, D) is called a **(KSBA) stable pair** if:

- (X, D) is semi-log-canonical (slc): X is S_2 ; (X, D) is slc as previously described, i.e., the normalization with the conductor plus pullbacks of D is log canonical, and $K_X + D$ is $\mathbf{Q}$ -Cartier,
- $K_X + D$ is ample,
- $\text{Aut}(X, D)$ is finite (which follows from ampleness and X reduced).

Definition 4.8.2.2 (Dual Complex of a Stable Degeneration). Let $\overline{X}$ be a reduced, finite-type, possibly reducible variety arising as the $\mathcal{X}_0$ of a degeneration of KSBA stable pairs. The **dual complex** $\Gamma(\overline{X})$ is the simplicial complex defined by:

- **Vertices:** Each irreducible component $\overline{V}_i$ of $\overline{X}$ corresponds to a vertex.
- **k -Simplices:** For every connected component of the intersection of $k + 1$ distinct irreducible components $\overline{V}_{i_0} \cap \overline{V}_{i_1} \cap \cdots \cap \overline{V}_{i_k}$ (with nonempty intersection), include a k -simplex whose vertices correspond to the involved components.
- **Faces and Gluing:** The simplices are glued according to inclusions of the corresponding strata.

The dual complex $\Gamma(\overline{X})$ thus encodes precisely the combinatorics of how the irreducible components of the degeneration $\overline{X}$ meet along their strata of higher codimension.

Definition 4.8.2.3 (KSBA Moduli Functor). Fix numerical invariants (such as dimension d , Hilbert polynomial h , coefficients $\{a_j\}$, and volume $v = (K_X + D)^d$). The **KSBA moduli functor** is

$$\mathcal{M}_{d,\vec{a},v}^{\text{KSBA}}: (\text{Sch}/k)^{\text{op}} \rightarrow \text{Sets}$$

assigning to each S the set of isomorphism classes of families of KSBA stable pairs $(\mathcal{X}, \mathcal{D}) \rightarrow S$ with the fixed invariants.

Definition 4.8.2.4 (KSBA Moduli Stack and Coarse Moduli Space). There exists a Deligne–Mumford stack $\mathcal{M}_{d,\vec{a},v}^{\text{KSBA}}$ of finite type over

k , whose geometric points parametrize KSBA stable pairs with the chosen invariants. The associated **coarse moduli space** $M_{d,\vec{a},v}^{\text{KSBA}}$ is an algebraic space, which is a projective scheme in many important cases.

Theorem 4.8.2.5 (Properness and Projectivity of the KSBA Moduli Space). *The stack $\mathcal{M}_{d,\vec{a},v}^{\text{KSBA}}$ is separated and proper, and its coarse moduli space $M_{d,\vec{a},v}^{\text{KSBA}}$ is projective. Any family of smooth (or slc) stable pairs over a punctured disk extends (after finite base change) to a family over the disk with a unique stable pair as $\mathcal{X}_0$; any isomorphism over the $\mathcal{X}_t$ extends uniquely. Thus, $M_{d,\vec{a},v}^{\text{KSBA}}$ is a modular compactification of the moduli of smooth pairs.*

4.8.3 The KSBA Stack of K -Trivial Pairs

Varieties with $K_X \sim 0$ numerically trivial, such as $K3$ and Enriques surfaces, are said to be **K -trivial**. They require special treatment in the theory of KSBA stable pairs and compactifications because the stability condition (ampleness of $K_X + R$) can not hold when $R = 0$. So one must *always* choose a nontrivial divisor R for such varieties, and the positivity must be entirely supplied by R in order to achieve any kind of stability. We are thus lead, as a first approximation, to consider pairs (X, R) . However, the MMP and KSBA compactification require pairs (X, D) where each component D_i of D appears with coefficient $a_i < 1$, noting that this must be a *strict* inequality. This ensures that limits have only slc singularities and that stability is preserved in families, and avoids the complications that arise in the $a_i = 1$ case – infinite stabilizers leading to Artin stacks instead of Deligne-Mumford stacks, more severe non-slc singularities in degenerations, a potential loss of separatedness, and so on. Thus, in the K -trivial setting, one “perturbs” the canonical class by achieve the necessary positivity, by considering pairs $(X, \varepsilon R)$ with a small rational coefficient $0 < \varepsilon \ll 1$. For sufficiently small ε , the sum $K_X + \varepsilon R$ becomes ample and thus $(X, \varepsilon R)$ is KSBA stable. With this setup, the moduli of stable K -trivial pairs is realized as a special locus in the general KSBA moduli stack described above, and all the foundational results (properness, separatedness, projectivity, etc.) apply directly.

Definition 4.8.3.1 (Stable K -Trivial Pair). Let X be a projective, Gorenstein, connected, reduced variety with $K_X \cong \mathcal{O}_X$ (that is, K_X is trivial; for example, a $K3$ or Enriques surface). Fix a discrete invariant $e > 0$ (e.g., $e = R^2$ for surfaces), and let $0 < \varepsilon \ll 1$ be a

(sufficiently small) rational number. A **stable K -trivial pair of type (e, ε)** is a KSBA stable pair $(X, \varepsilon R)$ such that:

- R is an effective Cartier divisor on X with $R^2 = e$,
- $(X, \varepsilon R)$ is semi-log-canonical,
- $K_X + \varepsilon R$ is ample,
- $\text{Aut}(X, \varepsilon R)$ is finite

Definition 4.8.3.2 (Family and Moduli Functor for Stable K -Trivial Pairs). Let $e > 0$ and $0 < \varepsilon \ll 1$. A **family of stable K -trivial pairs** over a scheme S is a flat, projective morphism $f: (\mathcal{X}, \varepsilon \mathcal{R}) \rightarrow S$ such that every geometric fiber $(\mathcal{X}_s, \varepsilon \mathcal{R}_s)$ is a stable K -trivial pair of type (e, ε) as above. The corresponding moduli functor $\mathcal{M}_e^{K\text{-triv}}(\varepsilon)$ assigns to S the set of isomorphism classes of such families over S .

The moduli functor for stable K -trivial pairs $\mathcal{M}_e^{K\text{-triv}}(\varepsilon)$ arises as a subfunctor of the previously discussed KSBA functor for stable pairs. Specifically, $\mathcal{M}_e^{K\text{-triv}}(\varepsilon)$ parametrizes those families where each fiber $(X, \varepsilon R)$ satisfies $K_X \cong \mathcal{O}_X$, the K -trivial condition, and R is an effective divisor of fixed primitive numerical class e appearing in the boundary with weight $\varepsilon \ll 1$. This is a closed locus in the moduli stack $\mathcal{M}_{d, \vec{a}, v}^{\text{KSBA}}$, where $d = \dim X$, the polarization type e is fixed for the divisor R , the boundary coefficients $\vec{a}$ have $a_j = \varepsilon$ at R and zero otherwise, and K_X is trivial as a line bundle or divisor. The relevance of this construction is that the general results of KSBA theory—separatedness, properness, finite automorphism group property, and the existence of projective coarse moduli spaces—are inherited by this subfunctor. Thus:

- Every family of stable K -trivial pairs admits unique stable limits in one-parameter degenerations, guaranteeing the moduli stack is proper.
- The moduli stack is of finite type, separated, and Deligne–Mumford, with a projective coarse moduli space $M_e^{K\text{-triv}}(\varepsilon)$.
- Extension and uniqueness of isomorphisms in families follow directly from the established machinery for stable pairs.

Thus by construction, the functor $\mathcal{M}_e^{K\text{-triv}}(\varepsilon)$ ensures that the moduli problem for stable K -trivial pairs is embedded as a closed substack in the general KSBA stack, inheriting all of the necessary geometric properties. The existence of universal stable limits in families, the finiteness and separatedness of isomorphism classes, and the projectivity of the coarse moduli space are immediate consequences of this embedding.

Lemma 4.8.3.3 (Independence from Small Boundary ε). *Given $e > 0$, there exists $\varepsilon_0(e) > 0$ such that for all $0 < \varepsilon \leq \varepsilon_0$, the stacks and spaces*

$$\mathcal{M}_e^{K\text{-triv}}(\varepsilon) \cong \mathcal{M}_e^{K\text{-triv}}(\varepsilon_0), \quad M_e^{K\text{-triv}}(\varepsilon) \cong M_e^{K\text{-triv}}(\varepsilon_0)$$

are canonically isomorphic; that is, passing to small boundary does not affect the structure of the moduli problem or its compactification.

Thus $\mathcal{M}_e^{K\text{-triv}}(\varepsilon)$ is a locus in a general KSBA moduli stack cut out by the condition $K_X \cong \mathcal{O}_X$ and a the specification of a single boundary divisor R of degree e with small coefficient ε .

4.8.4 Boundaries of KSBA compactifications

Remark 4.8.4.1 (Combinatorial Organization of the Boundary). For moduli spaces such as F_S of S -polarized K3 surfaces, and thus for spaces like $F_{(2,2,0)}$ and $F_{\text{En},2}$, the boundary of the KSBA compactification is stratified by SLC combinatorial type. Each stratum corresponds to a distinct type of geometric degeneration, classified via monodromy and combinatorial invariants which are encoded in dual complexes, fans, or semifans in related semitoroidal compactifications.

Definition 4.8.4.2 (Boundary Strata and Combinatorial Types). A **boundary stratum** in a KSBA compactification $\overline{X}$ of X is a locally closed subset parameterizing stable pairs (X, R) that are not smooth, i.e., those lying in the boundary $\partial\overline{X} := \overline{X} \setminus X$, where X is the locus of smooth KSBA stable pairs. The **slc combinatorial type** of a stable KSBA limit $(\overline{X}, \varepsilon\overline{R})$ is the discrete data given by the simplicial complex $\Gamma(\overline{X})$, along with the deformation type of the quasi-polarized minimal resolution (V_i, D_i, L_i) of each irreducible component $\overline{V}_i$ of $\overline{X}$, where $L_i = \mathcal{O}_{V_i}(R_i)$ is the line bundle associated to the pullback R_i of the boundary divisor to the resolution. An **slc stratum** is a boundary

stratum of $\overline{X}$ consisting of all stable pairs (X, R) with the same slc combinatorial type.

Given a nonzero vector λ in a lattice T , its **projective class** is $[\lambda] := \left\{ a\lambda \mid a \in \mathbf{R}_{>0} \right\}$, i.e. the ray it generates in $T_{\mathbf{R}}$. Letting T be the polarization lattice for a polarized moduli problem F_{Γ} of K3 surfaces, we consider degenerations at a cusp I in $\partial \overline{F}_{\Gamma}^{\text{BB}}$. The logarithmic monodromy N at I determines, up to the monodromy group and scaling, elements $\eta \in T$ and $\lambda \in \eta^{\perp T}$ by the explicit formula

$$N(\gamma) = (\gamma \cdot \eta)\lambda - (\gamma \cdot \lambda)\eta, \quad \eta^2 = 0, \quad \lambda^2 = \begin{cases} t > 0 & \text{if } I \text{ is a type III 0-cusp} \\ t = 0 & \text{if } I \text{ is a type II 1-cusp} \end{cases},$$

where t is the number of triple points in $\mathcal{X}_0$ in the type III case. Arcs in $\overline{F}_{\Gamma}^{\text{BB}}$ approaching a 0-cusp η are asymptotic to translates of co-characters determined by the class of $\lambda \in \overline{T}_{\eta} = \eta^{\perp T}/\eta$, and thus λ called the “monodromy invariant” of the degeneration. We finally arrive at the key results that make our combinatorial analysis of $F_{\text{En},2}$ possible:

Theorem 4.8.4.3 (Dependence of boundary strata on monodromy invariants [AE23a, Cor. 8.13]). *Suppose R is a recognizable divisor for F_S . Let $(\overline{X}^*, \varepsilon \overline{R}^*) \rightarrow C^*$ be a family of stable pairs over a punctured curve with monodromy invariant λ . Then the slc combinatorial type of the unique KSBA stable limit $(\overline{X}_0, \varepsilon \overline{R}_0)$ depends only on the projective class $[\lambda]$. Thus there is a well-defined **stratum function***

$$\mathbb{S}: \{\text{monodromy invariants } \lambda\} \longrightarrow \{\text{slc boundary strata in } \overline{X}\},$$

which assigns to each projective monodromy class the corresponding slc boundary stratum.

Proposition 4.8.4.4 (Normalization and Semitoroidal Strata [AE23a, Thm. 9.1, Cor. 9.2, Thm. 9.3]). *Let F_S denote the moduli space of S -polarized K3 surfaces, and suppose R is a recognizable divisor for F_S . Then there exists a KSBA compactification F_S^R by stable pairs associated to R , a unique semifan $\mathcal{F}_R$ with a morphism*

$$\Psi_R: \overline{F}_S^{F_R} \rightarrow \overline{F}_S^R$$

from a semitoroidal compactification realizing it as the normalization of $\overline{F}_S^R$. For each cusp $I \in \overline{F}_S^{\text{BB}}$, writing $\overline{T}_I$ for the corresponding

stable boundary lattice and $\mathfrak{C}_I$ for the corresponding positive cone, let

$$\mathfrak{C}_S^{\text{BB}} := \coprod_{I \in \partial \overline{F}_S^{\text{BB}}} (\mathfrak{C}_I \cap \overline{T}_I)$$

and let D be the polyhedral decomposition of $\mathfrak{C}_S^{\text{BB}}$ induced by the level sets of $\mathbb{S}$, i.e. whose tiles are the loci of all monodromy invariants λ on which $\mathbb{S}(\lambda)$ is constant. Then the maximal cones of D and $\mathcal{F}_R$ are in bijection, and Ψ_R sends each stratum in $\partial \overline{F}_S^{\mathcal{F}_R}$ to the corresponding slc stratum of $\partial \overline{F}_S^R$.

4.8.5 Conclusion

The main takeaway of this section is that the boundary of the KSBA compactifications of moduli spaces such as F_S of S -polarized K3 surfaces and related spaces like $F_{(2,2,0)}$ and $F_{\text{En},2}$ admit a precise, combinatorial description:

- The boundary $\overline{F}_S^R$ is stratified according by slc combinatorial type, which encodes both the deformation type of the minimal resolution (with its corresponding divisor) of a stable pair (X, R) and its simplicial dual complex,
- Each stratum corresponds to degenerations sharing the same monodromy data up to scaling, which is captured by the projective classes $[\lambda]$ of monodromy invariants in the stable boundary lattices $\overline{T}_I$. There is thus a well-defined stratum function $\mathbb{S}$ assigning to every monodromy class λ the associated slc boundary stratum in $\overline{F}_S$.

The stratification of $\overline{F}_S^R$ is thus canonically organized by combinatorial data: dual complexes, semifans, and monodromy invariants. In geometric terms:

- There exists a semitoroidal compactification $\overline{F}_S^{\mathcal{F}_R}$ whose normalization maps onto the KSBA compactification $\overline{F}_S^R$,
- Polyhedral decompositions and maximal cones of the semifan $\mathcal{F}_R$ correspond bijectively to the maximal slc strata of the KSBA boundary, and

- The normalization morphism sends each combinatorial stratum of the semitoroidal model to the corresponding slc stratum on the KSBA side.

Thus the boundary of $\overline{F}_S^R$ (and similar KSBA compactifications) is controlled by combinatorial invariants and is indexed purely in terms of discrete monodromy and combinatorial data.

The KSBA approach has a range of pros and cons. On the one hand, it yields a proper, separated, and projective moduli space and captures all stable degenerations (slc pairs). It is compatible with the MMP, and the corresponding coarse moduli space structure is always available. However, the approach is often abstract: explicit descriptions of the boundary and singularities are rare, and few algorithmic tools exist. The geometry of the boundary strata can be highly complicated, non-smoothable components and pathologies are generic in higher dimensions, and there is virtually no uniform combinatorial structure in those settings.

KSBA compactifications extend to surfaces of general type ($\kappa = 2$), but for $\kappa = 0$ surfaces in full generality, i.e. K3, Enriques, abelian, and bielliptic surfaces, as well as for $\kappa = -\infty$ (ruled and rational surfaces), these require significant modification. Moving to higher dimensions, compactifications for general K -trivial varieties like Calabi–Yau threefolds remains a major open problem. While the KSBA theory guarantees the existence of a good compact moduli space for varieties of log general type, the construction of projective and separated moduli spaces for K -trivial threefolds or Calabi–Yau varieties remains largely conjectural. The core technical obstacles here include the lack of smoothability results, the failure of Torelli-type theorems, and much more complex limiting varieties which may be non-reduced or have infinite automorphism groups. For varieties X of general type with $\dim(X) \geq 3$, many results for surfaces generalize via the MMP – however, explicit combinatorial classifications of the boundary are almost entirely missing.

Several parts of the KSBA theory have broad applicability. Universally valid points include the projectivity and properness of the moduli functor for slc varieties of (log) general type, yielding finite automorphism groups for stable pairs and thus a separated proper Deligne–Mumford stack. However, in higher dimensions, these stacks are generally expected to be non-irreducible and highly singular, and even the deformation spaces of smooth objects can possess arbitrarily

bad singularities – a phenomenon encapsulated in Vakil’s “Murphy’s law.” Unlike curves, where every stable limit is smoothable, smoothability fails in general: not every slc variety arises as a limit of smooth varieties. Combinatorial invariants such as dual complexes rapidly become complicated and lose the inductive or graph-like simplicity seen in the curve or surface cases as the dimension increases. For K3 and Enriques surfaces, the boundary components of $\overline{F}_S$ reflect geometric degenerations that align with period domain descriptions and semitoroidal constructions.

There is also an increasing interest in the relationship between K -stability and KSBA stability, particularly in the Fano and K -trivial settings. Heuristically, K -stability is the link between GIT and KSBA stability – for instance, any (GIT) K -semistable polarized variety has slc singularities, and any slc variety polarized by an ample canonical divisor is K -stable. Similarly, K -trivial, slc polarized pairs are K -semistable. In the Fano and Calabi-Yau cases, K -stability is necessary for establishing moduli that carry the expected differential-geometric invariants, namely, Kähler–Einstein metrics. However, K -stability is not an open or constructible condition in general, and so explicit construction of moduli stacks via K -stability remains conjectural for many classes. Recent work by Xu, Li, Wang, and others have constructed projective moduli spaces for Fano varieties using the K -stability criterion.

Some key open directions for future research include:

- Extending the success of explicit boundary classifications, as seen in K3 and Enriques surfaces, to Calabi–Yau threefolds and higher-dimensional Fano varieties;
- Exploiting deeper connections with period maps and Hodge theory to construct and better understand compactification in higher dimensions;
- Clarifying the exact relationships between KSBA and K -stability, and developing effective, algorithmic, or combinatorial tools for KSBA (or related) compactifications beyond the few well-understood special cases.

CHAPTER 5

INTEGRAL AFFINE GEOMETRY AND COMPARING COMPACTIFICATIONS

Sources for this chapter include [FS86; Kul77; PP81; Sca87; MM83; Fri84; FM83] Let T be an even lattice of signature $(2, n)$, and let $\Gamma \subset O(T)$ be an arithmetic subgroup. The period domain D_T parametrizes weight-two Hodge structures on T , and the modular varieties $F_\Gamma = D_T/\Gamma$ serve as coarse moduli spaces for (marked) polarized K3 surfaces. Given a one-parameter degeneration of K3 or Enriques surfaces, a **Kulikov model** arises after a ramified base change, and the possible types for $\mathcal{X}_0$ are:

- **Type I:** $\mathcal{X}_0$ is a smooth K3 surface.
- **Type II:** $\mathcal{X}_0 = V_0 \cup \dots \cup V_k$ is a chain of surfaces, where
 - V_0 and V_k are rational and $V_1, \dots, V_{k-1}$ are birational to $E \times \mathbf{P}^1$ and thus elliptically ruled,
 - The chain consists of components $V_0, \dots, V_k$ glued along elliptic curves $E_i = V_{i-1} \cap V_i$ ($1 \leq i \leq k$), all isomorphic to the same elliptic curve E , satisfying a compatibility condition on normal bundles, $\mathcal{N}_{E/V_i} \otimes \mathcal{N}_{E/V_{i+1}} \cong \mathcal{O}_E$
 - The dual complex $\Gamma(\mathcal{X}_0)$ is a simplicial $\mathbf{D}^1$, corresponding to a partition of the closed unit interval $[0, 1]$.
- **Type III:** $\mathcal{X}_0 = \bigcup_{i=0}^r V_i$ is a union of rational surfaces glued along rational curves such the dual intersection complex $\Gamma(\mathcal{X}_0)$ is a

simplicial S^2 . The union of the double curves on each component V_i forms an anticanonical divisor – that is, a (possibly reducible) cycle of smooth rational curves in $|-K_{V_i}|$

The dual intersection complex Δ of a Type III model is a triangulation of S^2 admitting the structure of a 2-dimensional **integral affine sphere with singularities** (IAS²) which correspond to a distinguished subset of vertices in Δ , after passing to a suitably complete triangulation. Away from this singular locus, the charts are locally modeled on open subsets of $\mathbf{R}^2$ with transition functions in the orientation-preserving affine linear group $\mathrm{SL}_2(\mathbf{Z}) \ltimes \mathbf{R}^2$. The possible singularities are modeled on the local structure of the quotient of $\mathbf{R}^2$ by the shearing matrices $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ for some integer $n \geq 0$; these are called I_n singularities, where I_1 are generic and I_0 is the trivial (toric) case. More general singularities arise as products of such shears, corresponding to collisions of the corresponding I_n singularities, see [AE22; Sym03]

By taking the star to obtain a fan, each vertex v in Δ yields an **anticanonical pair** (V, D) , where V is a rational surface and $D = \sum_j D_j \in |-K_V|$ is a cycle of rational curves in the anticanonical linear system. The dual polytope of this fan encodes a *semitoric* variety: varieties which arise from toric varieties via sequences of blowups and blowdowns. The data of these blowups is encoded in the following way: the **charge** at a vertex v , for the anticanonical pair (V, D) , is defined by $Q(V, D) := \frac{1}{2} \sum_j D_j^2 + 3$, where $D = \sum_j D_j$ is the (possibly reducible) anticanonical cycle, and D_j^2 is the self-intersection of component D_j in V . The charge is precisely designed to measure the defect from being toric: $Q(V, D) = 0$ precisely when (V, D) is toric. More fundamentally, [FS86, Prop. 2.11] asserts that $\sum_{v \in \Delta} Q(V_v, D_v) = 24$ for any Type III Kulikov model of a K3 surface, see [FM83; FS86].

Symington surgeries are local modifications of IAS² structures that manipulate the placement and type of singularities, and there are two main types: performing a *toric blowup* corresponds to the standard blowup in toric geometry attained by inserting rays into a fan or deleting triangles from its dual polytope. It increases the number of vertices in Δ , but preserves the total charge. A *non-toric* (interior) blowup is a surgery that increases the charge at an interior vertex and modifies the integral-affine structure locally by adding the data of a distinguished *monodromy ray*, where the transition functions across the ray are generically described by monodromy matrices as

described above. A toric blowup is thus a blowup at a node of the cycle, corresponding to a torus-invariant point in a toric model, while a nontoric/interior blowup is a blowup at a smooth point of the surface, either on or off the anticanonical cycle. Such moves are necessary to build arbitrary IAS^2 with prescribed collections of I_n singularities and total charge 24 from a standard simplicial sphere.

Symington's theory explicitly produces IAS^2 by sequences of such surgeries, along with *cuts* and *nodal slides*. Thus, the charge, and the combinatorial data of surgeries, encode all allowable dual complexes for Type III degenerations. Arbitrary IAS^2 s with prescribed $(I_{n_1}, \dots, I_{n_k})$ of total charge 24 are built by sequences of toric and nontoric starting from a convex polygon. This defines the base of a Lagrangian torus fibration, and we refer to the total space as a *semitoric variety* – a variety that is birationally equivalent to a toric variety, differing by only finitely many nontoric blowups. Whenever the data $(B(\lambda), Q)$ arise from such construction, there exists a corresponding Kulikov model $\mathcal{X}_0$ with $\Gamma(\mathcal{X}_0) = B(\lambda)$.

Any such IAS^2 satisfying a d -semistability condition $\text{Ext}_{\mathcal{O}_{\mathcal{X}_0}}^1(\Omega_{\mathcal{X}_0}, \mathcal{O}_{\mathcal{X}_0}) \cong \mathcal{O}_{\mathcal{X}_0^{\text{sing}}}$ admits a smoothing $\mathcal{X} \rightarrow \Delta$ to a (possibly singular) K3 surface. The d -semistability condition is necessary and sufficient for (local) smoothability; for global smoothings to K3 surfaces, the central fiber must also satisfy the additional conditions of being a Kulikov model. [GHK15], for example, relate this to the existence of Lagrangian torus fibrations on the nearby fibers, mirroring the affine geometry of $B(\lambda)$.

Every Type III Kulikov degeneration carries a *projective monodromy invariant* $[\lambda]$ in (projectivized) boundary lattice $\mathbf{P}(\overline{T}_\eta) = \mathbf{P}(\eta^\perp/\eta)$. This invariant records the monodromy around $\mathcal{X}_0$ and, crucially, determines the gluing data for reconstructing the degeneration from the IAS^2 and its singularities. A Eichler transvection relates $[\lambda]$, the vanishing cycle. The local system H^2 carries a monodromy operator T , computed via a primitive isotropic vector δ , the *vanishing cycle*, and the monodromy invariant λ :

$$T(x) = x + (x, \delta)\lambda - (\lambda, x)\delta, \quad x \in H^2(\mathcal{X}_t; \mathbf{Z})$$

where λ^2 counts the number of triple points (see [PP81; Sca87]).

Recall that a divisor R is **recognizable** [AE21; AEH21] for F_Γ if, for any K3 surface $X = \mathcal{X}_0$ and any smooth $\mathcal{X}$ to a Kulikov model, the flat limit R_0 on $\mathcal{X}_0$ is uniquely determined up to automorphism. Given a recognizable divisor R , the KSBA compactification $\overline{F}_\Gamma^R$ can

be formed, as well as a the corresponding normalizing semitoroidal compactification $\overline{F}_\Gamma^{\mathcal{F}_R}$. By way of the strata functions $\mathbb{S}(\lambda)$ from Chapter 4 for decorated intersection complexes $B(\lambda)$ constructed from monodromy invariants λ , boundary strata on both sides correspond to possible IAS^2 with singularities.

Mirror symmetry supplies a *Lagrangian torus fibration* over $B(\lambda)$, and the intersection complex $\Gamma(\mathcal{X}_0)$ of a Kulikov model coincides with the IAS^2 $B(\lambda)$ constructed from the monodromy data. The smoothability of the singularities then produces a family $\mathcal{X} \rightarrow \Delta$, whose general fiber is smooth, and an explicit contraction algorithm (mirroring the MMP) yields the KSBA stable model. For K3 surfaces with a nonsymplectic involution, the construction of IAS^2 s is mirrored by gluing a polygon P to its *opposite* P^{op} , forming an IAS^2 , $B(\lambda)$, with an induced involution.

Given a degeneration at a 0-cusp η , the cusp is classified by the boundary lattice $\overline{T}_\eta$ and the group Γ_η . The associated Coxeter polytope $P(\Gamma_\eta)$ specifies a Coxeter-Vinberg diagram $G(\Gamma_\eta)$. To build the intersection complex $\Gamma(\mathcal{X}_0)$ (for a degeneration with monodromy invariant $[\lambda]$),

- Compute $[\lambda] \in \mathfrak{C}_\eta$,
- Write barycentric coordinates $\ell_i = (\lambda, \alpha_i)$,
- Form the convex (planar) polygon $B_1(\lambda) = \text{Conv } v_i \subset \mathbf{R}^2$ where v_i are specified by the Coxeter diagram and ℓ_i govern their lengths,
- Obtain the full IAS^2 by gluing $B_1(\lambda)$ to its opposite: $B(\lambda) = B_1(\lambda) \cup -B_1(\lambda)$, yielding an IAS^2 .

We summarize this process in the following algorithm to construct Type III degenerations for F_Γ , which we in turn specialize to $F_{(2,2,0)}$ and $F_{\text{En},2}$ to construct dlt models for KSBA stable limits:

1. **Build the Coxeter polygon** $P(\Gamma_\eta)$ from the Coxeter–Vinberg diagram, using the configuration of simple roots α_i .
2. **Compute barycentric coordinates** $\ell_i = (\lambda, \alpha_i)$ from the monodromy invariant $[\lambda]$.
3. **Construct the monodromy polytope** $B_1(\lambda)$ in $\mathbf{R}^2$ using vertices v_i and these lengths.

4. **Form the IAS²** $B(\lambda) = B_1(\lambda) \cup -B_1(\lambda)$, with I_n singularities at prescribed locations, with total charge 24.
5. **Produce the corresponding Kulikov model** $\mathcal{X}_0$ where $\Gamma(\mathcal{X}_0)$ matches $B(\lambda)$, and smooth $\mathcal{X}_0$ to a family $\mathcal{X} \rightarrow \Delta$.
6. **Contract** $\mathcal{X}$ to the KSBA stable model $\overline{\mathcal{X}}$.

For Enriques surfaces, the above theory must be quotiented by a fixed-point-free involution on IAS^2 , reflecting the covering K3 surface structure. This mandates working with “half-divisor” models (??) and searching for IAS^2 s invariant under a group generated by two commuting involutions, one corresponding to ι_{dP} , corresponding to K3 surfaces X in $F_{(2,2,0)}$, and another fixed-point-free involution corresponding to ι_{En} , coming from the embedded locus $F_{\text{En},2}$, those X arising as K3 covers $X \rightarrow Z$ of Enriques surfaces. We impose the condition that the periods and dual complex of (X_0, R_0) are involution invariant – then the Torelli theorem for anticanonical pairs shows that (V_i, D_i, R_i) admits an involution $\iota_{\text{En},i}$ and the quotient $(V_i, D_i, R_i)/\iota_{\text{En},i}$ is a log Calabi–Yau pair. We note that ι_{En} is only a birational involution in general, and so we obtain half-divisor models for generic degenerations with a given monodromy invariant λ .

5.1 Kulikov Models

5.1.1 Degenerations

Let $\pi: \mathcal{X} \rightarrow C$ be a flat, proper morphism, with C a germ of a smooth complex curve (typically taken as a disk $\Delta = \{t \in \mathbf{C}: |t| < \epsilon\}$). The $\mathcal{X}_0 = \pi^{-1}(0)$ encodes the limiting geometry of the family as $t \rightarrow 0$. For $t \neq 0$, the fibers $\mathcal{X}_t$ are assumed smooth, often K3 or Enriques surfaces. The **punctured disk** $\Delta^* = \Delta \setminus \{0\}$ and the corresponding smooth locus $\mathcal{X}^* = \pi^{-1}(\Delta^*)$ are natural analytic settings for studying the variation of Hodge structures in the family. Over Δ^* , the fibers $\mathcal{X}_t$ are smooth and give rise to a variation of Hodge structure on the local system $R^2\pi_*\mathbf{Z}$. The behavior of $\mathcal{X}_0$ reflects the limiting geometry and possible singularities, providing key topological and moduli-theoretic invariants that determine both the topological degeneration type and the locus in the compactified moduli space.

Dual Complexes If $\mathcal{X}_0$ is a simple normal crossings (snc) divisor, its **dual complex** $\Gamma(\mathcal{X}_0)$ is the finite simplicial complex constructed as follows:

- vertices correspond to irreducible components of $\mathcal{X}_0$;
- k -simplices correspond to connected components of nonempty intersections of $k + 1$ distinct components.

This combinatorial invariant is stable under blowups and birational modifications that preserve the snc property. For degenerations with semi log canonical (slc) singularities, $\Gamma(\mathcal{X}_0)$ is defined up to homeomorphism and remains a meaningful measure of combinatorial complexity of $\mathcal{X}_0$. For degenerations of Enriques surfaces constructed as quotients by a biregular, fiberwise involution (the **Enriques involution**), the dual complex comes in one of two types:

- a topological closed disk $\mathbf{D}^2$, when the quotient preserves orientation, or
- a real projective plane $\mathbf{RP}^2$, when antipodal identification occurs.

This distinction can be read off from the intersection pairing on the lattice $\overline{T}_I$ attached to the relevant cusp I and the structure of $\mathcal{X}_0$.

5.1.2 Kulikov Models

A **Kulikov model** of a family of projective surfaces is a degeneration $\mathcal{X} \rightarrow C$ in which:

1. The total space $\mathcal{X}$ is regular, and thus smooth as a threefold);
2. The central fiber $\mathcal{X}_0$ is a reduced snc divisor;
3. The relative dualizing sheaf is trivial: $\omega_{\mathcal{X}/C} \cong \mathcal{O}_{\mathcal{X}}$.

This Calabi–Yau condition ensures that $\mathcal{X}$ is a Calabi–Yau threefold over C and provides precise control over the limiting MHS (see [Kul77; PP81]). The condition ensures that the limits of periods and Hodge structures are controlled, and $\mathcal{X}_0$ is as geometrically simple as feasible under semistable reduction.

The variation of Hodge structure on the local system $R^2\pi_*\mathbf{Z}$ equips the family with a locally constant sheaf whose fibers are $H^2(\mathcal{X}_t; \mathbf{Z})$ for $t \neq 0$. The monodromy transformation about $t = 0$ is given by the Picard-Lefschetz operator $T: H^2(\mathcal{X}_t; \mathbf{Z}) \rightarrow H^2(\mathcal{X}_t; \mathbf{Z})$. After passing to a finite ramified cover of Δ , one reduces to the case where T is unipotent and defines the logarithm $N = \log T$ which is nilpotent of order at most three. The index of nilpotency of N determines both the geometry of $\mathcal{X}_0$ and the combinatorial type of the degeneration.

Type I. If $N = 0$, then $\mathcal{X}_0$ is a smooth K3 surface. The degeneration is locally analytically trivial, and the dual complex is a single point.

Type II. If $N \neq 0$ and $N^2 = 0$, then $\mathcal{X}_0 = V_1 \cup \cdots \cup V_n$, where:

- V_1, V_n are rational surfaces;
- Each intermediate V_i ($2 \leq i \leq n-1$) is birational to $E \times \mathbf{P}^1$ for a fixed elliptic curve E ;
- The double locus $D_{i,i+1} = V_i \cap V_{i+1}$ is isomorphic to E ;
- The self-intersection numbers satisfy

$$D_{i,i+1}|_{V_i}^2 + D_{i,i+1}|_{V_{i+1}}^2 = 0.$$

The dual complex is a closed interval IAD^1 , i.e. a decomposition of the interval $[0, 1]$ into sub-segments.

Type III. If $N^3 = 0$ with $N^2 \neq 0$, then $\mathcal{X}_0$ is a union of smooth rational surfaces, with irreducible components meeting along rational curves, such that:

- On each V_i the sum of the double curves forms an anticanonical cycle:

$$\sum_{j \neq i} D_{ij} \sim -K_{V_i},$$

- The intersection numbers satisfy

$$d_{ij} + d_{ji} = -2, \quad d_{ij} = -2p_a(D_{ij}) - (D_{ij})^2,$$

where $p_a(D_{ij})$ is the arithmetic genus of D_{ij} .

The dual complex of $\mathcal{X}_0$ is a triangulation of the 2-sphere IAS^2 [FS86; Fri83a]. These types are naturally stratified by the rank of isotropic subspaces in the boundary lattice $\overline{T}_I$ and correspond to geometric monodromy vectors in the rational closure of $\mathfrak{C}_I$ in $\overline{T}_{I,\mathbf{R}}$:

- Type I degenerations correspond to interior points of $\mathfrak{C}_I$;
- Type II degenerations arising from rank 1 isotropic sublattices I correspond to rational boundary rays in $\mathfrak{C}_{I,\mathbf{Q}}$ and thus to parabolic/ $\widetilde{\text{ADE}}$ subdiagrams of the Vinberg-Coxeter diagram G_I and yield integral-affine segments,
- Type III degenerations arising from rank 2 isotropic sublattices I correspond to elliptic/ ADE subdiagrams of G_I and yield integral-affine spheres.

5.1.3 Picard-Lefschetz Theory

Picard–Lefschetz Transformations

Definition 5.1.3.1 (Picard–Lefschetz Transformation). Let $p: \mathcal{X} \rightarrow \Delta$ be a Kulikov model (or a semistable degeneration) with $\mathcal{X}_0 \subset \mathcal{X}$. The sheaf $\mathbf{R}^2 p_* \underline{\mathbf{Z}}_\Delta$ restricts to a locally constant system over the punctured disk Δ^* , whose fiber over $t \in \Delta^*$ is $H^2(\mathcal{X}_t; \mathbf{Z})$. After trivializing the pullback of this local system to the universal cover $\widetilde{\Delta}^*$, the fundamental group $\pi_1(\Delta^*, t)$ acts via monodromy $\pi_1(\Delta^*, t) \rightarrow \text{Aut}(H^2(\mathcal{X}_t; \mathbf{Z}))$. The image of a simple closed loop γ generating $\pi_1(\Delta^*, t)$ is the **Picard–Lefschetz transformation**:

$$T_\gamma: H^2(\mathcal{X}_t; \mathbf{Z}) \rightarrow H^2(\mathcal{X}_t; \mathbf{Z}),$$

represented by the **monodromy matrix** $T \in \text{GL}_n(\mathbf{Z})$ where $n = \text{Pic } H^2(\mathcal{X}_t; \mathbf{Z})$ [Kul77], [FS86].

Theorem 5.1.3.2 (Quasi-Unipotency and Log Monodromy). *If $\mathcal{X}_0$ is a simple normal crossings (semistable) degeneration, the monodromy T is **quasi-unipotent**: that is, there exist positive integers k, n such that $(T^k - \text{id})^n = 0$. For semistable degenerations, one can take $k = 1$. The logarithm of the unipotent part of T defines the **log monodromy operator**:*

$$N = \log(T) = (T - \text{id}) - \frac{1}{2}(T - \text{id})^2 + \frac{1}{3}(T - \text{id})^3 - \cdots + \frac{(-1)^{n+1}}{n}(T - \text{id})^n,$$

where n is the index of nilpotency [Kul77], [PP81].

Definition 5.1.3.3 (Monodromy Around Singularities). Let B be an integral affine manifold with singularities. The **monodromy** of B around a singularity p is the element of $\text{Aff}(\mathbf{Z}^n)$ defined by parallel transport around a loop encircling p . For an I_1 singularity in dimension 2, the monodromy matrix is $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. This affine monodromy encodes the classical Picard–Lefschetz transformation for such degenerations [KS06], [Sym03].

Remark 5.1.3.4 (Monodromy Invariant as Structural Parameter). For a Type III Kulikov degeneration with $\mathcal{X}_0$, the **monodromy invariant** $\lambda \in \overline{T}_\eta = \eta^\perp/\eta$ determines:

1. The number of triple points of $\mathcal{X}_0$, with $\lambda^2 = \#\{\text{triple points}\}$;
2. The *barycentric coordinates* $\lambda = (\ell_i)$, which determine $B(\lambda)$, an explicit IAS².

For more details, see [Ale+23; GHK15; Eng18].

Definition 5.1.3.5 (Weight Filtration on Cohomology). Let $N = \log(T)$ be the logarithm of the unipotent monodromy operator. The **weight filtration** $W^\bullet$ on $H^2(\mathcal{X}_t; \mathbf{C})$ is defined by:

$$W_k H^2 = \left\{ v \in H^2 \mid N^j v = 0 \text{ for } j > k \right\},$$

yielding a canonical increasing filtration

$$0 = W_{-1} \subset W_0 \subset W_1 \subset W_2 \subset W_3 \subset W_4 = H^2(\mathcal{X}_t; \mathbf{C}).$$

The **graded pieces** are $\text{Gr}_k^W H^2 := W_k/W_{k-1}$, and carry the limiting mixed Hodge structure by [Del71].

In the case of K3 surfaces, we can explicitly recover the filtration in terms of the order of nilpotency:

Theorem 5.1.3.6 (Weight Filtration and Kulikov Type). *Let N be the logarithm of the monodromy in a degeneration of K3 (or analogous) surfaces. Then:*

- **Type I** (trivial monodromy, $N = 0$): $W_2 = H^2$, and all higher pieces vanish.

- **Type II** ($N^2 = 0$, $N \neq 0$): $W_0 = \ker(N)$; $W_2 = H^2$; $W_4 = 0$.
- **Type III** ($N^3 = 0$, $N^2 \neq 0$): all steps in the filtration are nontrivial.

The geometric interpretations are as follows:

- Gr_0^W encodes contributions from singular or reducible components;
- Gr_2^W corresponds to the part of the cohomology preserved in the degeneration;
- Gr_4^W arises from new cycles present in Type III degenerations.

This filtration packages the mixed Hodge structure obtained from the Clemens–Schmid exact sequence and classifies degenerations according to their limit behavior by [Kul77].

Degenerations with unipotent monodromy give rise to *monodromy invariants*, which we now describe.

Definition 5.1.3.7 (Monodromy Invariant and Barycentric Coordinates). Let $\lambda \in \overline{T}_\eta := \eta^\perp/\eta$ denote the monodromy invariant associated to a degenerating family of K3 surfaces. Let $\{\alpha_i\} \subset L$ be a set of simple roots defining a rational polyhedral chamber $\mathfrak{C} \subset L_{\mathbf{R}}$, with wall hyperplanes $\alpha_i^\perp$. We define the **barycentric coordinates** of λ by:

$$\ell_i := \lambda \cdot \alpha_i.$$

The vector $\ell = (\ell_i)$ determines the affine geometry of the dual complex and the positions of singularities under the integral-affine structure.

Let $\pi: \mathcal{X} \rightarrow C$ be a flat projective family of K3 surfaces over a smooth curve C , endowed with a biregular involution ι_{En} acting on $\mathcal{X}$ such that for every $t \neq 0$, the restriction $\iota_{\mathrm{En}}|_{\mathcal{X}_t}$ is a fixed-point-free involution. Thus, for $t \neq 0$, the fibers $\mathcal{X}_t$ are K3 surfaces and the quotients $\mathcal{X}_t/\iota_{\mathrm{En}}$ are nonsingular Enriques surfaces. The specialization of ι_{En} to $\mathcal{X}_0$, denoted $\iota_{\mathrm{En},0}$, may acquire fixed points even if it acts freely on $\mathcal{X}_t$. This can result in additional singularities in $\mathcal{X}_0$, and consequently, in the quotient

$$\mathcal{Z}_0 := \mathcal{X}_0/\iota_{\mathrm{En},0}.$$

Here, $\mathcal{X}_0$ denotes the central fiber of the degeneration, and $\mathcal{Z}_0$ is its quotient by the specialized involution. The dual complex $\Gamma(\mathcal{Z}_0)$ of the central fiber $\mathcal{Z}_0$ is topologically determined by the action of $\iota_{\text{En},0}$ on the dual complex $\Gamma(\mathcal{X}_0)$ of the original Kulikov fiber. Specifically, if $\Gamma(\mathcal{X}_0)$ is a triangulated 2-sphere, then if $\iota_{\text{En},0}$ is free on $\Gamma(\mathcal{X}_0)$, the quotient dual complex $\Gamma(\mathcal{Z}_0)$ is homeomorphic to the real projective plane $\mathbf{RP}^2$, as occurs when the involution acts antipodally on the sphere. If $\iota_{\text{En},0}$ has fixed points on $\Gamma(\mathcal{X}_0)$ or preserves a region, then the quotient dual complex $\Gamma(\mathcal{Z}_0)$ is a disk $\mathbf{D}^2$, corresponding to a boundary stratum where the involution admits fixed locus or acts with boundary preserves.

5.2 Anticanonical Pairs and Kulikov Models

5.2.1 Charge

The components of certain singular fibers in degenerations are naturally described as log Calabi–Yau surfaces, also known as anticanonical pairs.

Definition 5.2.1.1 (Anticanonical Pairs). Let V be a smooth projective rational surface, and let $D = \sum_j D_j$ be a reduced divisor on V with simple normal crossings. The pair (V, D) is called an **anticanonical pair** if the log canonical divisor is $\mathbf{Q}$ -linearly trivial:

$$K_V + D \sim_{\mathbf{Q}} 0.$$

Such pairs are also called **log Calabi–Yau surfaces**. If V is a toric surface and $D = \partial V$ is its toric boundary divisor, (V, D) is called a **toric anticanonical pair**. Common examples include pairs where D is a smooth elliptic curve, a cycle of $n \geq 2$ smooth rational curves, or an irreducible rational nodal curve. For a classification of exceptional cases, see [GHK15, Lem. 4.2]. The following is a numerical invariant that measures the deviation of an anticanonical pair from being toric.

Definition 5.2.1.2 (Charge of an Anticanonical Pair). Let $(V, D = \sum_j D_j)$ be an anticanonical pair. The **charge** of the pair is defined as

$$Q(V, D) := 12 - \sum_j (D_j^2 + 3).$$

This quantity is a non-negative integer for all anticanonical pairs and is zero if and only if the pair (V, D) is toric.

The definition of charge can be restated using the adjunction formula, depending on the geometry of the boundary divisor D .

Definition 5.2.1.3 (Alternative Formulas for Charge). Let (V, D) be an anticanonical pair. If $D = \sum_j D_j$ is a nodal cycle of $n \geq 2$ rational components, the charge is $Q(V, D) = 12 + \sum_j (-D_j^2 - 3)$. If D is an irreducible nodal curve, the charge is $Q(V, D) = 11 - D^2$.

5.2.2 Toric Models and Blowups

Rational anticanonical pairs can be constructed from toric pairs via a sequence of two fundamental types of birational modifications.

Definition 5.2.2.1 (Corner and Interior Blowups). Let (V, D) be an anticanonical pair.

- A **corner blowup** is the blowup of V at a node of D . If $\pi : V' \rightarrow V$ is such a blowup and D' is the reduced total transform of D , then (V', D') is an anticanonical pair with its charge preserved: $Q(V', D') = Q(V, D)$.
- An **interior blowup** is the blowup of V at a smooth point of a component of D . If D'' is the union of the proper transform of D and the exceptional divisor, then (V'', D'') is an anticanonical pair whose charge increases by one: $Q(V'', D'') = Q(V, D) + 1$.

Any rational anticanonical pair can be obtained from a toric pair (which has charge 0) by a sequence of corner blowups followed by a sequence of interior blowups [GHK15]. In the context of K3 surface degenerations, these local invariants are subject to a global conservation law discovered by Friedman and Morrison.

Theorem 5.2.2.2 (Friedman–Morrison Charge Theorem for Type III Degenerations). *Let $\mathcal{X} \rightarrow \Delta$ be a Type III Kulikov degeneration of K3 surfaces, with $\mathcal{X}_0 = \bigcup_{i=1}^n V_i$. For each component V_i , let $D_i = V_i \cap \overline{(\mathcal{X}_0 \setminus V_i)}$ be its boundary divisor, so that (V_i, D_i) is an anticanonical pair. Then the sum of the charges of these pairs is constant:*

$$\sum_{i=1}^n Q(V_i, D_i) = 24.$$

This imposes the constraint that at most 24 of the components V_i can be non-toric, as toric pairs contribute zero to the sum. This result provides a rigidity condition on the combinatorics of K3 degenerations [FM83, Thm. 2.2], [Fri15].

Remark 5.2.2.3. The conservation property constrains the possible combinatorial types of $\mathcal{X}_0$ s and ensures that the total complexity of a degeneration, as measured by the charge, remains constant across the moduli space. This invariance establishes the relationship between the KSBA and Baily-Borel compactifications of the moduli space.

5.3 Integral Affine Structures

Definition 5.3.0.1 (Integral-Affine Structure). Let B be a topological manifold.

An **integral-affine structure** on B is specified by an atlas of charts

$$\{\phi_i: U_i \rightarrow \mathbf{R}^n\}$$

such that for each nonempty overlap $U_i \cap U_j$, the transition maps

$$\phi_j \circ \phi_i^{-1}: \phi_i(U_i \cap U_j) \rightarrow \phi_j(U_i \cap U_j)$$

are elements of the group $\text{Aff}(\mathbf{Z}^n) := \text{GL}_n(\mathbf{Z}) \ltimes \mathbf{R}^n$. That is, each transition function is of the form $x \mapsto Ax + b$, where $A \in \text{GL}_n(\mathbf{Z})$ and $b \in \mathbf{R}^n$, i.e., it is an affine transformation preserving the standard lattice $\mathbf{Z}^n \subset \mathbf{R}^n$. This defines a local system $\Lambda_B \subset \mathcal{T}_B$ of integral sections of the tangent sheaf on B : a subsheaf locally trivialized by those coordinate vector fields which whose transition maps act by the linear part A . The global monodromy of Λ_B records the obstruction to a global trivialization.

The integral-affine structure is said to be **singular** if there exists a discrete (or more generally, closed of codimension at least two) subset in B such that the charts cannot be completed, or equivalently, the monodromy of Λ_B is nontrivial around loops enclosing these points. These singularities model local monodromy defects of the affine structure and their presence is intimately related to topological data, encoding, for example, the global Euler characteristic or “total curvature” of the integral-affine manifold. In the context of degenerations, such

singularities mirror the loci in the total space responsible for nontrivial monodromy or the failure of smooth toric models.

Proposition 5.3.0.2 (Integral-Affine Structures on Kulikov Models).

Let $\pi: \mathcal{X} \rightarrow \Delta$ be a one-parameter degeneration of K3 surfaces over a disc, with reduced normal crossings $\mathcal{X}_0$

$$\mathcal{X}_0 = \bigcup_{i \in I} V_i$$

and denote by $\Gamma(\mathcal{X}_0)$ the dual complex of $\mathcal{X}_0$.

If $\mathcal{X}$ is a Type III Kulikov degeneration of K3 surfaces, then $\Gamma(\mathcal{X}_0)$ is canonically a triangulated 2-sphere. There exists a canonical singular integral-affine structure on $\Gamma(\mathcal{X}_0)$, defined as follows:

Each irreducible component is an anticanonical pair, which we recall admits a pseudofan. The pseudo-fans are glued along their boundaries via primitive integral-affine identifications determined by the cyclic order and intersection numbers of the components of D_i as they appear in $\mathcal{X}_0$. This construction defines an integral-affine structure on $\Gamma(\mathcal{X}_0)$.

The points of $\Gamma(\mathcal{X}_0)$ corresponding to non-toric anticanonical pairs, i.e. those of positive charge, become singularities for the integral-affine structure. The monodromy about these points records the nontriviality of the local structure. As shown in [FM83], the total number of such singularities equals: $\sum_{i \in I} Q(V_i, D_i) = 24$, where $Q(V_i, D_i)$ denotes the charge of the pair (V_i, D_i) .

Definition 5.3.0.3 (The Symington polytope $B(\lambda)$). Given a vector λ , define $P(\lambda)$ to be the **Symington polytope**, an integral-affine polygon uniquely determined by the entries of ℓ , constructed as specified in [Sym03]. The edge lengths and directions of $P(\lambda)$ are determined by the barycentric coordinates ℓ_i with respect to the roots α_i . Define the corresponding IAS² $B(\lambda)$ by $B(\lambda) := P(\lambda) \cup P(\lambda)^{\text{op}}$, where $P(\lambda)^{\text{op}}$ is a copy of $P(\lambda)$ with reversed orientation. The polytopes $P(\lambda)$ and $P(\lambda)^{\text{op}}$ are glued along their boundaries, which jointly form the equator of $B(\lambda)$.

Definition 5.3.0.4 (Integral affine polarizations). Let R_{IA} denote the **integral-affine polarization** on $B(\lambda)$, which is a distinguished equator in the integral-affine sphere $B(\lambda)$. This equator corresponds to the flat limit divisor $\mathcal{R}_0$ in the degeneration. For the cusp $(18, 2, 0)_1$, R_{IA} consists of edges along the equator with weights alternating between

2 and 1. For the cusp $(18, 0, 0)_1$, R_{IA} has no support on the bottom edge of $P(\lambda)$, and the associated components of $\mathcal{R}_0$ in $\mathcal{X}_0$ are empty.

The construction of $P(\lambda)$ and $B(\lambda)$ may require certain coordinates of ℓ to be even, depending on the configuration of the singularities and the structure of the underlying lattice. These divisibility conditions ensure, for example, that an I_2 -type singularity may be split into two I_1 -type singularities in the integral-affine structure. Coordinates may correspond to a multiple of the lattice distance between pairs of integral-affine singularities that are introduced via surgery or gluing operations in the polytope. These entries record a geometric “distance” between singularities (or the collisions thereof) in integral affine manifold. For example, the following shows the template for $B(\lambda)$ for $(18, 2, 0)_1$:

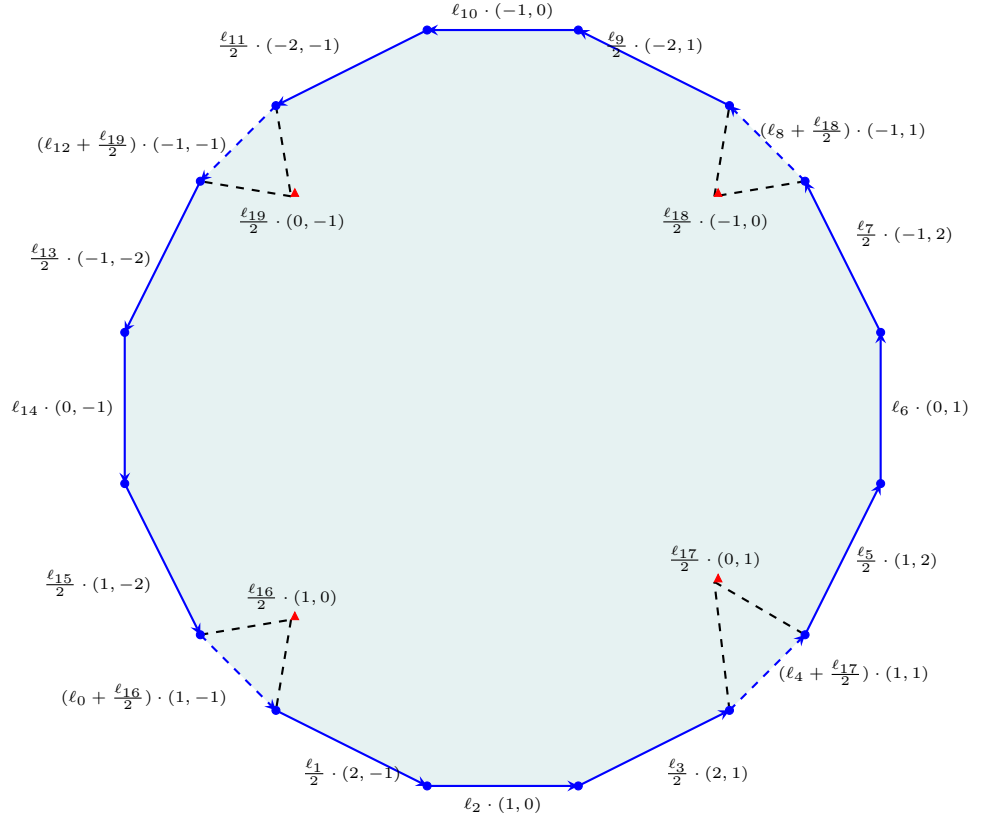


Figure 5.3.1: The template for building an $\text{IAS}^2 B(\lambda)$ for K3 surfaces in $F_{(2,2,0)}$, specified by the generalized barycentric coordinates $\ell_i := \lambda \cdot \alpha_i$ for the roots α_i in $(18, 2, 0)_1$. The IAS^2 for Enriques surfaces at cusps 1, 3, 4, and 5 will thus involve choices of ℓ_i which symmetric under an involution particular to each cusp.

Type III degenerations of Enriques surfaces inherit involutions from their K3 covers, which in turn act on the dual complexes of Kulikov models and hence on the IAS²s $B(\lambda)$, giving rise to an IAS² with commuting involutions $\iota_{\text{En}}, \iota_{\text{dP}}$. In the standard Euclidean embedding of S^2 into $\mathbf{R}^3$, these involutions act either by $(x, y, z) \mapsto (-x, -y, -z)$, the antipodal map with quotient $\mathbf{RP}^2$, or by $(x, y, z) \mapsto (x, y, -z)$, the “flip” whose quotient is a disc $\mathbf{D}^2$. These cases are distinguished by the fixed loci, and the quotients if the IAS²s $B(\lambda)$ correspond precisely to the dual complexes of Enriques stable degenerations, and thus determine the strata of the $\overline{F_{\text{En},2}}$. We refer to these as IAD² and IARP² respectively.

5.4 dlt Models

Let T be an even lattice of signature $(2, n)$, and let $\eta \in T$ be a primitive isotropic vector. The boundary lattice associated to η is

$$\overline{T}_\eta := \eta^\perp / \langle \eta \rangle$$

which has signature $(1, n-1)$ and admits a root system $\Phi(\overline{T}_\eta)$. For an arithmetic subgroup $\Gamma \leq \text{O}(T)$, the stable boundary group at η is

$$\Gamma_\eta := \text{Stab}_\Gamma(\eta) / U_\eta$$

where U_η is the maximal unipotent normal subgroup of $\text{Stab}_\Gamma(\eta)$ that acts trivially on $\overline{T}_\eta$. The group $W(\Gamma_\eta) \leq \text{O}(\overline{T}_\eta)$ generated by reflections in roots stabilized by Γ_η defines the stable reflection group, and the associated Coxeter chamber $\mathfrak{C}(\Gamma_\eta) \subset \overline{T}_{\eta, \mathbf{R}}$ is determined by the set of simple roots for $W(\Gamma_\eta)$ (see [Ale+23]).

Let $\mathfrak{C}_\eta$ the open positive cone associated to a cusp η . Any rational polyhedral fan $\Sigma(\eta)$ supported on the rational closure $\mathfrak{C}_{\eta, \mathbf{Q}}$ that is Γ_η -invariant determines the structure of the toroidal compactification of $F_\Gamma = D_T / \Gamma$, with the Coxeter fan subdividing $\mathfrak{C}_{\eta, \mathbf{Q}}$ into chambers indexed by $W(\Gamma_\eta)$ [Ale+23].

5.4.1 Boundary Models: Stable Involution Pairs and Quotients

Definition 5.4.1.1 (($K + D$)-Trivial Polarized Involution Pairs). A ($K + D$)-trivial polarized involution pair is a triple (X, D, ι) , where X

is a normal surface, $D \subset X$ a reduced effective divisor, and $\iota : X \rightarrow X$ is an involution satisfying $\iota(D) = D$. The canonical class and divisor are related by $K_X + D \sim 0$, and the fixed locus of ι consists of an ample Cartier divisor R , possibly with isolated fixed points. For all $0 < \varepsilon \ll 1$, the pair $(X, D + \varepsilon R)$ is log canonical. Each irreducible component occurring in stable degenerations of K3 surfaces with nonsymplectic involution is locally modeled by such a triple.

Given such a pair, the finite quotient $\pi : X \rightarrow Y = X/\iota$ has image divisor $C = \pi(D)$ and branch divisor $B = \pi(R) \in |-2(K_Y + C)|$. The pair (Y, C) is a log del Pezzo surface of index at most 2, and $(Y, C + \frac{1+\varepsilon}{2}B)$ is log canonical for $\varepsilon > 0$. There is an exact relation

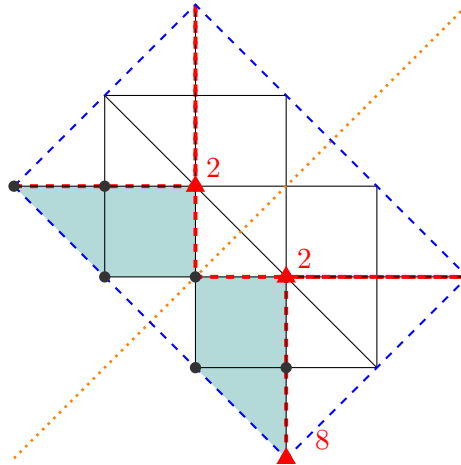
$$K_X + D + \varepsilon R = \pi^* \left(K_Y + C + \frac{1+\varepsilon}{2} B \right).$$

This induces a bijection between $(K + D)$ -trivial involution pairs and such log del Pezzo pairs with branch data. For instance, in the boundary analysis of $F_{\text{En},2}$, the image pairs (Y, C) arising in this way include the rational log del Pezzo surfaces of type A_n , D_n , E_n , and their foldings, as classified by the corresponding diagrams in [Ale+23].

Example 5.4.1.2. The following is an example corresponding to the third cusp η_3 in $F_{\text{En},2}$. A monodromy vector λ satisfying

$$\ell = (2, 0^{15}, 2, 4, 6, 4, 0, 4)$$

specifies gluing the vectors $(3, -3), (2, 2), (-3, 3), (-2, -2)$ in a sequence, forming a rectangular polygon, and then performing Symington surgeries of sizes 1, 2, 3, 2 respectively. The resulting polygon is as follows:



The full polygon $B(\lambda)$ is then obtained by gluing together two copies of this polygon along the blue equator, where a representation of the doubled polygon is depicted in [Ale+23, Fig. 13]. Taking the star at each vertex yields a semitoric surface.

5.5 ADE Surfaces

This section presents the theory of ADE surfaces [AT19], which provides a classification of log canonical log del Pezzo surfaces of index two with ADE boundary configurations. All definitions, constructions, and results in this section are from [AT19] unless otherwise noted.

5.5.1 ADE and $\widetilde{\text{ADE}}$ Surfaces: Combinatorics and Constructions

Let Y be a normal projective surface defined over an algebraically closed field of characteristic zero, and let $C = \sum_i C_i$ be a reduced, effective Cartier divisor on Y , with irreducible components C_i . The pair (Y, C) is called an ADE surface if the dual intersection graph of the C_i is isomorphic, as an abstract graph, to the Dynkin diagram of type A_n , D_n , or E_n in its simply-laced incarnation. This graph records the incidence structure of the boundary: each vertex corresponds to a component C_i , and an edge reflects a simple transverse intersection between C_i and C_j . These are log canonical log del Pezzo surfaces of index two: pairs (Y, C) such that $2K_Y + C$ is Cartier, $-(K_Y + C)$ is ample, and (Y, C) admits strictly log canonical singularities in the sense of the MMP. ADE configurations as boundary graphs result from surface birational geometry and log canonical threshold constraints, which restrict admissible degenerations and underlie the KSBA compactification for surfaces of general type [AT19, §3].

Pure ADE surfaces admit canonical toric models. For each simply-laced root system Φ , there exists a toric surface Y_Φ whose Coxeter fan reflects the arrangement of simple roots. The boundary divisor C is supported on certain torus-invariant irreducible divisors such that the dual intersection graph of C coincides with the diagram of Φ . The geometry of (Y_Φ, C) is encoded by the class of a convex lattice polytope $P \subset \mathbf{Z}^2$, with vertices explicitly given according to type: for instance, for type A_{2n-1} , the polytope is the triangle with vertices at $(0, 2)$, $(0, 0)$, $(2n, 0)$, and C equals the sum of the

divisors supported on the two edges meeting at $(0, 2)$. For type D_{2n} , the polytope is a quadrilateral with vertices $(2, 2), (0, 2), (0, 0), (2n - 2, 0)$; the configuration of toric divisors reflects the fork of the D -type diagram. The exceptional types E_6, E_7 , and E_8 correspond to quadrilaterals whose rightmost vertex differentiates the respective diagram [AT19, Table 1]. The polytope P has sides of length one or two as measured in the lattice $N \cong \mathbf{Z}^2$. An edge of length two is a long side (white node); an edge of length one is a short side (black node). Long and short sides determine which priming operations are allowed. Priming [AT19, §3C] blows up (potentially with weight) a smooth point on a boundary component C_j . A blowup at an edge of length two decreases this length to one and is marked by a prime at the corresponding node. Multiple sequential blowups (primings) are encoded by repeated primes or node-circling conventions, and a minus sign denotes a side that has become short. Allowed primings are classified in [AT19, Table 4], with only certain operations preserving normality, log canonicity, and toric structure. Pure shapes have simple undecorated boundary divisors; primed shapes arise from permissible priming sequences. *Affine* ADE surfaces have boundary graphs $\tilde{A}_n$ (cycle of $n + 1$ curves), $\tilde{D}_{2n}$, or $\tilde{E}_{7,8}$. Explicit models assign boundary configurations to toric (or stacky or nontoric) surfaces according to Coxeter fan data, combinatorial encodings of birational modifications, and MMP constraints.

Let Y be a normal projective surface as in Section 1, and C a reduced effective divisor whose dual intersection graph is of ADE type. These surfaces admit a canonical double cover encoding the relationship between boundary and branch data in stable pairs and moduli of surfaces of general type. Suppose (Y, C) is as above and admits additional data consisting of an effective divisor B , called the branch divisor, satisfying the numerical equivalence $B \sim 2K_Y + C$ in the Picard group $\text{Pic}(Y)$. The normalization of the double covering $\pi : X \rightarrow Y$ branched over B gives a normal surface X with $D = \pi_*^{-1}C$ the reduced preimage of C . The involution on X induced by the double cover is nontrivial and fixed by a ramification divisor $R \subset X$, equal to the reduced preimage of B . The canonical divisor satisfies

$$K_X + D + R = \pi^* \left(K_Y + C + \frac{1}{2}B \right),$$

When B is effective, Cartier, and disjoint from C , this gives a semi-log canonical surface pair of index two in the sense of classification theory [AT19, §4, Lemma 4.1].

The conditions on Y , C , and B are: The pair (Y, C) must be log canonical, with B an effective, reduced Cartier divisor whose support does not contain any component of C , and all singularities of Y lying along B must be Du Val singularities (rational double points). This determines the singularities of X and the positivity of canonical bundles.

Let $f : X \rightarrow Y$ be a double cover of a normal surface Y , branched over a reduced effective divisor $C \subset Y$, with $D \subset X$ the reduced branch divisor, i.e., the reduced preimage $D = f^{-1}(C)_{\text{red}}$. The classification of such double covers can be organized according to the possible configurations of the boundary divisor D :

- **Case I:** $D = \emptyset$ (empty boundary). Here $C = \emptyset$, so f is étale and X is an étale double cover of Y . If Y is canonical (e.g., a log Enriques surface or a K3 surface with isolated singularities), then X is canonical as well. In particular, when the étale cover is nontrivial (i.e., Y is an Enriques surface and X is its K3 canonical cover), the involution on X acts freely outside finitely many points; the fixed locus is empty, and the singularities of Y lift to corresponding canonical singularities on X .
- **Case II:** $D = f^{-1}(C)_{\text{red}}$ is a union of one or two smooth (not necessarily connected) elliptic curves. Here, $C \subset Y$ is a union of smooth elliptic curves, and the ramification is transverse and reduced. The pair (X, D) is strictly log canonical: at generic points of D , the pair is log smooth, and at intersection points, at worst strictly log canonical but not log terminal. The boundary reflects elliptic phenomena: both the covering X and the base Y can have log canonical singularities along D and C (respectively), but no worse. This case corresponds to degenerations where the branch locus acquires smooth elliptic components.
- **Case III:** $D = f^{-1}(C)_{\text{red}}$ is a cycle of smooth rational curves, i.e., $D = \cup_{i=1}^r D_i$ with $D_i \simeq \mathbf{P}^1$, and the dual graph of D is a cycle or a fork (a configuration star-shaped at a node with three branches). These configurations arise from branch curves C with normal crossings whose preimages under f arrange as such. The pair (X, D) is strictly log canonical: the discrepancies at

components of D are zero and at singularities of D (nodes), the log canonical threshold is one. Such arrangements correspond to semi-stable or boundary components in KSBA moduli, and the dual graphs of D reflect the intersection types classified in stable limits.

In all cases, the log canonical pair (X, D) is uniquely determined (up to isomorphism) by the branch data (Y, C) , and the configuration of D (empty, elliptic, or rational cycle/fork) determines both the geometric and singularity-theoretic type of the double cover and its place in the compactified moduli space. These are boundary strata in the KSBA or generalized stable pair compactification, determined by admissibility of (Y, C, B) by [AT19]. At a general point of B , if Y is smooth, the covering π yields ordinary double points (type A_1) on the ramified locus of X . More generally, at points of transverse intersection $p \in B \cap C$, the local equation near p can be written as $u^2 = f(x, y)$ where f defines the local analytic germ of B , and the preimage in X typically acquires a singularity of type A_n , reflecting the connection with the minimal resolution data of Du Val singularities.

Along the boundary, the intersection behavior is controlled by the combinatorics of the dual graph. Each component of C meets its neighbors transversely; self-intersection numbers are determined by the toric or decorated polytope but do not introduce further singularities. Ramification along C is precluded by requiring that B does not contain any component of C , so the preimage in X is unramified away from the finite set of intersection points $B \cap C$.

Singularities on Y (and correspondingly on X after normalization) for quadruples (Y, C, B, X) are restricted to ADE (Du Val) types, as local analytic models near the branch locus must yield rational double points. At each singularity of Y on B , the pair $(Y, C + \frac{1}{2}B)$ is log canonical if and only if the singularity of Y is Du Val and the point does not lie on C by [AT19]. All stable limit surfaces in the boundary strata of the KSBA or toroidal compactifications admit uniform description via double cover and divisor data, with explicit local models and combinatorial invariants from the arrangement of (Y, C, B) . These exhaust all arrangements under log canonicity and the positivity constraints on divisors.

If Φ is a simply-laced root system of ADE type, then *folding* passes from Φ to a (typically non-simply-laced) root system by quotienting the Dynkin diagram by a nontrivial automorphism. When the diagram

admits a symmetry of order two, the invariants of the automorphism define a new root system: for example, the outer automorphism of A_{2n-1} induces the root system C_n , and a similar involution on D_{n+1} yields B_n . The resulting diagrams are those of B_n , C_n , and BC_n , characterized by double bonds and merged nodes indicating multiple edge multiplicities in the root lattice.

Folding is depicted diagrammatically by arrows or symmetrization marks on the original diagram indicating when distinct nodes are combined into an orbit when passing to the folded root system. The quotient diagram thus acquires double bonds (for example, linking the unique short root in B_n or C_n to the adjacent node), and repeated nodes are merged.

Given an ADE or affine ADE surface (Y, C) , folding forms the quotient $Y' = Y/\tau$ by a finite group of automorphisms τ preserving the boundary divisor C . In the simplest cases, τ is the involution corresponding to the automorphism of the Dynkin diagram that exchanges symmetric nodes, and the geometric realization may require passing to a stacky (orbifold) quotient in the presence of fixed points.

Folding affects the toric polytope associated to Y : the involution acts as a lattice symmetry, identifying pairs of vertices and edges in a manner determined by the automorphism. The quotient polytope has fewer vertices and may possess edges of length one or two, mirroring the root length distinction in BC -type systems. In the divisor data, the new boundary C' is the quotient of the original C , and its components may now carry multiplicities, non-reduced structure, or even half-integral coefficients due to the branching behavior of the cover.

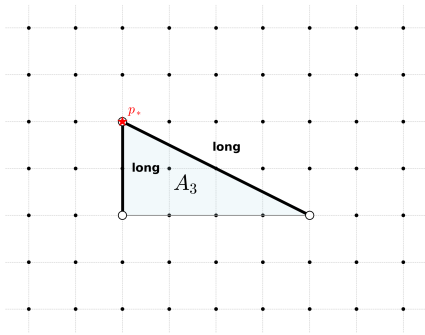
The boundary C' may consist of components of multiplicity two (resulting from merging distinct branches) or of non-reduced components. In the most general constructions, the quotient may possess reduced, non-reduced, or fractional boundary divisors, with the latter arising whenever the branch locus of the involution is ramified over the original boundary or when the quotient fails to be Cartier in codimension one. Such half-integral divisorial data play a crucial role in moduli and are absorbed into the theory of stable pairs by considering pairs with boundary coefficients in $\frac{1}{2}\mathbf{Z}$, as required by the $\mathbf{Q}$ -Gorenstein property of the singularities and divisor configuration.

If $\pi : Y \rightarrow Y'$ is the quotient morphism and C is the boundary on Y , then the pushforward $C' = \pi_*C$ may fail to be reduced, and the canonical divisor formula reflects half-divisors. Specifically, the

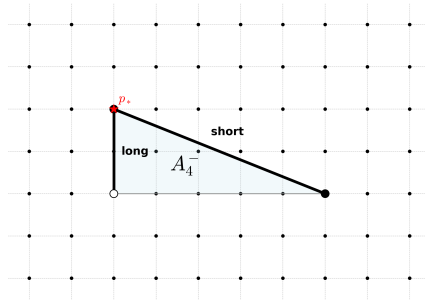
boundary on the quotient is typically expressible in the form $C' = \sum m_i C'_i$, where $m_i \in \{1, 2, \frac{1}{2}\}$ depending on the local ramification. The canonical double cover inevitably restores the ADE type: the pullback π^*C' decomposes into integral divisors on the cover, and the map is branched exactly along the non-reduced or half-integral components. The divisor relation under folding is

$$K_X + D + R = \pi^*(K_Y + C + \tfrac{1}{2}B),$$

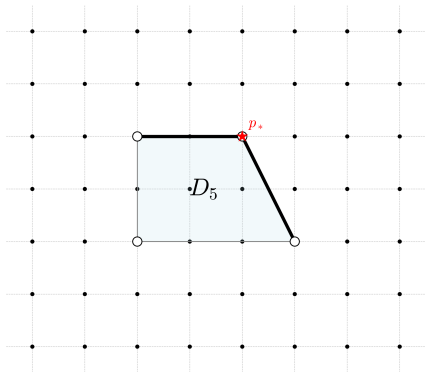
where K_X and K_Y denote the canonical class on the cover and base, D is the pullback of the boundary, R is the ramification divisor, and B is the branch divisor as described in [AT19]. In various degenerations – such as those of Enriques surfaces, realized via K3 surfaces with involution, or of log del Pezzo surfaces of index two – the boundary components are precisely the surfaces associated with ADE or folded BC -type Dynkin diagrams, and passing to a double cover eliminates half-integral phenomena: it yields a normalization of the associated quotient surface in which the boundary becomes reduced and Cartier. We give below several examples of the polytopes for such surfaces, indicating which sides are *long* and *short*. In particular, folded BC -type surfaces appear as irreducible components in the boundary of $\overline{F_{\text{En},2}}$.



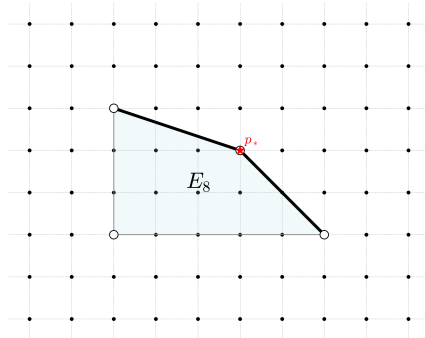
The A_3 surface.



The A_4^- surface.



The D_5 surface.



The E_8 surface.

Part III

The Main Theorem

CHAPTER 6

STATEMENT AND OUTLINE

This chapter is devoted to establishing the isomorphism $\overline{F_{\text{En},2}}^{\mathcal{F}_\bullet} \xrightarrow{\sim} \overline{F_{\text{En},2}}$:

Theorem 6.0.0.1 (Main Theorem [Ale+23, Theorem 1.1, Thm. 5.9]). *Let $\overline{F_{\text{En},2}}$ denote the KSBA compactification of the moduli space $F_{\text{En},2}$ of numerically polarized Enriques surfaces of degree 2, and let $\mathcal{F}_\bullet = \{\mathcal{F}_k\}_{k=1,2,3,4,5}$ be the explicit collection of semifans, one for each 0-cusp of $\overline{F_{\text{En},2}}^{\text{BB}}$. Then:*

1. *The normalization $\overline{F_{\text{En},2}}$ is isomorphic to the semitoroidal compactification $\overline{F_{\text{En},2}}^{\mathcal{F}_\bullet}$ for these semifans,*
2. *This compactification is toroidal over the 0-cusps 2 and 4, their adjacent 1-cusps, and cusp 35, and strictly semitoroidal over all other cusps.*
3. *There exists a toroidal compactification $\overline{F_{\text{En},2}}^{\Sigma_\bullet^{\text{Cox}}}$ associated to the collection of Coxeter fans $\Sigma^{\text{Cox}} = \{\Sigma_k^{\text{Cox}}\}_{k=1,2,3,4,5}$, such that there is a chain of morphisms*

$$\overline{F_{\text{En},2}}^{\Sigma_\bullet^{\text{Cox}}} \rightarrow \overline{F_{\text{En},2}}^{\mathcal{F}_\bullet} \rightarrow \overline{F_{\text{En},2}}$$

where the first morphism contracts cones corresponding to faces defined by irrelevant roots (see [Ale+23, §5.2]), and the second morphism is the normalization.

We summarize this entire situation in the following diagram:

$$\begin{array}{ccccc}
\overline{F_{\text{En},2}}^{\Sigma^{\text{Cox}}_{\bullet}} & & & & \overline{F_{(2,2,0)}}^{\Sigma^{\text{Cox}}_{\bullet}} \\
\downarrow & & & & \downarrow \\
\overline{F_{\text{En},2}}^{\mathcal{F}_{\bullet}} & \xlongequal{\quad := \quad} & \overline{B}^{\mathcal{F}_{\bullet}} & \hookrightarrow & \overline{F_{(2,2,0)}}^{\mathcal{F}_{\bullet}} \\
\Downarrow & & \Downarrow & & \Downarrow \\
\overline{F_{\text{En},2}}^{\nu} & \xrightarrow{\quad \exists \tilde{\Phi} \quad} & B^{\nu} & \hookrightarrow & \overline{F_{(2,2,0)}}^{\nu} \\
\downarrow \nu & \swarrow \Phi & \downarrow \nu & & \downarrow \nu \\
\overline{F_{\text{En},2}} & & B := \overline{j(F_{\text{En},2})} & \hookrightarrow & \overline{F_{(2,2,0)}} \\
\downarrow & & \uparrow & & \downarrow \\
F_{\text{En},2} & \xrightarrow{\quad j \quad} & & & F_{(2,2,0)} \\
& \searrow & \curvearrowright & \nearrow & \\
& & j(F_{\text{En},2}) & & x
\end{array}$$

We define the embedding j , restrict to its image, take the Zariski closure, and then normalize to produce B^η . Functoriality of normalization and the KSBA compactification gives an embedding of B^η into $\overline{F_{(2,2,0)}}^\nu$, which is identified with a specific semitoroidal compactification $\overline{F_{(2,2,0)}}^{\mathcal{F}_{\bullet}}$ for the ramification divisor. We pull back that semifan data to obtain an identification of B^ν with a semitoroidal compactification $\overline{B}^{\mathcal{F}_{\bullet}}$ by restriction, which we define to be $\overline{F_{\text{En},2}}^{\mathcal{F}_{\bullet}}$. We then construct a morphism $\Phi : B^\nu \rightarrow \overline{F_{\text{En},2}}$ by constructing a universal family, lifting it to an isomorphism to $\overline{F_{\text{En},2}}^\nu$ using the universal property of normalization, and obtain an isomorphism $B^\nu \cong \overline{F_{\text{En},2}}^\nu$ by Zariski's main theorem. Thus B^ν is used as an intermediate space which is isomorphic to both $\overline{F_{\text{En},2}}^{\mathcal{F}_{\bullet}}$ and $\overline{F_{\text{En},2}}^\nu$. The proof proceeds in the following steps:

1. Defining $F_{\text{En},2}$ and $\overline{F_{\text{En},2}}$:

The moduli stack $F_{\text{En},2}$ is defined to parameterize numerically polarized Enriques surfaces of degree two and their KSBA stable limits. The compactification $\overline{F_{\text{En},2}}$ is constructed using the theory of stable surface pairs with boundary divisor weighted by $\varepsilon \ll 1$. Each point corresponds to a pair $(Z, \varepsilon R_Z)$ where $K_Z + \varepsilon R_Z$ is ample and the total pair is semi-log-canonical.

2. $F_{\text{En},2}$ as a Noether–Lefschetz locus inside $F_{(2,2,0)}$:

By section 2.5.6, there is an immersion of $F_{\text{En},2}$ into the moduli stack $F_{(2,2,0)}$ of K3 surfaces with a del Pezzo involution ι_{dP} of type $(2,2,0)$. The image consists of K3 surfaces admitting an additional fixed-point-free involution ι_{En} , and is cut out by period-theoretic conditions. Its Zariski closure $B \subset \overline{F_{(2,2,0)}}$ in the KSBA compactification is the Noether–Lefschetz locus parameterizing K3 surfaces X whose quotients $Z := X/\iota_{\text{En}}$ define Enriques surfaces.

3. Extension of ι_{En} to the universal family of K3 surfaces over B inherited from $F_{(2,2,0)}$:

The moduli stack $\overline{F_{(2,2,0)}}$ admits a universal family $(\mathcal{X}, \varepsilon\mathcal{R})$, where $\mathcal{R}$ is the normalized ramification divisor of ι_{dP} . This family pulls back to a family over B whose general fiber admits an Enriques involution ι_{En} commuting with ι_{dP} . By the uniqueness of KSBA limits, ι_{En} extends over the total space.

4. Quotient to produce a family of stable Enriques pairs over B :

After extending ι_{En} , the quotient $(\mathcal{X}_B, \varepsilon\mathcal{R}_B)/\iota_{\text{En}}$ defines a family of stable Enriques pairs $(\mathcal{Z}, \varepsilon\mathcal{R}_Z) \rightarrow B$, extending the universal family over the interior. The quotient preserves KSBA stability and compatible degeneration data.

5. Pass to B^ν and descend the semifans from $\overline{F_{(2,2,0)}}$:

The normalization $B^\nu \rightarrow B$ is a proper, normal stack supporting the pullback of the universal family. The semifans associated to $\overline{F_{\text{En},2}}$, governed by the ramification semifan, restricts to induce a semifan on B^ν . The resulting combinatorics define a collection of folded semifans $\mathcal{F} = \{\mathcal{F}_k\}$ determining the boundary stratification.

6. Show finiteness of the classifying map using the geometry of Kulikov models of K3 surfaces:

The classifying morphism $\phi : B^\nu \rightarrow \overline{F_{\text{En},2}}$ is birational. To prove finiteness, one must exclude the existence of distinct K3 pairs giving the same Enriques quotient. This is achieved via explicit analysis of degenerations (see Sections 4 and 5): the Enriques quotient of a K3 degeneration cannot have a more singular $\mathcal{X}_0$ than its cover. Any strict coarsening of the semifans would contradict this.

7. Zariski's Main Theorem and identification of compactifications:

Since ϕ is finite and birational, it is an isomorphism. The normalization $\overline{F_{\text{En},2}}$ is thus identified with the semitoroidal compactification $\overline{F_{\text{En},2}}^{\mathcal{F}\bullet}$, as constructed via the restricted and folded ramification semifans for $F_{(2,2,0)}$.

6.1 Defining $F_{\text{En},2}$ and $\overline{F_{\text{En},2}}$

Definition 6.1.0.1 (Moduli Stack $F_{\text{En},2}$ of Numerically Polarized Enriques Surfaces of Degree Two). Let $F_{\text{En},2}$ be the Deligne–Mumford stack over $\mathbf{C}$ parameterizing isomorphism classes of pairs (Z, L) , where

- Z is a smooth Enriques surface,
- $L \in \text{Pic}(Z)$ is a nef and big divisor class satisfying

$$L^2 = 2, \quad L \cdot C > 0 \text{ for every } (-2)\text{-curve } C \subset Z.$$

Two pairs (Z, L) and (Z', L') are isomorphic if there exists an isomorphism $\varphi : Z \xrightarrow{\sim} Z'$ such that $\varphi^*L' \equiv L$. The numerical equivalence represents the action of the automorphism group of Z and preserves the polarization class up to numerical type.

Definition 6.1.0.2 (KSBA Compactification $\overline{F_{\text{En},2}}$ of $F_{\text{En},2}$). Fix a rational $0 < \varepsilon \ll 1$. The KSBA compactification $\overline{F_{\text{En},2}}$ is defined as the Deligne–Mumford stack whose objects over $\mathbf{C}$ -schemes are isomorphism classes of pairs $(Z, \varepsilon R_Z)$, satisfying:

- Z is a projective surface with numerically trivial canonical divisor, $K_Z \equiv 0$,
- R_Z is an effective $\mathbf{Q}$ -Cartier divisor such that the pair $(Z, \varepsilon R_Z)$ is semi-log-canonical (slc),
- The divisor $K_Z + \varepsilon R_Z$ is ample,
- On the open locus where Z is smooth, R_Z is numerically equivalent to $2L$ for some $(Z, L) \in F_{\text{En},2}$.

The open substack consisting of pairs with smooth Z and R_Z the branch divisor of the associated K3 double cover is canonically isomorphic to $F_{\text{En},2}$.

Theorem 6.1.0.3 (Basic Properties of $\overline{F_{\text{En},2}}$). *The stack $\overline{F_{\text{En},2}}$ is a proper, separated, normal Deligne–Mumford stack of finite type over $\mathbf{C}$. The inclusion $F_{\text{En},2} \hookrightarrow \overline{F_{\text{En},2}}$ realizes $F_{\text{En},2}$ as the maximal open substack whose geometric points correspond to smooth numerically polarized Enriques surfaces of degree two. Each one-parameter family in $F_{\text{En},2}$ admits a unique extension, up to unique isomorphism, to $\overline{F_{\text{En},2}}$. That is, the KSBA compactification provides unique stable limits for degenerating families.*

Pairs (Z, L) as above parameterize the interior of the moduli problem, encoding the data of smooth numerically polarized Enriques surfaces of degree two. The KSBA compactification $\overline{F_{\text{En},2}}$ systematically includes stable degenerations: objects $(Z, \varepsilon R_Z)$ where Z may acquire slc singularities and R_Z encodes the polarization determined by the ramification divisor of ι_{dP} on the K3 cover. We will develop further refinements of this construction – such as the explicit realization of $F_{\text{En},2}$ and its compactification as a Noether–Lefschetz locus in $F_{(2,2,0)}$, and the construction of the semitoroidal boundary.

6.2 $F_{\text{En},2}$ as a Noether–Lefschetz Locus in $F_{(2,2,0)}$

In this section, we establish that the moduli space $F_{\text{En},2}$ of degree-2 polarized Enriques surfaces is realized as a locally closed, irreducible sublocus of a Noether–Lefschetz locus inside $F_{(2,2,0)}$, the moduli space of degree-4 polarized K3 surfaces with a nonsymplectic involution of fixed type. Recalling that if $(Z, [L])$ is a smooth Enriques surface equipped with a numerical class of an ample line bundle L having self-intersection $L^2 = 2$. The canonical étale double cover $\pi : X \rightarrow Z$ is uniquely determined up to isomorphism. The deck transformations of X/Z are generated by a fixed-point-free involution $\iota_{\text{En}} : X \rightarrow X$ – the **Enriques involution**. The class $H := \pi^*L$ is a nef and big divisor on X with $H^2 = 4$. There is a unique involution $\iota_{\text{dP}} : X \rightarrow X$ distinct from ι_{En} , commuting with ι_{En} and fixing H . According to Nikulin’s classification, ι_{dP} is a nonsymplectic involution of type $(2, 2, 0)$.

Definition 6.2.0.1 (The Canonical Map j). The association

$$(Z, L) \mapsto (X, H = \pi^*L, \iota_{\text{dP}})$$

defines a functorial morphism

$$j : F_{\text{En},2} \rightarrow F_{(2,2,0)},$$

where $F_{\text{En},2}$ is the moduli of degree-2 polarized Enriques surfaces and $F_{(2,2,0)}$ the moduli of degree-4 polarized K3 surfaces with a nonsymplectic involution of type $(2, 2, 0)$. This construction is compatible with base change and extends to families.

Definition 6.2.0.2 (Noether–Lefschetz Locus $\text{NL}_{T_{\text{En}}}$). Let $\Lambda = H^2(X, \mathbf{Z}) \cong \Pi_{3,19}$, and let $T_{\text{En}} = U(2) \oplus E_8(-2)$ denote the anti-invariant lattice under the Enriques involution, reflecting both the required polarization and involution structure. Define the **Noether–Lefschetz locus**

$$\text{NL}_{T_{\text{En}}} = \left\{ [X] \in F_{(2,2,0)} \mid T_{\text{En}} \hookrightarrow \text{Pic}(X), [\iota_{\text{dP}}, \iota_{\text{En}}] = 0 \text{ for some } \iota_{\text{En}} \right\},$$

where we require the embedding to be primitive. This locus parameterize polarized quartic hyperelliptic K3 surfaces (X, H) with involutions so that $(X, H, \iota_{\text{dP}}, \iota_{\text{En}})$ descends to data of a (possibly degenerate) degree-2 polarized Enriques surface.

Proposition 6.2.0.3 (Lattice-Theoretic Description of the Image of j). *The image $j(F_{\text{En},2})$ consists precisely of those $(X, H, \iota_{\text{dP}}) \in F_{(2,2,0)}$ for which:*

- X admits a commuting involution ι_{En} ,
- ι_{En} is fixed-point-free,
- The anti-invariant lattice is $\text{Pic}(X)^{\iota_{\text{En}}} = T_{\text{En}}$.

Equivalently,

$$j(F_{\text{En},2}) = \left\{ (X, H, \iota_{\text{dP}}) \in F_{(2,2,0)} \mid \exists \iota_{\text{En}}, [\iota_{\text{En}}, \iota_{\text{dP}}] = 0, \text{Pic}(X)^{\iota_{\text{En}}=-1} \simeq T_{\text{En}} \right\}.$$

Proof. For $X = \pi^{-1}(Z)$ as above, the involution ι_{En} is fixed-point-free, ι_{dP} arises canonically from the polarization, and the anti-invariant lattice is explicitly computed as T_{En} using the cohomology of double covers and the configuration of the involutions. The presence of these two commuting involutions together with the polarization forces the Picard lattice to contain T_{En} primitively. Conversely, any $(X, H, \iota_{\text{dP}})$ with such a primitive embedding and compatible involutions arises

as the canonical cover of some Enriques surface with polarization of degree 2.

□

Proposition 6.2.0.4 (Period Compatibility and Linear Subdomain).

Let S_{dP} be the invariant lattice of ι_{dP} and S_{En} that of ι_{En} . Then

$$T_{\text{dP}} \oplus S_{\text{dP}}, T_{\text{En}} \oplus S_{\text{En}} \hookrightarrow H^2(X, \mathbf{Z}),$$

with $S_{\text{dP}} \cong U(2)$ and $S_{\text{En}} \cong U(2) \oplus E_8(2)$. For any $(Z, L) \in F_{\text{En},2}$, the period point for X lies in the orthogonal complement of S_{En} in the K3 period domain. Thus, the image $j(F_{\text{En},2})$ is described as the subset of the moduli space where the periods are orthogonal to the fixed lattice S_{En} .

Proof. This follows directly from the construction: the double cover, involutions, and polarization together prescribe the lattice embedding up to equivalence, forcing the period to be in the locus cut out by the orthogonality to S_{En} .

□

Definition 6.2.0.5 (Zariski Closure B). Define

$$B := \overline{j(F_{\text{En},2})} \subset \overline{F_{(2,2,0)}}$$

as the Zariski closure of $j(F_{\text{En},2})$ in the KSBA compactification. This B forms a closed Noether–Lefschetz locus, parameterizing all limits (in the KSBA sense) of K3 surfaces that are double covers of Enriques surfaces, possibly with semi-log canonical singularities and compatible involution structure.

Proposition 6.2.0.6 (Realization and Dimension). $F_{\text{En},2}$ is realized as a locally closed, irreducible component of the Noether–Lefschetz locus $\text{NL}_{T_{\text{En}}} \subset F_{(2,2,0)}$. The Zariski closure B in $\overline{F_{(2,2,0)}}$ parametrizes exactly all (possibly degenerate) degree-4 K3 surfaces (X, H) with a compatible pair of involutions $(\iota_{\text{En}}, \iota_{\text{dP}})$ arising from a numerically polarized Enriques surface (or its stable degeneration).

Proof. By the arguments above, the construction j embeds $F_{\text{En},2}$ into $\text{NL}_{T_{\text{En}}}$ defined by the period and lattice conditions. Local closedness follows because the period subdomain is defined by a linear orthogonality condition. The irreducibility and dimension count come from

the identification of the period domain for Enriques surfaces as a 10-dimensional linear section inside the ambient 20-dimensional K3 period domain by the global Torelli theorem.

□

Both B and $F_{\text{En},2}$ are 10-dimensional, as is determined by their period domains. The period domain for numerically degree-2 polarized Enriques surfaces is a linear subdomain (cut out by the orthogonality to S_{En}) of the 20-dimensional period domain for degree-4 K3s with involution, and its dimension is 10 by classical results. Every Enriques surface in $F_{\text{En},2}$ and every KSBA degeneration thereof in B is canonically covered by a K3 surface X (possibly degenerate). The full modular datum of the point corresponding to Z , including its possible degenerations, is encoded by the triple $(X, \iota_{\text{dP}}, \iota_{\text{En}})$: the K3 cover X and its two commuting involutions.

Proposition 6.2.0.7 (Compatibility of Period Map). *For any $(Z, L) \in F_{\text{En},2}$, the Hodge structure on its canonical K3 cover X lies in the orthogonal complement of S_{En} in the K3 period domain. The period map for $F_{\text{En},2}$ factors through the period map for $F_{(2,2,0)}$ as a linear subspace cut out by the linear condition of being orthogonal to S_{En} .*

Proof. The characterization of the period domain for Enriques surfaces (with fixed degree-2 polarization) as a linear section of the K3 period domain follows immediately from the splitting of the full lattice into invariant and anti-invariant parts for the involutions, and the polarization data forces a unique class in the orthogonal complement.

□

6.3 Extension of ι_{En} to the Universal Family over B

In this section, we construct and analyze the global extension of the Enriques involution ι_{En} to the universal KSBA family of K3 surfaces over the Noether–Lefschetz locus $B \subset \overline{F_{(2,2,0)}}$. Unless stated otherwise, all notation, functors, and structures are as established in the preceding sections.

6.3.1 The Universal Family in the KSBA Context

Let $\overline{F}_{(2,2,0)}$ be the KSBA compactification of the moduli space of degree-4 polarized K3 surfaces with a specified nonsymplectic involution ι_{dP} . There exists a universal family of stable pairs

$$\pi : (\mathcal{X}, \epsilon\mathcal{R}) \rightarrow \overline{F}_{(2,2,0)}$$

satisfying:

- The total space $\mathcal{X}$ is flat and proper over $\overline{F}_{(2,2,0)}$, specializing to smooth K3 surfaces over a dense open locus and possibly degenerating to semi-log-canonical surfaces in the boundary.
- The divisor $\mathcal{R} \subset \mathcal{X}$ is the ramification locus of ι_{dP} , defined as the fixed locus of the involution.
- The pair structure is preserved globally, including at all singular fibers.

Proposition 6.3.1.1 (Existence and Uniqueness of the Universal Family with Involution). *The existence, functoriality, and separatedness of the universal family $(\mathcal{X}, \epsilon\mathcal{R})$, including a prescribed pair of commuting involutions, are standard consequences of the general theory of KSBA moduli of stable surface pairs with finite automorphism group action acting fiberwise and preserving the pair. Explicitly, the automorphism scheme is proper and separated, and such an involution preserving the pair extends uniquely to the stable limit in families.*

6.3.2 Restriction to the Noether–Lefschetz Locus

Let $B \subset \overline{F}_{(2,2,0)}$ be the Zariski closure of the Noether-Lefschetz locus parameterizing degree 2 polarized Enriques surfaces constructed in the previous step. Consider the base change

$$\pi_B : (\mathcal{X}_B, \epsilon\mathcal{R}_B) \rightarrow B$$

where $\mathcal{X}_B = \mathcal{X} \times_{\overline{F}_{(2,2,0)}} B$ and $\mathcal{R}_B = \mathcal{R}|_{\mathcal{X}_B}$. This family retains the following structure:

- Each geometric point $b \in B$ parametrizes a (possibly degenerate) K3 surface equipped with a degree-4 polarization and a prescribed involution $(\iota_{\text{dP}})_b$.

- Over the open locus B° parameterizing smooth K3 surfaces, each fiber $\mathcal{X}_b$ admits a del Pezzo involution $(\iota_{\text{dP}})_b$ with ramification divisor $\mathcal{R}_b$, as well as a fixed-point-free Enriques involution $(\iota_{\text{En}})_b$.

For all $b \in B^\circ$, the involutions $(\iota_{\text{dP}})_b$ and $(\iota_{\text{En}})_b$ commute and both preserve the polarized pair $(\mathcal{X}_b, \mathcal{R}_b)$. These involutions are compatible with the moduli-theoretic structure and cover the appropriate automorphism data underlying the Enriques construction. We now globalize the involution ι_{En} over the entire family $(\mathcal{X}_B, \mathcal{R}_B)$, including all degenerate fibers.

Theorem 6.3.2.1 (Global Extension of the Enriques Involution). *There exists a unique global involution*

$$\iota_{\text{En}} : \mathcal{X}_B \rightarrow \mathcal{X}_B$$

with the following properties:

- ι_{En} restricts over B° to the fixed-point-free Enriques involution on each smooth fiber.
- $\iota_{\text{En}} \circ \iota_{\text{dP}} = \iota_{\text{dP}} \circ \iota_{\text{En}}$ fiberwise on all of $\mathcal{X}_B$.
- $\iota_{\text{En}}(\mathcal{R}_B) = \mathcal{R}_B$.
- For every $b \in B$, the restriction $(\iota_{\text{En}})_b$ is fixed-point-free on the smooth locus of the fiber $\mathcal{X}_b$.

Proof. Over B° the involution ι_{En} is part of the moduli data. Since the universal family in the KSBA moduli problem is separated and proper for automorphism group actions preserving the pair structure, any automorphism defined on the $\mathcal{X}_t$ extends uniquely to the $\mathcal{X}_0$ s, as in the valuative criterion for the separatedness of the moduli functor. The extension respects all divisor and commutativity data because these are closed conditions on the moduli stack of stable pairs with automorphism. Functoriality and the rigidity of automorphism schemes in the KSBA theory propagate these compatibilities to all fibers. $\square$

Once constructed, this global involution ι_{En} satisfies all required compatibilities:

- Commutativity $\iota_{\text{En}} \circ \iota_{\text{dP}} = \iota_{\text{dP}} \circ \iota_{\text{En}}$ holds everywhere. This is checked on B° and then globalized via the uniqueness of the extension.
- Preservation of the ramification divisor: $\iota_{\text{En}}(\mathcal{R}_B) = \mathcal{R}_B$. Again, this is a closed condition propagated by the structure of the moduli functor.
- On any smooth fiber, $(\iota_{\text{En}})_b$ is fixed-point-free, as dictated by the monodromy and period data of the Enriques surface. In degenerate fibers, fixed points may occur in higher codimension, but the geometric meaning of the involution is unambiguous due to the rigidity of the automorphism structure and the stability properties of the KSBA compactification.

For clarity in all subsequent arguments, we adopt the following notation:

- $(\mathcal{X}, \mathcal{R})$ is the universal KSBA family over $\overline{F}_{(2,2,0)}$;
- $(\mathcal{X}_B, \mathcal{R}_B)$ is the restriction of the family to the Noether–Lefschetz locus B ;
- ι_{dP} is the global del Pezzo involution acting on each fiber;
- ι_{En} is the global Enriques involution, commuting with ι_{dP} and preserving the pair;
- For each geometric point $b \in B$, the fiber $\mathcal{X}_b$ is equipped with involutions $(\iota_{\text{dP}})_b, (\iota_{\text{En}})_b$, and the ramification divisor is $\mathcal{R}_b$.

6.4 The Quotient Family over B

This section constructs, analyzes, and establishes the modular properties of the family of stable Enriques pairs obtained as the quotient of the universal KSBA-stable family of K3 pairs by the global Enriques involution, particularly over the Noether–Lefschetz locus B .

Proposition 6.4.0.1 (Existence and Finiteness of the Quotient Family). *Let $\iota_{\text{En}} : \mathcal{X}_B \rightarrow \mathcal{X}_B$ denote the globally defined Enriques involution acting fiberwise on the universal KSBA family $(\mathcal{X}_B, \varepsilon\mathcal{R}_B) \rightarrow B$. Define the quotient:*

$$\rho : (\mathcal{X}_B, \varepsilon\mathcal{R}_B) \rightarrow (\mathcal{Z}, \varepsilon\mathcal{R}_Z),$$

where $\mathcal{Z} = \mathcal{X}_B / \iota_{\text{En}}$ is the coarse space of the quotient stack $[\mathcal{X}_B / \langle \iota_{\text{En}} \rangle]$ and $\mathcal{R}_Z := \rho_*(\mathcal{R}_B)$ is the pushforward of the ramification divisor. Then:

- ρ is a morphism of degree 2 in the category of pairs over B ,
- The following square is Cartesian:

$$\begin{array}{ccc} (\mathcal{X}_B, \varepsilon \mathcal{R}_B) & \xrightarrow{\rho} & (\mathcal{Z}, \varepsilon \mathcal{R}_Z) \\ \downarrow & & \downarrow \\ B & \xlongequal{\quad} & B \end{array}$$

- The construction is well-defined: ι_{En} acts biregularly, freely in codimension 1, and preserves both the boundary divisor and the property of being a KSBA stable pair.

Proof. The quotient $\mathcal{Z}$ is constructed as the stack quotient $[\mathcal{X}_B / \langle \iota_{\text{En}} \rangle]$. Since ι_{En} is biregular and, by construction, acts freely on the smooth locus of each fiber and preserves the divisor $\mathcal{R}_B$, descent for group actions on surfaces with semi-log-canonical singularities applies. The finite morphism property, and the pullback structure on boundary divisors, follow from the theory of quotients of pairs and functoriality on the category of KSBA pairs with finite group actions. The claimed Cartesian property is a formal consequence. $\square$

Proposition 6.4.0.2 (Fiberwise Analysis of the Quotient). *Let $b \in B$ be a geometric point and $\mathcal{X}_b$, $\mathcal{Z}_b$ the corresponding fibers.*

1. *If $b \in B^\circ$ (open locus), $\mathcal{X}_b$ is a smooth K3 surface, ι_{En} has no fixed points, and $\rho_b : \mathcal{X}_b \rightarrow \mathcal{Z}_b$ is étale outside codimension at least 2.*
 - The quotient $\mathcal{Z}_b$ is a smooth Enriques surface.
 - The canonical bundle $K_{\mathcal{Z}_b}$ is numerically trivial up to 2-torsion: $2K_{\mathcal{Z}_b} \sim 0$.
2. *If $b \in B \setminus B^\circ$ is in the degenerate locus, $\mathcal{X}_b$ is a K3 surface with slc singularities and ι_{En} may have isolated fixed points in codimension ≥ 2 .*

- The quotient $\mathcal{Z}_b$ is again semi-log-canonical, as is the pair $(\mathcal{Z}_b, \varepsilon\mathcal{R}_{\mathcal{Z}_b})$.
- The canonical bundle $K_{\mathcal{Z}_b}$ remains numerically trivial.

Proof. In the smooth case, this is the standard construction of Enriques surfaces as fixed-point-free quotients of K3 surfaces by involution. In the presence of singularities, since ι_{En} is biregular and fixes only loci of codimension at least 2, the quotient remains slc by [KM98b]. The canonical bundle calculation follows from the adjunction formula and the behavior of ι_{En} on $K_{\mathcal{X}_b} \sim 0$; the pushforward identifies $K_{\mathcal{X}_b}$ with $\rho^*K_{\mathcal{Z}_b}$ so that $2K_{\mathcal{Z}_b} \sim 0$. $\square$

Proposition 6.4.0.3 (Preservation of KSBA Stability and Ample Boundary). *For every $b \in B$, consider the pair $(\mathcal{Z}_b, \varepsilon\mathcal{R}_{\mathcal{Z}_b})$. Then:*

- $(\mathcal{Z}_b, \varepsilon\mathcal{R}_{\mathcal{Z}_b})$ is KSBA-stable,
- The polarization (ample boundary) descends: $\rho^*(K_{\mathcal{Z}_b} + \varepsilon\mathcal{R}_{\mathcal{Z}_b}) = K_{\mathcal{X}_b} + \varepsilon\mathcal{R}_b$.

Proof. Since $(\mathcal{X}_b, \varepsilon\mathcal{R}_b)$ is KSBA-stable by construction, we use that ampleness is preserved under finite morphism: if L is ample and $f : X \rightarrow Y$ finite surjective, then f^*L ample $\implies L$ ample ([Hartshorne, III.Ex.5.7]). The quotient construction is functorial in the category of pairs, so all stability, numerical, and singularity conditions are satisfied in the target $(\mathcal{Z}, \varepsilon\mathcal{R}_{\mathcal{Z}})$. $\square$

Corollary 6.4.0.4 (Degenerations, Flatness, Dual Complex, and Monodromy). *Let Δ be a smooth curve with generic point η and special point 0, and $f : \Delta \rightarrow B$ a morphism. The base-changed family $\mathcal{X}_\Delta = \mathcal{X}_B \times_B \Delta$ carries a fiberwise involution ι_{En} and forms a family of KSBA-stable K3 pairs. The quotient family $\mathcal{Z}_\Delta = \mathcal{X}_\Delta / \iota_{\text{En}}$ satisfies:*

- The $\mathcal{X}_t$ is a smooth Enriques surface,
- The $\mathcal{X}_0$ is semi-log-canonical,
- The family $(\mathcal{Z}_\Delta, \mathcal{R}_{\mathcal{Z}_\Delta}) \rightarrow \Delta$ is flat, by flatness of the quotient and base change,

- The dual complex of $\mathcal{Z}_0$ is the quotient of the dual complex of $\mathcal{X}_0$ by the involution ι_{En} .

Monodromy operators on the local system $H^2(\mathcal{X}_\eta, \mathbf{Z})$ commuting with ι_{En}^* descend to $H^2(\mathcal{Z}_\eta, \mathbf{Z})$. Period data, lattice-theoretic structures, and cohomological invariants remain compatible under the quotient.

Proof. Flatness follows from the finiteness and flatness properties of group quotients on flat families. That semi-log-canonicity is preserved in $\mathcal{X}_0$ follows as above. The dual complex statement follows from the equivariant semistable reduction theory and the definition of the dual complex as depending only on the combinatorics of components and double curves modulo group action. Monodromy compatibility is a standard consequence of functoriality of Galois covers and representations. □

Remark 6.4.0.5 (Summary and Structural Consequences). The quotient family $(\mathcal{Z}, \varepsilon\mathcal{R}_Z) \rightarrow B$ constructed thus provides a complete, modular KSBA-stable compactification for the moduli space of degree-2 polarized stable Enriques surfaces as quotients of K3 pairs, with all necessary boundary, polarization, and singularity structures explicitly accounted for.

6.5 Descent of Semitoroidal Data

In order to construct a modular compactification of the moduli space of stable Enriques pairs – with explicit control of the boundary – it is necessary to analyze the singularities along the boundary of the Noether–Lefschetz locus $B \subset \overline{F_{(2,2,0)}}$ and to pass to the normalization. This normalization process allows the descent of the ramification semifans from $F_{(2,2,0)}$ to $F_{\text{En},2}$, which defines $\overline{F_{\text{En}}}^{\mathcal{F}\bullet}$.

Proposition 6.5.0.1 (Normalization and Pullback Family). *Let $\nu : B^\nu \rightarrow B$ be the normalization of B . Then:*

- B^ν is a normal projective variety,
- The morphism ν is finite, surjective, birational, and proper,
- The pullback family $\nu^*(\mathcal{Z}, \varepsilon\mathcal{R}_Z)$ is a family of KSBA-stable Enriques pairs of degree two over the normal base B^ν .

Proof. By construction, B is a closed subvariety of the projective stack $\overline{F}_{(2,2,0)}$; thus, the integral closure is finite, and normality follows. The KSBA stability of fibers is preserved under finite base change, giving the desired family. $\square$

Proposition 6.5.0.2 (Non-normality of the Noether–Lefschetz Locus). *Let $B \subset \mathcal{M}$ be a Noether–Lefschetz locus with compactification $\overline{B} \subset \overline{\mathcal{M}}$, and let $\Delta = \overline{\mathcal{M}} \setminus \mathcal{M}$ denote the boundary divisor. The failure of normality of $\overline{B}$ along Δ arises through three mechanisms:*

- (1) *There exist boundary divisors $D \subset \Delta$ such that the period map $\mathbf{P} : \mathcal{X} \rightarrow \mathcal{M}$ is not unramified over generic points of D ; precisely, there exist points $x \in D$ for which the fiber $\mathbf{P}^{-1}(\mathbf{P}(x))$ contains multiple distinct analytic branches, so the local ring of $\overline{B}$ at x is not integrally closed. For example, if D corresponds to a boundary component where the monodromy is not trivial, analytic local neighborhoods of $\overline{B}$ near x may be modeled by $\mathrm{Spec} \mathbf{C}[u, v]/(uv)$ modulo a nontrivial group action.*
- (2) *If irreducible components $D_i, D_j \subset \Delta$ meet nontransversely, so that the intersection $D_i \cap D_j$ is singular or not snc, the locus $\overline{B} \cap (D_i \cap D_j)$ acquires corresponding singularities, so that the local ring is not regular nor normal at such points, and normalization introduces ramification over this locus.*
- (3) *For monodromy representations $\rho : \pi_1(\mathcal{M}) \rightarrow \mathrm{Aut}(H_{\mathrm{prim}}^2)$ with nontrivial global monodromy group Γ acting on the vanishing cohomology parametrized by B , nontrivial identifications occur in the image $\overline{\mathrm{Im}}(B) \subset \overline{\mathcal{M}}$ along boundary strata, so that the normalization of $\overline{B}$ is a finite ramified cover branched along these loci, and the structure sheaf is not normal at points over such identifications.*

It follows that normalization of $\overline{B}$ corresponds to resolving the branching and non-normal crossing behavior caused by failure of injectivity of the period map, singularities in the intersection of boundary divisors, and ramification induced by the global monodromy representation.

Let $P : F_{\text{En},2} \rightarrow F_{(2,2,0)}$ denote the period map between the moduli stack of lattice-polarized K3 (or Enriques) surfaces and its image in the period domain, extended to suitable toroidal or semi-toric compactifications $\overline{F_{\text{En},2}} \rightarrow \overline{F_{(2,2,0)}}$ as established in [AEGS3]. Both source and target are Deligne–Mumford stacks, locally of finite type over $\mathbf{C}$.

Proposition 6.5.0.3. *The non-normality of the scheme-theoretic image of $\overline{F_{\text{En},2}} \rightarrow \overline{F_{(2,2,0)}}$ (and in particular for the closures of Noether–Lefschetz loci) along the boundary $\Delta = \overline{F_{(2,2,0)}} \setminus F_{(2,2,0)}$ is a consequence of failures of separatedness and unramifiedness of the period map at points of Δ , due to three mechanisms: (1) failure of injectivity of the period map at the boundary, (2) non-transversality of the intersection of irreducible components of Δ , and (3) identifications arising from monodromy action.*

Proof. **1. Failure of injectivity over boundary divisors.**

Let $x \in D \subset \Delta$ be a point lying on a boundary divisor. Consider the local behavior of $\overline{F_{\text{En},2}}$ and the period map over a small analytic neighborhood U of x . The period map may send distinct limit points in $\overline{F_{\text{En},2}}$ (corresponding to non-isomorphic degenerations with the same mixed Hodge structure or period data) to the same point in $\overline{F_{(2,2,0)}}$, particularly when the monodromy representation around D is nontrivial. Formally, there exist $y_1, y_2 \in \overline{F_{\text{En},2}}$ with $P(y_1) = P(y_2) = x$ but which are not identified scheme-theoretically in $\overline{F_{\text{En},2}}$. The completed local ring $\hat{\mathcal{O}}_{\overline{F_{\text{En},2}}, y_1} \times \hat{\mathcal{O}}_{\overline{F_{\text{En},2}}, y_2}$ then maps finitely (and possibly not surjectively) into $\hat{\mathcal{O}}_{\overline{F_{(2,2,0)}}, x}$, so the scheme-theoretic image is not normal at x : it has multiple analytic branches glued via the period map, and integral closure introduces a normalization that separates these branches.

2. Non-transversality of boundary divisor intersections.

Suppose $x \in D_1 \cap D_2$, where $D_1, D_2 \subset \Delta$ are distinct irreducible components and their intersection is non-transverse. Locally, the structure of $\overline{F_{(2,2,0)}}$ near x is modeled as $\text{Spec } \mathbf{C}[[u, v]]/(uv)$ or, for higher codimension intersections, as the vanishing locus of a product of local coordinates. If $\overline{F_{\text{En},2}}$ maps into $\overline{F_{(2,2,0)}}$ so that the scheme-theoretic fiber above x is reducible or singular, then the local ring at x fails Serre’s condition (R_1) or (S_2) for normality, as integral closure may add missing functions or resolve multiple

components. The normalization then corresponds to separating these intersection branches, producing a cover ramified along $D_1 \cap D_2$.

3. Monodromy identifications and stack-theoretic quotients.

The global monodromy group $\Gamma \leq \mathrm{O}(L)$ acts on the boundary components and may have nontrivial stabilizer orbits in the boundary. This manifests locally as a finite group action (coming from automorphisms in the degenerating family or stacky structure in the moduli) on the germ U of $\overline{F}_{(2,2,0)}$: the scheme-theoretic image is modeled by the quotient U/G , where G is a subgroup of Γ . The resulting singularities are quotient singularities, and the local ring of invariants is not integrally closed unless the action is free. Thus, normalization corresponds to passing to the cover U before forming the quotient, and non-normality reflects the presence of ramification or fixed points for the group action.

Each of these three mechanisms can be realized concretely in families of degenerating lattice-polarized K3 or Enriques surfaces (see [AEGS3], for explicit models). In each case, non-normality of the scheme-theoretic image of the period map along the boundary is a direct consequence of the existence of multiple branches, nontransverse intersections, or stacky (ramified) structure resulting from monodromy. The normalization resolves the non-normal behavior, yielding a finite (possibly ramified) cover of the image.

□

Theorem 6.5.0.4 (Universal Family and Moduli Classification). *The pulled-back family*

$$(\mathcal{Z}^\nu, \epsilon \mathcal{R}_Z^\nu) := \nu^*(\mathcal{Z}, \epsilon \mathcal{R}_Z) \rightarrow B^\nu$$

satisfies:

- *Every fiber is a KSBA-stable Enriques pair of degree two,*
- *The family is universal among stable families over normal bases mapping to $\overline{F}_{\mathrm{En},2}$,*
- *There exists a canonical classifying morphism $\phi : B^\nu \rightarrow \overline{F}_{\mathrm{En},2}$ compatible with the moduli functor.*

Proof. Stability is preserved by finite base change. The normalization B^ν is initial among all normalizations, so universality follows for any family over a normal base mapping to B . Existence of the classifying morphism is a consequence of functoriality and the representability of KSBA moduli stacks. $\square$

Proposition 6.5.0.5 (Restriction of Semifans and Semitoroidal Compactification). *The embedding $B \hookrightarrow \overline{F_{(2,2,0)}}$ induces on B^ν a semitoroidal structure as follows:*

- *The period domain D_B for B^ν embeds as a closed linear subdomain of the period domain for $F_{(2,2,0)}$, determined by additional constraints imposed by symmetry under the involution,*
- *The rational polyhedral cones from the ramification semifan $\mathcal{F}_{\text{ram}}$ on $\overline{F_{(2,2,0)}}$ restrict to produce a semifan $\mathcal{F}_B$ on B^ν ,*
- *The normalization B^ν is isomorphic, as a modular compactification, to the semitoroidal compactification associated to $\mathcal{F}_B$.*

Proof. The restriction of the period domain reflects the imposition of involution-invariant lattice conditions cutting out Enriques double covers. The restriction of the polyhedral structure from $\overline{F_{(2,2,0)}}$ follows functorially from intersecting the hyperplane arrangements of $\mathcal{F}_{\text{ram}}$ with the subdomain D_B , as the arrangement is defined by orthogonality to lattice roots/divisors that may be fixed or permuted by the involution. The unique semitoroidal compactification is then determined by the induced semifan structure on the normalization. $\square$

Construction 6.5.0.6 (Folded Semifans and Complete Boundary Stratification). The classifying morphism $\phi : B^\nu \rightarrow \overline{F_{\text{En},2}}$ transports the combinatorial structure of $\mathcal{F}_B$ to the boundary stratification on $\overline{F_{\text{En},2}}$. According to the explicit construction in [Ale+23, §2.2], the semitoroidal compactification $\overline{F_{\text{En},2}}$ is isomorphic to $\overline{F_{\text{En},2}}^{\mathcal{F}}$, where $\mathcal{F} = \{\mathcal{F}_k\}_{k=1}^5$ is the collection of semifans assigned to the five 0-cusps (maximal boundary components) of $\overline{F_{\text{En},2}}$. Each semifan $\mathcal{F}_k$ associated to the five 0-cusps of the moduli space $\mathcal{F}_{\text{En},2}$ is defined by intersecting the ambient ramification semifan $\mathcal{F}_{\text{ram}}$ for the K3 covering with the period subdomain corresponding to the cusp, followed by folding under the involution. This folding identifies cones related by involution-invariant

lattice automorphisms. For each k , the combinatorial structure is given by the following:

- The semifan $\mathcal{F}_k$ is a *generalized Coxeter semifan* in the sense that it is a coarsening of the corresponding folded Coxeter semifan. Its maximal cones are indexed by faces not in the orbits of *irrelevant roots*.
- For $\mathcal{F}_2$ and $\mathcal{F}_4$, the subgroup generated by the irrelevant roots is finite, and so the semifan is a finite rational polyhedral fan; these cusps correspond to genuinely toroidal strata in the compactification.
- For $\mathcal{F}_1$, $\mathcal{F}_3$, and $\mathcal{F}_5$, the subgroup generated by the irrelevant roots is infinite, and the resulting semifan is infinite but locally finite. The boundary strata corresponding to these cusps are strictly semitoroidal.
- The precise structure of all boundary strata is completely determined by the maximal cones of these semifans, together with the folding data by the involution.

Proposition 6.5.0.7 (Properties of the Classifying Morphism). *The classifying morphism $\phi : B^\nu \rightarrow \overline{F_{\text{En},2}}$ has the following properties:*

- ϕ is birational and an isomorphism over the interior moduli stack $F_{\text{En},2}$,
- ϕ is proper as a morphism between proper Deligne–Mumford stacks,
- ϕ respects the combinatorial boundary stratification induced by the semitoroidal structures: the combinatorial types of boundary strata, as encoded by cones of the folded semifans, correspond under ϕ , so that degeneration types are preserved.

Proof. Birationality and identification on the open locus is a consequence of the moduli interpretation and the universal property of normalization. Properness follows from the properness of the moduli stacks and the modular representability. Compatibility with the combinatorial stratification is a consequence of the construction of the semifans and their folding, and is treated in full detail in [Ale+23]. $\square$

6.6 Finiteness by Kulikov Models and a Coarsening Argument

We now prove a crucial structural property of the classifying morphism

$$\phi : B^\nu \rightarrow \overline{F_{\text{En},2}}$$

constructed in previous sections: namely, that ϕ is finite. This assertion is the final step needed for modular identification of the compactified moduli of degree-2 polarized stable Enriques pairs via the period map and semitoroidal construction.

6.6.1 Semifan Comparison and Finiteness: Addressing the Critical Gap

Lemma 6.6.1.1 (Semifan Comparison). *Let $\overline{F_{\text{En},2}}^{\mathcal{F}^\bullet}$ be the semitoroidal compactification defined using the five semifans $\mathcal{F} = \{\mathcal{F}_k\}_{k=1}^5$ as in [Ale+23]. There exist semifans $\mathcal{G} = \{\mathcal{G}_k\}_{k=1}^5$ associated with the normalization of the KSBA compactification $\overline{F_{\text{En},2}}$ such that:*

1. $\overline{F_{\text{En},2}} = \overline{F_{\text{En},2}}^{\mathcal{F}^\bullet}$ if and only if $\mathcal{G}_k = \mathcal{F}_k$ for all k .
2. Each $\mathcal{G}_k$ is a coarsening of $\mathcal{F}_k$.
3. The morphism ϕ is finite if and only if all $\mathcal{G}_k = \mathcal{F}_k$.

Proof. Any KSBA compactification with a recognizable divisors admits a semitoroidal structure determined by a tuple of semifans $\mathcal{G}_k$. By construction, these semifans are universal for the normalization and can only coarsen the initially defined Coxeter semifans $\mathcal{F}_k$. Explicitly, a cone in $\mathcal{F}_k$ may be identified in $\mathcal{G}_k$, corresponding to an identification of the associated boundary stratum in the KSBA moduli space.

The crux is that given any coarsening, there exists some codimension-one cone σ in $\mathcal{F}_k$ (for some k), a common face of two maximal cones τ_1, τ_2 , that is identified in $\mathcal{G}_k$. Points corresponding to distinct degenerations – specifically, configurations differing by the presence of a double curve in the dual complex – would thus be glued together in the target. Thus, ϕ is finite if and only if no such coarsening occurs, that is, all semifans agree. This reduces the global finiteness to

the combinatorial modularity of the boundary fans, with no possible positive-dimensional fibers away from the boundary: by normality, properness, and functoriality of the compactification morphisms, any such fiber must arise from a non-separated boundary stratum; but this is precisely what semifan agreement guarantees cannot occur. $\square$

6.6.2 Maximality, Moduli, and Injectivity: Scheme-Theoretic Proof

Definition 6.6.2.1 (Maximality of Degenerations). A degeneration $(X_0, \epsilon R_0)$ of K3 pairs is **maximal** if its dual complex realizes the largest possible number of vertices (components) and edges (double curves) among all degenerations with the same monodromy data. The analogous definition applies for Enriques degenerations $(Z_0, \epsilon R_{Z,0})$.

Proposition 6.6.2.2 (The Double Curve Constraint). *Let $(X_0, \epsilon R_0)$ be a degeneration of K3 pairs with a fixed-point-free Enriques involution ι_{En} , with quotient $(Z_0, \epsilon R_{Z,0})$. Then:*

- *The number of irreducible components of Z_0 is the number of ι_{En} -orbits of components of X_0 ;*
- *The number of double curves in Z_0 is the number of ι_{En} -orbits of double curves in X_0 ;*
- *$(Z_0, \epsilon R_{Z,0})$ is maximal if and only if $(X_0, \epsilon R_0)$ is maximal.*

Proof. This is established by [Ale+23] via an explicit analysis of half-divisor models and their quotients. The scenarios at each cusp of the compactification – Type III cusps (with dual complex $\mathbf{RP}^2$ or $\mathbf{D}^2$) and Type II (dual complex $\mathbf{D}^1$) – are treated explicitly:

- At cusp 1 (Type III), all components of $\mathcal{X}_0$ map to unique components of V_i .
- At cusps 2, 3, 4, 5 (Type III), the involution may act with isolated fixed points on components or double curves.
- In Type II degenerations, $\iota_{\text{En},0}$ may act by reflection, or as a fixed-point-free involution, or as an elliptic involution with explicit fixed locus on certain double curves.

By [Ale+23], the boundary degenerations (up to isomorphism of stable pairs) are fully classified by the monodromy and dual complex, which is entirely encoded in the semifan. The folding operations and passage to quotients by ι_{En} preserves this relation. $\square$

6.6.3 The Core Finiteness-Injectivity Argument

Theorem 6.6.3.1 (No Coarsening Occurs). *For each $k \in \{1, 2, 3, 4, 5\}$, the boundary semifans satisfy $\mathcal{G}_k = \mathcal{F}_k$.*

Proof. Suppose, for contradiction, that for some k the semifan $\mathcal{G}_k$ is a proper coarsening of $\mathcal{F}_k$, and let σ be a codimension-one cone which is a face of two maximal cones τ_1, τ_2 in $\mathcal{F}_k$ that become identified in $\mathcal{G}_k$. These maximal cones parameterize distinct maximal degeneration types, specifically boundary degenerations where the dual complex differs by exactly one double curve – maximality vs. non-maximality for a fixed monodromy. Under the period map, points of B^ν mapping to σ correspond to K3 degenerations $(X_0, \epsilon R_0)$ missing a single double curve from the maximal configuration; the quotient $(Z_0, \epsilon R_{Z,0})$ encodes this as well. But in the compactification, KSBA theory asserts that each dual complex arises as a distinct boundary stratum, and maximality is a complete moduli invariant – distinct configurations cannot be identified. Therefore, identification of such cones in a coarsened semifan would force positive-dimensional or non-separated fibers, contradicting the representability and separatedness of the moduli functor. Thus, no coarsening occurs, and $\mathcal{G}_k = \mathcal{F}_k$ for all k . $\square$

6.6.4 Properness, Quasi-finiteness, and Conclusion

Corollary 6.6.4.1 (Finiteness of the Classifying Map). *By the previous lemma, the boundary stratifications, as encoded by semifans, agree identically. Therefore, the morphism $\phi : B^\nu \rightarrow \overline{F_{\text{En},2}}$ is finite.*

Proof. It remains to ensure that no positive-dimensional fibers exist away from the boundary and that the map is proper. Since both B^ν and $\overline{F_{\text{En},2}}$ are normal, proper algebraic spaces (by the properness of the moduli of stable pairs, and semifan agreement guarantees finite fibers at the boundary, the only possible source of positive-dimensional fibers would be in the interior. However, in the open moduli, the period

map is finite (by Torelli for K3s, and the specific construction of $D_{T_{\text{En}}}$). Hence the morphism is quasi-finite and proper, and by Zariski's Main Theorem, ϕ is finite. $\square$

6.7 Application of Zariski's Main Theorem

In this section, we apply Zariski's Main Theorem to the classifying morphism $\phi : B^\nu \rightarrow \overline{F_{\text{En},2}}$ constructed in previous sections, obtaining a precise modular isomorphism between the KSBA compactification and the explicit semitoroidal model described via folded ramification semifans.

Theorem 6.7.0.1. *Let $f : X \rightarrow Y$ be a morphism of varieties. Suppose:*

1. *f is birational,*
2. *f is finite,*
3. *X and Y are normal,*
4. *Y is proper.*

Then f is an isomorphism.

Lemma 6.7.0.2 (Verification of Hypotheses for the Classifying Map). *The morphism $\phi : B^\nu \rightarrow \overline{F_{\text{En},2}}$ constructed above satisfies all hypotheses of Zariski's Main Theorem:*

1. *Birationality: By Proposition 5.6, ϕ restricts to an isomorphism over the dense open subset $F_{\text{En},2}$.*
2. *Finiteness: By Corollary 7.6.6, ϕ is finite.*
3. *Normality: B^ν is normal by definition as normalization; $\overline{F_{\text{En},2}}$ is normal as the normalization of the proper algebraic stack $\overline{F_{\text{En},2}}$.*
4. *Properness: $\overline{F_{\text{En},2}}$ is proper by the general theory of KSBA compactification for surfaces with numerically trivial canonical class (see Kollár[Theorem 1.2]).*

We are thus led to the following:

Theorem 6.7.0.3 (Main Isomorphism: KSBA and Semitoroidal Compactifications). *The classifying morphism induces canonical isomorphisms:*

$$\phi : B^\nu \xrightarrow{\sim} \overline{F_{\text{En},2}}$$

and hence,

$$\overline{F_{\text{En},2}} \cong \overline{F_{\text{En},2}}^{\mathcal{F}^\bullet}.$$

where $\mathcal{F} = \{\mathcal{F}_k\}_{k=1}^5$ denotes the system of folded semifans constructed previously.

Proof. We know that ϕ is an isomorphism of normal proper spaces. Surjectivity onto $\overline{F_{\text{En},2}}$ follows from the construction of ϕ itself. Finally, B^ν is canonically identified, as a semitoroidal compactification via $\mathcal{F}$, with $\overline{F_{\text{En},2}}^{\mathcal{F}^\bullet}$. □

Remark 6.7.0.4 (Toroidal vs. Semitoroidal Structure). We finally remark that the resulting compactification exhibits hybrid toroidal/semitoroidal structures:

- It is toroidal over the 0-cusps 2 and 4 (where $\mathcal{F}_2$ and $\mathcal{F}_4$ are honest fans),
- It is toroidal over the 1-cusps adjacent to the 0-cusps 2 and 4,
- It is strictly semitoroidal over the 1-cusp labeled 35, - It is strictly semitoroidal (i.e., not toroidal but modeled on an infinite semifan) over the remaining cusps ($\mathcal{F}_1, \mathcal{F}_3, \mathcal{F}_5$).

This completes the identification of the KSBA compactification with the explicit semitoroidal model given by folding and restricting the ambient K3 ramification semifans. All boundary and degeneration structures in $\overline{F_{\text{En},2}}$ are thus characterized in terms of folded data for $F_{(2,2,0)}$.

CHAPTER 7

COMPUTATIONAL EXAMPLES

7.1 DLT Models

This section gives a precise structural and combinatorial description of divisor models (for degenerations of K3 pairs) and half-divisor models (for Enriques quotients), connecting KSBA limits explicitly to integral-affine data on dual complexes. Let $\pi : \mathcal{X} \rightarrow C$ be a degeneration of complex surfaces with $\mathcal{X}_0 \subset \mathcal{X}$. The **dual complex** $\Gamma(\mathcal{X}_0)$ encodes the topology of $\mathcal{X}_0$: each vertex of $\Gamma(\mathcal{X}_0)$ corresponds to an irreducible component V_i , each edge to a double curve $D_{ij} = V_i \cap V_j$. When $\mathcal{X}_0$ has normal crossings, $\Gamma(\mathcal{X}_0)$ is a finite graph. Given a Cartier divisor $\mathcal{R} \subset \mathcal{X}$ disjoint from the singular strata of $\mathcal{X}_0$ (i.e., $\mathcal{R}$ does not meet points where two or more components meet), the combinatorial geometry of $(\mathcal{X}_0, \mathcal{R}_0)$ is encoded by an **integral-affine divisor** $R_{\text{IA}} \subset \Gamma(\mathcal{X}_0)$. This means:

- Each edge of $\Gamma(\mathcal{X}_0)$ (corresponding to a double curve D_{ij}) is assigned an integer weight n_{ij} , expressing the degree of intersection of $\mathcal{R}_0$ with D_{ij} .
- To each vertex v_i (component V_i) one assigns a line bundle $L_i \in \text{Pic}(V_i)$ so that, for any edge v_{ij} , $\deg(L_i|_{D_{ij}}) = n_{ij}$. The weighting must be compatible on edges and satisfy global compatibility conditions.

These weights obey a **balancing condition** at every vertex v_i . For a toric vertex, resp. a non-toric vertex arising from an internal blowup in the direction $\vec{e}$, we have

$$\sum_j n_{ij} \vec{e}_{ij} = 0, \quad \sum_j n_{ij} \vec{e}_{ij} \in \mathbf{Z} \vec{e}$$

where $\vec{e}_{ij}$ is the primitive integral direction associated to the corresponding edge. These constraints ensure that the line bundles patch together along double curves and that the data collectively define a global Cartier divisor structure in the smoothing.

Definition 7.1.0.1 (Divisor Model). A **divisor model** for a degeneration $\pi : \mathcal{X} \rightarrow C$ of K3 (or Enriques) surfaces is a degeneration of pairs $(\mathcal{X}, \mathcal{R}) \rightarrow C$ such that:

- $\mathcal{R}$ is a Cartier divisor, with $\mathcal{R}_t = \mathcal{R} \cap \mathcal{X}_t$ effective for all $t \in C$,
- For $t \neq 0$, $\mathcal{R}_t$ is semiample,
- $\mathcal{R}$ does not meet the singular strata of $\mathcal{X}_0$ (i.e., is disjoint from double/triple intersections).

Given a divisor model, the isomorphism class of $\mathcal{O}_{\mathcal{X}_0}(\mathcal{R}_0)$ is encoded by its corresponding integral-affine divisor R_{IA} on $\Gamma(\mathcal{X}_0)$. The dual complex, together with R_{IA} , captures all line bundle glueing data and allows for explicit calculation of limit objects.

Proposition 7.1.0.2 (Classification via Integral-Affine Data). *Given a fixed Picard–Lefschetz monodromy invariant λ , the combinatorial type $(\Gamma(\mathcal{X}_0), R_{\text{IA}})$ —that is, the dual complex with its weighted, balanced subgraph—uniquely determines the KSBA stable limit $(\overline{\mathcal{X}}_0, \epsilon \overline{\mathcal{R}}_0)$. Furthermore, this combinatorial type is locally constant in families with fixed Picard–Lefschetz form.*

Proposition 7.1.0.3 (Semitoroidal Compactification via Recognizable Divisors). *If R is a recognizable divisor (such as the fixed locus of a nonsymplectic involution), then there exists a unique semifan $\mathcal{F}_R$ whose semitoroidal compactification normalizes the KSBA compactification of the relevant moduli space $F_{\text{En},2}[6, \text{Sec. } 5C]$.*

Theorem 7.1.0.4 (Explicit Construction and Type Determination). *Given a polarized integral-affine structure $(B(\ell), R_{\text{IA}})$ (with $\ell = (\lambda \cdot \alpha_i)_{i \in G}$), and an appropriate triangulation, one obtains*

$$(B(\ell), R_{\text{IA}}) = (\Gamma(\mathcal{X}_0), \Gamma(\mathcal{R}_0))$$

as the dual complex of the $\mathcal{X}_0$ of a divisor model with monodromy invariant λ .

Definition 7.1.0.5 (Half-Divisor Model). Suppose $(\mathcal{X}, \mathcal{R}) \rightarrow (C, 0)$ is a divisor model for a family of K3 surfaces admitting an involution ι_{En} that preserves $\mathcal{R}$. The **half-divisor model** is the quotient

$$(\mathcal{Z}, \mathcal{R}_{\mathcal{Z}}) := (\mathcal{X}, \mathcal{R}) / \iota_{\text{En}}.$$

These models realize degenerations of Enriques pairs as quotients of K3 divisor models, and in generic settings, the quotient inherits slc singularities, and the divisor structure matches the normalization of the image of $\mathcal{R}$ [2, Prop. 4.5].

Proposition 7.1.0.6 (Geometric Types and Boundary Strata). *Let $(\mathcal{Z}, \mathcal{R}_{\mathcal{Z}}) \rightarrow (C, 0)$ be a half-divisor model for $F_{\text{En},2}$ as constructed above. Then the following properties hold:*

- *The fibers of $\mathcal{Z}$ have semi-log canonical (slc) singularities.*
- *The divisor $K_{\mathcal{Z}} + \epsilon \mathcal{R}_{\mathcal{Z}}$ is relatively big and nef over C .*
- *The divisor $\mathcal{R}_{\mathcal{Z}}$ contains no log canonical centers.*

More precisely:

- *In the case of Type III degenerations at cusp 1, the dual complex $\Gamma(\mathcal{Z}_0)$ is homeomorphic to $\mathbf{RP}^2$; each irreducible component V_i of the $\mathcal{X}_0$ $\mathcal{Z}_0$ is, after normalization, isomorphic to one of the preimage components of $\mathcal{X}_0$.*
- *For Type III degenerations at cusps 2–5, the dual complex $\Gamma(\mathcal{Z}_0)$ is homeomorphic to $\mathbf{D}^2$. If a component V_i lifts to two distinct components of $\mathcal{X}_0$, the normalized copies are isomorphic; if it arises from a single irreducible component, the involution $\iota_{\text{En},0}$ acts on V_i with precisely four fixed points.*
- *In the case of Type II degenerations, $\Gamma(\mathcal{Z}_0)$ is a segment, and the quotient is described by the action of $\iota_{\text{En},0}$ as detailed in [2, Proposition 4.5].*

Proof. See detailed analyses in [Ale+23, Prop. 4.5], which classify Enriques degenerations by their dual complex and describe the induced slc structure and divisor support in every case.

□

Corollary 7.1.0.7 (Computability of the KSBA Stable Limit). *Given a degeneration $(\mathcal{Z}^*, \epsilon\mathcal{R}_{\mathcal{Z}}^*) \rightarrow C^*$, the KSBA-stable limit is*

$$\mathrm{Proj}_C \bigoplus_{n \geq 0} H^0(\mathcal{Z}, n\mathcal{R}_{\mathcal{Z}}),$$

where the right-hand side is computed from the data of the half-divisor model $(\mathcal{Z}, \mathcal{R}_{\mathcal{Z}}) \rightarrow (C, 0)$.

Remark 7.1.0.8. The quotient $\Gamma(\mathcal{Z}_0) = \Gamma(\mathcal{X}_0)/\iota_{\mathrm{En}, \mathrm{IA}}$ always inherits a natural integral-affine structure, encoding both the combinatorics and divisor data of the degeneration [2, Prop. 4.5]. In Type III, certain boundary components are the images of the “Enriques equator” and are characterized by four A_1 singularities.

Remark 7.1.0.9. For general monodromy invariant λ , half-divisor models exist only generically: the involution ι_{En} may be only birational on $\mathcal{X}_0$ s. After contracting exceptional loci to resolve indeterminacies, one obtains only a dlt pair. This supports the broader philosophy (see ,) that dlt models, rather than strictly semistable ones, are the correct analogues of Kulikov models for K -trivial surface degenerations.

7.2 The Five Cusps

This section provides an explicit, case-by-case analysis of the five 0-cusps of $F_{\mathrm{En}, 2}$, describing the structure of the corresponding semifans, dual complexes, integral-affine data, and involution symmetries arising in the boundary of the KSBA compactification, as described in [Ale+23]. The semifans $\mathcal{F}_k$ are defined by intersecting the ambient ramification semifan $\mathcal{F}_{\mathrm{ram}}(T_{\mathrm{dP}})$ with the Enriques lattice subspace T_{En} , for each lattice polarization T_{dP} occurring at a cusp. By ??, the resulting collection $\mathcal{F}_{\bullet} = \{\mathcal{F}_k\}_{k=1}^5$ consists of generalized Coxeter semifans governing the semitoroidal structure at each cusp. Each semifan encodes the stratified boundary behavior of the compactification via its rays (corresponding to degenerations of Types II and III) and the folding symmetries inherited from involutions on the integral-affine structures. Let $(\mathcal{Z}, \mathcal{R}_{\mathcal{Z}}) \rightarrow (C, 0)$ be a half-divisor model for $F_{\mathrm{En}, 2}$ as in Proposition 4.5. The following facts hold for each such degeneration:

- The fibers of $\mathcal{Z}$ have semi-log-canonical (slc) singularities.
- $K_{\mathcal{Z}} + \epsilon\mathcal{R}_{\mathcal{Z}}$ is relatively big and nef over C .

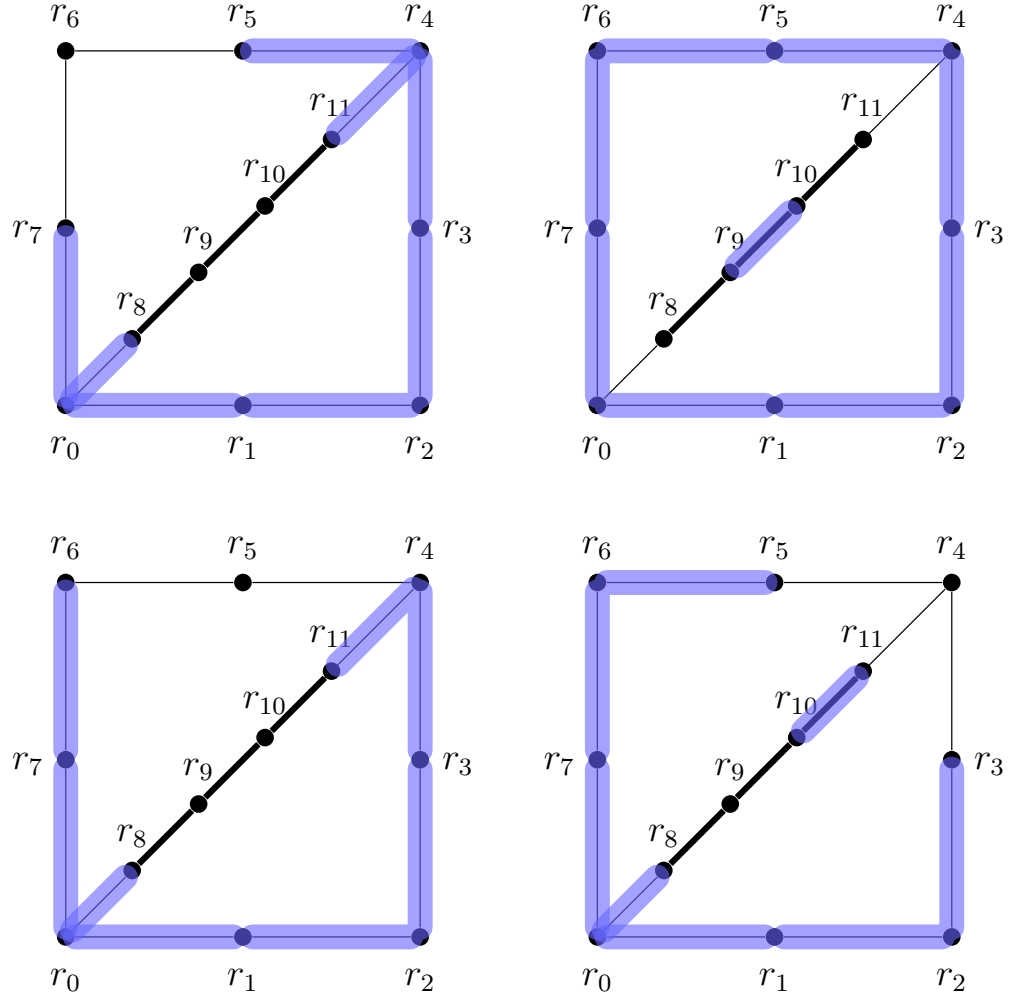
- The boundary $\mathcal{R}_{\mathcal{Z}}$ contains no log canonical centers.

We recall some notation:

- ι_{En} : the fixed-point-free Enriques involution of the total space $\mathcal{X}$ in a K3 surface degeneration;
- $\iota_{\text{En},0}$: the restriction to the $\mathcal{X}_0$; when $\iota_{\text{En},0}$ acts freely, $\mathcal{Z}_0 = \mathcal{X}_0/\iota_{\text{En},0}$ is an Enriques surface;
- $\iota_{\text{En,IA}}$: the involution acting on the integral-affine sphere $\text{IAS}^2 B(\ell)$ or the dual complex $\Gamma(\mathcal{X}_0)$ by folding.

A dual complex $\Gamma(\mathcal{X}_0)$ or sphere $B(\ell)$ is said to be of *Enriques type* if and only if it admits such an involution $\iota_{\text{En,IA}}$. Let $(\mathcal{Z}, \mathcal{R}_{\mathcal{Z}})$ be a half-divisor model. If a component $V_i \subset \mathcal{Z}_0$ is covered by two irreducible components of $\mathcal{X}_0$, then (up to normalization) V_i is isomorphic to both components. If V_i is instead covered by a single irreducible component $\tilde{V}_i \subset \mathcal{X}_0$, then $\iota_{\text{En},0}$ acts on $\tilde{V}_i$ with exactly four fixed points, occurring as two pairs on double curves $\tilde{D}_{ij}, \tilde{D}_{ik}$. We conclude this chapter with an analysis of the actual combinatorial data at each of the five zero cusps. In particular, we exhibit orbit representatives of the the maximal elliptic subdiagrams of each Coxeter diagram under diagram symmetries, which are in bijection with Type III cones:

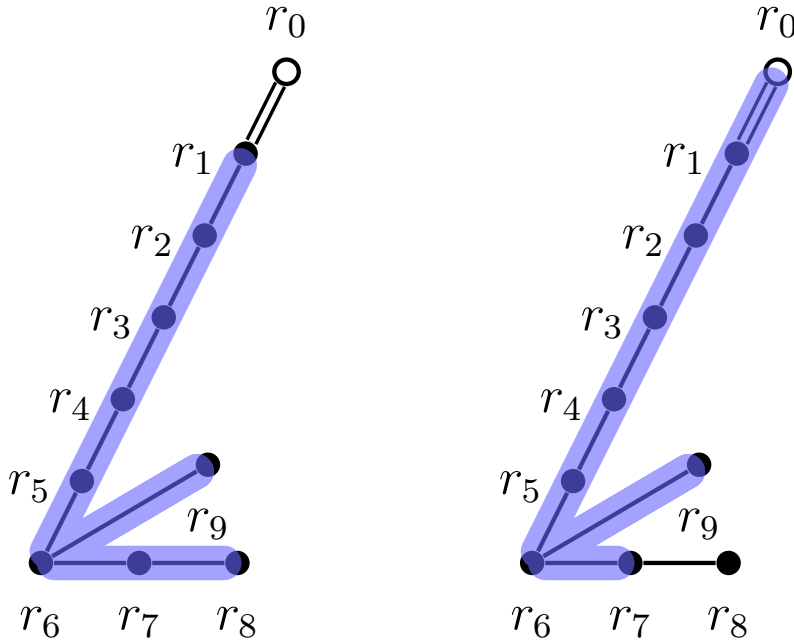
7.2.1 Cusp 1



- **Boundary Type:** Maps to cusp $(10, 10, 0)_1$ of F_{En} , and $\Gamma(\mathcal{Z}_0) = \mathbb{RP}^2$.
- **Lattice Data:** $\overline{T_{\text{dP}}} = (18, 2, 0)_1 = U(2) \oplus E_8^2$, with symmetry J given by 180° rotation; invariant sublattice $\overline{T_{\text{dP}}}^{J=1} = U(2) \oplus E_8(2)$.
- **Semifan Structure:** The irrelevant subgroup is infinite, noting that it contains the Weyl group of $\tilde{A}_1$ due to the presence of r_9, r_{10} which are images of irrelevant roots in the corresponding K3 diagram, so $\mathcal{F}_1$ is a strict semifan and the compactification at this 0-cusp is strictly semitoroidal.

- **Rays in the Coxeter compactification:** 4 Type II rays corresponding to the 4 elliptic subdiagrams above; 4 type III rays corresponding to maximal parabolic subdiagrams; a total of 8 rays.
- **Rays in the Semitoroidal compactification:** 2 Type II rays, 0 Type III rays; a total of 2 rays. These correspond to subdiagrams for the Coxeter compactification above, where any subdiagram which has a connected component of irrelevant roots is removed.
- **IAS Symmetry:** $\lambda \in \mathfrak{C}^J$ if and only if $\ell_i = \ell_{8+i}$ ($i = 0, \dots, 7$), $\ell_{16} = \ell_{18}$, $\ell_{17} = \ell_{19}$; involution acts as 180° rotation on each hemisphere, swapping P and P^{op} .

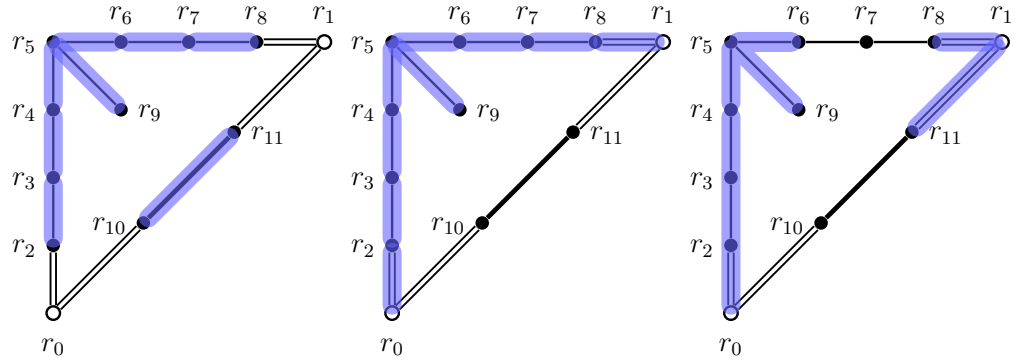
7.2.2 Cusp 2



- **Boundary Type:** Maps to cusp $(10, 8, 0)_1$ of F_{En} ; $\Gamma(\mathcal{Z}_0) = \mathbf{D}^2$.
- **Lattice Data:** $\overline{T_{\text{dP}}} = (18, 0, 0)_1 = U \oplus E_8^2$, J is a vertical reflection; the invariant sublattice is $U \oplus E_8(2)$.
- **Semifan Structure:** The irrelevant subgroup is S_2 , thus Σ_2 is a fan, giving a toroidal compactification over this cusp.

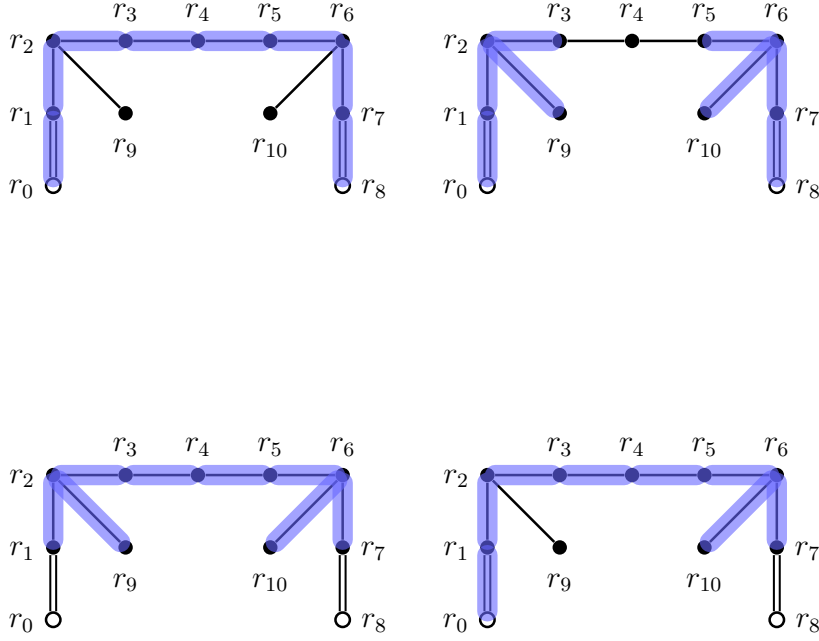
- **Rays in the Coxeter compactification:** 2 Type II, 8 Type III; 10 total.
- **Rays in the Semitoroidal compactification:** 2 type II, 7 type III; 9 total.
- **IAS Symmetry:** λ in $\mathfrak{C}^J$ iff $\ell_i = \ell_{20-i}$ ($i = 1, \dots, 9$); involution flips both hemispheres about the vertical axis.

7.2.3 Cusp 3



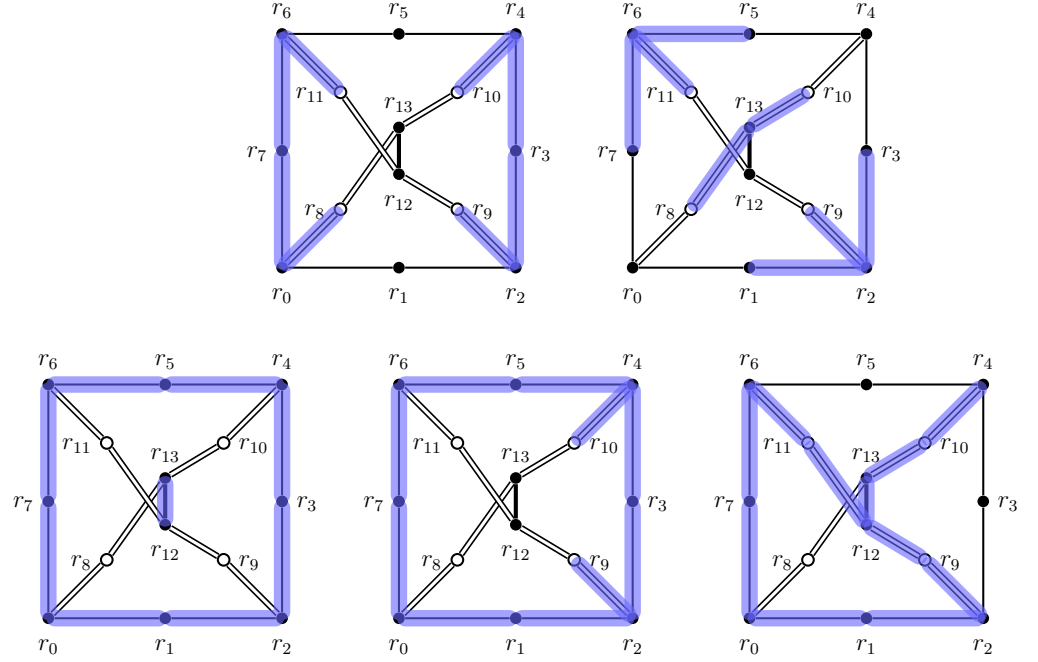
- **Boundary Type:** Maps to cusp $(10, 8, 0)_1$; $\Gamma(\mathcal{Z}_0) = \mathbf{D}^2$.
- **Lattice Data:** $\overline{T}_{\text{dP}} = (18, 2, 0)_1 = U(2) \oplus E_8^2$, J is diagonal reflection followed by α_{20} reflection; $\overline{T}_{\text{dP}}^{J=1} = U(2) \oplus E_8(2)$.
- **Semifan Structure:** The Irrelevant subgroup is infinite since it again contains $W(\tilde{A}_1)$ due to the presence of r_{10}, r_{11} , so $\mathcal{F}_3$ is strictly semitoroidal.
- **Rays in the Coxeter compactification:** 3 Type II, 15 Type III; 18 total.
- **Rays in the Semitoroidal compactification:** 2 Type II, 7 Type III; 9 total.
- **IAS Symmetry:** λ in $\mathfrak{C}^J$ iff $\ell_i = \ell_{16-i}$ ($i = 1, \dots, 7$), $\ell_{17} = \ell_{19}$, $\ell_{20} = 0$ (folding symmetry requires $\ell_{20} = 0$); the involution acts as diagonal flip.

7.2.4 Cusp 4



- **Boundary Type:** Maps to cusp $(10, 8, 0)_1$; $\Gamma(\mathcal{Z}_0) = \mathbf{D}^2$.
- **Lattice Data:** $\overline{T_{\text{dP}}} = (18, 2, 0)_1 = U(2) \oplus E_8^2$, J is horizontal reflection; $\overline{T_{\text{dP}}}^{J=1} = U(2) \oplus E_8(2)$.
- **Semifan Structure:** The irrelevant subgroup is $W(A_1) \times W(A_1) \cong S_2^2$ which is finite, and thus Σ_4 is a fan and the compactification is toroidal.
- **Rays in the Coxeter compactification:** 4 Type II, 12 Type III; 16 total.
- **Rays in the Semitoroidal compactification:** 4 Type II, 7 Type III; 11 total.
- **IAS Symmetry:** λ in $\mathfrak{C}^J$ iff each hemisphere of $B(\ell)$ is symmetric under a horizontal flip; the involution reflects across this axis.

7.2.5 Cusp 5



- **Boundary Type:** Maps to cusp $(10, 8, 0)_1$; $\Gamma(\mathcal{Z}_0) = \mathbf{D}^2$.
- **Lattice Data:** $\overline{T}_{\text{dP}} = (18, 2, 0)_1 = U(2) \oplus E_8^2$, J is composition of eight commuting reflections $\alpha_1, \dots, \alpha_{15}$; $\overline{T}_{\text{dP}}^{J=1} = U(2) \oplus E_8(2)$.
- **Semifan Structure:** The irrelevant subgroup contains $W(\tilde{A}_1)$ due to r_{12}, r_{13} , thus $\mathcal{F}_5$ is a semifan and this yields a semitoroidal compactification.
- **Rays in the Coxeter compactification:** 5 Type II, 17 Type III; 22 total.
- **Rays in the Semitoroidal compactification:** 3 Type II, 0 Type III; 3 total.
- **IAS Symmetry:** $\lambda \in \mathfrak{C}^J$ iff $\ell_{2i+1} = 0$ for $i = 0, \dots, 7$; the involution flips hemispheres of $B(\lambda)$.

We conclude with the following summary:

η	$\Gamma(\mathcal{Z}_0)$	W_{irr}	$\mathcal{F}$	Type II	Type III	Total
1	$\mathbf{RP}^2$	Infinite	Semi.	2	0	2

η	$\Gamma(\mathcal{Z}_0)$	W_{irr}	$\mathcal{F}$	Type II	Type III	Total
2	$\mathbf{D}^2$	S_2	Tor.	2	7	9
3	$\mathbf{D}^2$	Inf.	Semi.	2	7	9
4	$\mathbf{D}^2$	S_2^2	Tor.	4	7	11
5	$\mathbf{D}^2$	Inf.	Semi.	3	0	3

APPENDIX A

ROOT SYSTEMS

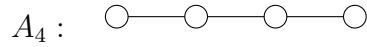
A.0.1 Examples of Specific Root Systems

A_4

The relevant Euclidean space is $\mathbb{E}_{A_4} = \{x \in \mathbf{R}^5 : x_1 + x_2 + x_3 + x_4 + x_5 = 0\}$ with

$$\Phi(A_4): \begin{cases} \alpha_1 = e_1 - e_2 \\ \alpha_2 = e_2 - e_3 \\ \alpha_3 = e_3 - e_4 \\ \alpha_4 = e_4 - e_5 \end{cases} \quad G_{A_4(-1)} = \begin{pmatrix} -2 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -2 \end{pmatrix}$$

By convention, we take A_n to mean $A_n(-1)$. This has Coxeter diagram



B_4

Roots live in $\mathbb{E}_{B_4} = \mathbf{R}^4$, and

$$\Phi(B_4): \begin{cases} \alpha_1 = e_1 - e_2 \\ \alpha_2 = e_2 - e_3 \\ \alpha_3 = e_3 - e_4 \\ \alpha_4 = e_4 \end{cases} \quad G_{B_4(-1)} = \begin{pmatrix} -2 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

Because we typically work with roots with $v^2 = -2$ or -4 , we often replace B_n with $B_n(2)$, where for example

$$G_{B_4(-2)} = \begin{pmatrix} -4 & 2 & 0 & 0 \\ 2 & -4 & 2 & 0 \\ 0 & 2 & -4 & 2 \\ 0 & 0 & 2 & -2 \end{pmatrix}$$

By convention, we thus take B_n to mean $B_n(-2)$, and identify the following Coxeter diagram:

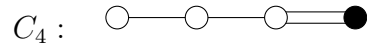


C_4

Set $\mathbb{E}_{C_4} = \mathbf{R}^4$, then

$$\Phi(C_4): \begin{cases} \alpha_1 = e_1 - e_2 \\ \alpha_2 = e_2 - e_3 \\ \alpha_3 = e_3 - e_4 \\ \alpha_4 = 2e_4 \end{cases} \quad G_{C_4(-1)} = \begin{pmatrix} -2 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 2 & -4 \end{pmatrix}$$

We again take C_n to mean $C_n(-1)$ by convention, with Coxeter diagram



D_4

Roots live in $\mathbb{E}_{D_4} = \mathbf{R}^4$ and

$$\Phi(D_4): \begin{cases} \alpha_1 = e_1 - e_2 \\ \alpha_2 = e_2 - e_3 \\ \alpha_3 = e_3 - e_4 \\ \alpha_4 = e_3 + e_4 \end{cases} \quad G_{D_4(-1)} = \begin{pmatrix} -2 & 1 & 0 & 0 \\ 1 & -2 & 1 & 1 \\ 0 & 1 & -2 & 0 \\ 0 & 1 & 0 & -2 \end{pmatrix}$$

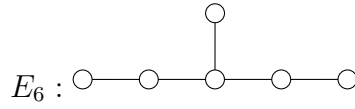
We take D_n to mean $D_n(-1)$, and identify th

E_6

Roots live in $\mathbb{E}_{E_6} = \{x \in \mathbf{R}^8 : x_1 + \cdots + x_8 = 0\}$ with

$$\Phi(E_6): \begin{cases} \alpha_1 = e_1 - e_2 \\ \alpha_2 = e_2 - e_3 \\ \alpha_3 = e_3 - e_4 \\ \alpha_4 = e_4 - e_5 \\ \alpha_5 = e_5 - e_6 \\ \alpha_6 = \frac{1}{2}(e_1 + e_2 + e_3 + e_4 - e_5 - e_6 - e_7 - e_8) \end{cases}, G_{E_6(-1)} = \begin{pmatrix} -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & -1 \\ 0 & 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & -1 & 0 & -2 \end{pmatrix}$$

We take E_n to mean $E_n(-1)$, and identify the Coxeter diagram as:



F_4

We take the simple roots

$$\begin{cases} \alpha_1 = e_2 - e_3 \\ \alpha_2 = e_3 - e_4 \\ \alpha_3 = e_4 \\ \alpha_4 = \frac{1}{2}(e_1 - e_2 - e_3 - e_4) \end{cases}$$

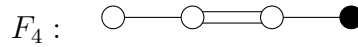
with the standard Gram matrix

$$G_{F_4} = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 1 & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & 1 \end{pmatrix}$$

To work integrally, we rescale by -2 :

$$G_{F_4(-2)} = \begin{pmatrix} -4 & 2 & 0 & 0 \\ 2 & -4 & 2 & 0 \\ 0 & 2 & -2 & 1 \\ 0 & 0 & 1 & -2 \end{pmatrix}$$

and thus take F_4 to mean $F_4(-2)$, with Coxeter diagram



Roots live in $\mathbb{E}_{G_2} = \{x \in \mathbf{R}^3 \mid x_1 + x_2 + x_3 = 0\}$ with simple roots and Gram matrix

$$\Phi(G_2): \begin{cases} \alpha_1 = e_1 - e_2 \\ \alpha_2 = -e_1 + 2e_2 - e_3 \end{cases} \quad G_{G_2(-1)} = \begin{pmatrix} -2 & 3 \\ 3 & -6 \end{pmatrix}.$$

 $G_2 :$  -6

To fix conventions, we record here Bourbaki's conventions for these Dynkin diagrams and the corresponding simple roots. The Euclidean embeddings can be used to compute the following invariants in all types, which we show below. Note that for D_n , $A_L = \mathbf{Z}_{2^2}$ when n is odd, and $A_L = (\mathbf{Z}_2)^2$ when n is even.

	A_n	D_n	E_6	E_7	E_8
$\text{rank}(L) = \Phi(L) $	n	n	$2 \cdot 3$	7	2^3
$ R(L) $	$n(n+1)$	$2n(n-1)$	$2^3 \cdot 3^2$	$2 \cdot 3^2 \cdot 7$	$2^4 \cdot 3 \cdot 5$
$W(L)$	S_{n+1}	$(\mathbf{Z}_2)^{n-1} \rtimes S_n$	$W(E_6)$	$W(E_7)$	$W(E_8)$
$ W(L) $	$(n+1)!$	$2^{n-1} \cdot n!$	$2^7 \cdot 3^4 \cdot 5$	$2^{10} \cdot 3^4 \cdot 5 \cdot 7$	$2^{14} \cdot 3^5 \cdot 5^2 \cdot 7$
A_L	$\mathbf{Z}_{n+1}$	$\mathbf{Z}_{2^2}$ or $\mathbf{Z}_2^2$	$\mathbf{Z}_3$	$\mathbf{Z}_2$	$\{0\}$
$\text{disc}(L)$	$n+1$	2^2	3	2	1

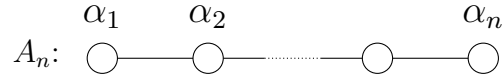
205

	B_n	C_n	F_4	G_2
$\text{rank}(L) = \Phi(L) $	n	n	2^2	2
$ R(L) $	$2n^2$	$2n^2$	$2^4 \cdot 3$	$2^2 \cdot 3$
$W(L)$	$\mathbf{Z}_2^n \rtimes S_n$	$\mathbf{Z}_2^n \rtimes S_n$	$W(F_4)$	$W(G_2)$
$ W(L) $	$2^n \cdot n!$	$2^n n!$	$2^7 \cdot 3^2$	$2^2 \cdot 3$
A_L	$\mathbf{Z}_2$	$\mathbf{Z}_2$	$\{0\}$	$\mathbf{Z}_3$
$\text{disc}(L)$	2	2	1	3

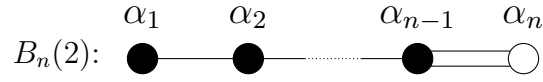
We now record explicit representatives for the simple roots of each lattice, their Gram matrix, and the associated Coxeter diagrams:

A.0.3 Types A,B,C, D

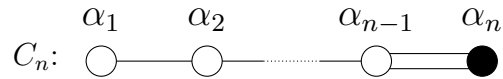
$$A_n : \begin{cases} \alpha_1 &= e_1 - e_2 \\ \alpha_2 &= e_2 - e_3 \\ \vdots &\vdots \\ \alpha_n &= e_n - e_{n+1}, \end{cases} \quad G_{A_n} = \begin{pmatrix} 2 & -1 & \cdot & \cdots & \cdot \\ -1 & 2 & \ddots & \cdots & \cdot \\ \cdot & \ddots & \ddots & \ddots & \vdots \\ \vdots & \cdots & \ddots & 2 & -1 \\ \cdot & \cdots & \cdot & -1 & 2 \end{pmatrix}$$



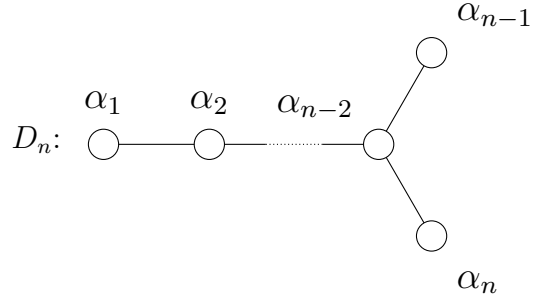
$$B_n : \begin{cases} \alpha_1 &= e_1 - e_2 \\ \alpha_2 &= e_2 - e_3 \\ \vdots &\vdots \\ \alpha_{n-1} &= e_{n-1} - e_n \\ \alpha_n &= e_n \end{cases}, \quad G_{B_n} = \begin{pmatrix} 2 & -1 & \cdot & \cdot & \cdots & \cdot & \cdot \\ -1 & 2 & \ddots & \cdot & \cdots & \cdot & \cdot \\ \cdot & \ddots & \ddots & \ddots & \cdots & \cdot & \cdot \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ \cdot & \cdot & \cdots & \ddots & 2 & -1 & \cdot \\ \cdot & \cdot & \cdots & \cdot & -1 & 1 & \cdot \end{pmatrix}$$



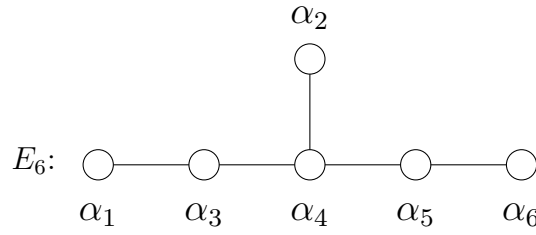
$$C_n : \begin{cases} \alpha_1 &= e_1 - e_2 \\ \alpha_2 &= e_2 - e_3 \\ \vdots &\vdots \\ \alpha_{n-1} &= e_{n-1} - e_n \\ \alpha_n &= 2e_n, \end{cases} \quad G_{C_n} = \begin{pmatrix} 2 & -1 & \cdot & \cdot & \cdots & \cdot \\ -1 & 2 & \ddots & \cdot & \cdots & \cdot \\ \cdot & \ddots & \ddots & \ddots & \cdots & \cdot \\ \vdots & \vdots & \ddots & 2 & -1 & \vdots \\ \cdot & \cdot & \cdots & -1 & 2 & -2 \\ \cdot & \cdot & \cdots & \cdot & -2 & 4 \end{pmatrix}$$



$$D_n : \begin{cases} \alpha_1 &= e_1 - e_2 \\ \alpha_2 &= e_2 - e_3 \\ \vdots &\vdots \\ \alpha_{n-2} &= e_{n-2} - e_{n-1} \\ \alpha_{n-1} &= e_{n-1} - e_n \\ \alpha_n &= e_{n-1} + e_n, \end{cases} \quad G_{D_n} = \begin{pmatrix} 2 & -1 & \cdot & \cdot & \cdots & \cdot & \cdot & \cdot \\ -1 & 2 & \ddots & \cdot & \cdots & \cdot & \cdot & \cdot \\ \cdot & \ddots & \ddots & \ddots & \cdots & \cdot & \cdot & \cdot \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ \cdot & \cdot & \cdots & \ddots & 2 & -1 & -1 \\ \cdot & \cdot & \cdots & \cdot & -1 & 2 & \cdot \\ \cdot & \cdot & \cdots & \cdot & -1 & \cdot & 2 \end{pmatrix}$$

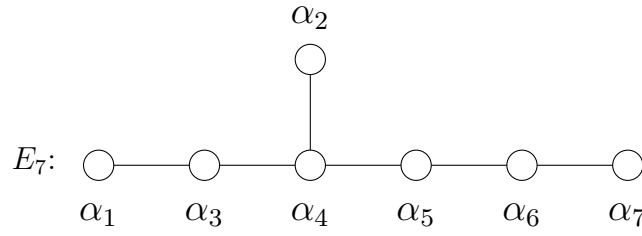


$$E_6 : \begin{cases} \alpha_1 &= \frac{1}{2}(e_1 - e_2 - e_3 - e_4 - e_5 - e_6 - e_7 + e_8) \\ \alpha_2 &= e_1 + e_2 \\ \alpha_3 &= -e_1 + e_2 \\ \alpha_4 &= -e_2 + e_3 \\ \alpha_5 &= -e_3 + e_4 \\ \alpha_6 &= -e_4 + e_5, \end{cases} \quad G_{E_6} = \begin{pmatrix} 2 & \cdot & -1 & \cdot & \cdot & \cdot \\ \cdot & 2 & \cdot & -1 & \cdot & \cdot \\ -1 & \cdot & 2 & -1 & \cdot & \cdot \\ \cdot & -1 & -1 & 2 & \ddots & \cdot \\ \cdot & \cdot & \cdot & \ddots & \ddots & -1 \\ \cdot & \cdot & \cdot & \cdot & -1 & 2 \end{pmatrix}$$

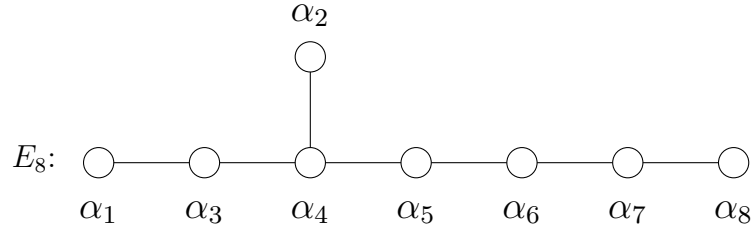


A.0.4 Type E

$$E_7 : \begin{cases} \alpha_1 &= \frac{1}{2}(e_1 - e_2 - e_3 - e_4 - e_5 - e_6 - e_7 + e_8) \\ \alpha_2 &= e_1 + e_2 \\ \alpha_3 &= -e_1 + e_2 \\ \alpha_4 &= -e_2 + e_3 \\ \alpha_5 &= -e_3 + e_4 \\ \alpha_6 &= -e_4 + e_5 \\ \alpha_7 &= -e_5 + e_6 \end{cases}, \quad G_{E_7} = \begin{pmatrix} 2 & \cdot & -1 & \cdot & \cdot & \cdots & \cdot \\ \cdot & 2 & \cdot & -1 & \cdot & \cdots & \cdot \\ -1 & \cdot & 2 & -1 & \cdot & \cdots & \cdot \\ \cdot & -1 & -1 & 2 & -1 & \cdots & \cdot \\ \cdot & \cdot & \cdot & -1 & 2 & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & -1 \\ \cdot & \cdot & \cdot & \cdot & \cdots & -1 & 2 \end{pmatrix}$$

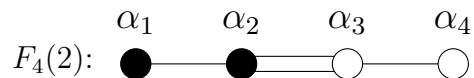


$$E_8 : \begin{cases} \alpha_1 &= \frac{1}{2}(e_1 - e_2 - e_3 - e_4 - e_5 - e_6 - e_7 + e_8) \\ \alpha_2 &= e_1 + e_2 \\ \alpha_3 &= -e_1 + e_2 \\ \alpha_4 &= -e_2 + e_3 \\ \alpha_5 &= -e_3 + e_4 \\ \alpha_6 &= -e_4 + e_5 \\ \alpha_7 &= -e_5 + e_6 \\ \alpha_8 &= -e_6 + e_7 \end{cases}, \quad G_{E_8} = \begin{pmatrix} 2 & \cdot & -1 & \cdot & \cdot & \cdots & \cdot \\ \cdot & 2 & \cdot & -1 & \cdot & \cdots & \cdot \\ -1 & \cdot & 2 & -1 & \cdot & \cdots & \cdot \\ \cdot & -1 & -1 & 2 & -1 & \cdots & \cdot \\ \cdot & \cdot & \cdot & -1 & 2 & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & -1 \\ \cdot & \cdot & \cdot & \cdot & \cdots & -1 & 2 \end{pmatrix}$$

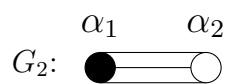


A.0.5 Types F_4 and G_2

$$F_4 : \begin{cases} \alpha_1 = e_2 - e_3 \\ \alpha_2 = e_3 - e_4 \\ \alpha_3 = e_4 \\ \alpha_4 = \frac{1}{2}(e_1 - e_2 - e_3 - e_4), \end{cases} \quad G_{F_4} = \begin{pmatrix} 2 & -1 & \cdot & \cdot \\ -1 & 2 & -1 & \cdot \\ \cdot & -1 & 1 & -\frac{1}{2} \\ \cdot & \cdot & -\frac{1}{2} & 1 \end{pmatrix}$$



$$G_2 : \begin{cases} \alpha_1 = e_2 - e_3 \\ \alpha_2 = e_1 - 2e_2 + e_3, \end{cases} \quad G_{G_2} = \begin{pmatrix} 2 & -3 \\ -3 & 6 \end{pmatrix}$$



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